# Maintenance Manual

## Models 8210 and 8250 Pallet Trucks


| Model | Serial Numbers |
| --- | --- |
| 8210 | 821-15-00100 & Up |
| 8250 | 825-17-00001 & Up |

Publication: 1146945/001D

Revised: 25 Sep 2018

RAYMOND

This publication applies to Model 8210 pallet trucks S/N 821-15-00100 and Up and Model 8250 pallet trucks S/N 825-17-00001 and Up, and to all subsequent releases of these products until otherwise indicated in new editions or bulletins. Changes occur periodically to the information in this publication.

For a list of changes, see "Page Revision Record" on page xiii.

Technical changes or additions to the text and illustrations are indicated by a vertical line to the left of the change.

If you need assistance with your pallet truck, contact your local authorized Raymond Sales and Service Center. To locate a Sales and Service Center, go to www.raymondcorp.com.

# Table of Contents

# Page Revision Record

This page is a record of the revision of all pages in this manual. Whenever a page is revised, this section is updated and included in the revision.

Pages are revised due to technical and non-technical changes identified as follows:

- Technical changes – These changes are identified by a vertical line (change bar) in the left margin next to the change. Pages affected by technical changes are identified with “Revised: Month Year” in the footer. These pages may also be available on the Raymond Portal.
- Non-technical changes – These changes consist of typographical and grammatical corrections, paragraph renumbering, repagination, and so on. Non-technical changes are not identified with a change bar, however, affected pages are identified with “Revised: Month Year” in the footer.

## Document Revision History

- 1146945A, Original Issue Date, Issued: 30 Sept 2015
- 1146945B, Revisions/Updates/Additions, Issued: 12 Jan 2017
- 1146945C, Addition of material to cover lithium-ion battery powered Model 8250, Issued: 21 Jul 2017
- 1146945/001D, Addition of material to cover 2-CMA lithium-ion batteries, update schematic, and miscellaneous updates, Issued: 25 Sep 2018

# Service Information Documents Included

The following Service Information documents were incorporated into this manual.

Use the blank rows below to log Service Information documents when they are added to this manual.

| Document Number | Subject | Date |
|---|---|---|
| RSI PLT-15-015 | E250 Code with Delta-Q Charger | January 5, 2016 |
| RSI PLT-15-016 | Delta-Q Charger Troubleshooting | February 29, 2016 |
| RSI PLT-16-003 | Code C70 Correction | April 7, 2016 |
| RSI PLT-16-004 | Parameter Settings when Using Nexus™ Battery Pack | September 12, 2016 |
| RSI PLT-16-005 | New Software Release | September 13, 2016 |
| | | |
| RSI MUL-17-001 | Scheduled Maintenance Changes | August 15, 2017 |
| RSI PLT-17-007 | Lower Link Bushing Spacer | October 19, 2017 |
| RSI PLT-17-008 | Main Harness Wear | November 7, 2017 |
| RSI PLT-17-010 | NEXSYS Battery Parameter Settings | December 19, 2017 |
| RSI MUL-18-001 | Cold Storage Hydraulic Fluid Part Number Change | March 12, 2018 |
| RSI MUL-18-002 | FlashWare 7.8 Functionality and Enhancements | March 9, 2018 |
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# Section 1. How To Use This Manual

# Manual Design

This manual is designed to give personnel, with an expected level of expertise, the technical information necessary to maintain, troubleshoot, and repair a *Raymond* product.

The two-line running page header at the top of each page contains the name of the manual, the title of the current section, and the topic of the current page.

This manual includes the following sections:

- **1. How to Use This Manual** explains the manual format and design as well as abbreviations and symbols used.
- **2. Safety** explains warning and caution notes, general safety rules, and (as applicable) specific safety rules for batteries, jacking, static electricity, towing, transport, and welding.
- **3. Systems Overview** includes general lift truck specifications, modes of operation, and setup/configuration information.
- **4. Scheduled Maintenance** identifies the recommended maintenance tasks and intervals necessary to keep the lift truck working most efficiently.
- **5. Troubleshooting** provides information used to isolate a problem or failing component based on the lift truck's symptoms.
- **6. Messages and Codes** gives (as applicable) operator messages, fault codes, and procedures for running diagnostic tests.
- **7. Component Procedures** contains component locator photos and step-by-step procedures for the testing, removal, installation, and adjustment of individual truck components. Components are grouped by system. A detailed List of Component Procedures can be found at the beginning of the section
- **8. Theory of Operation** explains signal flow within the electrical and hydraulic systems for various conditions of lift truck operation. This section also contains a detailed connection point table (Pinout Matrix) designed to assist in testing and troubleshooting the truck.
- **Appendix** contains reference information such as torque values, lubricants, and standard/metric conversions.
- **Index** alphabetically lists subject matter with applicable page references.

# Interactive Electronic Technical Manuals

The electronic version of this document is an Interactive Electronic Technical Manual (IETM). IETMs provide quick access to specific service and parts information and are available on iManuals or from your local authorized Raymond Sales and Service Center.

**NOTE:** IETMs require Adobe Reader 9.0 or higher.

Blue text in the Maintenance Manual is linked to a location within the manual. Clicking on blue text takes you to the linked location.

# Abbreviations & Symbols

These abbreviations, acronyms, and symbols are used in this manual.

| Term/Symbol | Definition |
|---|---|
| **A** | |
| A | Ampere |
| AC | Alternating Current |
| amp | Ampere or amplifier |
| ANSI | American National Standards Institute |
| API | American Petroleum Institute |
| approx | approximately |
| aux | auxiliary |
| AWG | American Wire Gauge |
| **B** | |
| BSOC | Battery State-of-Charge |
| BMC | Battery Management Controller |
| BMU | Battery Monitoring Unit |
| **C** | |
| C | Celsius or Centigrade |
| CAN | Controller Area Network |
| CCW | counterclockwise |
| cm | centimeter |
| CMA | Cell Module Assembly |
| COP | Computer Operating Program |
| CS | Cold Storage |
| CV | check valve |
| CW | clockwise |
| **D** | |
| DC | Direct Current |
| DGND | digital ground |
| diam. | diameter |
| DMM | Digital Multi-Meter |
| DOT | US Department of Transportation |
| DVM | Digital Volt-Meter |
| **E** | |
| E | UL Electric Truck Type Certification Rating with safeguards against inherent fire and electrical shock hazards |
| EE | UL Electric Truck Type Certification Rating where electrical equipment is completely enclosed |
| ESD | Electrostatic Discharge |
| ESDS | Electrostatic Discharge Sensitive |
| ETAC | Electronic Tiller Arm Card (See Vehicle Manager) |
| **F** | |
| F | Fahrenheit |
| ft. | foot or feet |
| ft. lb. | foot pound(s) |
| FU | Fuse |
| **G** | |
| GA | gauge |
| gal. | gallon or gallons |
| gm | grams |
| Gnd | ground |
| GPM | Gallons Per Minute |
| **H** | |
| HD | hours on deadman |
| HUD | heads up display |
| **I** | |
| IETM | Interactive Electronic Technical Manual |
| in. | inch or inches |
| in. lb. | inch pound(s) |

| Term/Symbol | Definition |
|---|---|
| **J** | |
| JP | jack and pin connector |
| **K** | |
| K | thousand |
| kg | kilogram(s) |
| km/cm$^2$ | kilometers per square centimeter |
| km/h | kilometers per hour |
| kPa | kilo Pascal |
| **L** | |
| l | liter(s) |
| lb. | pound or pounds |
| LED | Light Emitting Diode |
| LIB | Lithium-ion Battery |
| L/H | Load Holding |
| L/L | Lift/Lower |
| **M** | |
| m | meter(s) |
| mA | milliampere |
| mm | millimeter |
| mph | miles per hour |
| ms | millisecond(s) |
| **N** | |
| N | newton |
| N/A | Not Applicable or Not Available |
| Nm | newton meter |
| NVM | Non-Volatile Memory |
| **O** | |
| OD | Operator's Display |
| OSHA | Occupational Safety and Health Administration |
| oz. | ounce |

| Term/Symbol | Definition |
|---|---|
| **P** | |
| P | pump or lift contactor |
| PMT | Programmable Maintenance Tool |
| pot | potentiometer |
| psi | pounds per square inch |
| PWM | Pulse Width Modulation |
| P/N | Part Number |
| **R** | |
| RAM | Random Access Memory |
| RCFP | Relay Control Fuse Panel |
| ROM | Read Only Memory |
| rpm | revolutions per minute |
| R/R | Remove and Replace |
| **S** | |
| S or SW | Switch |
| SAE | Society of Automotive Engineers |
| SG | Specific Gravity |
| S/N | Serial Number |
| SoH | State of Health |
| SOL | Solenoid |
| spec | specification |
| SPI | Service Port Interface |
| SPL | Splice |
| Std | Standard |
| SWM | Supplier Wireless Module |
| **T** | |
| TA | Traction Amplifier |
| temp | Temperature |
| TM | Traction Motor |
| TP | Tie Point |
| TS | Terminal Strip |
| T/S | Troubleshoot |

Abbreviations & Symbols

| Term/Symbol | Definition |
|---|---|
| **U** | |
| UL | Underwriters Laboratories, Inc. |
| UNC | Unified Coarse thread |
| UNF | Unified Fine thread |
| USB | Universal Serial Bus |
| **V** | |
| V | Volt or Volts |
| VAC | Volts Alternating Current |
| VDC | Volts Direct Current |
| VM | Vehicle Manager (ETAC) |
| **W** | |
| wrt | with respect to |
| **Symbol** | |
| @ | at |
| ™ | trademark |
| ® | registered trademark |
| © | copyright |
| + | plus or positive |
| – | minus or negative |
| ± | plus or minus |
| ° | degrees |
| °F | degrees Fahrenheit |
| °C | degrees Celsius |
| < | less than |
| > | greater than |
| % | percent |
| = | equals |

# Section 2. Safety

# Definitions

Throughout this manual, you will see two kinds of safety reminders:

**⚠ WARNING**

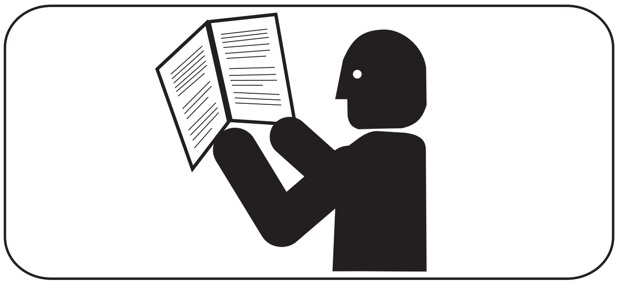
**Warning means a potentially hazardous situation exists, which, if not avoided, could result in death or serious injury.**

**⚠ CAUTION**

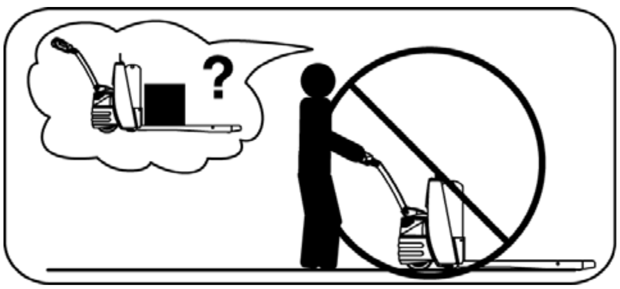
**Caution means a potentially hazardous situation exists, which, if not avoided, may result in minor or moderate injury or in damage to the lift truck or nearby objects. It may also be used to alert against unsafe practices.**

# General Safety

Do **not** operate or work on this truck unless you are trained, qualified, authorized, and have read the Owner and Operator Manuals.

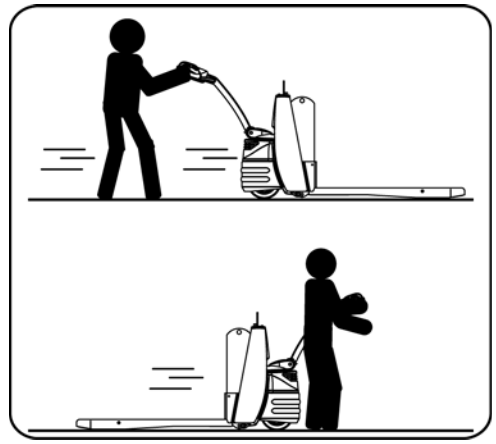


Know the truck's controls and what they do.

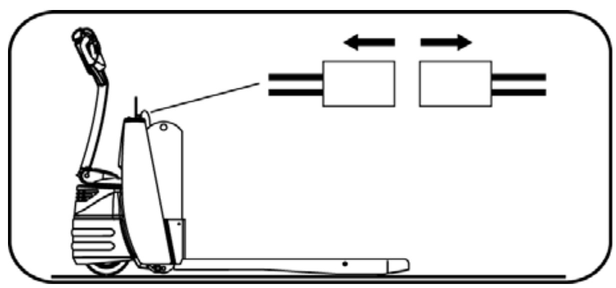


Report any malfunction or unsafe condition to your supervisor immediately. Do **not** operate this truck if it needs repair or if it is in any way unsafe.

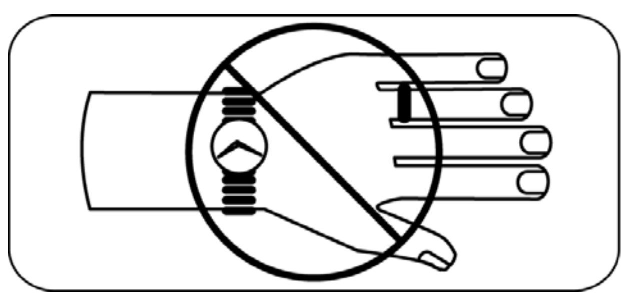
Operate this truck only from the operator's position.



Before working on this truck, if equipped with the optional keypad press the red OFF ( **O** ) key, place the Main ON/OFF Switch in the OFF position, and disconnect the truck's battery connector (unless this manual tells you otherwise).

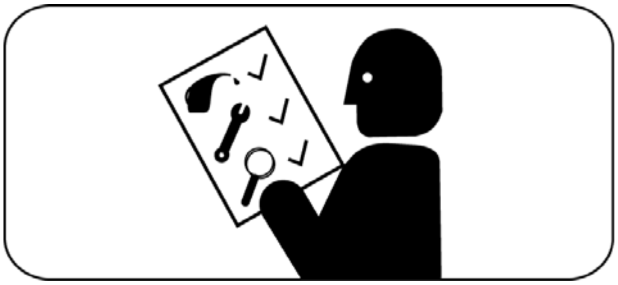


Do **not** wear watches, rings, or jewelry when working on this truck.

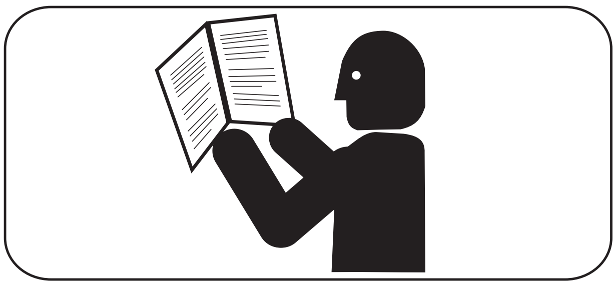


General Safety

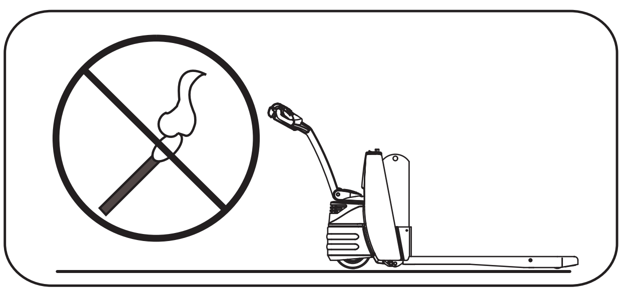
Obey the scheduled lubrication, maintenance, and inspection steps.



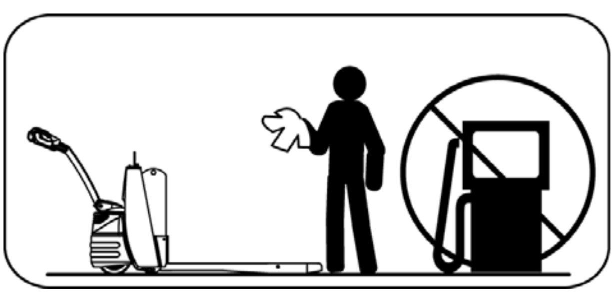
Obey exactly the safety and repair instructions in this manual. Do **not** take “shortcuts.”



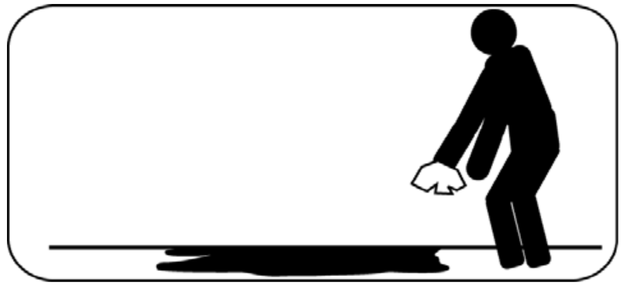
Do **not** use an open flame near the truck.



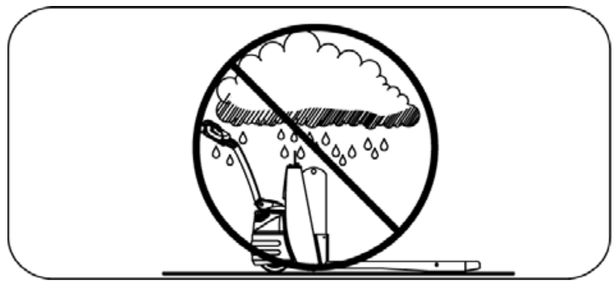
Do **not** use gasoline or other flammable liquids for cleaning parts.



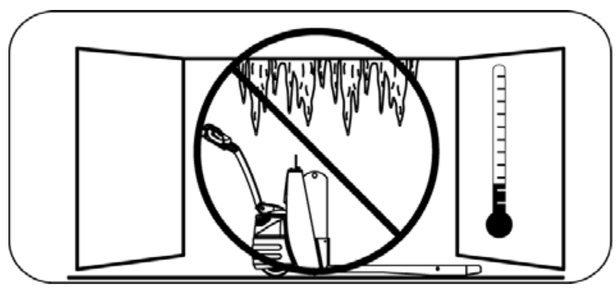
Clean up any hydraulic fluid, oil, or grease that has leaked or spilled on the floor.



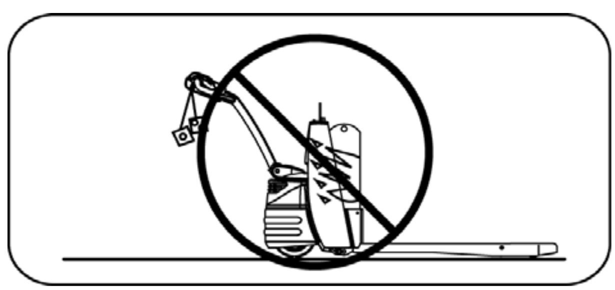
Always park this truck indoors.



Do **not** park this truck in a cold storage area overnight.



Do **not** add to or modify this truck until you contact your local authorized Raymond Sales and Service Center to receive written manufacturer approval.



# Lead Acid Battery Safety

**⚠ WARNING**

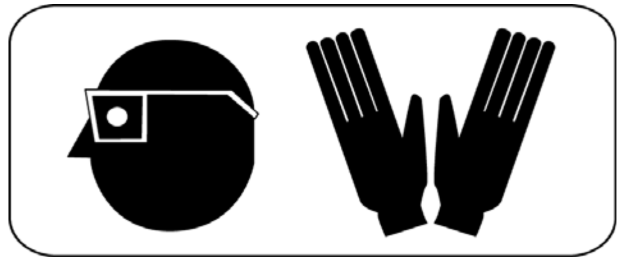
**As a battery is being charged, an explosive gas mixture forms within and around each cell. If the area is not correctly ventilated, this explosive gas can remain in or around the battery for several hours after charging. Make sure there are no open flames or sparks in the charging area. An open flame or spark can ignite this gas, resulting in serious damage or injury.**

**⚠ WARNING**

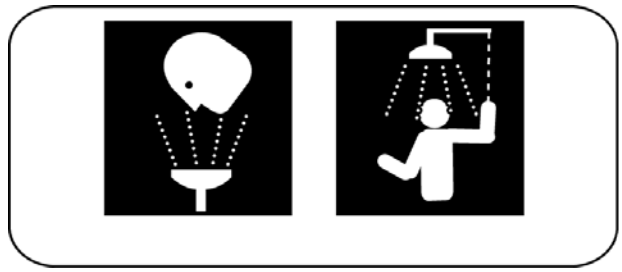
**Battery electrolyte is a solution of sulfuric acid and water. Battery electrolyte causes burns. If any electrolyte comes in contact with your clothing or skin, flush the area immediately with cold water. If the solution gets on your face or in your eyes, flush the area with cold water and get medical help immediately.**

Read, understand, and follow procedures, recommendations, and specifications in the battery and battery charger manufacturer's manuals.

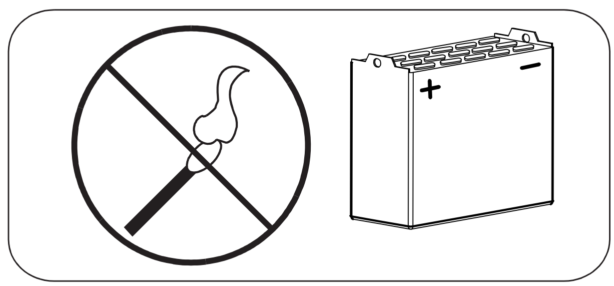
Wear personal protective equipment to protect eyes, face, and skin when checking, handling, or filling batteries. This equipment includes goggles or face shield, rubber gloves (with or without arm shields), and a rubber apron.



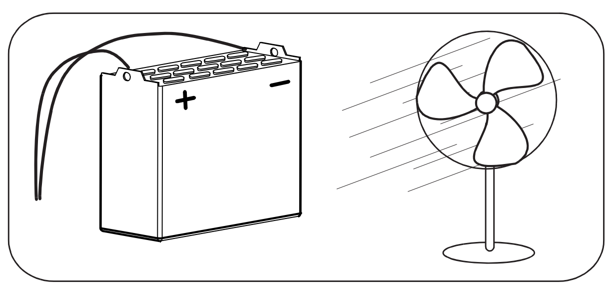
Make sure a shower and eyewash station are nearby in case there is an accident.



A battery gives off explosive gases. **Never** smoke, use an open flame, or use anything that gives off sparks near a battery.

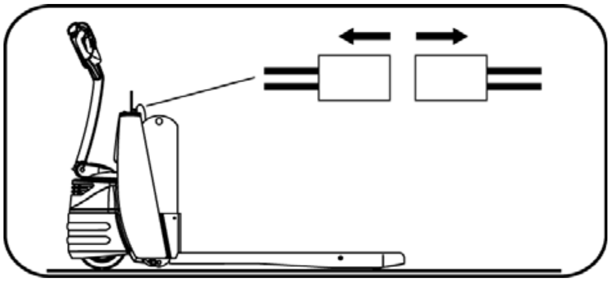


Keep the charging area well-ventilated to avoid hydrogen gas concentration.

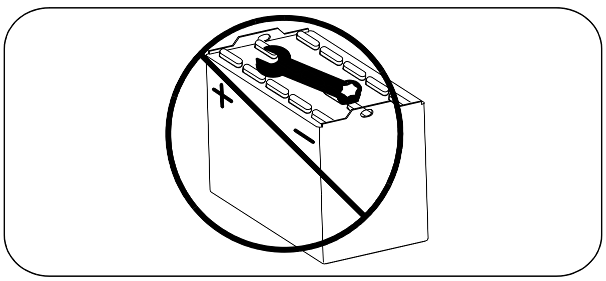


If equipped with the optional keypad, press the red OFF ( **O** ) key and place the Main ON/OFF

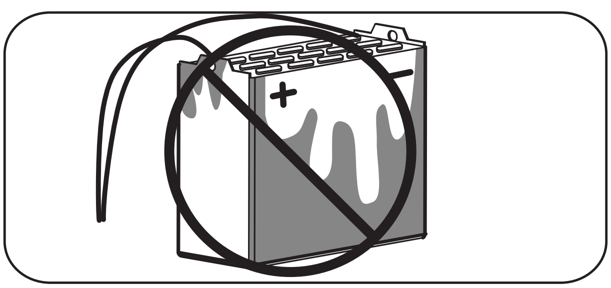
Switch in the OFF position *before* disconnecting the battery from the truck at the battery connector. Do **not** break live circuits at the battery terminals. A spark often occurs at the point where a live circuit is broken.



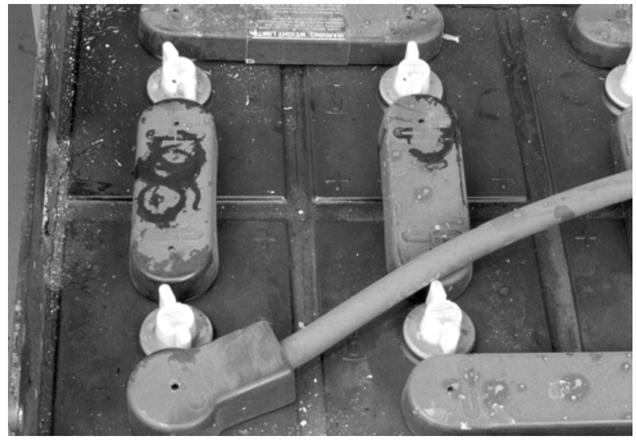
Do **not** lay tools or metal objects on top of the battery. A short circuit or explosion could result.



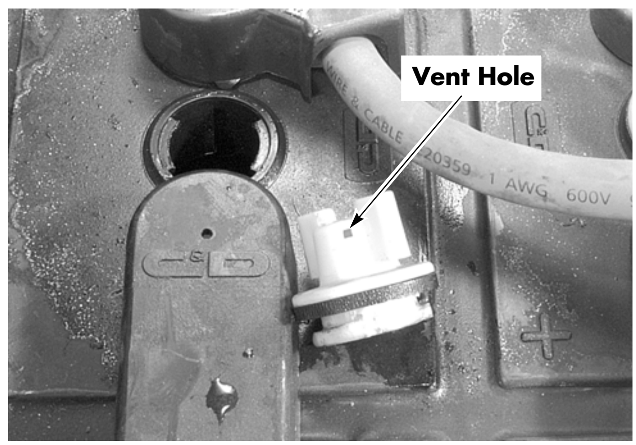
Keep batteries clean. Corrosion causes shorts to the frame and possibly sparks.



Keep plugs, terminals, cables, and receptacles in good condition to avoid shorts and sparks.

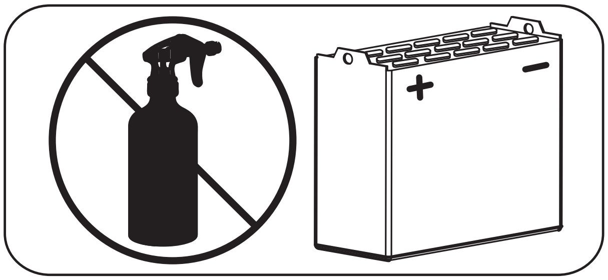


Keep filler plugs firmly in position at all times when not checking the electrolyte level, adding to the cells, or checking the specific gravity.

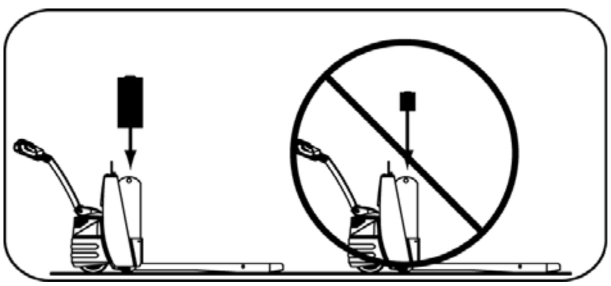



Make sure the vent holes in the filler plugs are open to permit the gas to escape from the cells.

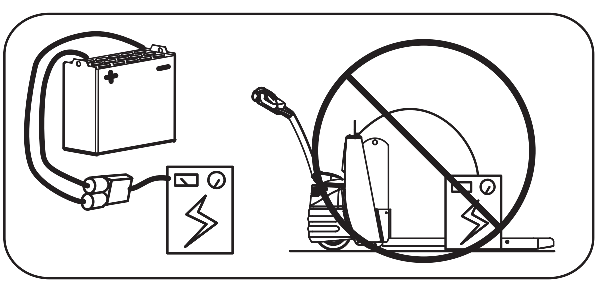
Do **not** permit cleaning solution, dirt, or any foreign matter to enter the cells.



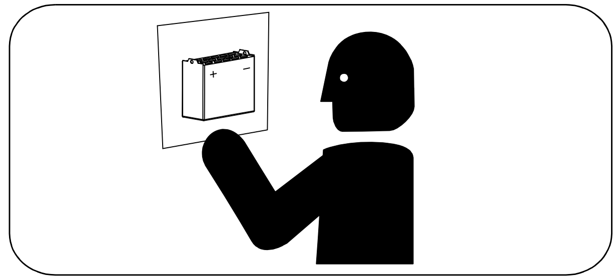
Make sure you install the correct size and voltage battery. A smaller or lighter weight battery could seriously affect truck stability. An incorrect voltage battery could damage the truck's electrical system. See the truck's specification tag for more information.



**Never** plug a battery charger into the truck's battery connector. Plug the battery charger only into the battery connector from the battery.



Obey the charging procedures in the Battery Instruction Manual and in the Battery Charger Instruction Manual.



# Lithium-Ion Battery Safety

Read, understand, and obey all instructions and warnings on the battery and in the battery charger manufacturer's manuals. Follow these precautions when working with or around lift trucks with lithium-ion batteries.

Do not over-discharge the battery. Over-discharging accelerates the aging process and decreases the life of the battery. It may also be cause to void the Warranty.

Do not store or operate lithium-ion batteries at higher than recommended temperatures. Higher temperatures accelerate the aging process and decrease the life of the battery.

Thermal runaway is the build-up of heat and pressure in the battery. Thermal runaway can cause venting of flammable gases, and the release of energy in the form of heat, smoke, and/or fire.

**⚠CAUTION**

**In a Thermal Runaway condition, turn the lift truck off, evacuate the area immediately, and activate the fire alarm system and call 911 or the local fire department. Do NOT try to extinguish flames or remedy the situation.**

Do not attempt to replace fuses within the lithium-ion battery unless you are trained and authorized to do so. Contact your local authorized Raymond Sales and Service Center for assistance.

Use caution when using a high pressure washer to clean the exterior of the lithium-ion battery. Make sure battery compartment drain holes are clear to prevent liquid buildup in the lithium-ion battery.

The battery pack must be secured to the truck prior to truck operation and servicing the battery pack. The battery pack may tip over if the battery pack is not secured.

Do not crush, drop, cut, puncture, or subject the lithium-ion battery to excessive physical impact that may cause damage.

Transport the battery in accordance with all applicable laws.

Never smoke, use open flames, or use devices that produce sparks near lithium-ion batteries.

Never lay tools or metal objects on top of the battery or any part of the battery circuit. A short circuit or explosion could result.

Do not use the ventilation ports as handles.

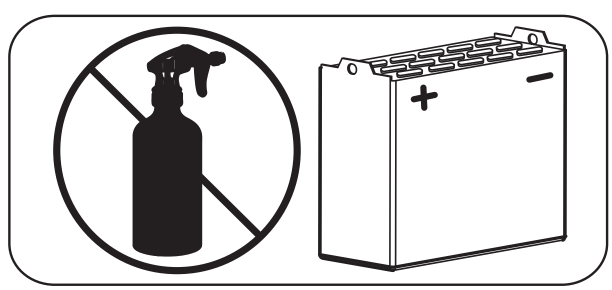
Do not remove the battery cable at the battery terminal.

Do not attempt to lift the lithium-ion battery without a hoist.

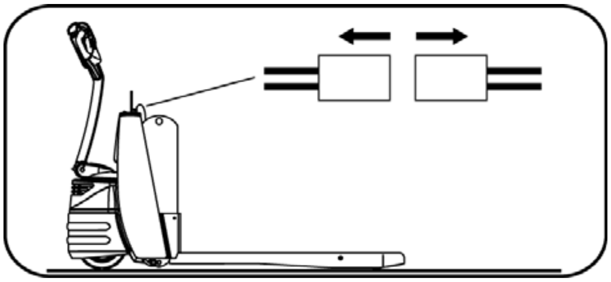
Never plug a battery charger into the truck's battery connector. Plug the battery charger only into the battery connector from the battery.

Keep the charging area well-ventilated to avoid hydrogen gas concentration.

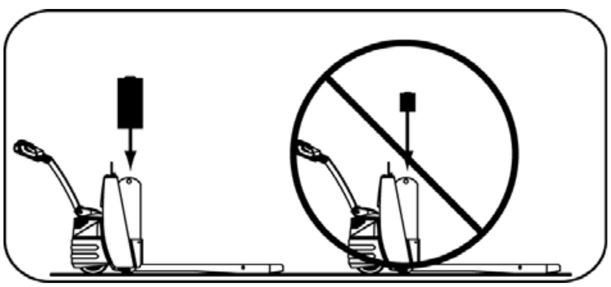
Do not permit cleaning solution, dirt, or any foreign matter to enter the cells.



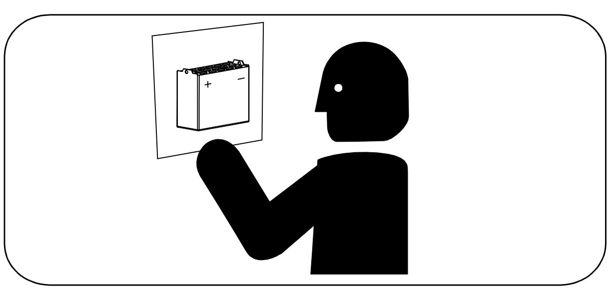
If equipped with the optional keypad, press the red OFF ( O ) key and place the Main ON/OFF Switch in the OFF position before disconnecting the battery from the truck at the battery connector. Do not break live circuits at the battery terminals. A spark often occurs at the point where a live circuit is broken.



Make sure you install the correct size and voltage battery. A smaller or lighter weight battery could seriously affect truck stability. An incorrect voltage battery could damage the truck's electrical system. See the truck's specification tag for more information.



Charge batteries at specified intervals. Do not overcharge. Obey the charging procedures in the Battery Instruction Manual and in the Battery Charger Instruction Manual.

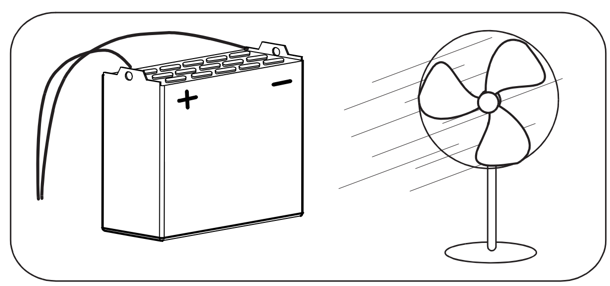


When storing a lithium-ion battery, follow these guidelines:

- Place the lithium-ion battery in an environment that resides within the recommended Storage Temperature Range. The following table identifies maximum ambient temperature for specific durations of time.

| Duration | Maximum Ambient Temperature |
|---|---|
| Less than 1 day | 130°F (54°C) |
| Less than 1 week | 115°F (46°C) |
| Less than 1 month (30 days) | 95°F (35°C) |
| Less than 6 months (180 days) | 75°F (24°C) |
| Greater than 6 months (180 days) | 40°F (4°C) |

- Place the lithium-ion battery in a location that will not be exposed to direct sunlight.
- Place the lithium-ion battery in a location that will not be exposed to falling objects.
- Place the lithium-ion battery underneath or within close proximity to a fire/smoke detection system.
- Place the lithium-ion battery in a well ventilated area.



- Place the lithium-ion battery in a designated area that is not subjected to potential flooding.
- Position the lithium-ion battery in an orientation that minimizes the chances of tipping over.

# Jacking Safety

When it is necessary to jack the truck off the floor to perform maintenance procedures, observe the following safety precautions:

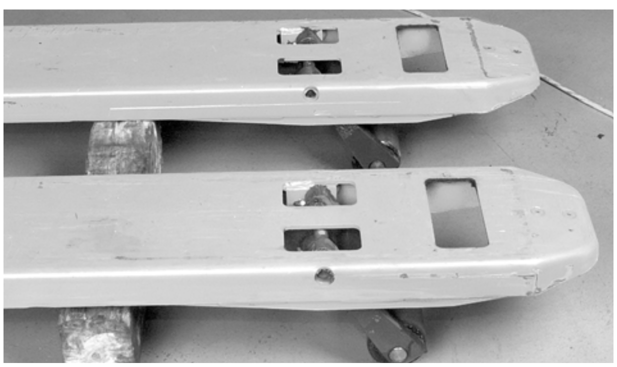
**⚠ WARNING**

**Use extreme care whenever the truck is jacked up. Keep hands and feet clear from the vehicle while jacking the truck. After the truck is jacked, put solid blocks beneath it to support it. DO NOT rely on the jack alone to hold the truck.**

## Fork Section

1. Park the lift truck on a level surface.
2. Using the lift button, raise the forks to maximum height.
3. Block the fork section by placing a block behind the load wheels as shown in Figure 2-1. The tractor section remains on the floor.

*Figure 2-1. Blocking the Fork Section*

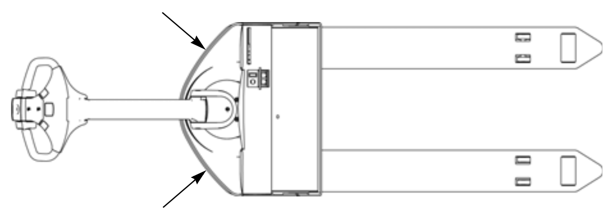


4. Lower the forks on the blocks.
5. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.

## Tractor Section

1. Park the lift truck on a level surface.
2. Lower the forks completely. Remove any load.
3. Position all controls in neutral.
4. Block the wheels to prevent movement of the vehicle.
5. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
6. Position the jack in the designated jacking position.

*Figure 2-2. Tractor Jacking Points*



7. Jack one side of the truck so that the drive tire is no more than 1 in. (25.4 mm) off the floor.
8. Block that side of the truck in position.
9. Jack the other side of the truck level with the first side.
10. Block that side of the truck in position.

**NOTE:** After working on a vehicle, test all controls and functions to make sure operation is correct.

# Towing

To safely tow this truck:

1. Lower the forks and remove any load.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
3. Using a suitable towing vehicle, lift the tractor end of the truck at the jacking points until the drive tire is no more than 1 in. (25.4 mm) off the floor.
4. Tow the truck slowly in the tractor-first direction.

**NOTE:** If a suitable towing vehicle is not available, the electromagnetic brake must be disabled to move this truck. See “Electric Brake Release” on page 7-37.

# Transport

To transport your *Raymond* walkie truck in an over-the-road vehicle or rail car, perform these steps:

1. Lower the forks and put the truck in the center of the transport vehicle.
2. Place the control handle in the full upright position to set the brakes.

# Static Safety

Electronic circuit boards can contain Electrostatic Discharge Sensitive (ESDS) devices.

Static charges can accumulate from normal operation of the truck as well as movement or contact between non-conductive materials (plastic bags, synthetic clothing, synthetic soles on shoes, styrofoam coffee cups, and so forth).

Accumulated static can be discharged through human skin to a circuit board or component by touching the parts. Electrostatic Discharge (ESD) is also possible through the air when a charged object is put close to another surface at a different electrical potential. *Static discharge can occur without you seeing or feeling it.*

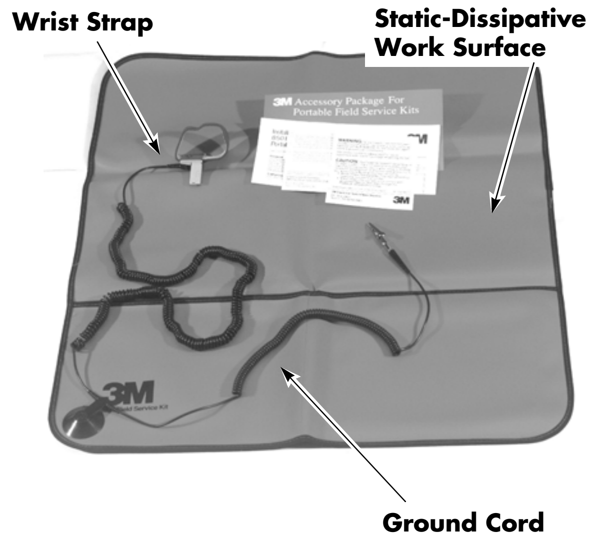
Whenever working on or near static-sensitive electronics, always use static discharge precautions.

1. Put a static discharge wrist strap around your wrist. Connect the ground cord to the wrist strap connector.
2. Connect the ground cord to an unpainted, grounded surface on the truck frame.
3. If you are removing or installing static-sensitive components, put them on a correctly grounded static mat.
4. To transport static-sensitive components, including failed components being returned, put the components in an anti-static bag or box (available from your Raymond Sales and Service Center).

The wrist strap and associated accessories should be tested monthly to verify they are working correctly. The wrist strap contains a one megohm resistor in the strap cord that acts as a fuse for personal protection. If this resistor is open, the strap becomes ineffective.

Figure 2-3 shows the components of the *Raymond* anti-static field service kit, part number 1-187-059. The kit includes a wrist strap, ground cord, and static-dissipative work surface (mat). Follow the instructions packaged with this kit.

*Figure 2-3. Anti-Static Kit (P/N 1-187-059) With Wrist Strap and Mat*




Replacement wrist straps (P/N 1-187-058/001) are available. Contact your authorized Raymond Sales and Service Center for information.

# Welding Safety

**⚠ WARNING**

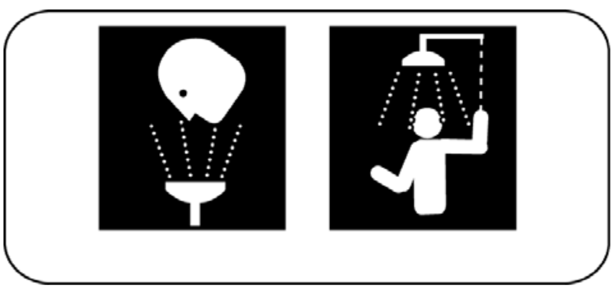
**Flame cutting or welding on painted surfaces may produce potentially harmful fumes, smoke, and vapors. Remove any coating in the vicinity where the operation(s) will be performed prior to performing flame cutting or welding operations.**

**⚠ WARNING**

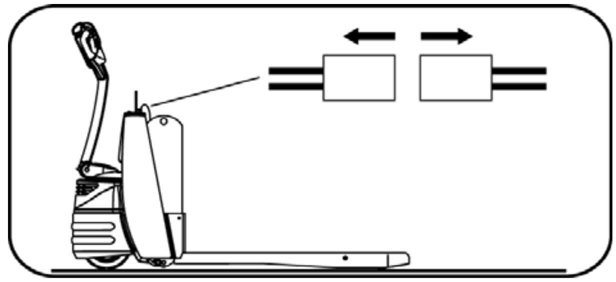
**Coating removal may be by mechanical methods, chemical methods, or a combination of methods. Perform flame cutting and/or welding operations only in well ventilated areas. Use local exhaust if necessary.**

Before working on this truck, make sure that:

- Fire protection equipment is nearby.
- You know where the nearest eyewash station is located.



- You always press the red OFF key ( O ) on the keypad (if equipped), place the Main ON/OFF switch in the OFF position, and disconnect the battery connector before you attempt to inspect, service, or repair the lift truck. Discharge residual charge in the power amplifier by connecting a load across the power amplifier’s B+ and B– (such as a contactor coil).



- Check for shorts to frame as identified on page 5-5. If any shorts are found, remove them before you proceed with the welding operation.
- Clean the area to be welded.
- Protect all truck components from heat, weld spatter, and debris.
- Attach the ground cable to a clean, unpainted surface as close to the weld area as possible.
- Do not perform any welding operations near the electrical components.
- Do not attach the ground cable to fasteners or other removable components.
- If you must weld near the battery compartment, remove the battery from the truck. See “Lead Acid Battery” on page 7-40.
- When you are finished welding, perform all ground tests and electrical inspections before operating the vehicle.

# Section 3. Systems Overview

# Introduction

This manual provides information for maintenance and repair of the *Raymond* Model 8210 pallet trucks.

This manual contains the most current and accurate procedures, drawings, and photographs available at the time of publication. Subsequent releases of this product may differ slightly from that shown here. Accordingly, some changes in parts, layout, or procedures may not be reflected in this manual.

For the latest information on your *Raymond* lift truck, contact your local authorized Raymond Sales and Service Center.

# Truck Model Identification

*Figure 3-1. Model 8210 Pallet Truck*

# Lift Truck Specifications

This lift truck is rated for performance by evenly distributed load weight with forks elevated. Load weight includes the weight of the pallet, container, or load-holding attachment.

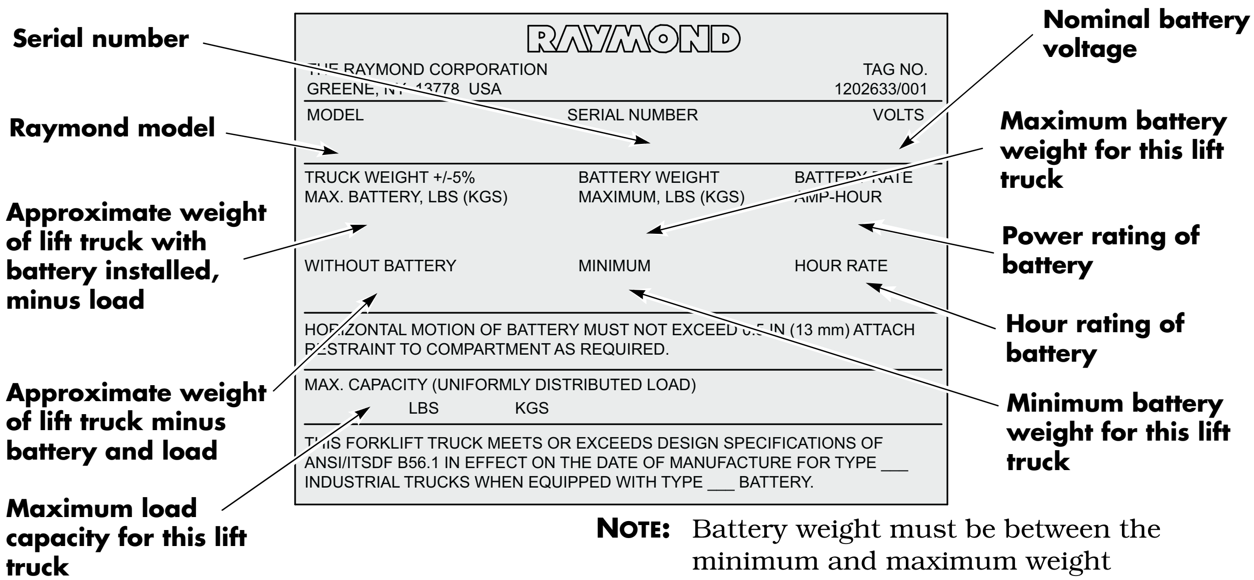
The specification tag is located on the left side of the upper cover when facing the truck from the operating position.

A second specification tag is located inside the tractor compartment on the right side bulkhead plate as you face the tractor compartment. See Figure 3-2.

A battery specification tag is located on the left side of the top of the battery when facing the truck from the operating position. See Figure 3-3.

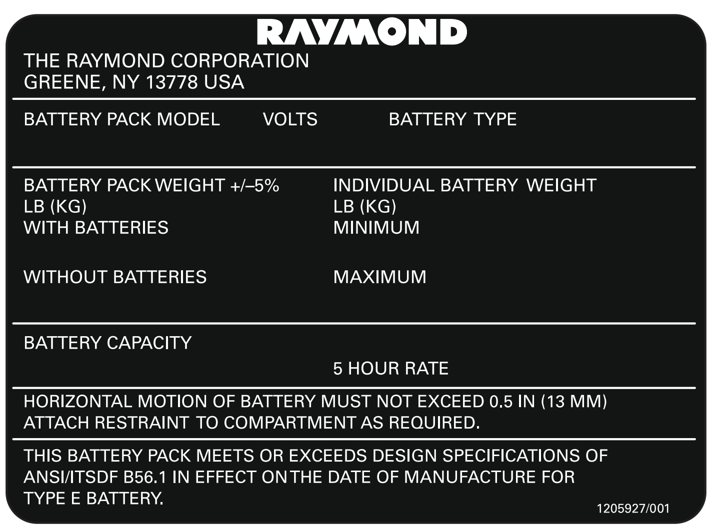
Due to continuous product improvement, specifications are subject to change without notice or obligation.

*Figure 3-2. Lift Truck Specification Tag - Model 8210 Pallet Truck*




**NOTE:** Battery weight must be between the minimum and maximum weight

*Figure 3-3. Battery Specification Tag - Model 8210 Pallet Truck*



# Operator Display and Programming

The Operator Display is installed on the top of the control handle and displays operational and servicing information.

*Figure 3-4. Operator Display*

**Explanation of Symbols:**

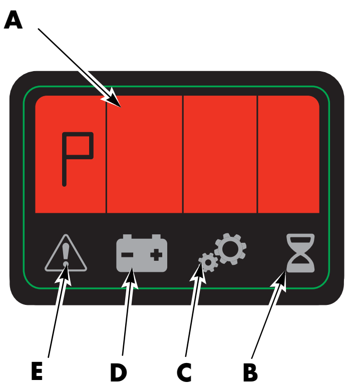
**A = Alpha-Numeric Field**

**B = Hour Meter Indicator**

**C = Parameter Indicator**

**D = Battery Indicator**

**E = Error Indicator**




## Special Truck Mode

**NOTE:** Installation of a Service Key is required to make some changes. See "Programming Service Parameters" on page 3-8.

To enter a Special Truck Mode:

1. Press and hold the horn button (S18). At the same time, place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.
2. Continue to hold the horn button while the display (see Figure 3-4) cycles through the following functions:
   - (H)our meters: use the speed control to choose. See parameter 20 on page 3-10 for hour meter options (hours shown are rounded down, for example, 3 hours and 50 minutes will display as 3).
   - (E)rror codes log: the most recent error code will be displayed, followed by the hour meter (activity time) it occurred. Use the speed control to scroll through the last 25 error codes (if no error code is logged, dashes will be displayed).
   - (P)arameters: use the speed control to choose, and the horn button to access a parameter. The parameter light blinks (a service key is required to change some parameters). See page 3-6. When the service key is used to change driver parameters and parameter 10, the driver number appears to the right of the normal parameter number display. All driver parameters and drivers are cycled through before Service Parameters, 1 - 9, 0 (0 = 10th driver).
   - (P)art (n)umbers: use the speed control to scroll through the following:
     - (S)oftware (P)art (n)umber (of VM)
     - (H)ardware (P)art (n)umber (of VM)
     - (H)ardware (S)erial (n)umber (of VM)
     - Type = Model number
     - PA = Curtis Power Amplifier Operating System
     - IPT = iPORT / RDP capable
     - bnc = Battery Management Controller
     - bnu = Battery Management Unit
     - LiUi = Lithium User Interface (HUD)
   - (d)isplay Test: use the horn button to access the display test. See page 3-16. This test verifies that all display segments and icons are functioning.
   - Lib: Lithium-ion battery: use the speed control to scroll through the following:
     - CCnt: Lithium-ion battery Cycle count
     - SOH: Lithium-ion battery State of Health
     - r ct: number of times the Lithium-ion battery has entered Reserve Mode
     - b ct: number of times the Lithium-ion battery has entered Manual Mode
     - d ct: number of times the Lithium-ion battery has entered Dormant Mode
     - bHrS: number of warranty hours accrued on the Lithium-ion battery

3. Release the horn button at the selected display. If the horn button is released at the wrong time, turn the truck OFF and start over.

## Hour Meter (H)

To change Hour Meter settings:

1. Enter the Special Truck Mode. See "Special Truck Mode" on page 3-5.
2. Press and hold the horn button until "H" is displayed. The hour meter indicator (hourglass icon) is illuminated. See Figure 3-4. If the horn button is released at the wrong time, turn the truck OFF and start over.
3. Choose the desired time to be displayed by pressing the thumb controls:
   - A = Key Time: number of hours the Main ON/OFF switch (and optional keypad if equipped) is turned ON ( **I** ). These hours include key ON and an operator logged in hours.
   - b = Activity Time: number of hours the truck has traveled, lifted, or lowered.
   - c = Travel Time: number of hours the truck has traveled (hours are counted any time travel is requested).
4. Press the thumb control to step between the different functions. The display shows the number of hours for each function.
5. If equipped with the optional keypad, press the red OFF ( **O** ) key, and place the Main ON/OFF Switch in the OFF position to end the hour meter display.

## Error Codes (E)

When an error occurs, a code is displayed. See "Messages and Caution Codes" on page 6-5.

### Error Code History

To access Error Code History:

1. Enter the Special Truck Mode. See "Special Truck Mode" on page 3-5.
2. Press and hold the horn button until "E" is displayed. The error indicator (attention icon) is illuminated. See Figure 3-4. If the horn button is released at the wrong time, turn the truck OFF and start over.
   - Use the thumb control to scroll through the last 25 error codes. The latest fault is shown first. The display first shows the error code and then the hour the fault occurred.
   - The error codes are divided into two groups C = caution and E = error.
   - The C error codes are not stored. It is only a caution and is highlighted on the display as long as the fault exists.
   - For example, if you only have one fault registered in the memory, the second empty place is shown like this:
     Fault: E - - - and time: - - - -.
3. To end display of the error code history, press the red OFF ( **O** ) key on the keypad (if equipped), and place the Main ON/OFF switch in the OFF position.

## Changing Truck Parameters (P)

**⚠ WARNING**

**Modifying specific truck parameters changes the driving characteristics of the truck.**

### Programming Truck Parameters

To program the truck's operating parameters, follow the directions below (also see "Setting Individual PIN-key Codes" on page 3-8):

1. Enter Special Truck Mode. See "Special Truck Mode" on page 3-5.
2. Press and hold the horn button until "P" is displayed. The parameter indicator (wheel icon) is illuminated. If the horn button is released at the wrong time, turn the truck OFF and start over.
3. Use the thumb control to scroll to the desired truck parameter you want to change or view. See "Truck Driver Parameters" on page 3-7. Release the thumb control to neutral when the desired parameter number is shown.

4. Press the horn button once to access the parameter. The parameter symbol on the display starts flashing.
5. Change the parameter value by rotating the thumb control up or down. Release the thumb control to neutral when the desired parameter number is shown.
6. Press the horn button again to confirm the change. The parameter symbol on the display stops flashing and remains illuminated.
7. To end programming mode, press the red OFF ( **O** ) key on the keypad (if equipped), and place the Main ON/OFF switch in the OFF position.
   - The parameter change is complete. The next time the truck is started, the new parameter is in effect.

*Table 3-1. Truck Driver Parameters*

| **Parameter** | **Name** | **Unit** | **Range (Step)** | **Step** | **Default** | **Description** |
|---|---|---|---|---|---|---|
| 01 | Max Speed | % | 0 to 40 | 2 | 40 | Maximum speed value sent to the traction power amplifier. This parameter controls the scaling of the throttle request. A setting of 40% corresponds to approximately 3.6 MPH. |
| 02 | Not used | | | | | |
| 03 | Not used | | | | | |
| 04 | Neutral Braking (Deceleration) | % | 0 to 6 | 1 | 5 | Defines the truck's neutral braking characteristic when the throttle is returned to neutral.<br>A higher number results in a less aggressive deceleration.<br>6: low reduction force<br>0: high reduction force |
| 05 | Truck OFF Delay (with optional keypad only) | Min | 0 to 20 | 1 | 0 | Sets the truck off delay (energy saving feature). If this amount of time passes while the tiller arm is in a brake position and the truck is idle (no outputs requested), the VM powers the truck OFF. When this parameter is set to 0, the truck never powers OFF. If parameter 39 is 1 or 2 (truck does not have an optional keypad), the truck never powers OFF.<br>0: never turns off<br>20: 20 minutes. |
| 06 | Not used | | | | | |
| 07 | Acceleration Rate | | 1 to 5 | 1 | 3 | Selects from a table for acceleration rate. A higher number results in a less aggressive acceleration.<br>1 = fastest,<br>5 = gentlest. |
| 08 | Not used | | | | | |
| 09 | Not used | | | | | |

## Programming Service Parameters

1. Connect the service key to the service key connector J5. Make sure the Main ON/OFF switch (SW1) is in the ON position. See “Special Tools” on page 3-21.
2. Turn the truck ON while holding the horn button until “P” is displayed. Release the horn button at “P” to enter parameter mode. The Parameter indicator (wheel icon) on the display illuminates when in parameter mode. See “Special Truck Mode” on page 3-5. If the horn button is released at the wrong time, turn the truck OFF and start over.
3. Rotate the thumb control in the fork direction to progress forward through the parameters. While the thumb control is held and briefly after it is released, both the parameter number and the driver number are displayed. Stop at parameter 10 (PIN code) for the first driver; the default value of “1” is displayed. See Figure 3-4.
4. Press the horn button once to access this parameter. The parameter symbol on the display flashes. Now you can make changes to the value of the first driver’s PIN code. See Table 3-5.

**NOTE:** All parameters can be viewed; however, not all parameters can be changed. Driver parameters may or may not be accessed, depending on the value of parameter 39.

## Parameter Displays

When parameters 1 through 5, and 10 are viewed with the service key, both the parameter number (left side of display) and the 1$^{st}$ operator number (right side of display) are shown.

As the parameter number display progresses through parameters 1 through 10, the display rolls over to the next operator (the operator number increases). If a parameter is changed, it is valid only for the operator shown. After all operators and operator parameters are scrolled through, Service Parameters 11 through 49 are displayed.

To quickly display/access a Service Parameter, rotate the thumb control away from the forks (sequence backwards through the parameters/operators).

## Setting Individual PIN-key Codes

1. Connect the service key at J5. See “Special Tools” on page 3-21. Make sure that the Main ON/OFF switch is ON.
2. Enter Special Truck Mode. See “Special Truck Mode” on page 3-5.
3. Press and hold the horn button until “P” is displayed. The parameter indicator (wheel icon) is illuminated. If the horn button is released at the wrong time, turn the truck OFF and start over.
4. Use the thumb control to scroll through the service parameters. While the thumb control is held and briefly after it is released, both the parameter number (on the left side of the display) and the operator number (on the right-this only occurs if the service key is connected) are displayed. Stop (release the thumb control) at parameter 10 (PIN-key code) for the first operator; the default value of “1” is displayed.
5. Press the horn button once to access this parameter. The parameter indicator on the display flashes. Now you can make changes to the value of the first operator’s PIN-key code.

**⚠CAUTION**

**Changing the 1st operator’s PIN-key code alters default operation. You will not be able to press “1” and green ON ( I ) to power the truck unless an operator’s PIN-key code is set to “1.”**

6. Rotate the thumb control in the fork direction to increase the PIN-key code value, or in the operator direction to decrease the value. Holding the thumb control in a direction for a time helps quickly advance to larger numbers (for example, codes greater than 100). The PIN-key code can be from one to four digits

long. Release the thumb control when the desired value is reached.

7. Press the horn button again to confirm the new PIN-key code value. The parameter indicator on the display stops flashing.
8. To enable additional operators; rotate the thumb control in the fork direction to roll over to the second operator, and stop at parameter 10. The default value of “0” is displayed. Follow the previous steps for the desired number of different PIN-key codes you want to set up. There are a total of ten operators.

**NOTE:** Disabling all operators (all PIN-key code values = 0) prohibits truck operation without the service key.

**NOTE:** Use unique PIN-key codes to avoid confusion. If two operators have the same PIN-key code, the higher number operator’s parameters are used. For example if both the first and second operator PIN-key code is 1111, and the first operator’s max speed is 80% and the second operator’s max speed is 90%, when someone logs in with 1111 the max speed used is 90%.

*Table 3-5. Service Parameters*

| Parameter | Name | Unit | Range | Step | Default | Description |
|---|---|---|---|---|---|---|
| 10 | PIN code (keypad only) | | 0 to 9999 | 1 | 1 | Capability to assign up to 10 PIN-key codes (for 10 different operators). A Service Key is required to assign up to 10 operators their own PIN-key code (one to four digits) for access to the truck. See "Setting Individual PIN-key Codes" on page 3-8.<br>**NOTE:** 0 disables a driver (the 2nd through 10th driver's default PIN is 0). If all drivers are disabled, the truck cannot be driven without the service key.<br>**NOTE:** If two drivers have the same PIN number, the parameters of the driver with the highest settings are used for both drivers. |
| 11 | Not used | | | | | |
| 12 | Throttle Map | % | 5 to 80 | 5 | 50% | Alters the slope of the calculated throttle. 50% is a linear throttle response.<br>>50% provides a more rapid increase in speed at lower throttle requests.<br><50% provides a slower increase in speed at lower throttle request.<br>Maximum speed is achieved at full throttle for all response curves. |
| 13 | Deceleration (Plugging) Rate | | 1 to 5 | 1 | 3 | Selects from a table for deceleration rate. A higher number results in a less aggressive deceleration.<br>1 = hardest feel<br>5 = gentlest feel |
| 14 | Run-Time Display | | 0 or 1 | 1 | 0 | 0: BSOC is the default run-time display. At power-up the Hour Meter briefly displays, then the BSOC percent is displayed during run-time<br>1: Hour Meter (as selected by parameter 20) is the run-time display. At power-up the BSOC percent briefly displays, then the Hour Meter is displayed during run-time.<br>**NOTE:** A setting of "1" is useful for trucks with fuel cells. |
| 15 | Not used | | | | | |

| Parameter | Name | Unit | Range | Step | Default | Description |
|---|---|---|---|---|---|---|
| 16 | Travel Alarm Type | | 0 to 7 | 1 | 0 | 0 = No Alarm<br>1 = Tractor-First Alarm<br>2 = Forks-First Alarm<br>3 = All Travel Alarm<br>4 = Low Battery Alarm<br>5 = Low Battery and Forks-First Travel Alarm<br>6 = Low Battery and Tractor-First Travel Alarm<br>7 = Low Battery and All Travel Alarm |
| 17 | Not used | | | | | |
| 18 | iPORT Init Speed (only with iWAREHOUSE option) | | 0.1 to 3.9 repre-senting 0 MPH to FULL SPEED | 0.1 | 3.8 | Prior to communication with the SWM, iPORT Initial Speed sets the maximum allowable truck speed.<br>**NOTE:** A 0 MPH setting disables truck functions.<br>**NOTE:** Raymond recommends that Init Speed be set to 0.5 (0.8 km/h) to prevent operation without being logged in. |
| 19 | iPORT Error Speed (only with iWAREHOUSE option) | | 0.1 to 3.9 repre-senting 0 MPH to FULL SPEED | 0.1 | 3.8 | The Error Speed limit is activated when communication is interrupted and the truck comes to a stop. The truck is limited to Error speed while traveling down to 1.0 MPH (1.6 km/h). Once stopped, the lower speed is activated if Error Speed is set to less than 1.0 MPH (1.6 km/h). When communication is re-established, and throttle is brought back to neutral, normal operation is restored.<br>**NOTE:** Raymond recommends that Error Speed be set to 1.0 MPH (1.6 km/h). |
| 20 | Hour Meter Selection | | 1 to 3 | 1 | 2 | 1 (A) = key time<br>2 (b) = activity time (drive, lift, or lower)<br>3 (c) = travel time |
| 21 | Battery Type | | 1 to 4 | 1 | 2 | Defines battery type for correct BSOC (battery state-of-charge) configuration.<br>1 = Industrial battery (8.5 inch)<br>2 = AGM battery<br>3 = Wet Cell battery and 7.5 inch NEXSYS battery pack<br>4 = 5 inch NEXSYS battery Slim Pack<br>**Model 8250:** This setting has no effect with the CAN based lithium-ion battery. |
| 22 | Not used | | | | | |
| 23 | Not used | | | | | |

| Parameter | Name | Unit | Range | Step | Default | Description |
|---|---|---|---|---|---|---|
| 24 | Battery Reset Level | | 50 to 100% | 1 | 75% | Reset value for BSOC function. The displayed BSOC value must be below the selected level at key OFF before a fully charged battery will reset BSOC to 100% at key ON or reset to 100% when the voltage at key OFF is above the reset level. When battery BSOC level drops below this level, the truck will allow the BSOC to reset once Reset Voltage (Parameter 44) is reached.<br>**Model 8250:** This setting has no effect with the CAN based lithium-ion battery. |
| 25 | Not used | | | | | |
| 26 | Sequencing Delay | | 100 to 500 ms | 20 | 200 ms | Creates a sequencing delay to allow for throttle engagement just before the brake is released. The Sequencing Delay timer is enabled when the throttle is engaged while the brake is applied. If the timer expires before the brake is released, error code 'Hpd' is generated by the VM. |
| 27 | Not used | | | | | |
| 28 | Not used | | | | | |
| 29 | Fork Height / Travel Speed Configuration (Lift and Go) | | 0 to 15 | 1 | 0 | Monitors lift/lower requests. The travel speed is 'X' until the 'Y' condition is met.<br>0: No restrictions<br>1: X = 1 MPH, Y = Lift for 0.25 seconds<br>2: X = 1 MPH, Y = Lift for 0.50 seconds<br>3: X = 1 MPH, Y = Lift for 1.0 seconds<br>4: X = 1 MPH, Y = Lift for 1.5 seconds<br>5: X = 1 MPH, Y = Lift timeout not activated<br>6: X = 2 MPH, Y = Lift for 0.25 seconds<br>7: X = 2 MPH, Y = Lift for 0.50 seconds<br>8: X = 2 MPH, Y = Lift for 1.0 seconds<br>9: X = 2 MPH, Y = Lift for 1.5 seconds<br>10: X = 2 MPH, Y = Lift timeout not activated<br>11: X = 3 MPH, Y = Lift for 0.25 seconds<br>12: X = 3 MPH, Y = Lift for 0.50 seconds<br>13: X = 3 MPH, Y = Lift for 1.0 seconds<br>14: X = 3 MPH, Y = Lift for 1.5 seconds<br>15: X = 3 MPH, Y = Lift timeout not activated |
| 30 | iPORT Basic (only with iWAREHOUSE option) | | 0 or 1 | 1 | 0 | Requires the operator to login before truck operation is allowed.<br>0 = OFF<br>1 = ON<br>**NOTE:** This parameter is Read Only except when using FlashWare. |

| Parameter | Name | Unit | Range | Step | Default | Description |
|---|---|---|---|---|---|---|
| 31 | iALERT (only with iWAREHOUSE option) | | 0 or 1 | 1 | 0 | Requires the operator to login before truck operation is allowed. Enables caution and error codes to be sent to the iWAREHOUSE system.<br>0 = OFF<br>1 = ON<br>**NOTE:** This parameter is Read Only except when using FlashWare. |
| 32 | iCONTROL (only with iWAREHOUSE option) | | 0 or 1 | 1 | 0 | Requires the operator to login before truck operation is allowed. Enables remote setting of operator parameters.<br>0 = OFF<br>1 = ON<br>**NOTE:** This parameter is Read Only except when using FlashWare. |
| 33 | Not used | | | | | |
| 34 | iBATTERY (only with iWAREHOUSE option) | | 0 to 2 | 1 | 0 | Identifies the source of BSOC display information.<br>0 = OFF<br>1 = ON and displayed BSOC is from the TA<br>2 = ON and displayed BSOC is from the BSM<br>**NOTE:** This parameter is Read Only except when using FlashWare. |
| 35 | Not used | | | | | |
| 36 | Not used | | | | | |
| 37 | Not used | | | | | |
| 38 | Not used | | | | | |
| 39 | Access Code | | 1 to 4 | 1 | 1 | Defines whether truck has a Main ON/OFF switch or a keypad and what access level service personnel have to parameters.<br>1 = ON/OFF switch (no key pad) and driver parameters are open,<br>2 = ON/OFF switch (no key pad) and driver parameters are closed,<br>3 = keypad and driver parameters are open,<br>4 = keypad and driver parameters are closed.<br>**NOTE:** When driver parameters are closed, a Service Key or FlashWare is required to access driver parameters. |
| 40 | Not used | | | | | |

| Parameter | Name | Unit | Range | Step | Default | Description |
|---|---|---|---|---|---|---|
| 41 | Click-to-Creep | | 0 to 4 | 1 | 1 | Enables creep travel with the tiller arm in the fully raised or lowered position.<br>0: Disable<br>1: Approximately 0.5 MPH<br>2: Approximately 1 MPH<br>3: Approximately 1.5 MPH<br>4: Approximately 2 MPH<br>Activate click-to-creep by pressing the turtle button on the control head with the tiller in the brake position. “SLO” blinks on the display when in click-to-creep mode.<br>Deactivate by pressing the turtle button again. Also, the mode deactivates automatically after 10 seconds or immediately if the emergency reverse (belly button) switch is pushed. |
| 42 | Not used | | | | | |
| 43 | Not used | | | | | |
| 44 | Reset Voltage | Volts | 23.16 to 25.20 | 0.01 | 24.84 | Changing this value will automatically change the “Full Volts” voltage.<br>The Reset voltage will always be greater than the “Full Volts” voltage by at least 0.24 volts.<br>**Model 8250:** This setting has no effect with the CAN based lithium-ion battery. |
| 45 | Not used | | | | | |
| 46 | CAN Enabled HF Charger | | 0 to 2 | 1 | 0 | Enables CAN communication for Delta-Q charger with charger profile selection.<br>0 = Disabled (default)<br>1 = Discover AGM Profile<br>2 = Trojan Wet Cell Profile |
| 47 | Not used | | | | | |
| 48 | Not used | | | | | |
| 49 | Alternate Default Selection | | 0 to 1 | 1 | 0 | Provides a separate profile with custom default settings for seven specific parameters.<br>**Model 8250:** This setting has no effect with the CAN based lithium-ion battery. |
| 56 | Alternate Energy Source | | 0 to 2 | 1 | 0 | Only setting 2 will be allowed while the lithium-ion battery is connected to the truck via CAN.<br>0 = Disabled (default)<br>1 = Fuel Cell (Not Available)<br>2 = Lithium-ion Battery |
| 60 | Not Used | | | | | |

| Parameter | Name | Unit | Range | Step | Default | Description |
|---|---|---|---|---|---|---|
| 61 | Li-Ion Battery - SOC Threshold | 5% | 1 to 16 | 1 | 4 | Charge interval notification. State of Charge range is from 5 to 80%.<br>1 = 5%<br>2 = 10%<br>3 = 15%<br>4 = 20%<br>5 = 25%<br>6 = 30%<br>7 = 35%<br>8 = 40%<br>9 = 45%<br>10 = 50%<br>11= 55%<br>12 = 60%<br>13 = 65%<br>14= 70%<br>15 = 75%<br>16 = 80% |
| 62 | Li-Ion Battery - Alarm Interval | min. | 1 to 99 | 1 | 30 | Audible alarm sounds once every 10 seconds and the beacon flashes. Range is 1-99 minutes with the alarm active.<br>Alarm triggers are:<br>• low state of charge<br>• exceeded time for scheduled charge<br>• imminent shutdown<br>• travel request while attached to AC power source<br>• entering sleep mode<br>• exiting sleep mode (waking up) |
| 63 | Li-Ion Battery - Sleep Delay Timer (Key On) | | 0 to 48 | 1 | 2 | Verifies min/max/default timing for this parameter. Sleep delay starts when the amperage is below 6 amps, resets when amperage goes above 6 amps with key ON. Range is 0 to 48 in 30 minute increments.<br>0 = Sleep Delay disabled<br>2 = 1 hour |
| 64 | Li-Ion Battery - Sleep Delay Timer (Key Off) | | 0 to 48 | 1 | 2 | Verifies min/max/default timing for this parameter. Sleep delay starts when the key is OFF, disables when the key is ON. 0 causes immediate power OFF with key ON. Range is 0 to 48 in 30 minute increments. (for example 2 = 1 hour) |
| 65 | Li-Ion Battery - Dormant Mode | | 20 to 100 | 1 | 20 | Set state of charge between 20 and 100. Dormant mode error persists until state of charge is higher than this parameter. |
| 66 | Li-Ion Battery - Low SOC Threshold | | 5 to 10% | 1 | 10% | Set the low state of charge warning SOC%. This is the value where “b_Lo” error code appears on the Operator Display. Truck top speed is limited. |

# Display Part Numbers (Pn)

1. Enter the Special Truck Mode. See “Special Truck Mode” on page 3-5.
2. Press and hold the horn button until “Pn” is displayed. If the horn button is released at the wrong time, turn the truck OFF and start over.
3. Use the thumb control to scroll and display the following information:
   - HPn = Hardware part number (of VM)
   - HSn = Hardware serial number (of VM)
   - SPn = Software part number (of VM)
   - Type = Model number
   - PA = Curtis Power Amplifier Operating System
   - IPT = iPORT / RDP capable
   - bnc = Battery Management Controller
   - bnu = Battery Management Unit
   - LiUi = Lithium User Interface (HUD)
4. End displaying the part number by pressing the red OFF ( **O** ) key on the keypad (if equipped) and placing the Main ON/OFF switch in the OFF position.

# Display Test (d)

1. Enter the Special Truck Mode. See “Special Truck Mode” on page 3-5.
2. Press and hold the horn button until “d” is displayed. If the horn button is released at the wrong time, turn the truck OFF and start over.
3. During Display Test, all portions of each segment (as well as the icon below the segment) are illuminated. This occurs for each segment and icon pair. The sequence repeats until a re-key is performed.

Figure 3-6.

**Segment 1 and icon pair illuminated**

**Segment 2 and icon pair illuminated**

**Segment 3 and icon pair illuminated**

**Segment 4 and icon pair illuminated**

4. To end the Display Test, press the red OFF ( O ) key on the keypad (if equipped) and place the Main ON/OFF switch in the OFF position.

# Service Display

- Connect the service key (see "Special Tools" on page 3-21) in connection point J5, place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key to start the lift truck.
- The battery status is displayed in the alpha-numerical field (A) and the battery indicator symbol (D) is illuminated.
- Press horn button (S18) to toggle between Service Display modes. See Table 3-1.

*Figure 3-7. Operator Display*

**Explanation of Symbols:**

**A = Alpha-Numeric Field**

**B = Hour Meter Indicator**

**C = Parameter Indicator**

**D = Battery Indicator**

**E = Error Indicator**

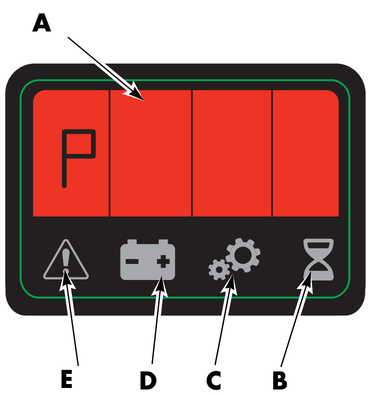



*Table 3-1. Service Display Modes*

| Flashing Symbol | Displayed Data |
|---|---|
|  | Throttle Request (Speed reference value) sent to traction amplifier (TA). -127 represents full throttle with forks-leading +127 represents full throttle with forks-trailing |
|  | Digital inputs/outputs from traction amplifier (TA) and vehicle manager (VM). * See explanation following |
|  | Battery voltage (V) at vehicle manager (VM) |
|  | Traction Motor RPM (increases as travel is requested) |
|  | Phase Current RMS |
|  | TA Temp. °C |
|  | Motor Temp. °C |

* Digital inputs/outputs are displayed by highlighting the figure segments of four of the alpha-numerical field figures (A) in the operator display. See "Digital Inputs/Outputs from Traction Amplifier and VM".

## Digital Inputs/Outputs from Traction Amplifier and VM

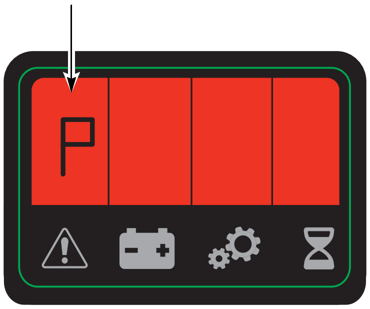
Select service display mode "C" detailed in the section "Service Display" on page 3-17. Also see Table 3-1, "Service Display Modes," on page 3-17.

The digital input/output functions are displayed by illuminating the figure segments of four of the numerical field figures in the operator display.

Service Display

## Traction Amplifier Inputs

The first most significant figure active (marked with arrow).



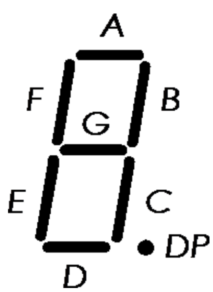
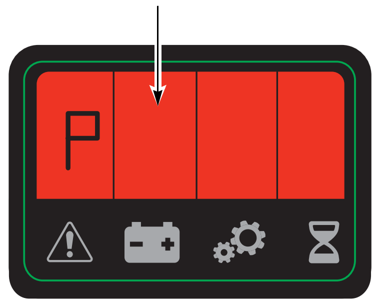




**Figure Segment**

| Figure Segment | Function |
|---|---|
| A | Brake Switch |
| B | |
| C | |
| D | Lift Switch |
| E | |
| F | |
| G | |
| DP | Brake Override Switch |

## Traction Amplifier Outputs

The second most significant figure active (marked with arrow).



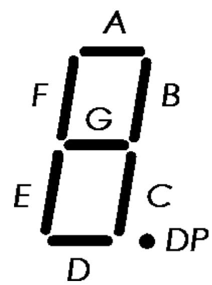
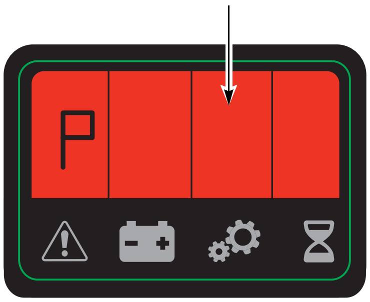




**Figure Segments**

| Figure Segment | Function |
|---|---|
| A | Main Contactor |
| B | Brake Switch |
| C | Lift Contactor |
| D | |
| E | |
| F | Travel Alarm |
| G | Lower |
| DP | |

## Digital Input from Vehicle Manager Control Sensors

The third most significant figure active (marked with arrow).



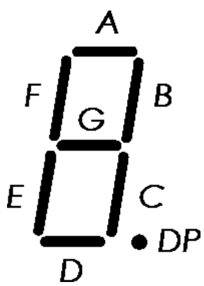
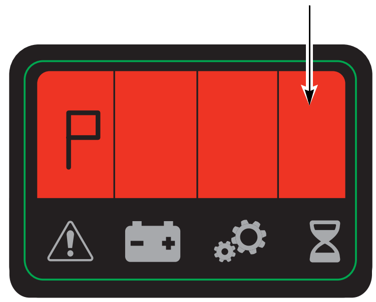




**Figure Segments**

| Figure Segment | Function |
|---|---|
| A | |
| B | |
| C | |
| D | Lift Switch |
| E | Lower Switch |
| F | Click-to-Creep |
| G | |
| DP | VM Input: Horn |

## Digital Input from Vehicle Manager Control Sensors

The fourth most significant figure active (marked with arrow).



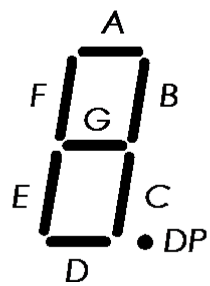




**Figure Segments**

| Figure Segment | Function |
|---|---|
| A | |
| B | |
| C | |
| D | |
| E | |
| F | |
| G | |
| DP | |

# Typical Power Amplifier Status

| Status | Comments |
|---|---|
| 1 | Motor drive, plug, or active neutral braking |
| 4 | Field reversal (transitional) |
| 6 | Disable (major fault, such as severe overvoltage) |
| 10 | Regen |
| 11 | Regen taper (transitional) |
| 13 | No Activity: Main contactor open because no output requested for 30 seconds or an error occurred. |
| 16 | Passive restraint |
| 17 | Active restraint while driving (overspeed) |
| 24 | Passive restraint (transitional) |
| 65 | Emergency reverse (drive) |
| 74 | Emergency reverse (regen) |
| 129 | Emergency shutdown while in drive |
| 144 | Emergency shutdown while in passive restraint |

1. The system mode shows details of traction amplifier activity, including during normal operation. It is different than, and does not correspond to, the code from the traction amplifier Status LED.
2. Throttle requests are not allowed for any system mode status of 32 and over.
3. Truck is in an emergency shutdown for any system mode status of 128 and over.

# Special Tools

The following tools are available from your local authorized Raymond Sales and Service Center.

*Table 3-2. Special Tools*

| Tool | Part Number | Purpose |
|---|---|---|
| Anti-static Field Kit | 1-187-059 | ESD Protection |
| Anti-static Wrist Straps | 1-187-058/001 | |
| Anti-static Wrist Strap Tester | 1-187-060/100 | |
| Crimp Tool | 1069861 | Crimp power cable lugs |
| Surge Protector | 154-010-801 | ESD/voltage surge protection for serial type FlashWare connections |
| USB/CAN Interface | 230489-001 | FlashWare connection to truck. See “FlashWare” on page 3-25 for details. |
| USB Cable | 163793BT | USB cable used with FlashWare connection. See “FlashWare” on page 3-25 for details. |

## Service Key

Use the optional Service Key (P/N 851-201-500) directly on the truck to troubleshoot and program the truck service parameter settings. See “Programming Service Parameters” on page 3-8.

*Figure 3-8. Service Key*

## Programmable Maintenance Tool

**NOTE:** A serial breakout harness is required to use the programmable maintenance tool on this lift truck.

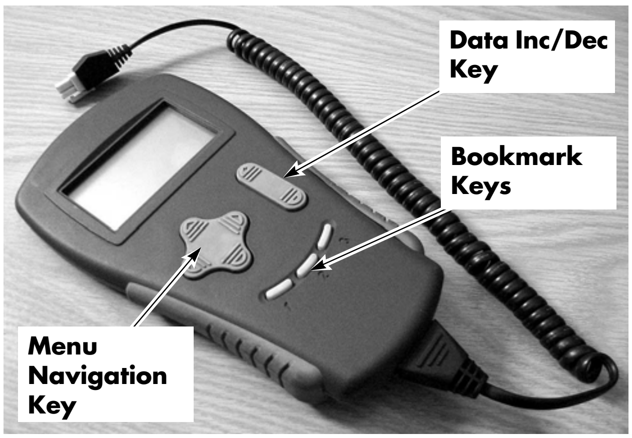
The optional Programmable Maintenance Tool (PMT) permits you to test and diagnose the power amplifier in the truck. See Figure 3-9.

*Figure 3-9. Programmable Maintenance Tool*

The PMT is powered by the traction amplifier through a four-pin connector at the bottom of the power amplifier.

The Menu Navigation Key moves the screen cursor up or down through the Menu list (top or bottom arrow), and opens or closes Sub-Menus (right or left arrows).

*Figure 3-10. Programmable Maintenance Tool*




The Data Inc/Dec Key changes the value of the parameter indicated by the cursor.

The Bookmark Keys allow you to quickly go back to your favorite selections without having to navigate back through the Menu.

- To select a position in the menu, hold a Bookmark Key down for three seconds until the bookmark set screen is displayed.
- To jump to a selected Bookmark position, press the appropriate Bookmark Key.

**NOTE:** The Bookmarks are not permanently stored in the PMT. They are cleared when the PMT is unplugged.

The Main Menu is the starting point for the PMT. The main menu displays menu titles:

- Program (not used)
- Monitor
- Faults
- Functions (not used)
- Information
- Programmer Setup

If there are no entries within a Menu, then the Menu title is not displayed.

## Monitor Mode

Real-time status information is displayed in the Monitor Mode for various inputs, outputs, temperature, and so forth. This information can be helpful in troubleshooting many problems and is useful for checking out the operation of the controller during initial installation.

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position.
2. Remove the tractor covers.
3. Connect the PMT to the power amplifier. Wait for the PMT to “boot up” before proceeding to the next step.
4. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.
5. Use the Menu Navigation Key to select Monitor.
6. Press the right arrow on the Menu Navigation Key to enter the sub-menu.
7. Use the Up and Down arrows on the Menu Navigation Key to scroll through the sub-menu list of monitor variables.
8. Use the right arrow to select and view a single variable.
9. To change the value of the parameter, use the Data Inc/Dec Key. Alternately, you can press the right arrow Menu Navigation Key once more and enter the detail screen. A bar graph appears as well as minimum and maximum data points. Change the parameter value by pressing the Data Inc/Dec Key. The new value is set as soon as the Data Inc/Dec Key is released.

To close a Monitor Menu, sub-menu, or detail screen, press the left arrow on the Navigation Key.

## Faults Mode

In System Faults mode, currently active faults detected by the controller are displayed.

In Faults History mode, the controller's diagnostic history file is displayed. This field includes a list of all faults observed and recorded by the controller since the history was last cleared.

**NOTE:** Each fault is listed only once, regardless of the number of times it occurred.

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position.
2. Remove the tractor covers.
3. Connect the PMT to the power amplifier. Wait for the PMT to "boot up" before proceeding to the next step.
4. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.
5. To view the present status of the unit, use the Menu Navigation Key to select Faults -> System Faults
6. To access the log, use the Menu Navigation Key to select Faults -> Fault History.
7. Use the Up and Down arrows on the Menu Navigation Key to scroll through the list of multiple faults.

### Clear

After you have diagnosed and corrected the problem, clear the diagnostic history file. This permits the power amplifier to accumulate a new file of faults. By checking the new history file at a later date, you can easily determine whether the problem was completely fixed.

1. To clear the fault history of the unit, use the Menu Navigation Key to select Faults -> Clear Fault History.
2. Press the increment arrow (+) for yes and the decrement arrow (-) to cancel and not clear the fault history.

"Test Mode Menu" on page 3-24 lists possible messages you may see displayed when the PMT is operating in either System Faults or Fault History mode.

## Information Mode

The Information Menu provides access to product information describing the basic revision level of the PMT.

To view, use the Menu Navigation Key to select Information in the Main Menu. Remember to press the right arrow to select a Menu. Press the left arrow to exit.

## Programmer Mode

Programmer mode permits you to perform a variety of tasks.

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position.
2. Remove the tractor covers.
3. Connect the PMT to the power amplifier. Wait for the PMT to "boot up" before proceeding to the next step.
4. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.
5. Use the Menu Navigation Key to select Programmer.
6. Press the right arrow on the Menu Navigation Key to enter the sub-menu.
7. Use the Up and Down arrows on the Menu Navigation Key to scroll through the sub-menu list of Programmer displays.
8. Use the right arrow to select and view a single variable.
9. Change the parameter value by pressing the Data Inc/Dec Key. The new value is set as soon as the Data Inc/Dec Key is released.

Special Tools

*Table 3-3. Test Mode Menu*

| Display | Explanation |
| --- | --- |
| THROTTLE% | Throttle reading, in % of full throttle |
| ARM CURRENT | Motor armature current, in amps |
| FIELD CURRENT | Motor field current, in amps |
| ARM PWM | Armature duty cycle (PWM) - applied as % |
| FIELD PWM | Motor field duty cycle (PWM) - applied as % |
| BATT VOLTAGE | Battery voltage (B+ and B– connection) at key switch input |
| CAP VOLTAGE | Voltage of power amplifier B+ (bus bar) |
| HEAT SINK TEMP | Heatsink temperature - °C |
| FORWARD INPUT | Forward switch: ON/OFF |
| REVERSE INPUT | Reverse switch: ON/OFF |
| MODE | Power Amplifier operating mode: 1 to 4 |
| INTRLCK INPUT | Interlock (brake/deadman) switch: ON/OFF |
| PEDAL INPUT | (not used) |
| EMR REV INPUT | Emergency reverse button: ON/OFF |
| MOTOR RPM | (not used) |
| MAIN CONT | Main contactor: ON/OFF |
| AUX CONT | (not used) |
| REV OUTPUT | (not used) |
| BRAKE OUTPUT | (not used) |
| FAULT 1 OUTPUT | (not used) |
| FAULT 2 OUTPUT | (not used) |
| CONTROL STATE | (not used) |
| MODSEL 1 | Mode select 1 switch: ON/OFF |
| MODSEL 2 | Mode select 2 switch: ON/OFF |

# FlashWare

## Overview

FlashWare allows you to interface with the truck software and update and change settings through the following features:

- Update Vehicle Manager software
- Erase Vehicle Manager software
- Reset Factory Default Settings
- Update Power Amplifier Software
- Diagnostics
- Display Error Log

For more detailed FlashWare information, click on Help and select Help Topics from the menu bar.

## Requirements

FlashWare can be installed on any IBM-compatible PC with Windows XP, Vista, or Windows 7 operating system. The PC communicates with the lift truck software through a 4-pin serial cable, standard USB cable, and dongle (USB to CAN translator) (P/N 230489/001).

## Installing FlashWare on PC

If you are a customer service technician, obtain FlashWare from your Raymond Sales and Service Center.

If you are a Raymond Sales and Service Center technician, obtain FlashWare from the iNet software download site.

To install FlashWare on the PC, double-click the installation file and follow the instructions on the screen. The software package is a self-extracting executable file. Read the "Read me" file in the software package for the latest detailed installation instructions.

## Connecting PC to Truck

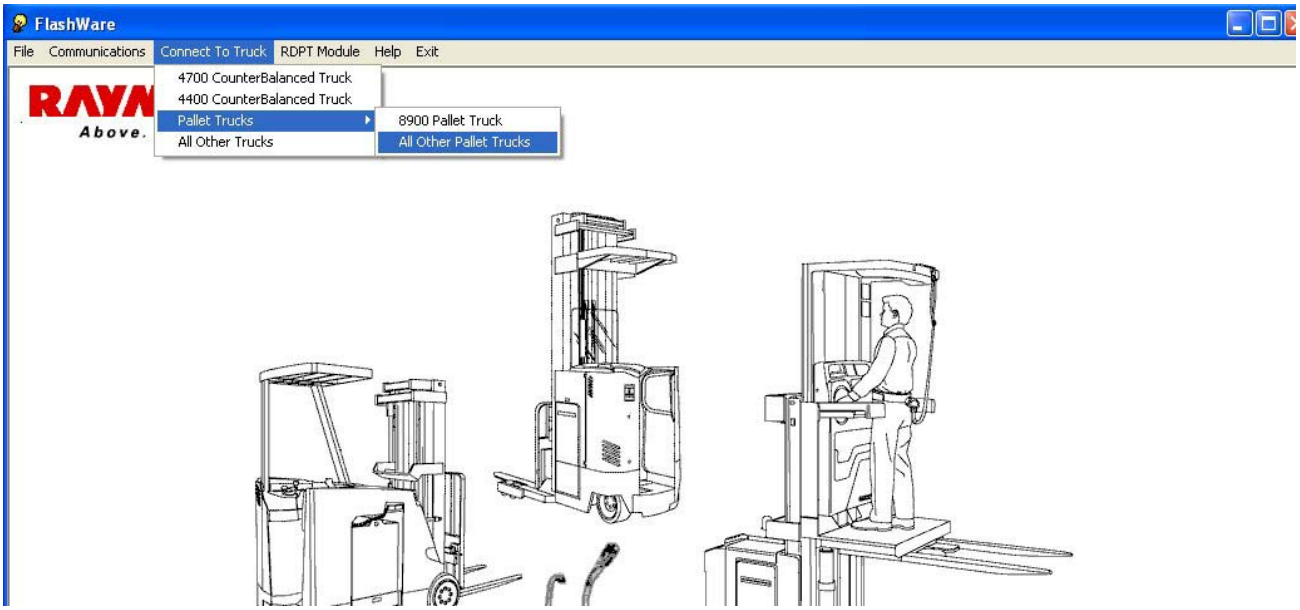
1. Turn the Main ON/OFF Switch to the OFF position.
2. Remove tractor covers.
3. Connect cable (P/N 163793A) to the USB/CAN Interface (dongle P/N 230489-001).
4. Connect the dongle to the PC with a standard USB cable.
5. Connect cable (P/N 163793A) to the J5 CAN Service Port connector.

## Starting FlashWare

1. Make sure the Main ON/OFF switch is ON.
2. Double-click the FlashWare icon on the main desktop screen or navigate via Start > Programs > FlashWare. The lift truck opening screen appears.
3. From the menu bar, click "Connect to Truck", then select "Pallet Trucks", then click "All Other Pallet Trucks". See Figure 3-11. If the PC is connected to a correctly functioning Vehicle Manager, the Truck Setup screen appears on your PC screen.
4. For detailed instruction on how to use FlashWare, follow the on-screen instructions in FlashWare Help.

FlashWare

*Figure 3-11. Truck Opening Screen with Menu*



# Section 4. Scheduled Maintenance

# Maintenance Guidelines

Following a regularly scheduled maintenance program:

- Promotes maximum truck performance
- Prolongs maximum truck life
- Reduces costly down time
- Prevents unnecessary repairs

Scheduled maintenance includes:

- Lubrication
- Cleaning
- Inspection
- Service

Perform all of the scheduled inspections and maintenance during the suggested intervals. The time intervals given in this guide are based on Deadman Hours (HD) under normal operating conditions.

When operating under Severe or Extreme conditions, perform these services more often as indicated in Table 4-1.

Refer to the “Lubrication Equivalency Chart” on page A-2 in the Appendix for lubrication equivalents. Refer to the manufacturer’s supplements for components not identified in this manual.

*Table 4-1. Maintenance Frequency Table*

| Operating Conditions | Working Environment | Service Frequency |
|---|---|---|
| Light to Moderate | An eight hour shift of basic material handling | First inspection at 90 days or 250 hours and then every 180 days or 500 hours, whichever comes first |
| Severe | • Extended heavy duty operation<br>• Freezer operation<br>• Sudden temperature changes such as going from freezer to room temperature | 60 days or 250 hours, whichever comes first |
| Extreme | • All UL Type EE rated lift trucks<br>• Dusty or sandy conditions such as in cement plants, lumber or flour mills, coal dust or stone-crushing areas<br>• High temperature areas such as in steel mills, foundries, enclosed (Type EE) applications<br>• Outdoor environments<br>• Corrosive chemical atmosphere, such as:<br>  • fish, meat or poultry processing plants, tanneries, or any other similar applications<br>  • chlorine or salt-sea air environments<br>• Adverse high humidity, wet, damp, or moist conditions | 30 days or 100 hours, whichever comes first |

# Initial 90 Day/250 Deadman Hours (HD) Maintenance

| **Perform the following maintenance tasks 90 days or 250 HD after the truck was put into service, whichever comes first.** | |
|---|---|
| **Component** | **Task** |
| Drive Unit | Clean magnet on drain plug. See “Drive Unit Housing Lubrication” on page 7-33. |
| Hydraulic Reservoir | Change hydraulic fluid. Clean screen and magnet. See “Filter Screen and Inlet Tube” on page 7-101. |
| Forks, Pull Rod, and Lift Linkage | Check for wear on the pull rod. Check all lift linkage bushings for wear. When replacing bushings, inspect the pins for wear. Replace the pins if any wear is evident. Check for wear at the lower link or fork heel. Replace any broken or missing roll pins. Adjust downstops as necessary. See “Downstop Adjustment” on page 7-116. |
| Electrical Cables | Inspect all cables for nicks or cuts (including battery pack and charger cables). Give special attention to those cables that are not stationary, for example, cables to the traction motor. Replace any cable that is damaged or shows signs of excessive heat. Failure to do so may cause intermittent system shutdowns and/or electronic failures. |
| Arm Angle Switch | Check the adjustment. See “Arm Angle Proximity Switch Adjustment” on page 7-81. |
| Traction Amplifier | Check bolt torque of all mounting and terminal hardware. |

# Every 180 Days or 500 Deadman Hours (HD)

For Severe or Extreme operating condition service intervals, see Table 4-1 on page 4-2 .

| Perform the following maintenance tasks every 180 days or 500 HD, whichever comes first. | |
|---|---|
| **Component** | **Task** |
| Battery | Check the weight labeled on the battery in the pallet truck against the minimum and maximum allowable weights on the spec tag for the pallet truck. Report any pallet trucks that are running with batteries below the minimum or over the maximum allowable weight. Inspect all battery connectors and leads for damage and cuts in protective coatings. Make sure the battery has no more than 0.5 inch (13 mm) free play in any direction. |
| Brakes | Traveling in an open area at walking speed or less, the empty lift truck should stop in 2 to 4 ft. (0.6 to 1.2 m), Test the brake in both the upper and lower brake positions. Traveling with a rated load and at top speed, the pallet truck should stop within approximately six feet. Stopping distance depends on the load, floor, and tire condition.<br>Examine for signs of oil on the friction disc and mating surfaces. If oil is present, disassemble the brake and replace the friction disc. Clean the mating surfaces.<br>Measure for correct gap between the coil housing and pressure plate. When applied, the gap should be between 0.001 to 0.013 in. (0.03 to 0.33 mm).Replace the friction disc when the air gap is greater than 0.0157 in. (0.40 mm). See “Air Gap Inspection” on page 7-36.<br>Make sure the rubber boot is correctly seated. |
| Control Handle Assembly | Make sure steering function is smooth and responsive, without binding or excess play. Verify lift/lower function is smooth and controllable. Verify travel function is smooth and responsive through full range of acceleration and plugging. Verify no codes on display. Verify function of all switches. Verify tiller arm returns to neutral when released. |
| Arm Angle Switch | Check the adjustment. See “Arm Angle Proximity Switch Adjustment” on page 7-81. |
| Drive Unit | Inspect for leaks. Check fluid level. Check drive unit axle for play. See “Drive Unit Housing Lubrication” on page 7-33. |
| Electrical Cables | Inspect all cables for nicks or cuts (including battery pack and charger cables). Give special attention to those cables that are not stationary, for example, cables to the traction motor. Replace any cable that is damaged or shows signs of excessive heat. Failure to do so may cause intermittent system shutdowns and/or electronic failures. |
| Forks, Pull Rod, and Lift Linkage | Check for wear on the pull rod. Check all lift linkage bushings for wear. When replacing bushings, inspect the pins for wear. Replace the pins if any wear is evident. Check for wear at the lower link or fork heel. Replace any broken or missing roll pins. Adjust downstops as necessary. See “Downstop Adjustment” on page 7-116.<br>Check upper link bolts for rotation. |
| Frame and Tractor Checks | General visual inspection of structural members for cracks, including but not limited to the tractor and forks. |

## Every 180 Days or 500 Deadman Hours (HD)

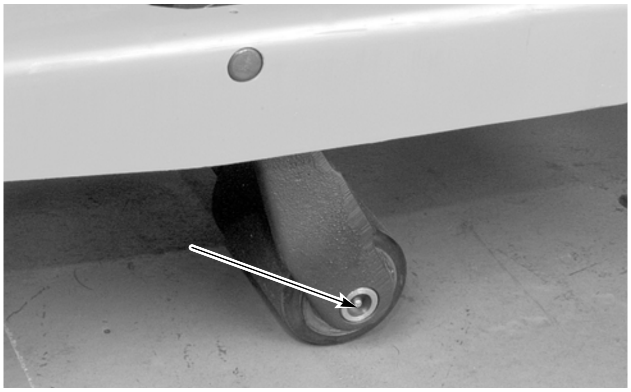
| Perform the following maintenance tasks every 180 days or 500 HD, whichever comes first. | |
|---|---|
| **Component** | **Task** |
| Hardware | Check bolt torque of major components (motors, brake, bumper, drive unit, and hydraulic unit). Tighten any loose hardware. Replace any broken or missing hardware. See "Appendix" on page A-1. |
| Horn | Check that the horn operates when you press the horn button. |
| Hydraulic Hoses | Inspect all hydraulic hoses for leaks, nicks, cut, chafing, and bulges. Replace damaged hoses as soon as possible. Inspect all fittings for leaks. Repair any leaks immediately. |
| Hydraulic Reservoir | Check fluid level. See "Hydraulic Fluid" on page 7-95. Add fluid if necessary. |
| Lubrication | Lubricate all grease points. Lubricate all pivoting shafts on the control handle with spray lube. See "Grease Fittings" on page 4-7. |
| Motors | **AC** (traction) - Check the cable lugs to make sure they are tight to the terminal studs. The outside nut should be torqued to the values listed on page 7-85. Replace any cable that shows signs of excessive heat. Check sensor wires for sound connection and condition.<br>DC (lift) - Visually inspect brushes for excessive heat (discoloration of the pigtails). If excessive heat is evident, inspect the armature circuit for loose connections. Check condition of commutator per photos on page 5-9. Find the shortest brush in the holder. Remove the brush and check overall dimension as per the chart on page 7-83. Inspect the brush for even wear over the full surface of the brush. If the brush is not contacting the complete surface, replace the brushes. Inspect the brush rigging for damage or loose brush holders. Make sure that the connections on the brush leads are tight. Check brush spring tension per the chart on page 7-84. Blow out the inside of the motor with compressed air. Check the cable lugs to make sure they are tight to the terminal studs. Both the inside and outside nut should be torqued to the values listed on page 7-85. Replace any cable that shows signs of excessive heat.<br>See "DC Motors, General" on page 7-83. |
| Shorts to Frame | Check for electrical shorts to frame. See "Shorts to Frame Test" on page 5-6. Wipe down the inside of the compartment. |
| Switches | Check all switches for correct operation and adjust as needed. |
| Ventilation Slots | Make sure ventilation slots in the tractor grille cover are clear of obstructions and debris. |
| Warning Decals | Replace any unreadable or damaged decals. |
| Load Wheels and Drive Tires | Examine for bond failure, chunking, and excessive or uneven wear. Inspect the load wheel bearings for binding or excessive play. Apply grease. |
| Casters (Optional) | Examine tire for condition. Verify that tires rotate freely. |

# Every 360 Days or 2000 Deadman Hours (HD)

| **Perform all 180 day/500 deadman hour maintenance tasks plus the following every 360 days/2000 HD.** | |
|---|---|
| **Component** | **Task** |
| Drive Unit | Change fluid. Clean magnet on drain plug. See “Drive Unit Housing Lubrication” on page 7-33. |
| Hydraulic Reservoir | Change hydraulic fluid.<br>Clean screen and magnet. See “Filter Screen and Inlet Tube” on page 7-101. |

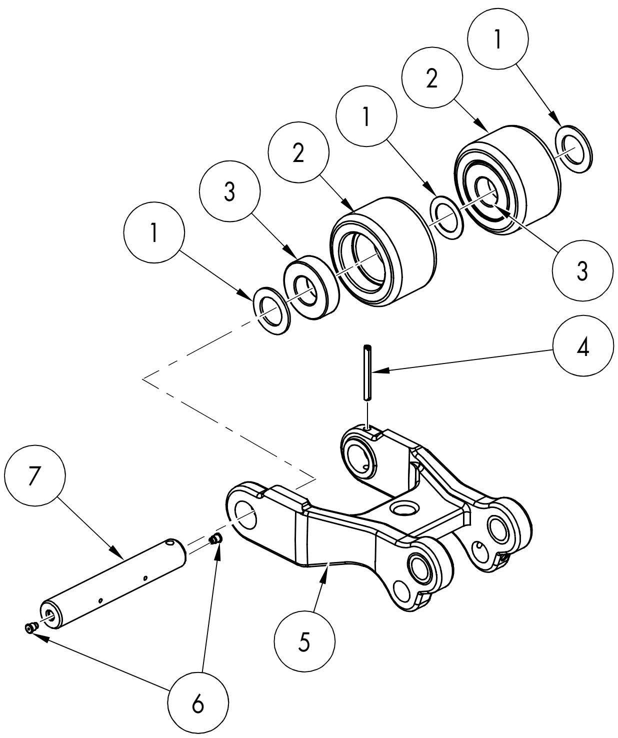
# Grease Fittings

*Figure 4-1. Load Wheel Axle (Single Wheel)*



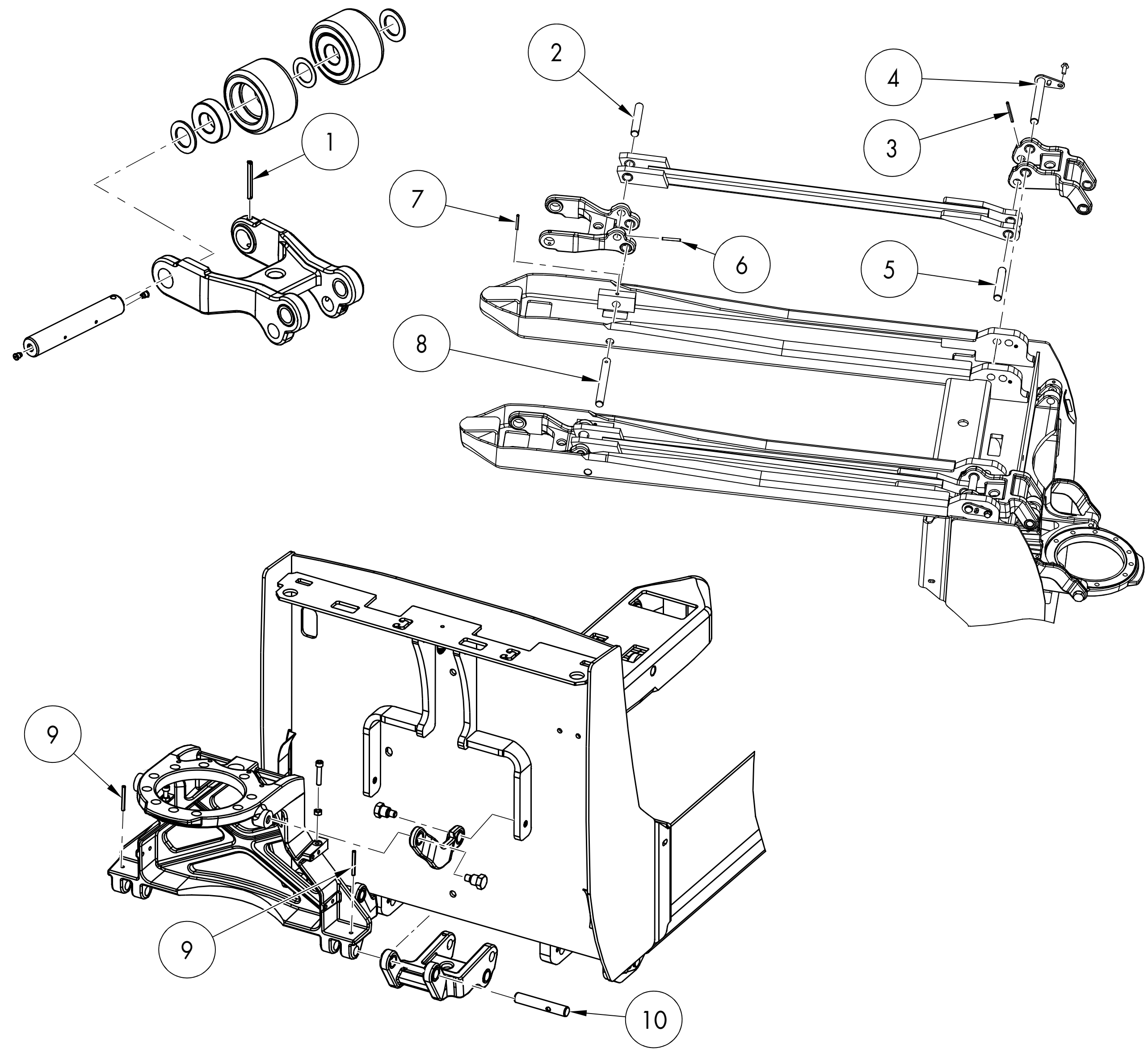
Dual load wheels have a grease fitting [6] in each end of the wheel axle [7].

*Figure 4-2. Dual Load Wheel (item 6)*



# Pin Locations

*Figure 4-3: Pin Locations*




| Legend: | | | |
|---|---|---|---|
| 1 | Load Wheel Roll Pin | 6 | Roll Pin |
| 2 | Wheel Fork Shaft | 7 | Roll Pin |
| 3 | Roll Pin | 8 | Wheel Fork/Tractor Frame Pin |
| 4 | Lower Link/Tractor Frame Pivot Pin | 9 | Roll Pin |
| 5 | Pull Rod/Lower Link Pin | 10 | Lower Link Pin |

# Section 5. Troubleshooting

# List of Troubleshooting Charts and Tables

# Electrical Troubleshooting Guidelines

## General

**⚠ WARNING**

**Block the lift truck so that the drive tire is off the floor whenever a troubleshooting procedure requires turning the truck ON. This prevents accidents caused by unexpected lift truck travel.**

**Use extreme care when the truck is jacked up for any reason. Keep hands and feet clear while jacking the lift truck. After the lift truck is jacked, place solid blocks or jack stands beneath it to support it. *Do not* rely on the jack alone.** **See "Jacking Safety" on page 2-10.**

**⚠CAUTION**

**Unless otherwise directed, disconnect the battery connector when you check electrical circuits or components with an ohmmeter. Electrical current can cause damage to the ohmmeter.**

Many problems are caused by a faulty or dirty battery. Make sure the battery is clean. Check the electrolyte level and state-of-charge. See "Lead Acid Battery" on page 7-40.

Save time and trouble by looking for simple causes first.

Visually inspect all wiring and electrical components for:

- Loose connections or connectors
- Loose or broken terminals
- Damaged terminals, blocks, or strips
- Broken wiring and shorted conditions (especially those that are close to metal edges or surfaces)

Use an ohmmeter to check for wiring continuity and shorts.

Use a Digital MultiMeter (DMM), such as a Fluke meter, for all measurements. Analog meters can give inaccurate readings and load down sensitive electronic circuits enough to cause failure. Make sure meter cables are connected to the correct meter jacks and that the correct function and scale are selected.

When measuring voltage, connect the positive meter lead to the connector or probe point marked (+) in the test. Connect the negative meter lead to the connector or probe point marked (–).

Whenever measuring resistance, turn the truck OFF and disconnect the battery connector. Battery current can damage an ohmmeter. Isolate the component from the circuit.

For troubleshooting DC electric motors, see "DC Electric Motors" on page 5-8.

For troubleshooting AC electric motors, see "AC Electric Motors" on page 5-14.

For information on pin, connector, and harness connections, see "Wiring Harness" on page 7-68.

## Shorts to Frame

Shorts to frame is an industry term for unintentional current leakage paths between normally isolated electrical circuits and their metal enclosures.

Shorts to frame may be metallic connections, such as a wire conductor contacting metal through worn insulation. More often, shorts to frame are resistive "leakage" paths caused by contamination and/or moisture.

These leakage paths can result in unwanted electrical noise on the metallic pallet truck structure and can cause incorrect operation.

Shorts to frame are caused by:

- Accumulation of dirt
- Battery electrolyte leakage
- Motor brush dust
- Motor brush leads touching the housing
- Breakdown in insulation
- Bare wires
- Pinched wiring harness
- Incorrect mounting of circuit cards

Shorts to frame can occur at numerous locations on a pallet truck, including:

- Batteries
- Motors
- Cables, wiring, and harnesses
- Heatsinks
- Bus bars
- Solenoids
- Contactors
- Terminal strips
- Switches
- Power panel insulation
- Circuit card mounts

## Shorts to Frame Test

To test for shorts to frame:

1. Turn the Main ON/OFF switch OFF and disconnect the battery connector.
2. To test the battery for shorts to case, connect a 12V test light to the battery case from battery B+, and then to the battery case from battery B–. If the light illuminates at all, even momentarily, there is a **serious** problem with the battery, either external contamination or internal damage. Do *not* continue until this condition is corrected. The meter may be damaged if you proceed before correcting this condition.
   a. Install another battery in the truck and repeat this procedure from Step 1.
   b. If the test light does not light, continue to the next step.
3. Use a DMM set on the ampere function to measure the current leakage from the battery case to battery B+ and from the battery case to battery B– (*not truck B+ and B–*). Begin measuring at the highest ampere scale and work toward the lowest. A reading of more than 0.001A (1mA) indicates a **serious** short. Do *not* continue until this condition is corrected. The meter may be damaged if you proceed before correcting this condition.
   a. Install another battery in the pallet truck and repeat this procedure from Step 1.
   b. If the current is less than 0.0002A (0.2mA), go to Step 4. If the current is greater than 0.0002A (0.2mA) and less than 0.001A (1mA), remove the battery from the truck, then continue with Step 4. Make sure the battery case does not touch the truck frame during the remaining tests.
4. With the battery disconnected (or removed and disconnected) from the truck, use a DMM to measure the resistance from truck frame to truck B+, to truck B– (*not battery B+ and B–*), and to all fuses and motors. A reading of less than 1000 ohms indicates a **serious** short. Do *not* continue until this condition is corrected. The meter may be damaged if you proceed before correcting this condition. To correct the condition, follow steps a through e.
   a. To identify the cause of the short to frame, disconnect circuit components until the low resistance condition disappears. Do not reconnect components one at a time, but leave them disconnected until the low resistance reading disappears. Prevent disconnected terminals or connectors from touching the truck frame or other conductive surfaces.
   b. The most likely areas to check are:
      - Motors
      - Heatsinks
      - Power cables
      - Power circuit components
      - Control circuit components

c. Repair or replace the component causing the low resistance condition. Repeat Step 4.

d. Reconnect all other components previously disconnected, one at a time, measuring resistance between steps. If a reading is less than 1000 ohms when reconnecting a component, that component or its wiring is faulty; repair or replace as appropriate.

e. When, after all components are reconnected, and you get readings greater than 1000 ohms, continue with the next step.

5. Reconnect the battery connector and turn the Main ON/OFF switch ON. If the battery was previously removed, make sure the battery case does not touch the truck frame.

**NOTE:** The functions being checked must be energized. Example: to check for shorts to frame in the travel circuit, travel must be requested.

6. Use a DMM set to the current function to measure current leakage to the truck frame from truck B+, B– (*not battery B+ and B–*), and all fuses and DC motor terminals. Begin measuring at the highest ampere scale and work toward the lowest. If the current is less than 0.001A (1mA), go to step 7. If the current is more than 0.001A (1mA), continue with the following steps.

   a. To identify the cause of the short to frame, disconnect circuit components until the leakage current reads less than 0.001A (1mA). Do not reconnect components one at a time, but leave them disconnected until the leakage current reads less than 0.001A (1mA). Prevent disconnected terminals or connectors from touching the truck frame or other conductive surfaces.

   b. The most likely areas to check are:
      - Motors
      - Heatsinks
      - Power cables
      - Power circuit components
      - Control circuit components

   c. Repair or replace the component(s) causing the leakage current. Repeat Step 6.

   d. Reconnect all other components previously disconnected, measuring current between steps. If a reading is more than 0.001A (1mA) when reconnecting a component, that component or its wiring is faulty. Repair or replace as appropriate.

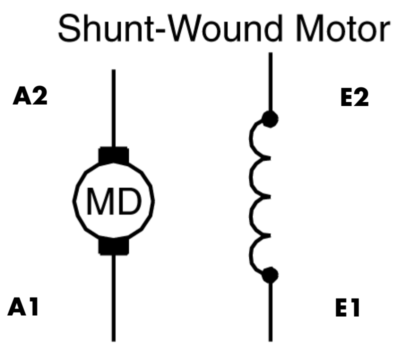
7. When, after all components are reconnected, you get a reading less than 0.001A (1mA) there is no short to frame condition with the truck or the battery. If you previously removed the battery from the truck, re-install the battery.

# DC Electric Motors

## DC Motor Types

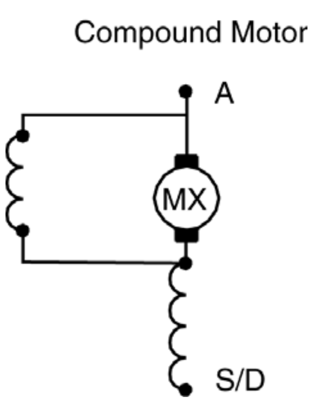
- A shunt-wound motor has four external connections: two armature (A) and two field (E). See Figure 5-1.

*Figure 5-1. Shunt-Wound Motor Circuits*




- A compound motor, such as the lift motor, has only two external connections because the armature and field windings are connected internally. See Figure 5-2.

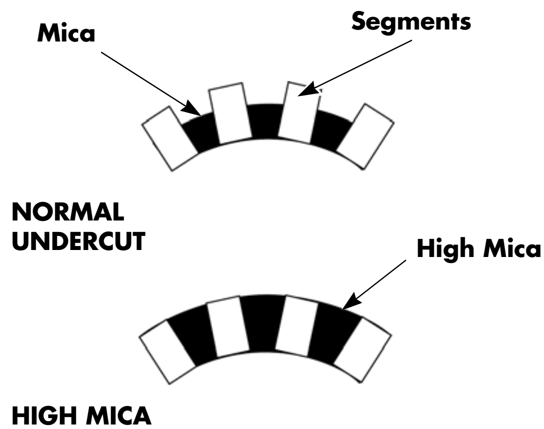
*Figure 5-2. Compound Wound Motor Circuits*




## Inspection

Inspect the commutator for surface condition and high mica. Most armatures have the mica undercut. If the armature on your motor does not, do not attempt to cut it.

*Figure 5-3. Mica Inspection*




The commutator must be smooth and clean to provide maximum brush life. When commutators are not correctly maintained, carbon dust can collect in the grooves between the segments. This can lead to a short circuit in the armature.

Remove the brushes from their box. Inspect the contact surface and brush length. If the brush surface has groove(s) or pit marks, it indicates the presence of a burr on the commutator surface.

Good commutation is indicated by a dark brown polished commutator and an evenly polished brush wearing surface. See Table 5-1, “Commutator Surfaces.”

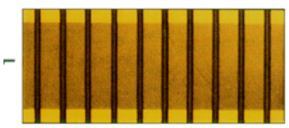
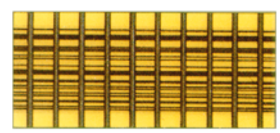
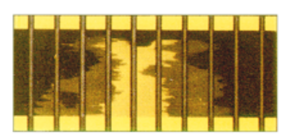
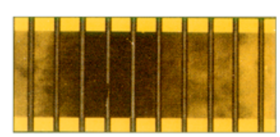
If the commutator appears rough, pitted, or has signs of burning or heavy arcing between the commutator bars, remove the motor for servicing.

## Service

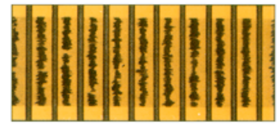
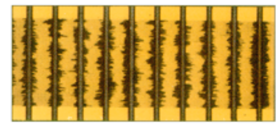
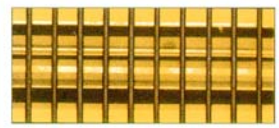
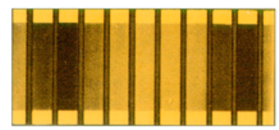
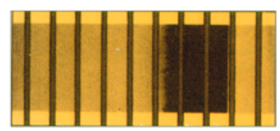
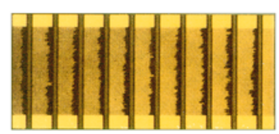
Some conditions, identified in Table 5-1, may be resolved by cleaning the commutator with a special polishing stone. Polishing the commutator should only be attempted for the specified conditions. Refer to “Polishing the Commutator” on page 7-85 for instructions.

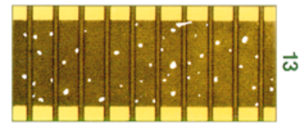
Some conditions can be resolved by recutting the commutator on a lathe. This requires special equipment and training and should only be attempted at a qualified DC motor repair facility.

*Table 5-1. Commutator Surfaces*

| Condition | Probable Cause | Commutator Surface |
|---|---|---|
| Good Condition-Light Film | Uniform coloring indicates satisfactory operation of the machine and brushes. Acceptable film color may vary from light to dark due to film thickness. | 1  |
| Satisfactory Condition-Light and Dark Pattern | This condition can appear in alternating bars as shown or every 3rd or 4th bar. This is caused by imbalances in the windings or other motor variances. It is not an indication of commutator problems. The motor should have no issues running, but the condition may indicate a need for more frequent inspections. |  3 |
| Unsatisfactory Condition-Streaky Film With No Commutator Wear | Threading or streaked film on the commutator surface. May be accompanied by rattling or loud operation. This condition is frequently due to incorrect brush grade. Environmental conditions, such as freezer or low humidity, can be a contributing factor. This condition may be repaired with a polishing stone if caught soon enough. Refer to “Polishing the Commutator” on page 7-85. |  4 |
| Unsatisfactory Condition-Uneven Film | Patchy colors of varying densities and shape. This condition is generally due to unclean operating conditions and does not indicate damage to the commutator. Clean the commutator with a polishing stone and compressed air (see “Polishing the Commutator” on page 7-85). Consider more frequent scheduled maintenance intervals. |  5 |
| Unsatisfactory Condition-Film With Dark Areas | Dark areas can be isolated or regular. This condition indicates a commutator is out-of-round. Continued use may result in failure and damage to other systems. The motor should be replaced or recut at a qualified repair facility. |  6 |

DC Electric Motors

| Condition | Probable Cause | Commutator Surface |
|---|---|---|
| Unsatisfactory Condition-Example of Incorrect Commutator Machining | Bars are low on entry and leaving edges causing the brushes to ride on the middle of the bars. Continued use may result in failure and damage to other systems. The motor should be replaced or recut at a qualified repair facility. Consider more frequent scheduled maintenance intervals. |  7 |
| Unsatisfactory Condition-Example of Incorrect Commutator Machining | Bars are low in the middle causing the brushes to ride on the entry and leaving bar edges. Continued use may result in failure and damage to other systems. The motor should be replaced or recut at a qualified repair facility. Consider more frequent scheduled maintenance intervals. |  8 |
| Unsatisfactory Condition-Streaky Film With Commutator Wear | This is a further development of the third example. Earlier corrective action should have been taken. Continued use may result in failure and damage to other systems. The motor should be replaced or recut at a qualified repair facility. Consider more frequent scheduled maintenance intervals. |  9 |
| Unsatisfactory Condition-Double Pole Pitch | Darkening of commutator in sequences two pole pitches apart. This condition may result from an armature fault or some other fault. Continued use may result in failure and damage to other systems. The motor should be replaced or recut at a qualified repair facility. |  10 |
| Unsatisfactory Condition-Brush Contact Mark | This condition may result from storage of the machine for long periods with brushes in position. It may also result from operation in prolonged stall conditions. Storage marks are easily removed with a polishing stone, see "Polishing the Commutator" on page 7-85. Stall marks should be cleaned and inspected for damage or shorts. The motor should be replaced if damage has occurred. |  11 |
| Unsatisfactory Condition-Bar Edge Burning-Cause High Mica | This condition is caused by high mica in every slot and can be caused by incorrect or excessive stoning. The same effect can occur on one bar only. Chattering or loud operation may also be noticed. Continued operation will result in increased brush wear and may result in commutator damage. The motor should be replaced or recut at a qualified repair facility. |  12 |

| Condition | Probable Cause | Commutator Surface |
|---|---|---|
| Unsatisfactory Condition-Small Bright Spots | This condition is due to sparking under the brush. This is caused by overloaded machines or low brush pressure. Check the brush spring tension, see ““Motor Brush Spring Tension” on page 7-84. If not corrected, this condition will result in damage to the commutator. This condition may be repaired with a polishing stone if caught early and no scoring has occurred, see “Polishing the Commutator” on page 7-85. If damaged, the motor should be replaced or recut at a qualified repair facility. |   |

# Open Circuit Motor Test

An open circuit is an electrical circuit within the motor that is broken. This can be caused by:

- bad brushes or brush springs
- a broken wire in the field or armature winding
- a loose or bad connection

To test a motor for an open circuit:

1. Isolate the motor from the truck circuit by removing the power cables. Use two wrenches to avoid twisting the terminal stud.
2. With the motor at room temperature, connect the leads of a digital ohmmeter between the individual circuits in the motor.
3. Observe the measurements given in Table 5-2.
4. If the meter indicates high resistance in the armature, check the condition of the brushes before replacing the motor.
5. If an open circuit is found in a motor, the motor must be disassembled by a motor rebuilding facility to isolate the problem to the field or armature circuit.

*Table 5-2. Lift Motor Resistance Readings*

| Terminals | Acceptable Resistance Readings | Unacceptable Resistance Readings |
|---|---|---|
| A to S | 0.020 to 0.030 ohms | Greater than 0.1 ohms |
| **NOTE:** Measure all readings at room temperature. See Figure 5-2 on page 5-8. | | |

# Grounded Motor Test

In a grounded motor, an unintentional electrical connection exists between the current-carrying conductors and the motor housing. This can be caused either by direct contact or through conductive foreign material.

The ground may be caused by:

- insulation breakdown
- brush leads touching the motor housing
- build-up of carbon dust or other materials

To test a motor for grounds:

1. Isolate the motor from the truck circuit by removing the power cables. Use two wrenches to avoid twisting the terminal studs.
2. Attach one lead of a megohm meter or a digital ohmmeter to a motor terminal and the other lead to an unpainted surface of the motor housing. Set the ohmmeter to the highest scale.
3. If the ohmmeter reads resistance of less than 100,000 ohms, the motor is grounded. Clean, repair, or replace the motor as necessary.

# Short-Circuited Armature

A short circuit in the armature causes heating, and can result in burning of:

- Armature coil
- Brush wires
- Commutator segments

Visual inspection may reveal this condition.

Positive determination of a short-circuited armature requires special equipment at a motor rebuilding facility.

## Short-Circuited Winding

A short-circuited winding is one where the insulation on the field or armature has broken down at two or more points. The breakdown creates a low resistance path, permitting current to flow from one turn of the coil to another adjacent coil turn, without actually flowing through the coil wire. The result is a decrease in the total resistance of the motor winding and an increase in the current flow. The severity of the short circuit depends on its location.

A shorted motor may be indicated by:

- Slow or sluggish operation
- Running faster than normal
- Overheating
- Blowing a power fuse
- On DC motors, severe burning or discoloring on one or two commutator segments every 90° of rotation

These symptoms can also be caused by problems other than the motor itself, such as:

- Brake too tight or dragging
- Wheel bearings too tight
- Faulty transmission
- Binding in an attached pump

Testing a motor for short-circuited windings requires special equipment at a motor rebuilding facility.

# AC Electric Motors

## AC Motor Type

The traction motor is a brushless, 3-phase, variable speed AC motor.

The AC motor has a rotor (in place of the DC armature) and a stator (in place of the DC field). There is no electrical connection to the rotor; current is induced in the rotor. The stator has three windings staggered 120° apart, and three external connections labeled U, V, and W. See Figure 5-4 and Figure 5-5.

*Figure 5-4. AC Traction Motor circuit - Phase A*

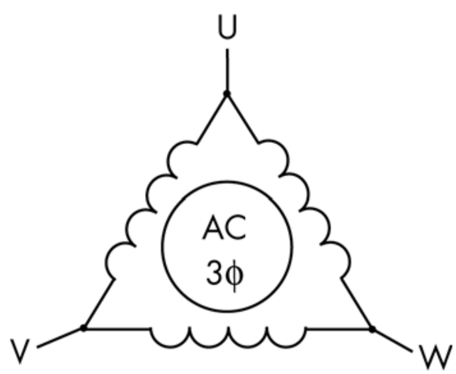
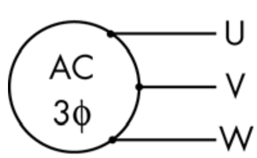



*Figure 5-5. AC Traction Motor circuit - Phase B*




## Open Winding

If the AC motor fails with an open winding, the motor moves erratically, as if hunting/searching, and there is a ticking sound. Rotation is much slower than normal.

Using a clamping ammeter, measure current in each of the motor power cables. The open phase reads significantly lower than the other two phases.

## Shorted Winding

If the AC motor fails with a shorted winding, the motor speed fluctuates up and down, as if hunting/searching, and there is a high-pitched sound.

Using a clamping ammeter, measure current in each of the motor power cables. The shorted phase reads significantly higher than the other two phases.

# Hydraulic Troubleshooting Guidelines

When you measure the voltage at solenoids, make sure the hydraulic lines and components are fully installed.

Use an ohmmeter to measure wiring continuity to solenoids. Use an ammeter to measure for the correct current to the solenoids and contactor coils.

**⚠CAUTION**

**Unless otherwise directed, disconnect the battery connector when you examine electrical circuits or components with an ohmmeter. Electrical current can damage to the ohmmeter.**

Visually inspect all hydraulic lines and components for:

- Leaking connections or connectors
- Loose or broken fittings
- Damaged tubing, hoses, vents, or seals

Inspect the hydraulic system for the correct pressure and that the relief valve is functioning correctly. See "Hydraulic Pump Pressure Relief Valve Adjustment" on page 7-103.

Check the hydraulic reservoir fluid level. If necessary, add fluid; fill until fluid is visible to within 1 in. (25.4 mm) below the fill port elbow. The usable reservoir capacity is 0.9 qt. (0.85 L). See "Lubrication Equivalency Chart" on page A-2.

**NOTE:** Always check hydraulic fluid level with forks fully lowered and when hydraulic fluid is cold.

Cap any open hydraulic lines to prevent contamination.

# Symptom Table: Electrical System

**NOTE:** Reference electrical schematics.

## BDI Does Not Reset to 100% (Lead Acid Battery Only)

| Possible Cause | Action |
|---|---|
| Truck not configured correctly | Verify the truck is configured for the type of battery installed. Verify parameter 21 setting. |
| Incorrect wiring | Verify that the wire from the battery connector is attached to the right terminal of the Main Contactor (MPC), that the Traction System Fuse (FU1) cable connects to the left MPC terminal, and the Lift Motor Fuse (FU2) connect the left MPC terminal to the right LPC terminal, and that the wire from the right terminal of the Traction System Fuse (FU1) is attached to the Traction Amplifier at the B+ terminal. |
| Low static voltage | If the voltage is below 25.2 volts after a charging cycle, charge the battery pack 5 times consecutively. |
| **NOTE:** The voltage must be at or above 25.2 volts for the truck to reset and stay at 100%. Parameter settings can also be a factor here. Verify parameter 24 (battery reset voltage) is set higher than 75%. | |
| Low voltage during travel | Make sure that the voltage is above 25.2 volts at key ON. |
| Main contactor wired incorrectly | Verify that the wire from the battery connector is attached to the right terminal of the Main Contactor (MPC), that the Traction System Fuse (FU1) cable connects to the left MPC terminal, and the Lift Motor Fuse (FU2) connect the left MPC terminal to the right LPC terminal, and that the wire from the right terminal of the Traction System Fuse (FU1) is attached to the Traction Amplifier at the B+ terminal, |
| Low voltage at key ON | If voltage is below 25.2 volts at key ON after charging the battery, troubleshoot the battery and charger. |
| Poor wiring connection or bad VM | 1. Check the battery voltage displayed on the VM using the “Service Display” on page 3-17.<br>2. Compare the displayed voltage to the voltage read at the battery using a volt meter.<br>3. If there is more than a 0.5 volt difference, measure the voltage at the VM, J11-1 with respect to J11-3.<br>4. If the voltage is the same as what was read at the battery, replace the VM.<br>5. If the voltage is different, troubleshoot the wiring between the VM and the battery.<br>6. If no poor connections are found, replace the VM. |

## Horn Does Not Sound When Horn Button Pushed. No Fault Codes.

| Possible Cause | Action |
|---|---|
| Faulty horn switch/wiring in the handle head | Repair or replace as necessary. |
| Faulty Horn | Replace the horn. |
| Faulty traction amplifier | Replace if necessary. |

## Green/Red LED on Keypad Not Illuminated When Key Pressed.

| Possible Cause | Action |
|---|---|
| No power supplied to the vehicle manager | Check the control wires in the control handle arm harness. |
| Faulty Connectors | Repair and replace the faulty connectors (check connector J11 at the control handle arm base, the connector into the vehicle manager (JPX3), and connector J1 to the motor controller). |
| FU3 | Make sure the control fuse has not opened. |
| Key switch jumper not connected | Connect the key switch jumper. |
| Faulty LED(s) | Check the LED(s) if the truck functions normally but the LED(s) are not lighting correctly. |

# Symptom Tables: Lift/Lower System

**NOTE:** If it is determined that a component failed as a result of hydraulic fluid contamination, replace the failed component and flush, fill, and bleed the hydraulic system. See “Hydraulic Fluid” on page 7-95.

**NOTE:** Use these symptom tables in conjunction with the electrical schematics and hydraulic schematics.

## No Lift, Lift Motor Does Not Run, Travel is OK

| Possible Cause | Action |
|---|---|
| Bad lift pump contactor | With the main ON/OFF switch ON and the lift switch depressed, do the main contactor (MPC) and the pump contactor (LPC) energize? If not, measure the voltage between the pump contactor coil terminals. If B+ voltage is present, replace the pump contactor. |
| Bad VM | With the main ON/OFF switch ON, the PIN-key code entered, the green ON ( ❙ ) key on the keypad pressed, and the lift switch depressed, does the pump contactor (LPC) energize? If not, check for lift input at the CAN (see “Service Display” on page 3-17). If no input is detected, replace the VM.<br>**NOTE:** Check for mechanical problems in the handle head before replacing the VM. |
| Bad wiring continuity or wiring shorts | With the main ON/OFF switch OFF and the battery disconnected, check for continuity and wiring shorts in the wiring harness or cables. |
| Bad lift pump motor brushes... | With the main ON/OFF switch ON, the PIN-key code entered, the green ON ( ❙ ) key on the keypad pressed, and the lift switch depressed, does the pump contactor (LPC) energize? If YES, measure the voltage between B– and the lift pump motor terminal. If OK, check the lift motor brushes... |
| ...or bad lift pump motor... | ...if the brushes are OK, test the pump motor. |

## No Lift or Slow Lift, Lift Motor Does Run

| Possible Cause | Action |
|---|---|
| Battery problems | Lead acid battery: Replace the battery with a fully-charged battery.<br>Lithium-ion battery: Check charge status LEDs for charge level. |
| Mechanical binding in the lift mechanism | Repair the lift mechanism. |
| Low hydraulic pressure or relief valve setting | Check the hydraulic pressure setting and adjust if necessary. See "Hydraulic Pump Pressure Relief Valve Adjustment" on page 7-103. |
| Relief valve contaminated or bad. (Unable to get the correct hydraulic pressure) | With the main ON/OFF switch OFF and the battery disconnected, remove the pressure relief valve. Check for damage or contamination. Replace the valve or flush the system and replace the hydraulic fluid. See "Hydraulic Fluid" on page 7-95. |
| Bad lift pump motor or brushes | Check the lift pump motor brushes. If OK, replace the pump motor. |
| Contamination found in the pump | Flush the system and replace the hydraulic fluid. |
| Bad pump | If the lift motor is OK, remove the motor from the pump and rotate the pump by hand. Replace the pump if not OK. |

## No Lower, Lift and Travel OK

| Possible Cause | Action |
|---|---|
| Bad lower solenoid valve... | With the main ON/OFF switch ON, cycle the lower switch. Does the lower solenoid valve energize? If YES, disconnect the battery, then slowly loosen, but DO NOT REMOVE, the hydraulic line at the hydraulic pump assembly. Permit the forks to slowly lower. Have rags and a drain pan ready to catch the fluid. |
| ... if the forks do not lower, check for mechanical binding in the lift cylinder or the fork mechanism... | Replace the lift cylinder or repair the lift mechanism. |
| ...if the forks lower, remove and inspect the solenoid valve... | Inspect for free movement and for signs of contamination. Clean the solenoid valve or flush the system, clean the filter, and replace the hydraulic fluid. See "Hydraulic Fluid" on page 7-95.<br>If no contamination is found, repair or replace the solenoid valve or the pump assembly. |
| ...check for a bad VM | With main ON/OFF switch ON, the PIN-key code entered, the green ON ( \| ) key on the keypad pressed, and the lower button depressed, check for lower signal input at the CAN (see "Service Display" on page 3-17). If no signal is detected, replace the VM.<br>**NOTE:** Check for mechanical problems in the handle head before replacing the VM. |

Symptom Tables: Lift/Lower System

## Unable to Pick Up a Load

| Possible Cause | Action |
|---|---|
| Load too large for the lift truck capacity | Check the load weight. Check the lift pressure setting. |
| Battery problems | Lead acid battery: Replace the battery with a fully-charged battery.<br>Lithium-ion battery: Check charge status LEDs for charge level. |
| Incorrect lift pressure adjustment | Check and adjust the lift pressure setting. |
| Contaminated or bad pressure relief valve | Check for contamination in the hydraulic fluid.<br>If contamination is present, flush, fill, and bleed the hydraulic system. Clean the relief valve. fluid. See “Hydraulic Fluid” on page 7-95.<br>If no contamination is found, replace the relief valve. |
| Bad lift pump or motor | Replace the lift pump or motor. |

## Slow Lower

| Possible Cause | Action |
|---|---|
| Bad lift/lower solenoid valve | Repair or replace the solenoid valve. |
| Plugged or malfunctioning flow control valve | Check for contamination in the hydraulic fluid.<br>If contamination is present, flush, fill, and bleed the hydraulic system. Clean the relief valve. See “Hydraulic Fluid” on page 7-95.<br>If no contamination is found, replace the flow control valve. |
| Contamination in the lowering solenoid valve | Inspect the solenoid for contamination or binding. |
| Mechanical binding in the lift mechanism | Adjust or replace the mechanism. |
| **NOTE:** If the pump hits relief and lower is attempted, lowering is slow. Release the lower button momentarily, then continue to lower. Normal lower speed should resume. | |

## Load Drifting/Settling

| Possible Cause | Action |
|---|---|
| Contaminated or bad check valve | Check for contamination in the hydraulic fluid.<br>If contamination is present, flush, fill, and bleed the hydraulic system. Clean the check valve. See “Hydraulic Fluid” on page 7-95.<br>If no contamination is found, replace the check valve. |
| Leaking or contaminated lowering solenoid valve | Check for contamination in the hydraulic fluid.<br>If contamination is present, flush, fill, and bleed the hydraulic system. Clean the solenoid valve. See “Hydraulic Fluid” on page 7-95.<br>If no contamination is found, replace the solenoid valve. |
| Load too large for the lift truck capacity | Check the load weight. Check the lift pressure setting. |

# No Lift or Lower. No Fault Codes.

| Possible Cause | Action |
| --- | --- |
| Throttle magnet-actuator arm (in the control handle) out of adjustment | Check with the service display and repair as needed.<br>Bad VM. If no other problems are noted but the VM does not detect the presence of the magnet, replace the VM. |
| Incorrect VM software installed. | **NOTE:** This is only possible if the VM was recently replaced.<br>Lowering may not occur if the incorrect software was inadvertently installed (the drive function will also not function correctly). Check for the correct software part number and replace the VM if not correct. |

# Symptom Tables: Travel (Forward/Reverse) System

**NOTE:** Use these symptom tables in conjunction with the electrical schematics and hydraulic schematics.

## Slow Travel, Lift/Lower OK. No Fault Codes

| Possible Cause | Action |
|---|---|
| Truck configured for slow speed | Check the truck configuration. |
| Battery problems | Lead acid battery: Replace the battery with a fully-charged battery.<br>Lithium-ion battery: Check charge status LEDs for charge level. |
| Binding drive wheel assembly/traction motor | Jack and block the truck. Disable the brake. See “Mechanical Brake Release” on page 7-37. Check that the drive wheel spins freely. If not, repair or replace. |
| Binding brake disc | Check the “Air Gap Inspection” on page 7-36.<br>Check for dragging brake caused by poor or incomplete brake release. Listen for an audible click to indicate full brake release. |
| Bad traction motor<br>(may test OK with no load, but fail under load) | Repair or replace the traction motor. |
| Open Arm Angle Switch | Check for correct operation of the arm angle proximity switch (SW2) and correct if necessary. Check the wiring, the traction amplifier, and the switch adjustment. The arm angle proximity switch (SW2) is open-circuited. See “Arm Angle Proximity Switch” on page 8-2. |

## Truck Does Not Accelerate Correctly

| Possible Cause | Action |
|---|---|
| Truck configured for slow acceleration | Check the truck configuration. |
| Battery problem | Lead acid battery: Verify a correct, fully-charged, good battery (24V) is installed in the truck.<br>Lithium-ion battery: Check charge status LEDs for charge level. |
| Binding or mechanical problem | Jack and block the truck. Turn the main ON/OFF switch OFF. Check the brake, the wheel bearings, and the load wheels to see if there is a mechanical problem. Repair or replace. |
| Bad motor | With the main ON/OFF switch OFF and the battery disconnected, check the traction motor. Repair or replace. |
| Bad traction amplifier | If all other possible causes check out, replace the traction amplifier. |

## No Travel Mode. No Fault Codes.

| Possible Cause | Action |
|---|---|
| Arm Angle Proximity Switch Failure | Check the arm angle proximity switch adjustment, wiring, and the traction amplifier. See "Arm Angle Proximity Switch" on page 8-2. |
| Throttle magnet-actuator arm (in the control handle) out of adjustment | Check with the service display and repair as needed.<br>Bad VM. If no other problems are noted but the VM does not detect the presence of the magnet, replace the VM. |

## No Travel or Slow Travel. TA Flash Code 2,2, (Thermal Cutback) Heatsink Temperature Exceeded 185°F (85°C). Operator Display May Indicate Hot2 (C45)

| Possible Cause | Action |
|---|---|
| Battery Problem | Verify the correct, (24V) fully-charged, battery is installed in truck. |
| Binding drive unit or dragging brake | Measure current draw on U, V, and W cables. Traveling at about 1 mph (1.6 km/h) with no load, current should be about 35 to 50 Amps and the same at each cable. If current is high, inspect the brake for dragging and the drive unit for binding. If both are OK, replace traction motor. |
| Bad TA | Examine mounting of TA. It must be mounted securely and in contact with the truck frame (that acts as a heatsink). If TA is mounted correctly, replace TA. |
| Truck does not respond to Keypad inputs | Enter Parameter 39 Access code. Use FlashWare and change value to either 3 or 4. Refer to "Programming Service Parameters" on page 3-8. |

## No Travel, No Lift/Lower. TA Flash Code 3,1. Operator Display May Indicate Error Code E106

| Possible Cause | Action |
|---|---|
| Short in M1 coil or wiring | With the truck OFF and the battery disconnected, remove connector JP1 from the TA. Measure resistance between JP1-6 and JP1-13. Resistance should be approx. 22 to 24.2 Ohms for standard contactor (76.5 to 93.5 Ohms for cold storage contactor). If not, troubleshoot wires and coil for shorts. If resistance is OK, replace TA. |

## No Travel, No Lift/Lower. TA Flash Code 1,3. Operator Display Indicates Error Code E202

| Possible Cause | Action |
|---|---|
| Bad wiring between TA and traction motor | With the truck OFF and the battery disconnected, disconnect cables W, V, and U from TA and traction motor. Check for continuity and shorts between cables and truck frame. |
| Bad TA | With the truck OFF and the battery disconnected, disconnect the cables at the TA. Reconnect the battery and turn the truck ON. If code 1,3 is still displayed, replace the TA. If code 1,3 is not displayed, replace the traction motor. |

## No Travel, Main Contactor Does Not Close. TA Flash Code 3,9. Operator Display Indicates Error Code E107

| Possible Cause | Action |
|---|---|
| Loss of B+ at the TA | Troubleshoot B+ from the battery to B+ at the TA. Check the wiring, connections, and fuses.<br>Lithium-ion battery: Check charge status LEDs for charge level. |
| Open in M1 coil or wiring | With the truck OFF and the battery disconnected, remove connector JP1 from the TA. Measure resistance between JP1-6 and JP1-13. Resistance should be approx. 22 to 24.2 Ohms for standard contactor (76.5 to 93.5 Ohms for cold storage contactor). If not, troubleshoot wires and coil for opens. If resistance is OK, replace TA. |

## No Travel, No Lift/Lower. TA Flash Code 1,2. Operator Display Indicates Error Code E201

| Possible Cause | Action |
|---|---|
| Bad wiring between TA and traction motor | With the truck OFF and the battery disconnected, disconnect cables W, V, and U from TA and traction motor. Measure for continuity and shorts between cables and to truck frame. |
| Bad TA | With the truck OFF and the battery disconnected, disconnect motor cables at TA. Reconnect the battery and turn the truck ON. If code is still displayed, replace the TA. If code goes away, replace the traction motor. |

## No Truck Functions Active. TA Flash Code 1,7, (Low Battery Voltage). Operator Display May Indicate E221

| Possible Cause | Action |
|---|---|
| Battery Problem | Verify the correct, (24V) fully-charged, battery is installed in truck. |
| Bad traction amplifier | Verify battery voltage at key switch input. Measure voltage at JP1-1 wire with JP1 disconnected. If 24V, replace TA. |

## No Truck Functions Active. TA Flash Code 1,8, (Excessive Battery Voltage). Operator Display May Indicate E222

| Possible Cause | Action |
|---|---|
| Battery Problem | Verify the correct, (24V) fully-charged, battery is installed in truck.<br>Lithium-ion battery: Check charge status LEDs for charge level. |
| Bad traction amplifier | Verify battery voltage at key switch input. Measure voltage at JP1-1 wire with JP1 disconnected. If 24V, replace TA. |

# Charger Troubleshooting (Lead Acid Batteries)

| Symptom | Action |
|---|---|
| No indicator lights | Check AC power and connection to batteries. |
| Only blue AC light ON | Charger has AC power and is waiting for connection to batteries or CAN remote control commands. Battery voltage must rise over 0.2 volts per cell before charging begins. |
| Solid red Fault/Error/USB Indicator | Read fault code number on the charger display and refer to "Delta-Q Charger Codes" on page 6-46 for cause and corrective action. |
| Flashing amber Fault/Error/USB Indicator | Read fault code number on the charger display and refer to "Delta-Q Charger Codes" on page 6-46 for cause and corrective action. |

# Lithium-ion Battery Troubleshooting

If you have a problem with the lithium-ion battery on your Model 8250 lift truck, use the table below to decide what to do. Do not try to service this lift truck unless you are properly trained and authorized to do so. Contact your Service Department or local authorized Raymond Sales and Service Center.

| Problem | Procedure |
|---|---|
| The lithium-ion battery does not turn ON. | The battery connector might not be completely installed. Press the battery connector firmly into the truck-side connector and verify a proper connection has been made.<br>The lithium-ion battery could be in one of its low-voltage modes. Press and hold the Multi-function Button.<br>The lithium-ion battery display could be disconnected, which would remove the ability to wake the system up using the Multi-function Button. Check the lithium-ion battery display plug in the cubby to verify a proper connection.<br>The battery communication cable could be damaged. Inspect for any signs of open/exposed conductors.<br>The Cell Module Assembly (CMA) could require service. A replacement CMA might be necessary. Contact a service technician for further investigation. |
| The alarms on the lithium-ion battery are constant | A reoccurring fault might be present. Toggle the truck ON/OFF switch once to see if the fault clears. Refer to the alarm patterns in Table 5-3 on page 27 to see what fault is indicated. Verify that the lithium-ion battery is not in a high temperature environment. |
| No power is being transferred between the lithium-ion battery and truck | Contactor could be open due to a temperature constraint. Let the lithium-ion battery thermally equalize within the operating temperature range before attempting to drive or charge again.<br>Contactor could be open due to a voltage constraint. Plug the lithium-ion battery into an energized outlet to charge and perform active balancing on the cells to get the cells within an appropriate voltage range.<br>Contactor could be open due to an overvoltage condition where hardware protection activates. This will require a service technician to inspect.<br>One of the lithium-ion battery fuses could be blown. Contact a service technician for further investigation. |
| Not as much energy appears to be available | Lithium-ion battery could be reaching the end of its life. Capacity starts to diminish once the warranty conditions have been exceeded.<br>The lithium-ion battery could be cold. Internal heaters maintain a proper operating temperature, but drain the battery if they are being heated from the battery instead of a charging source.<br>The lithium-ion battery might not have completed a full charge routine. Allow the lithium-ion battery to reach full charge when it is performing the charge routine.<br>The lithium-ion battery could be self-discharging due to damage. Contact a service technician for further investigation. |
| Charging is taking longer than specified | The lithium-ion battery could be too warm to charge. Charging is not allowed at high temperatures. Move the lithium-ion battery to a cooler area for charging.<br>The lithium-ion battery could be too cold to charge. Internal heaters will need to heat the cells up to an appropriate charging temperature. If the internal heaters are activated in the charge routine, energy is shared between the cells and the internal heaters in the lithium-ion battery and slows down the charging process. |

*Table 5-3. Status Indicator LED Patterns and Tones*

| Operating Condition | Status Indicator LEDs | Beacon LEDs | Audible Tone |
|---|---|---|---|
| Ready for Operation or Operating | G _ | None | None |
| Charger Plugged In | G _ A * | None | None |
| Low Charge | A * | Blinking | Loud & Obnoxious |
| Charge Scheduled | G ~ A * * ~ | None | Alert Beeps |
| Time To Charge Alert (See Note) | G, A * * ~ (G is OFF when A is ON) | None | Alert Beeps every 15 seconds |
| High Temperature | G ~ A * * * | None | Alert Beeps |
| Low Temperature | G ~ A * ~ ~ | None | Alert Beeps |
| Charger Fault | R ~ * * | None | Alert Beeps |
| Battery Fault | R ~ ~ * | None | Alert Beeps |
| Sleep Mode Engaged | A _ | None | None |
| Shutdown Pending | R * * * | Blinking | Loud & Obnoxious |
| Reserve Mode | A _ R ~ | None | Alert Beeps every 15 seconds |
| Manual Travel Mode | A _ R * ~ | None | Alert Beeps every 15 seconds |
| Dormant Mode | A _ R * * ~ | None | Alert Beeps every 15 seconds |

| Status Indicator LED Key | | |
|---|---|---|
| _ | = | Steady Light |
| * | = | Short Blink |
| ~ | = | Long Blink |
| G | = | Green |
| A | = | Amber |
| R | = | Red |

**NOTE:** Three conditions that will discontinue the Charge Schedule Alert are:

1. The alert times out.
2. The multi-function button is pressed.
3. The charger is plugged in.

# Section 6. Messages and Codes

# List of Messages and Codes

# Messages and Caution Codes

**NOTE:** Codes C001 through C012 are iBATTERY related error codes. Refer to iBATTERY Battery Monitoring System Installation and Maintenance Instructions.

## Code 'GATE'

| | |
|---|---|
| Code Title | Battery Gate Switch Open |
| Operator Display | 'GATE' |
| System Response | Travel limited to 1 MPH (1.6 km/h) |
| Reason | Battery gate missing and operator moved handle out-of-neutral. The TA detected that Battery Gate Interlock switch S21 or S22 is open. |
| Tests to Run | Troubleshoot wiring to and from switches. See "Pinout Matrix" on page 8-9. |
| How to Clear | Make sure the battery gate is installed correctly. Adjust or replace switches. |

## Code 'TEST'

| | |
|---|---|
| Code Title | TEST |
| Operator Display | 'TEST' |
| System Response | Truck functions are not allowed. |
| Alarm Sounds | No |
| Reason | New VM was installed that was not loaded with software. |
| Tests to Run | None |
| How to Clear | Code is cleared after the software is loaded in the VM. |

## Code 'SLO'

| | |
|---|---|
| Code Title | Creep Speed |
| Operator Display | 'SLO'<br>**NOTE:** This can be mistaken for a code 075 if the display is read upside-down. There is no code 075. |
| System Response | Travel Limited to Creep Speed |
| Reason | Click-to-Creep feature activated. |
| Tests to Run | None |
| How to Clear | With the handle in brake position, quickly click the speed control twice to deactivate creep speed. |

## Code 'Sro' (C14)

| | |
|---|---|
| Code Title | SRO (Static Return to Off) |
| Operator Display | 'Sro'<br>Error ( ⚠ ) and Parameter ( ⚙ ) Indicators Blinking. |
| System Response | Truck is disabled (Emergency Reverse is Active). |
| Alarm Sounds | No |
| Reason | Brake is not applied when the truck is turned ON. Handle is in the brake released range prior to the system being powered ON. |
| Tests to Run | Confirm arm angle proximity switch function. |
| How to Clear | Apply the brake. |

## Code 'LoGn' (C15) (iWAREHOUSE® Only)

| | |
|---|---|
| Code Title | Operator not Logged ON the 3rd Party Device (applies only to trucks with iWAREHOUSE option) |
| Operator Display | 'LoGn' |
| System Response | Limited to most restrictive speed if iCONTROL is enabled, otherwise no limits. |
| Reason | Operator is not logged ON to the 3rd party supplier wireless module (SWM) device. |
| Tests to Run | None |
| How to Clear | Log into the iWAREHOUSE display unit. |

## Code br_o (C16)

| | |
|---|---|
| Code Title | Brake Override |
| Operator Display | br_o<br>TA Flash Code "5,6" |
| System Response | Travel, lift, and lower are disabled. |
| Alarm Sounds | No |
| Reason | Brake Override switch is activated to manually move the truck. |
| Tests to Run | None |
| How to Clear | Move the tiller arm to the upper brake position and switch the brake override switch back to the OFF position. |

## Code C19

| | |
|---|---|
| Code Title | Default Parameter Warning |
| Operator Display | C19 |
| System Response | No effect on performance. |
| Alarm Sounds | No |
| Reason | New default parameters have been loaded into the RAM. |
| Tests to Run | None |
| How to Clear | Self clears after the timer expires. |

## Code HPd (C20)

| | |
|---|---|
| Code Title | HPD – Directional/Speed Control Warning (High Pedal Disable) |
| Operator Display | 'HPd'<br>Error ( ⚠ ) and Parameter ( ⚙ ) Indicators Blinking. |
| System Response | Travel is disabled, all other systems active. Emergency reverse is active. |
| Alarm Sounds | No |
| Reason | Directional/Speed Controls were not in the neutral position at power ON or when the brake is released, or failure of the Hall Effect sensor in the speed control circuit. |
| Tests to Run | Confirm the directional/speed controls are in neutral, and the handle sensors and wiring are correct. |
| How to Clear | Return the directional/speed controls to the neutral position. |

## Code C26

| | |
|---|---|
| Code Title | Lift Switch Stuck |
| Operator Display | C26 |
| System Response | System ignores stuck input. Turns off Lift Contactor. Stuck switch is disabled. If truck is equipped with an alternate switch, the alternate switch is still active. Lift contactor responds to input from alternate switch if equipped. |
| Reason | Lift request ON longer than time out value (10 seconds) and the lift cut-out time out did not activate. The lift system is prevented from lifting (jammed or too heavy a load) to a point of activating the lift cut-out time out. |
| Tests to Run | Examine for stuck switch or shorted wiring harness. See "Pinout Matrix" on page 8-9. |
| How to Clear | Release the Lift switch. |

Messages and Caution Codes

# Code C27

| | |
|---|---|
| Code Title | Lower Switch Stuck |
| Operator Display | C27 |
| System Response | System ignores stuck input. Turns off Lower Solenoid. Stuck switch is disabled. If truck is equipped with an alternate switch, the alternate switch is still active. Lower solenoid responds to input from alternate switch if equipped. |
| Reason | Lower request ON longer than time out value (10 seconds). Lower switch is jammed or the operator is holding the switch for longer than is required to lower the forks. |
| Tests to Run | Examine for stuck switch or shorted wiring harness. See “Pinout Matrix” on page 8-9. |
| How to Clear | Release the Lower switch. |

# Code C30

| | |
|---|---|
| Code Title | Emergency Reverse Activated |
| Operator Display | C30 |
| System Response | Normal travel controls are disabled. The truck responds only to activations of the Emergency Reverse switch. |
| Reason | The Emergency Reverse switch has been pressed and released. |
| Tests to Run | Cycle the brake (deadman) switch by engaging the brake. |
| How to Clear | Clears after the brake is applied. |

# Code C31

| | |
|---|---|
| Code Title | Lost Brake Pot VR1 Input (Brake Pot Out-of-Range) |
| Operator Display | C31<br>TA Flash Code “3,4” |
| System Response | Travel is disabled |
| Reason | Brake pot input at JP1-17 is open. See “Pinout Matrix” on page 8-9. |
| Tests to Run | Troubleshoot wiring to and from the Brake pot. |
| How to Clear | Adjust or replace pot or repair wiring. Cycle the truck OFF/ON. |

## Code C32

| | |
|---|---|
| Code Title | Emergency Reverse Before Brake Release |
| Operator Display | C32 |
| System Response | Travel is disabled |
| Reason | The emergency reverse switch is activated (pressed) prior to the brake being released. |
| Tests to Run | Examine for stuck emergency reverse switch. |
| How to Clear | Clears after the Emergency Reverse switch is released. |

## Code C33

| | |
|---|---|
| Code Title | Horn Switch Stuck |
| Operator Display | C33 |
| System Response | Ignores stuck input. Turns off Horn. Horn responds to input from alternate source if equipped (this does not include the second horn switch on the control handle). |
| Reason | Horn request ON longer than time out value (10 seconds). Stuck switch is disabled; if equipped with an alternate switch, the alternate switch is still active. |
| Tests to Run | Examine horn switch and related wiring. See “Pinout Matrix” on page 8-9. |
| How to Clear | Release or repair Horn switch. |

## Code C35

| | |
|---|---|
| Code Title | Brake Switch (Arm Angle Proximity Switch) Error |
| Operator Display | C35 |
| System Response | Normal travel controls are disabled. The truck responds only to activations of the Emergency Reverse switch if the traction motor is rotating in the tractor-first direction. |
| Alarm Sounds | Yes (3 beeps, once, when error code is generated) |
| Reason | The brake switch (arm angle proximity switch) is out of adjustment or damaged giving a false indication that the brake is engaged. At the same time, the traction motor is detected rotating. |
| Tests to Run | Test the brake switch (arm angle proximity switch) and wires. |
| How to Clear | Apply the brake and have the switch adjusted or replaced. |

Messages and Caution Codes

# Code Lo (C41)

| | |
|---|---|
| Code Title | Battery Undervoltage Warning |
| Operator Display | 'Lo'<br>Battery ( ) Indicator Blinking.<br>TA Flash Code “2,3” |
| System Response | Travel performance may be limited due to low voltage. |
| Alarm Sounds | No |
| Reason | Battery voltage is below approximately 17V. The battery is discharged or there is excessive load on the battery. |
| Tests to Run | Measure the battery voltage at JT1-1. Use the service key to read the voltage. See “Service Display” on page 3-17. |
| How to Clear | The battery must be charged. Examine the battery and battery connections. The fault is cleared when the battery voltage increases above 17V. |

# Code Hi (C42)

| | |
|---|---|
| Code Title | Battery Overvoltage Warning |
| Operator Display | 'Hi'<br>Battery ( ) Indicator Blinking.<br>TA Flash Code “2,4” |
| System Response | Travel performance may be limited. |
| Alarm Sounds | No |
| Reason | Battery voltage is over approximately 32V. This can be caused by the wrong type of battery installed in the truck, bad battery connections/cables, or an overcharged battery. |
| Tests to Run | Measure the battery voltage at JPT1-1. See “Service Display” on page 3-17. See also “Code Lo (C41)”. |
| How to Clear | Check battery and battery connections. Make sure that the correct size battery is installed in the truck. The fault is cleared when the battery voltage drops below 32V. |

## Code Cold (C43)

| | |
|---|---|
| Code Title | Traction Amplifier Cold Thermal Cutback |
| Operator Display | Cold<br>TA Flash Code “2,1” |
| System Response | Travel performance may be limited. |
| Alarm Sounds | No |
| Reason | Low temperature (below –13°F (–25°C)) at the TA heatsink. Operation in extreme cold environment or incorrect TA mounting. |
| Tests to Run | 1. Verify the Traction Amp is cold. If it is, take action to bring the temperature back within normal operating range.<br>2. Check for correct traction amplifier mounting.<br>3. Bad traction amplifier. Check TA for error code. |
| How to Clear | Code is cleared when the traction amplifier heatsink temperature returns to within normal operating range. |

## Code Hot1 (C44)

| | |
|---|---|
| Code Title | Motor Temperature Hot Cutback |
| Operator Display | ‘Hot1’<br>TA Flash Code “2,8” |
| System Response | Travel performance may be limited. |
| Alarm Sounds | No |
| Reason | Traction motor temperature is above 248°F (120°C). Resistance of the temperature sensor should be 591 Ohms at room temperature. |
| Tests to Run | Use an accurate temperature measuring device to verify the temperature reading on the service display. See “Service Display” on page 3-17. |
| How to Clear | Permit the motor to cool down. Replace the temperature sensor. |

# Code Hot2 (C45)

| | |
|---|---|
| Code Title | Traction Power Amp Warm Thermal Cutback |
| Operator Display | 'Hot2'<br>TA Flash Code "2,2" |
| System Response | Travel performance may be limited |
| Reason | TA internal heatsink temperature is above 185°F (85°C). Operation in extreme hot environment, excessive load on vehicle, or incorrect TA mounting. |
| Tests to Run | Verify the Traction Amp is hot. If it is, determine the cause by checking for mechanical drag such as brakes dragging, gear box issue, or load wheel binding. Check the current through U, V, and W and verify that they are even. See "Service Display" on page 3-17. Also see "No Travel or Slow Travel. TA Flash Code 2,2, (Thermal Cutback) Heatsink Temperature Exceeded 185°F (85°C). Operator Display May Indicate Hot2 (C45)" on page 5-23. |
| How to Clear | Take action to bring heatsink temperature into normal operating range. |

# Code C46 (iWAREHOUSE® Only)

| | |
|---|---|
| Code Title | RDP-T Module CAN Communication Failure |
| Operator Display | C46 |
| System Response | No effect on performance unless the "Error Speed" is configured for less than full speed. |
| Reason | VM not receiving messages from the RDP-T module. |
| Tests to Run | Check for heartbeat on the RDP-T module. Run FlashWare RDP Communication Test at the service port. Test wiring to RDP-T module. |
| How to Clear | Cycle the truck OFF/ON. |

# Code C47 (iWAREHOUSE® Only)

| | |
|---|---|
| Code Title | RDP-T Module Not Communicating With The 3rd Party SWM |
| Operator Display | C47 |
| System Response | No effect on performance unless the "Error Speed" is configured for less than full speed. |
| Reason | Cable problem between the translator module and the 3rd party Supplier Wireless Module (SWM) or the SWM has failed. |
| Tests to Run | Check harness. Run FlashWare RDP Communication Test at the RDP-T module. |
| How to Clear | Clears automatically once communication is again established. |

# Code C48

| | |
|---|---|
| Code Title | Click-to-Creep Button is Stuck Closed |
| Operator Display | C48 |
| System Response | No restriction on normal travel. Click-to-Creep travel disabled. Input from Click-to-Creep button ignored. |
| Reason | Click-to-Creep button is stuck closed for more than 10 seconds and is closed when truck is being powered ON. Click-to-Creep is disabled. |
| Tests to Run | Check the brake switch (arm angle proximity switch) and wires. |
| How to Clear | Release the Click-to-Creep button. Apply the brake and have the proximity switch adjusted or replaced. |

**NOTE:** Caution Error Codes 50 and above are logged. See the following tables for how to clear the code and the allowed activity during the error. Some codes can be cleared by cycling the truck OFF/ON.

# Code C57

| | |
|---|---|
| Code Title | Brake Interlock Proximity Switch Input Signal Mismatch |
| Operator Display | C57 |
| System Response | Hard Plug to 1 MPH (1.6 km/h) |
| Reason | The TA detected that the input from the Brake Switch at JP1-9 and the Redundant Brake Switch input at JP1-24 do not match. |
| Tests to Run | Troubleshoot wiring to and from the Brake Switch. Troubleshoot wiring to and from the Redundant Brake Switch. See "Pinout Matrix" on page 8-9. |
| How to Clear | Investigate cause of open or short. |

# Code C60

| | |
|---|---|
| Code Title | Lift Contactor Coil (M2) Driver Shorted or Open |
| Operator Display | C60<br>TA Flash Code "3,3" |
| System Response | Lift function not available. Lower functions normally. |
| Reason | Lift coil driver open or shorted. |
| Tests to Run | Verify Parameter 38 is set correctly. See "Service Parameters" on page 3-10. Examine lift contactor coil and wiring. Resistance of coil should be approximately 17 Ohms. See "Pinout Matrix" on page 8-9. |
| How to Clear | Investigate cause of open or short and repair or replace as needed. |

Messages and Caution Codes

## Code C61

| | |
|---|---|
| Code Title | Lower Solenoid Coil (Sol 1) Driver Shorted or Open |
| Operator Display | C61<br>TA Flash Code “3,5” |
| System Response | Lower function not available. Lift functions normally. |
| Reason | Lower coil driver open or shorted. |
| Tests to Run | Examine for short or open for lower coil, or related wiring. Resistance of coil should be approximately 39 Ohms. See “Pinout Matrix” on page 8-9. |
| How to Clear | Investigate cause of open or short and repair or replace as needed. |

## Code C62

| | |
|---|---|
| Code Title | Horn Coil Driver Shorted or Open |
| Operator Display | C62<br>TA Flash Code “3,4” |
| System Response | Horn function not available. |
| Reason | Horn coil driver open or shorted. |
| Tests to Run | Examine horn and related wiring. See “Pinout Matrix” on page 8-9. |
| How to Clear | Investigate cause of open or short and repair or replace as needed. |

## Code C64

| | |
|---|---|
| Code Title | Digital Output Overcurrent |
| Operator Display | C64<br>TA Flash Code “2,6 or 2,7” |
| System Response | Travel Alarm Inoperable. Travel limited to walking speed. |
| Reason | Digital output driver overcurrent |
| Tests to Run | Examine for shorted travel alarm, or related wiring. See “Service Display” on page 3-17. See also “Pinout Matrix” on page 8-9. |
| How to Clear | Investigate cause of short and repair or replace as needed. |

# Code C66

| | |
|---|---|
| Code Title | Brake Coil Driver Open or Shorted |
| Operator Display | C66<br>TA Flash Code “3,2” |
| System Response | No Travel |
| Reason | The TA detected an open or short in the Brake Coil Driver 2 circuit. |
| Tests to Run | Troubleshoot wiring to and from the brake and the brake coil for opens or shorts. The brake coil should have approximately 14 ohms resistance. See “Pinout Matrix” on page 8-9. |
| How to Clear | Determine cause of short or open circuit and correct. |

# Code C67

| | |
|---|---|
| Code Title | EM Brake Not Set |
| Operator Display | C67<br>TA Flash Code “9,2” |
| System Response | No Travel |
| Reason | Triggered if the brake switch is open (handle up) and the brake is applied and no motion is detected and then motion is detected without releasing the brake. A weak brake or excessive skidding and then gripping during a deadman braking event could cause this. |
| Tests to Run | Troubleshoot wiring to and from the brake and the brake coil for opens or shorts. The brake coil should have approximately 14 ohms resistance. See “Pinout Matrix” on page 8-9. Check brake gap. See “Air Gap Inspection” on page 7-36. |
| How to Clear | Return throttle to neutral. Replace the brake or rotor as needed. |

# Code C68

| | |
|---|---|
| Code Title | Brake SelfTest Failed |
| Operator Display | C68 |
| System Response | Traction performance limited to 1 MPH (1.6 km/h) |
| Reason | Brake did not hold the truck stationary during SelfTest. |
| Tests to Run | Check for worn brake. Check brake gap. See “Air Gap Inspection” on page 7-36. Verify brake release bolts are not installed. |
| How to Clear | Replace the brake or rotor as needed. Cycle the truck OFF/ON. |

# Code C70

| | |
|---|---|
| Code Title | Motor Temperature Sensor Fault |
| Operator Display | C70<br>TA Flash Code “2,9” |
| System Response | Traction performance reduced. |
| Reason | Failure of motor temperature sensor. Sensor or wiring to traction motor open or shorted. Resistance of temperature sensor should be from 990 to 1100 Ohms at room temperature. |
| Tests to Run | Examine component and related wiring for shorts or open condition. See “Service Display” on page 3-17. See also “Pinout Matrix” on page 8-9. |
| How to Clear | Determine cause of short or open circuit and correct. |

# Code C71

| | |
|---|---|
| Code Title | Sensor Bearing Defective (Speed Sensor) |
| Operator Display | C71<br>TA Flash Code “3,6” |
| System Response | Traction performance limited. |
| Reason | TA detects the loss of traction motor encoder input (missing phase(s), power, and so forth). |
| Tests to Run | Use service key and test the traction motor encoder by scrolling to the RPM display (D) in diagnostic displays and command the motor to rotate. The value displayed is the motor RPM as reported from the TA. Examine wiring to traction motor encoder. See “Service Display” on page 3-17. See also “Pinout Matrix” on page 8-9. |
| How to Clear | Troubleshoot and repair wiring to Traction Motor Encoder. If still not clear, replace the encoder. |

# Code C72

| | |
|---|---|
| Code Title | Traction Motor Stalled |
| Operator Display | C72<br>TA Flash Code “7,3” |
| System Response | Traction performance limited until code cleared. Brake is activated on motor stall. |
| Reason | TA detects that the traction motor has stalled for 5 seconds. |
| Tests to Run | Use service key and test the traction motor encoder by scrolling to the RPM display (D) in diagnostic displays and command the motor to rotate. The value displayed is the motor RPM as reported from the TA. See “Service Display” on page 3-17. See also “Pinout Matrix” on page 8-9. |
| How to Clear | Cycle the truck OFF/ON. |

## Code 075

There is no Code 075, this is a misinterpretation of the SLO code. Refer to Code 'SLO' for causes and how to clear.

## Code AC (C256)

| | |
|---|---|
| Code Title | Charger is Charging |
| Operator Display | AC |
| System Response | No truck function active |
| Reason | Charger is plugged into an AC outlet. |
| Tests to Run | None |
| How to Clear | Unplug AC cord once charging is completed. |

## Code C257

| | |
|---|---|
| Code Title | Charger is Reporting that it has a Fault |
| Operator Display | C257 |
| System Response | No truck function active |
| Reason | Internal issue with the AC charger. Charger reports fault over the CAN bus. |
| Tests to Run | None |
| How to Clear | Refer to the charger's display for the charger specific fault code. Troubleshoot based on that code. |

## Code C258

| | |
|---|---|
| Code Title | Lift Button Stuck |
| Operator Display | C258 |
| System Response | Lift and lower disabled. No restriction on normal travel. |
| Reason | Lift button was held during login/startup. |
| Tests to Run | None |
| How to Clear | Release lift button. |

# Code C259

| | |
|---|---|
| Code Title | Lower Button Stuck |
| Operator Display | C259 |
| System Response | Lift and lower disabled. No restriction on normal travel |
| Reason | Lower button was held during login/startup. |
| Tests to Run | None |
| How to Clear | Release lower button. |

# Code C380

| | |
|---|---|
| Code Title | Lift Contactor Tips Not Closed |
| Operator Display | C380 |
| System Response | None |
| Reason | Open connection from the TA to the lift contactor coil or faulty contactor. |
| Tests to Run | None |
| How to Clear | Check the wiring from the TA to the lift contactor. If wiring okay, replace faulty contactor. |

# Code C382

| | |
|---|---|
| Code Title | Delta-Q Charger or Traction Amplifier is Not Reading the Battery Voltage Correctly |
| Operator Display | C382 |
| System Response | None |
| Reason | The charger battery voltage is more than +/– 1V in difference from the TA key switch voltage for 1 second. |
| Tests to Run | None |
| How to Clear | Determine which device is incorrectly reading the battery voltage. Repair or replace as necessary. |

## Code C383

| | |
|---|---|
| Code Title | 5 Volt Rail on the VM (ETAC) is Out-of-Range |
| Operator Display | C383 |
| System Response | The 5 volt rail supplies power to the horn button. The horn may not activate when requested. |
| Reason | VM (ETAC) issue or low voltage. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, replace the VM. |

## Code C384

| | |
|---|---|
| Code Title | Traction Amplifier BDI SDO Parameter Failure |
| Operator Display | C384 |
| System Response | None |
| Reason | TA BDI parameter (5) settings related to configuring the BDI did not get set correctly within the TA after three attempts during start-up. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, replace the TA. |

## Code C385

| | |
|---|---|
| Code Title | VM Not Calibrated |
| Operator Display | C385 (for several seconds after each power ON) |
| System Response | None |
| Reason | VM analog reference voltage not calibrated by VM supplier. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, replace the VM. |

## Code C395

| | |
|---|---|
| Code Title | Unexpected Battery Charger Communication at Startup |
| Operator Display | C395<br>TA Flash Code “2,7” |
| System Response | None |
| Reason | At startup the VM detects the charger and the CAN enable parameter is not enabled. Parameter 46 not set to 0 and must be set correctly. |
| Tests to Run | None |
| How to Clear | Set parameter 46 to a valid charging profile. |

## Code C405

| | |
|---|---|
| Code Title | Load Hold Coil Driver Open or Shorted |
| Operator Display | C405 |
| System Response | If open, truck cannot lower. If shorted, truck lowers when the short occurs. |
| Reason | Load Hold coil driver open or shorted. Resistance of coil should be approximately 37 Ohms. |
| Tests to Run | Examine for open or shorted Load Hold Coil driver circuit and related harness. See “Pinout Matrix” on page 8-9. |
| How to Clear | Investigate cause of open or short and repair or replace as needed. |

# Error Codes

**NOTE:** Error Codes are 100 and above. Error codes are logged. No truck activity is allowed during error. Some codes can be cleared when the truck is cycled OFF/ON (unless otherwise noted).

## Code E101

| | |
|---|---|
| Code Title | Traction Amp Type Error |
| Operator Display | E101 |
| System Response | No truck function active. |
| Alarm Sounds | No |
| Reason | Incorrect Traction Amplifier installed on the truck. |
| Tests to Run | Determine correct TA P/N. Verify TA and VM software versions. |
| How to Clear | Replace with the correct TA. |

## Code E106

| | |
|---|---|
| Code Title | Main Contactor Driver Overcurrent Error |
| Operator Display | E106<br>TA Flash Code "3,1" |
| System Response | No truck function active |
| Reason | Short in main contactor coil. Resistance of coil should be 22 to 24.2 Ohms for standard contactor (76.5 to 93.5 Ohms for cold storage contactor). |
| Tests to Run | Examine main contactor for shorts to related wiring harness. See "No Travel, No Lift/Lower. TA Flash Code 3,1. Operator Display May Indicate Error Code E106" on page 5-23. |
| How to Clear | Examine for shorted component or related wiring. Repair or replace as needed. |

# Code E107

| | |
|---|---|
| Code Title | Contactor Off Error |
| Operator Display | E107<br>TA Flash Code “3,9” |
| System Response | No truck function active |
| Reason | Main contactor did not close. Open in main contactor coil. Open fuse. |
| Tests to Run | Examine contactor circuit for opens in related wiring harness. See “No Travel, Main Contactor Does Not Close. TA Flash Code 3,9. Operator Display Indicates Error Code E107” on page 5-24. See “Service Display” on page 3-17. Examine for open Main Contactor or related wiring. Check for open fuse. |
| How to Clear | Repair or replace as needed. |

# Code E108

| | |
|---|---|
| Code Title | Contactor Weld Error |
| Operator Display | E108<br>TA Flash Code “3,8” |
| System Response | No truck function active |
| Reason | Main Contactor welded (detected on start-up). Open in U phase of traction motor. |
| Tests to Run | Inspect contact tips for welding. Check for B+ voltage at the B+ connection of the traction amplifier. See “Service Display” on page 3-17.<br>**NOTE:** Check the U cable for correct connection and continuity. An open in only the U circuit causes this code. An open in the V and W circuit causes codes E201 and E203.<br>Examine for welded contact tips. |
| How to Clear | Repair or replace as needed. |

# Code E109

| | |
|---|---|
| Code Title | CAN Bus Communication Error |
| Operator Display | E109 |
| System Response | No truck function active. |
| Alarm Sounds | No |
| Reason | Indicates the failure of a device on the truck to respond to a CAN SDO message sent by the VM. There is a 1 second time-out. |
| Tests to Run | None |
| How to Clear | Examine the VM, TA, and related wiring. Repair or replace as needed. |

# Code E140

| | |
|---|---|
| Code Title | Checksum Error |
| Operator Display | E140 |
| System Response | No truck function active. |
| Alarm Sounds | No |
| Reason | Software check sum not the same as the stored value. |
| Tests to Run | 1. Bad software.<br>a. If applicable, download new software.<br>2. Bad Vehicle Manager. |
| How to Clear | Code may clear on cycling the key switch OFF/ON. Reload software in the VM. If still not cleared, replace the VM. |

# Code E141

| | |
|---|---|
| Code Title | Software Problem |
| Operator Display | E141 |
| System Response | No truck function active |
| Reason | VM internal error, or missing or faulty software due to a failure or problem while downloading. VM determined that software files were corrupt. |
| Tests to Run | 1. Missing or bad software.<br>a. If applicable, download new software.<br>2. Bad Vehicle Manager. |
| How to Clear | Reload the software as allowed. |

# Code E142

| | |
|---|---|
| Code Title | Parameter Corruption During Start-up |
| Operator Display | E142 |
| System Response | No truck function active |
| Reason | VM EE issue. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, replace the VM. |

# Code E150

| | |
|---|---|
| Code Title | CAN Bus Communication Error |
| Operator Display | E150 |
| System Response | No truck function active. Code may be displayed briefly then goes out. |
| Alarm Sounds | No |
| Reason | PDO messages are not being returned by the TA. VM has not detected any response from the TA to CAN messages. Bad connection or terminal crimp or a damaged wire between the TA and VM. Check for low battery voltage. Check that battery parameters are set correctly. |
| Tests to Run | None |
| How to Clear | Check the battery voltage and connections. Verify battery parameters are set correctly. Examine the VM, TA, and related wiring. Repair or replace as needed. |

# Code E151

| | |
|---|---|
| Code Title | CAN Bus Communication Error |
| Operator Display | E151 |
| System Response | No truck function active. |
| Alarm Sounds | No |
| Reason | VM PDO messages have not successfully sent after three attempts. Bad connection or terminal crimp or a damaged wire between the TA and VM. Check for low battery voltage. Check that battery parameters are set correctly. |
| Tests to Run | None |
| How to Clear | Check the battery voltage and connections. Verify battery parameters are set correctly. Examine the VM, TA, and related wiring. Repair or replace as needed. |

# Code E152

| | |
|---|---|
| Code Title | CAN Bus Communication Error |
| Operator Display | E152 |
| System Response | No truck function active. |
| Alarm Sounds | No |
| Reason | VM has not seen TA heartbeat, TA heartbeat has gone into pre-operational mode more than 3 times in 5 seconds, or TA has been unresponsive since power ON. Bad connection or terminal crimp or a damaged wire between the TA and VM. Check for low battery voltage. Check that battery parameters are set correctly. |
| Tests to Run | None |
| How to Clear | Check the battery voltage and connections. Verify battery parameters are set correctly. Examine the VM, TA, and related wiring. Repair or replace as needed. |

# Code E157

| | |
|---|---|
| Code Title | CAN Bus Off Error |
| Operator Display | E157 |
| System Response | No truck function active. |
| Alarm Sounds | No |
| Reason | TA did not reply to the CAN messages. |
| Tests to Run | Examine the VM, TA, and related wiring. |
| How to Clear | Repair or replace as needed. |

# Code E159

| | |
|---|---|
| Code Title | CAN Bus Overrun Error |
| Operator Display | E159 |
| System Response | No truck function active. |
| Alarm Sounds | No |
| Reason | TA did not reply to the CAN messages. |
| Tests to Run | Examine VM, TA, and related wiring. |
| How to Clear | Repair or replace as needed. |

# Code E160

| | |
|---|---|
| Code Title | Emergency Reverse (Belly Button)/Hall Supply Error |
| Operator Display | E160 |
| System Response | No truck function active. |
| Alarm Sounds | No |
| Reason | Faulty hall effect sensor in the handle (VM). |
| Tests to Run | None |
| How to Clear | Determine the cause of the faulty sensor and repair or replace as needed. The VM is the likely cause. Replace the VM. |

# Code E201

| | |
|---|---|
| Code Title | Overcurrent Fault |
| Operator Display | E201<br>TA Flash Code “1,2” |
| System Response | No truck function active |
| Reason | TA faulty or motor short circuit. External short of U, V, or W traction motor connections. |
| Tests to Run | See “No Travel, No Lift/Lower. TA Flash Code 1,2. Operator Display Indicates Error Code E201” on page 5-24. See “Service Display” on page 3-17. |
| How to Clear | Investigate cause of apparent short. Start with cables, then troubleshoot the traction motor. If cables and traction motor are okay, replace the TA. If still not clear, replace the traction motor. |

# Code E202

| | |
|---|---|
| Code Title | TA Current Sensor Error |
| Operator Display | E202<br>TA Flash Code “1,3” |
| System Response | No truck function active |
| Reason | Phase current sensor fault. |
| Tests to Run | See “No Travel, No Lift/Lower. TA Flash Code 1,3. Operator Display Indicates Error Code E202” on page 5-23. See “Service Display” on page 3-17. |
| How to Clear | Investigate cause of apparent short. Start with cables, then troubleshoot the traction motor. If cables and traction motor are okay, replace the TA. If still not clear, replace the traction motor. |

# Code E203

| | |
|---|---|
| Code Title | Motor Open |
| Operator Display | E203<br>TA Flash Code “3,7” |
| System Response | No truck function active |
| Reason | Motor phase V or W detected open. Bad power amplifier.<br>**NOTE:** An open in the U phase causes a code E108. |
| Tests to Run | Examine motor and related wiring for open circuit. See “Service Display” on page 3-17. |
| How to Clear | Repair cause of open circuit. |

# Code E220

| | |
|---|---|
| Code Title | Default Parameters DNL |
| Operator Display | E220<br>TA Flash Code “4,6” |
| System Response | No truck function active |
| Reason | TA default parameters did not load. |
| Tests to Run | None |
| How to Clear | Check battery. Reload software. If unable to reload software, replace the TA. |

# Code E221

| | |
|---|---|
| Code Title | Severe Battery Undervoltage |
| Operator Display | E221<br>Battery ( ) Indicator Blinking<br>TA Flash Code “1,7” |
| System Response | No truck function active |
| Reason | Severe battery undervoltage; battery voltage under 12V. |
| Tests to Run | Measure battery voltage and specific gravity. See “No Truck Functions Active. TA Flash Code 1,7, (Low Battery Voltage). Operator Display May Indicate E221” on page 5-24. See “Service Display” on page 3-17.<br>**NOTE:** Check for other codes and, if others are present, troubleshoot following those codes. (Failure of the M1 contactor circuit during operation causes this code to be shown along with E106 and E107.) |
| How to Clear | Lead acid battery: Replace battery with fully-charged battery.<br>Lithium-ion battery: Check charge status LEDs for charge level.<br>A bad TA or bad wiring could also be the cause. |

# Code E222

| | |
|---|---|
| Code Title | Severe Battery Overvoltage |
| Operator Display | E222<br>Battery ( ) Indicator Blinking<br>TA Flash Code “1,8” |
| System Response | No truck function active |
| Reason | Severe battery overvoltage; TA detects over 42V during run time or VM detects above 32V at key ON. |
| Tests to Run | Measure battery voltage. See “No Truck Functions Active. TA Flash Code 1,8, (Excessive Battery Voltage). Operator Display May Indicate E222” on page 5-24. See “Service Display” on page 3-17. |
| How to Clear | Lead acid battery: Install battery of correct voltage.<br>Lithium-ion battery: Check charge status LEDs for charge level.<br>A bad TA or bad wiring could also be the cause. |

# Code E223

| | |
|---|---|
| Code Title | TA Severe Undertemperature Cutback |
| Operator Display | E223<br>TA Flash Code “1,5” |
| System Response | No truck function active |
| Reason | TA heatsink temperature below minimum operating temperature (below –40°C/–40°F). |
| Tests to Run | Test the temperature sensor. See “Service Display” on page 3-17. |
| How to Clear | Warm truck to normal operating temperature and cycle the truck OFF/ON.<br>Investigate to determine the cause of the undertemperature condition. |

# Code E224

| | |
|---|---|
| Code Title | TA Severe Overtemperature Cutback |
| Operator Display | E224<br>TA Flash Code “1,6” |
| System Response | No truck function active |
| Reason | TA heatsink temperature over maximum operating temperature (+95°C/203°F). |
| Tests to Run | Test the temperature sensor. See “Service Display” on page 3-17. |
| How to Clear | Allow truck to cool. Investigate to determine the cause of the overheating. |

# Code E225

| | |
|---|---|
| Code Title | Precharge Fault |
| Operator Display | E225<br>TA Flash Code “1,4” |
| System Response | No truck function active |
| Reason | TA precharge circuit failed, internal short in TA, SOL1 shorted to the lift motor, blown power fuse, contactor failed to close, or contactor sense line shorted. |
| Tests to Run | None |
| How to Clear | Check for shorts, blown power fuse, and contactor failure. Try cycling the truck OFF/ON. If that does not solve the problem, replace the TA. |

# Code E228

| | |
|---|---|
| Code Title | Parameter Change Fault |
| Operator Display | E228<br>TA Flash Code “4,9” |
| System Response | No truck function active |
| Reason | TA parameters have changed. Critical TA parameter changed while operating. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, replace the TA. |

# Code E230

| | |
|---|---|
| Code Title | External Supply Out-Of-Range |
| Operator Display | E230<br>TA Flash Code “6,9” |
| System Response | No truck function active |
| Reason | External supply current out-of-range (+12V, +5V). Devices connected to these outputs on the TA are drawing too much current (> 250 mA). |
| Tests to Run | Check all wiring and components connected to the 12 volts from the TA for shorts. If OK, check the wiring and encoder connected to JP1-26 for the traction motor for shorts. |
| How to Clear | Cycle the truck OFF/ON. If not cleared by cycling the truck OFF/ON, try to determine what device caused the problem. |

# Code E232

| | |
|---|---|
| Code Title | VCL Runtime |
| Operator Display | E232<br>TA Flash Code “6,8” |
| System Response | No truck function active |
| Reason | Internal TA software error. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. Check battery. Reload software. If unable to reload software, replace the TA. |

# Code E233

| | |
|---|---|
| Code Title | +5V Supply Failure |
| Operator Display | E233<br>TA Flash Code “2,5” |
| System Response | No truck function active |
| Reason | TA +5V supply failed. |
| Tests to Run | Check all wiring and components connected to the 12 volts from the TA for shorts. If OK, check the wiring and encoder connected to JP1-25 for the traction motor for shorts. See Code E230. See “Pinout Matrix” on page 8-9. |
| How to Clear | Cycle the truck OFF/ON. |

# Code E235

| | |
|---|---|
| Code Title | NVM Reset Error |
| Operator Display | E235 |
| System Response | No truck function active |
| Reason | TA NVM (Non-Volatile Memory) error. TA Non-volatile memory corrupted, unable to reset. |
| Tests to Run | None |
| How to Clear | Try cycling the truck OFF/ON. If that does not solve the problem, replace the TA. |

# Code E236

| | |
|---|---|
| Code Title | Traction Amp OS Fault |
| Operator Display | E236 |
| System Response | No truck function active |
| Reason | Internal TA fault |
| Tests to Run | None |
| How to Clear | Reload software via FlashWare. If unable to reload software, replace the TA. |

# Code E248

| | |
|---|---|
| Code Title | Lift Contactor Tips Did Not Open |
| Operator Display | E248 |
| System Response | Main contactor (MPC) opens |
| Reason | Welded contactor tips or shorted output driver from the TA. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, check the contactor or replace the TA. |

# Code E249

| | |
|---|---|
| Code Title | Charger Voltage is Incorrect |
| Operator Display | E249 |
| System Response | No truck function active |
| Reason | The voltage sense line from the on-board HF charger to the truck is either at 0 volts or more than 5 volts difference from what the TA is reading. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, check the charger and the charger cables. |

# Code E250

| | |
|---|---|
| Code Title | Communication Error with the HF Charger |
| Operator Display | E250 |
| System Response | No truck function active |
| Reason | HF Charger CAN communication issue. |
| Tests to Run | None |
| How to Clear | Verify the CAN cable to the battery pack is connected.<br>Code will self clear after plugging the charger into AC power to reset the charger from sleep mode. |

# Code E251

| | |
|---|---|
| Code Title | VM Memory Fault |
| Operator Display | E251 |
| System Response | No truck function active |
| Reason | VM memory corruption. |
| Tests to Run | None |
| How to Clear | Try cycling the truck OFF/ON. If that does not solve the problem, replace the VM. |

# Code E254

| | |
|---|---|
| Code Title | Invalid Encoder Pulse (Valid only for VM versions below 23) |
| Operator Display | E254 |
| System Response | No truck function active |
| Reason | Encoder Pulse Error. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. |

## Code E690

| | |
|---|---|
| Code Title | Software Mismatch |
| Operator Display | E690 |
| System Response | No truck function active |
| Reason | Invalid TA Operating System version. |
| Tests to Run | None |
| How to Clear | Reload software via FlashWare. |

## Code E691

| | |
|---|---|
| Code Title | TA Current Cutback |
| Operator Display | E691<br>TA Flash Code “6,1” |
| System Response | No truck functions active after moving the handle to the brake position or pressing the emergency reverse switch. |
| Reason | Voltage is higher than 30 Volts, traction amplifier in thermal limit, or traction motor at thermal limit. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear after several attempts, replace the TA. |

## Code E700

| | |
|---|---|
| Code Title | Unexpected Battery Charger Detection During Runtime |
| Operator Display | E700 |
| System Response | No truck functions active. |
| Reason | Parameter 46 is set to 0. |
| Tests to Run | None |
| How to Clear | Set parameter 46 to a valid charging profile. |

# Code E901

| | |
|---|---|
| Code Title | SDO Verification Failure |
| Operator Display | E901 |
| System Response | No truck functions active. |
| Reason | TA did not save the requested parameter value. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear after several attempts, replace the TA. |

# Code E996

| | |
|---|---|
| Code Title | Upload and Download Failure Timeout |
| Operator Display | E996 |
| System Response | No truck functions active. |
| Reason | If just loaded software, then is a boot loader issue.<br>If did not just load software, bad handle head. |
| Tests to Run | None |
| How to Clear | Try loading software again. Check service port and cables to VM. If okay, replace handle head. |

# Code E997

| | |
|---|---|
| Code Title | Internal VM Software/Hardware Fault or CAN Bus Issue |
| Operator Display | E997 |
| System Response | No truck functions active. |
| Reason | Faulty VM, CAN line, or CAN adapter issue while trying to load/download software to or from the VM. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear after several attempts, replace the VM. |

# Code E998

| | |
|---|---|
| Code Title | Internal VM Software/Hardware Fault |
| Operator Display | E998 |
| System Response | No truck functions active. |
| Reason | Faulty VM. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear after several attempts, replace the VM. |

# Code E999

| | |
|---|---|
| Code Title | Internal VM Software/Hardware Fault |
| Operator Display | E999 |
| System Response | No truck functions active. |
| Reason | Faulty VM. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear after several attempts, replace the VM. |

# Code BTLR

| | |
|---|---|
| Code Title | Boot loader Error |
| Operator Display | BTLR |
| System Response | No truck function active. |
| Reason | VM Software invalid or missing on startup. |
| Tests to Run | None |
| How to Clear | Load VM software. If unable to load, replace the VM. |

# 8250 Messages and Caution Codes

The following messages and caution codes apply only to Model 8250 lift trucks with a Lithium-ion Battery (LIB).

## Code AC

| | |
|---|---|
| Code Title | AC is Present while Truck is turned ON |
| Operator Display | AC |
| System Response | No truck function active |
| Reason | Charger is plugged into an AC outlet. |
| Tests to Run | None |
| How to Clear | Unplug AC cord once charging is completed. |

## Code b_Lo (C500)

| | |
|---|---|
| Code Title | Very Low State-of-Charge |
| Operator Display | b_Lo |
| System Response | Travel limited to 1 MPH (1.6 km/h). Lift disabled. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery in reserve mode. Truck has up to 500 feet (152.4 m) of travel. |
| Tests to Run | None |
| How to Clear | Charge the Lithium-ion battery. |

## Code bYE (C360)

| | |
|---|---|
| Code Title | Sleep in 5 Seconds Warning |
| Operator Display | bYE |
| System Response | |
| Alarm Sounds | No |
| Reason | Lithium-ion battery will go into Sleep Mode in approximately 5 seconds. |
| Tests to Run | None |
| How to Clear | |

## Code C501

| | |
|---|---|
| Code Title | LIB Contactor Will Not Open |
| Operator Display | C501 |
| System Response | Travel limited to 1 MPH (1.6 km/h). Lift disabled. |
| Alarm Sounds | No |
| Reason | Contactor closed when commanded open. Lithium-ion battery contactor coil bad. |
| Tests to Run | None |
| How to Clear | Repair lithium-ion battery contactor. |

## Code C502

| | |
|---|---|
| Code Title | LIB Charge Current Warning |
| Operator Display | C502 |
| System Response | Travel limited to 1 MPH (1.6 km/h). Lift disabled. Warning sounds after 5 seconds. |
| Alarm Sounds | No |
| Reason | Charge current limit exceeded. |
| Tests to Run | None |
| How to Clear | Limit truck's performance. |

## Code C503

| | |
|---|---|
| Code Title | LIB Discharge Current Warning |
| Operator Display | C503 |
| System Response | Travel limited to 1 MPH (1.6 km/h). Lift disabled. Warning sounds after 5 seconds. |
| Alarm Sounds | No |
| Reason | Discharge current limit exceeded. |
| Tests to Run | None |
| How to Clear | Limit truck's performance. |

# Code C504

| | |
|---|---|
| Code Title | LIB Current Measurement Error |
| Operator Display | C504 |
| System Response | Travel limited to 1 MPH (1.6 km/h). Lift disabled. |
| Alarm Sounds | No |
| Reason | Internal measurement error, current. Potentially bad FRU Component. Code will be sent in conjunction with a FRU fault code. |
| Tests to Run | None |
| How to Clear | Limit truck's performance. |

# Code C505

| | |
|---|---|
| Code Title | LIB High Cell Voltage |
| Operator Display | C505 |
| System Response | Travel limited to 1 MPH (1.6 km/h). Lift disabled. Contactor will open if exceeded for 5 seconds. |
| Alarm Sounds | No |
| Reason | Cell voltage out-of-range. Greater than 4.15 volts on any cell. |
| Tests to Run | None |
| How to Clear | Limit truck's performance. |

# Code C506

| | |
|---|---|
| Code Title | LIB Low Cell Voltage |
| Operator Display | C506 |
| System Response | Travel limited to 1 MPH (1.6 km/h). Lift disabled. Contactor will open if exceeded for 5 seconds. |
| Alarm Sounds | No |
| Reason | Cell voltage out-of-range. Less than 3.00 volts on any cell. |
| Tests to Run | None |
| How to Clear | Limit truck's performance. |

## Code C507

| | |
|---|---|
| Code Title | LIB Cell Temperature Measurement Error |
| Operator Display | C507 |
| System Response | Travel limited to 1 MPH (1.6 km/h). Lift disabled. |
| Alarm Sounds | No |
| Reason | Internal measurement error - temperature. Cell temperature out of range (greater than 65 degrees C/149 degrees F or less than –30 degrees C/–22 degrees F). |
| Tests to Run | None |
| How to Clear | Limit truck's performance. |

## Code Hot3 (C508)

| | |
|---|---|
| Code Title | LIB Too Hot to Charge |
| Operator Display | Hot3 |
| System Response | Disabled while on charge. Charging will not start until temperature decreases. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery too hot. |
| Tests to Run | None |
| How to Clear | Allow Lithium-ion battery to cool down. |

## Code Cold (C509)

| | |
|---|---|
| Code Title | LIB Too Cold to Charge |
| Operator Display | Cold |
| System Response | Disabled while on charge. Charging will not start until temperature increases. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery too cold. |
| Tests to Run | None |
| How to Clear | Allow Lithium-ion battery to warm up. |

8250 Messages and Caution Codes

# Code C510

| | |
|---|---|
| Code Title | LIB Internal Failure |
| Operator Display | C510 |
| System Response | Truck performance is limited. |
| Alarm Sounds | No |
| Reason | Within the Lithium-ion battery: Heater, fan, internal cell balancing failure. BMC real time clock error, shock/vibration limit exceeded, or BMC humidity limit exceeded. Refer to the lithium-ion battery error history for the exact cause. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, the lithium-ion battery requires service. |

# Code C511

| | |
|---|---|
| Code Title | LIB HUD Failure |
| Operator Display | C511 |
| System Response | Truck performance is limited. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery HUD potential failure. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, replace the HUD. |

# Code C512

| | |
|---|---|
| Code Title | LIB BMC Failure |
| Operator Display | C512 |
| System Response | Truck performance is limited. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery BMC potential failure. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, replace the BMC. |

# Code C513

| | |
|---|---|
| Code Title | LIB BMU Failure |
| Operator Display | C513 |
| System Response | Truck performance is limited. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery BMU potential failure. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, replace the BMU. |

# Code C514

| | |
|---|---|
| Code Title | LIB Charger Failure |
| Operator Display | C514 |
| System Response | Truck performance is limited. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery charger potential failure. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, replace the Charger. |

# Code C515

| | |
|---|---|
| Code Title | LIB Not Commissioned |
| Operator Display | C515 |
| System Response | Truck performance is limited. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery needs to be commissioned. |
| Tests to Run | None |
| How to Clear | Commission the lithium-ion battery. |

# Code C516

| | |
|---|---|
| Code Title | LIB Detected |
| Operator Display | C516 |
| System Response | No limitations. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery is detected and the truck is not configured for that option. |
| Tests to Run | None |
| How to Clear | No action require, parameter is set correctly by the VM. |

# Code C517

| | |
|---|---|
| Code Title | LIB Reserve Mode |
| Operator Display | C517 |
| System Response | Truck performance is limited. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery state-of-charge is low. |
| Tests to Run | None |
| How to Clear | Charge the Lithium-ion battery. |

# Code C518

| | |
|---|---|
| Code Title | LIB Manual Mode |
| Operator Display | C518 |
| System Response | Truck performance is limited. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery state-of-charge is low. |
| Tests to Run | None |
| How to Clear | Charge the Lithium-ion battery. |

## Code C519

| | |
|---|---|
| Code Title | LIB Dormant Mode |
| Operator Display | C519 |
| System Response | Truck performance is limited. |
| Alarm Sounds | No |
| Reason | Lithium-ion battery has reached its final state-of-charge state. |
| Tests to Run | None |
| How to Clear | Charge the Lithium-ion battery. |

## Code C521

| | |
|---|---|
| Code Title | Multi-CMA Loss |
| Operator Display | C521 |
| System Response | Truck performance is limited to reduced speed and no lift. |
| Alarm Sounds | No |
| Reason | 2-CMA Lithium-ion battery has loss of communication. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, check the wiring. |

## Code E751

| | |
|---|---|
| Code Title | CAN Bus Communication Error with the LIB |
| Operator Display | E751 |
| System Response | No truck function active. |
| Alarm Sounds | No |
| Reason | The VM has not detected any response on the CAN Bus from the lithium-ion battery. Possible bad connection or damaged wire/crimp between the VM and lithium-ion battery. |
| Tests to Run | None |
| How to Clear | Cycle the truck OFF/ON. If still not clear, check the wiring. |

# Traction Amplifier Flash Codes

An internal microcontroller automatically examines the function of the Traction Amplifier (TA). When this microcontroller finds a fault, the LEDs on the bottom of the TA flash the appropriate code. See Table 6-1. Correct faults by following the "Messages and Caution Codes" beginning on page 6-5 and the "Error Codes" beginning on page 6-21.

**NOTE:** When displaying a fault code, the red LED flashes once for the first number, followed by the orange LED for the first number of the code itself. The red LED then flashes twice, followed by the orange LED flashing the second number of the code.

*Table 6-1. TA Flash Codes*

| TA Flash Code | Error Code | Description | Action |
|---|---|---|---|
| 1,2 | E201 | Overcurrent Fault | See "Code E201" on page 6-26. |
| 1,3 | E202 | TA Current Sensor Error | See "Code E202" on page 6-26. |
| 1,4 | E225 | Precharge Fault | See "Code E225" on page 6-29. |
| 1,5 | E223 | TA Severe Under-Temperature Cutback | See "Code E223" on page 6-28. |
| 1,6 | E224 | TA Severe Over-Temperature Cutback | See "Code E224" on page 6-28. |
| 1,7 | E221 | Severe Battery Undervoltage | See "Code E221" on page 6-27. |
| 1,8 | E222 | Severe Battery Overvoltage | See "Code E222" on page 6-28. |
| 2,1 | Cold | TA in Cold Thermal Cutback | See "Code Cold (C43)" on page 6-11. |
| 2,2 | Hot2 | TA in Warm Thermal Cutback | See "Code Hot2 (C45)" on page 6-12. |
| 2,3 | Lo | Battery Undervoltage Warning | See "Code Lo (C41)" on page 6-10. |
| 2,4 | Hi | Battery Overvoltage Warning | See "Code Hi (C42)" on page 6-10. |
| 2,5 | E233 | +5V Supply Failure | See "Code E233" on page 6-30. |
| 2,6 or 2,7 | C64 | Digital Output Overcurrent | See "Code C64" on page 6-14. |
| 2,7 | C405 | Load Hold Coil Driver Open or Shorted | See "Code C405" on page 6-20. |
| 2,8 | Hot1 | Motor Temperature Hot Cutback | See "Code Hot1 (C44)" on page 6-11. |
| 2,9 | C70 | Motor Temperature Sensor Fault | See "Code C70" on page 6-16. |
| 3,1 | E106 | Main Contactor Driver Overcurrent Error | See "Code E106" on page 6-21. |
| 3,2 | C66 | Brake Coil Driver Open or Shorted | See "Code C66" on page 6-15. |
| 3,3 | C60 | Lift Contactor Coil Open or Shorted | See "Code C60" on page 6-13. |
| 3,4 | C31 | Lost Brake Pot Input | See "Code C31" on page 6-8. |
| | C62 | Horn Coil Driver Open or Shorted | See "Code C62" on page 6-14. |
| 3,5 | C61 | PD Driver Open/Shorted - Lower | See "Code C61" on page 6-14. |
| 3,6 | C71 | Sensor Bearing Faulty | See "Code C71" on page 6-16. |
| 3,7 | E203 | Motor Open | See "Code E203" on page 6-26. |
| 3,8 | E108 | Contactor Weld Error | See "Code E108" on page 6-22. |
| 3,9 | E107 | Contactor OFF Error | See "Code E107" on page 6-22. |

| TA Flash Code | Error Code | Description | Action |
|---|---|---|---|
| 4,6 | E220 | Default Parameters DNL | See “Code E220” on page 6-27. |
| 4,9 | E228 | Parameter Change Fault | See “Code E228” on page 6-29. |
| 5,6 | br_o | Brake Override | See “Code br_o (C16)” on page 6-6. |
| 6,1 | E691 | TA Current Cutback | See “Code E691” on page 6-33. |
| 6,8 | E232 | VCL Runtime | See “Code E232” on page 6-30. |
| 6,9 | E230 | External Supply Out-of-Range | See “Code E230” on page 6-29. |
| 7,3 | C72 | Traction Motor Stalled | See “Code C72” on page 6-16. |
| 9,2 | C67 | EM Brake Not Set | See “Code C67” on page 6-15. |

# Delta-Q Charger Codes

The Charge Profile / Error Display shows one of four possible codes to indicate different conditions:

- An 'F' code means that an internal fault condition has caused charging to stop.
- An 'E' code means that an external error condition has caused charging to stop.
- A 'P' code means that the charger programming mode is active.
- A 'USB' code means that the USB interface is active, and the USB flash drive should not be removed

The 'E,' 'F' and 'P' codes will appear, then are followed by three numbers and a period to indicate different conditions (for exemple: E-0-0-4). See the "Charger Fault Codes" or "Charger Error Codes" tables for details on these conditions and their solutions.

*Table 6-2. Charger Fault Codes*

| Code | Description | Solution |
|---|---|---|
| F-0-0-1 | DC-DC failure: LLC excessive leakage fault. | Internal charger fault. Remove AC and battery for minimum 30 seconds and retry charger. If it fails again, please contact the manufacturer of your vehicle or machine. |
| F-0-0-2 | PFC failure: PFC excessive leakage fault. | |
| F-0-0-3 | PFC has taken too long to boost. | |
| F-0-0-4 | The charger has been unable to calibrate the current offset. | |
| F-0-0-5 | The voltage drop across the DC relay is too high while the relay is closed. | |

*Table 6-3. Charger Error Codes*

| Code | Description | Solution |
|---|---|---|
| E-0-0-1 | Battery voltage over limit in software. Typically 2.5V/cell. | • Check the battery voltage and cable connections.<br>• Verify the charger voltage model is appropriate for the batteries.<br>• This error will automatically clear once the condition has been corrected. |
| E-0-0-2 | Battery voltage too low to start a charge cycle. Algorithm dependent – typically 0.1V/cell. | • Check the battery voltage and cable connections.<br>• Check the battery size and condition. Batteries may be over-discharged. Use another charger to bring the batteries above the minimum voltage.<br>• This error will automatically clear once the condition has been corrected. |

| Code | Description | Solution |
|---|---|---|
| E-0-0-3 | Charge time limit reached. Algorithm dependent. | • Charger output is reduced due to high temperatures. Operate at lower ambient temperature.<br>• Charger output reduced due to low AC voltages, Check AC voltage.<br>• Check for shorted or damaged cells.<br>• Poor battery health. Replace battery.<br>• Very deeply discharged battery. Retry charge.<br>• Poorly conected battery. Check connections.<br>• Extra loads. Turn off other devices running on the battery.<br>This error will automatically clear once the charger is reset by cycling DC or by loss of AC for over 10 minutes. |
| E-0-0-4 | Battery could not be trickle charged up to the minimum voltage. May also be used for other battery-related errors, depending on the algorithm. | • Check for shorted or damaged cells.<br>• Poor battery health. Replace the battery pack.<br>• Check the DC connections.<br>This error will automatically clear once the charger is reset by cycling DC or by loss of AC for over 10 minutes. |
| E-0-0-7 | Charge amp-hour Limit reached. Algorithm dependent. | • Charger output is reduced due to high temperatures. Operate at lower ambient temperature.<br>• Charger output reduced due to low AC voltages, Check AC voltage.<br>• Check for shorted or damaged cells.<br>• Poor battery health. Replace battery.<br>• Very deeply discharged battery. Retry charge.<br>• Poorly connected battery. Check connections.<br>• Extra loads. Turn off other devices running on the battery.<br>This error will automatically clear once the charger is reset by cycling DC or by loss of AC for over 10 minutes. |
| E-0-0-8 | Battery temperature is out-of-range. Algorithm dependent. | • Cool or warm batteries as needed.<br>• Check temperature sensor and connections.<br>This error will automatically clear once the condition has been corrected. |
| E-0-1-2 | Reverse polarity | Battery is connected to the charger incorrectly. Check the battery connections.<br>This error will automatically clear once the condition has been corrected. |

Delta-Q Charger Codes

| Code | Description | Solution |
|---|---|---|
| E-0-1-3 | Battery does not take current | • Check for an electrical device connected between the charger and the battery which passes through voltage but not current (for example: Diode).<br>• Some lithium algorithms try to charge without detecting battery voltage and will show this fault if it is not connected. Make sure the charger is correctly connected to approved equipment.<br>This error will automatically clear once the charger is reset by cycling DC or by loss of AC for over 10 minutes. |
| E-0-1-6, E-0-1-8 | Software upgrade failed | • Make sure the USB flash drive is properly formatted and is not corrupted.<br>• Make sure the USB flash drive does not draw excessive current.<br>• Copy the install files to the USB flash drive again.<br>• Retry the update by reinserting the USB flash drive into the charger.<br>• If software updates continue to fail, please contact the manufacturer of your vehicle or machine. |
| E-0-1-7 | USB mount/unmount error | • Remove and reinsert the USB flash drive into the charger.<br>• Make sure the USB flash drive is properly formatted and is not corrupted.<br>• Make sure the USB flash drive does not draw excessive current.<br>• If the condition persists, remove AC and battery power for minimum of 30 seconds and retry charger.<br>• If the problem continues to exist, please contact the manufacturer of your vehicle or machine. |
| E-0-1-9 | Hardware build does not support software version | • The charger hardware does not support the new software version trying to be programmed. Existing software is left running. Please contact the manufacturer of your vehicle or machine. |
| E-0-2-0 | No active algorithm selected. | • Reprogram the charger with its original software, algorithms, and settings.<br>• Use the wrench button to select the correct algorithm if it is still available on the charger.<br>This problem will automatically clear when an available algorithm is set on the charger as default. |
| E-0-2-1 | High battery voltage while charging. Algorithm dependent - typically 2.8V/cell. | • When already full, some new batteries may exhibit this error. Cycle the batteries and see if it reoccurs.<br>• Check battery size and condition. Resistive batteries in poor condition may cause this. If charged when already full, some new batteries will also cause this. Cycle the batteries a few times.<br>• Check the battery voltage and cable connections.<br>This error will automatically clear once the condition has been corrected. |

| Code | Description | Solution |
|---|---|---|
| E-0-2-2 | Low battery voltage while charging. Algorithm dependent - typically 0.1V/cell. | • Another device may be drawing current from the battery.<br>• Check the battery voltage and cable connections.<br>• Check battery size and condition. Batteries may be over discharged. Use another charger to bring the batteries above the minimum voltage.<br>This error will automatically clear once the condition has been corrected. |
| E-0-2-3 | High AC voltage error (>270VAC) | • AC voltage is too high. Connect the charger to an AC source that provides stable AC between 85 and 270 VAC (45-65 Hz).<br>This error will automatically clear once the condition has been corrected. |
| E-0-2-4 | Charger failed to turn on correctly | • Disconnect AC input and battery for 30 seconds before retrying.<br>• If the problem continues to exist, please contact the manufacturer of your vehicle or machine. |
| E-0-2-5 | AC voltage has dipped below 80 VAC three (3) times in 30 seconds | • AC source is unstable. This could be caused by an undersized generator and/or input cables that are too long or too short. Connect the charger to an AC source that provides stable AC between 85 and 270 VAC (45-65 Hz).<br>This error will automatically clear once the condition has been corrected. |
| E-0-2-6 | One or more USB script commands failed | • Make sure the USB flash drive is properly formatted and is not corrupted.<br>• Make sure the right update package is being used.<br>• Copy the install files to the USB flash drive again.<br>• Retry the update by reinserting the USB flash drive into the charger.<br>• If software updates continue to fail, please contact the manufacturer of your vehicle or machine. |
| E-0-2-7 | USB overcurrent fault | • USB hardware overcurrent protection has been tripped. Remove and reinsert USB flash drive.<br>• If condition persists, try using a different USB flash drive. |
| E-0-2-8 | Attempt to select algorithm incompatible with this software | • Update charger software, continue to use existing algorithm* or select a different charging algorithm that is compatible.<br>**NOTE:** If a different algorithm is selected, then the existing algorithm will remain in the charger.<br>**NOTE:** If upgrading existing algorithm then existing algorithm will be deleted, please contact the manufacturer of your vehicle or machine for a software upgrade to run the new algorithm. |

Delta-Q Charger Codes

| **Code** | **Description** | **Solution** |
|---|---|---|
| E-0-2-9 | Cannot transmit on CAN bus | • Check the physical CAN connector, electrical bus conditions and other CAN modules for correct functioning. For example, check termination resistance is approximately 60 ohms. |
| E-0-3-0 | CAN heartbeat timeout on battery module | • May be caused by a missing heartbeat message. Check the CAN Bus battery module for correct function.<br>This error will automatically clear once the condition has been corrected. |
| E-0-3-1 | The Vref for the ADC measurements has triggered an alarm | • Internal charger error. Remove AC and battery for minimum 30 seconds and retry charger.<br>• If the problem continues to exist, please contact the manufacturer of your vehicle or machine.<br>This error will automatically clear once the condition has been corrected. |
| E-0-3-6 | Battery temperature sensor is missing or shorted | • Check if sensor is connected correctly<br>• The charger behavior when this fault occurs can be configured.<br>This error will automatically clear once the condition has been corrected. |
| E-0-3-7 | CANOpen reprogramming failed | • Re-try CANOpen download or re-program using the USB.<br>This error will automatically clear once reprogramming has completed successfully. |
| E-0-3-8 | Fan will not turn | • Check fan connections for loose wires.<br>• Check rotor is not locked, or fan is not obstructed. Inspect the fan and clear the blockage.<br>This error will automatically clear once the condition has been corrected. |
| E-0-4-0 | Fan voltage pulled low | • Make sure the fan is not stuck, sticking, or otherwise overloaded. |

# Section 7. Component Procedures

# List of Component Procedures by Component System

List of Component Procedures by Component System

# Component Location Photos

*Figure 7-1. Tractor Interior*

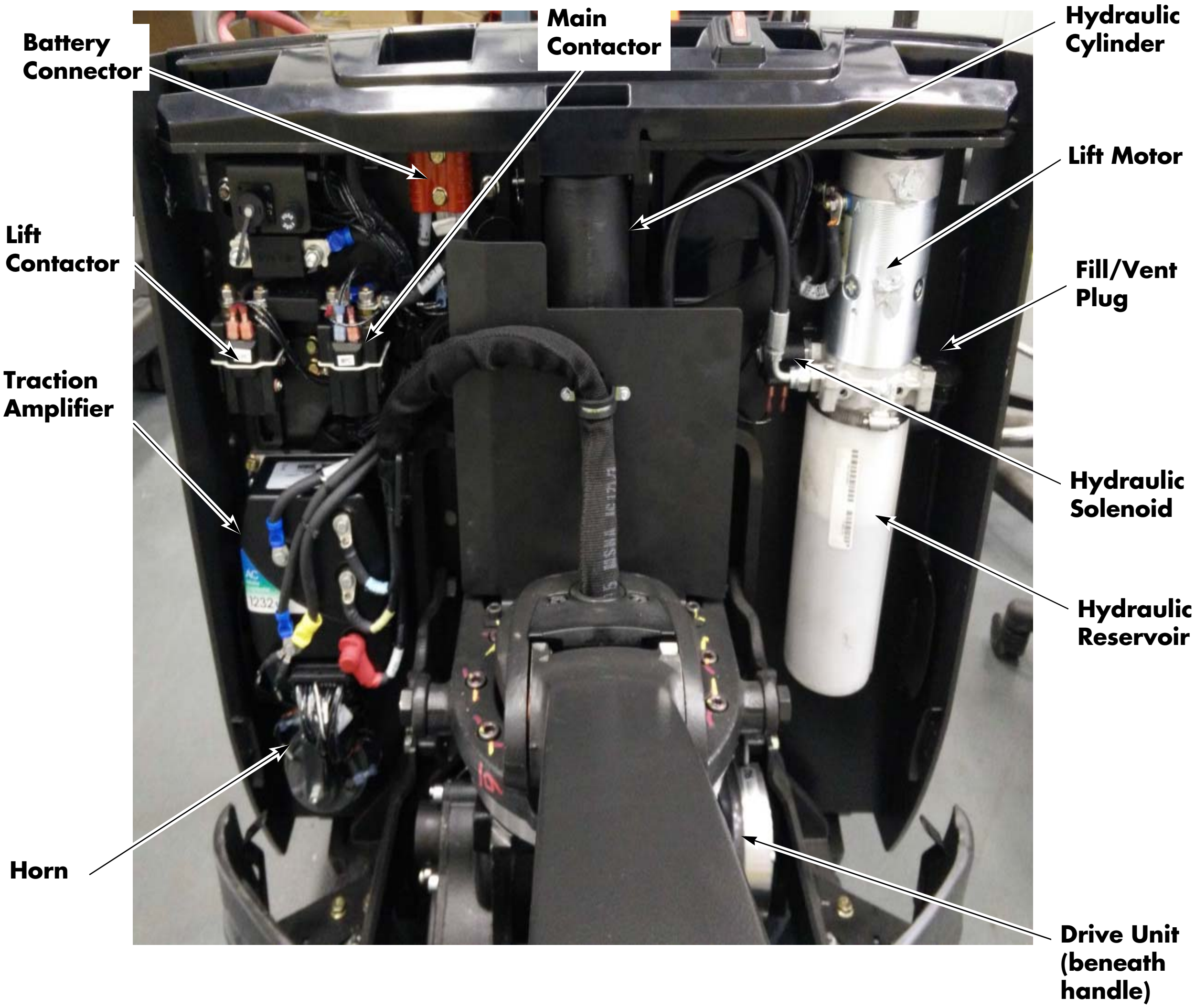


*Figure 7-2. Model 8210 Major Components*

# Finish and Accessories

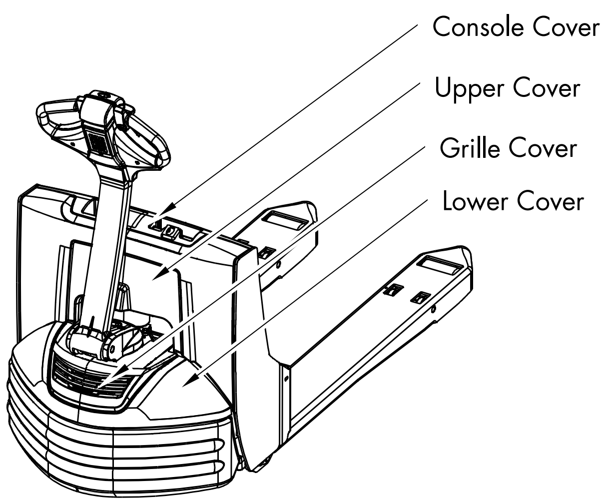
# Tractor Covers

**NOTE:** For replacement parts information refer to the Parts Manual.

## Tractor Cover Removal

The tractor covers are designed to be easily removed and reinstalled for service and maintenance.

*Figure 7-3. Tractor Cover Identification*



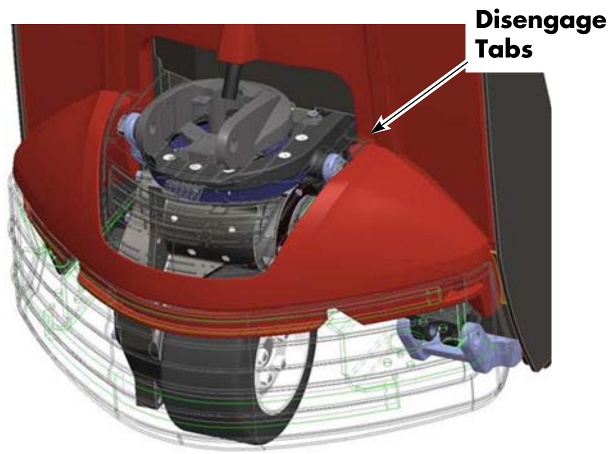

Remove covers in this order:

1. Lower Cover
2. Grille Cover
3. Upper Cover
4. Console Cover

### Lower Cover Removal

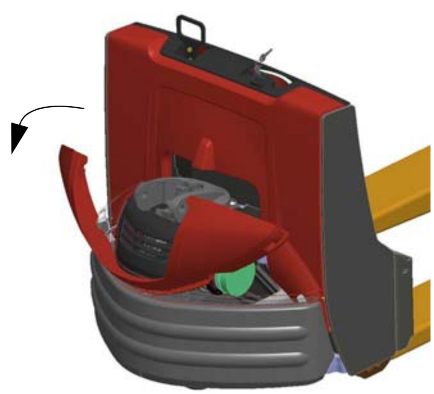
1. Place the steer tiller in the centered (straight ahead) position.
2. Disengage the tabs (first one side and then the other) from the grille cover.

*Figure 7-4. Disengage Tabs at Grille Cover*




3. Lift the cover while tilting it toward the operating position to disengage it from the bumper.

*Figure 7-5. Tilt outward while lifting*



## Grille Cover Removal

1. Grasp one side of the grille cover and pull straight out (toward the operating position) to release the snaps from the main harness bracket.

*Figure 7-6.*

## Upper Cover Removal

1. Place hand(s) under the center of the upper cover. Lift upward to disengage the upper cover from brackets on the side plates.

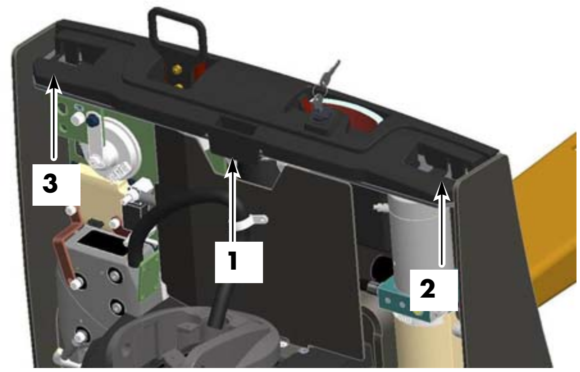
*Figure 7-7.*

2. Tilt the top of the upper cover away from the truck while lifting the cover off.

## Console Cover Removal

1. Disengage the center snap from the top plate and lift upwards.

*Figure 7-8. Console Cover Removal Sequence*




2. While lifting the center of the cover, disengage the snap on the right side of the cover.
3. Continue lifting after the right side of the cover is released and disengage the snap on the left side of the cover.
4. Lift the cover straight up to remove it completely from the lift truck.

# Tractor Cover Installation

Install the tractor covers in this order:

1. Console Cover
2. Upper Cover
3. Grille Cover
4. Lower Cover

## Console Cover Installation

1. Align the console cover with the top plate. Use the side plates and bulkhead as guides.

*Figure 7-9. Align Console with side plates and bulkhead*

2. Press the console cover straight down onto the top plate.
3. Make sure all three snaps are engaged fully to retain the console cover.

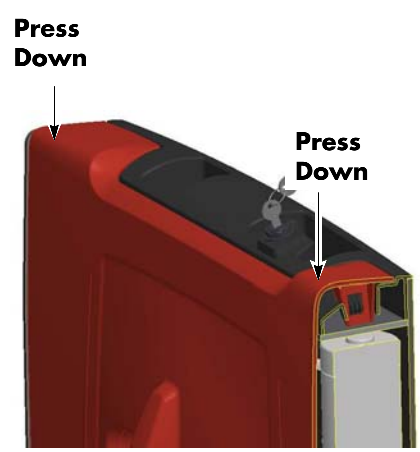
## Upper Cover Installation

1. Rest the upper cover on the top of the bottom upper cover bracket on the side plate.

*Figure 7-10.*


2. Tilt the top of the upper cover toward the forks until it is vertical.
3. Press down on both shoulders of the upper cover until you feel the snaps engage.

*Figure 7-11.*

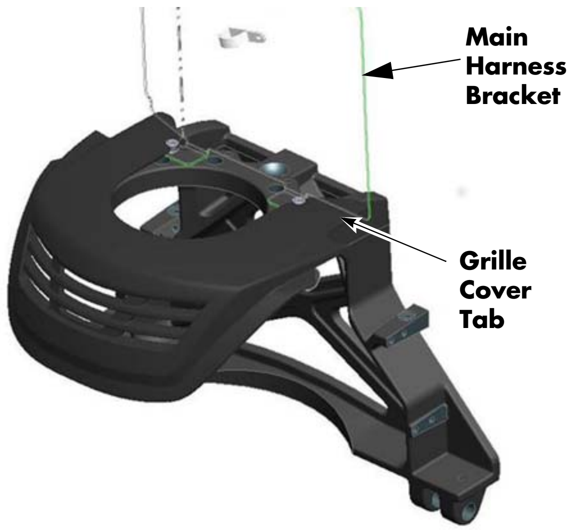



4. Gently pull up on the upper cover to verify the snaps are engaged.
5. Inspect the gap between the console cover and the upper cover to make sure it is uniform along the entire surface.

## Grille Cover Installation

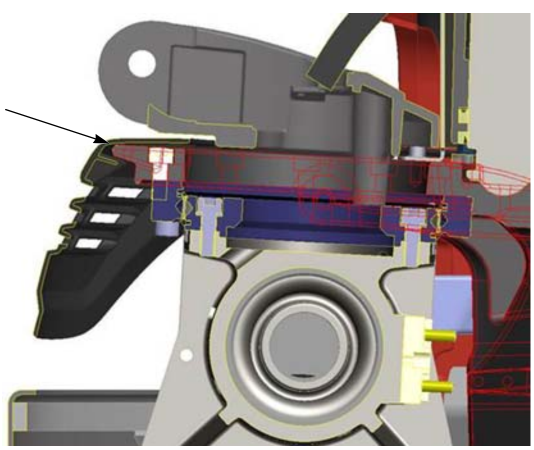
1. Place the grille cover flat on the top surface of the drive unit frame.
2. Slide the grille cover toward the forks of the trucks. The grille cover tabs will slide underneath the main harness bracket and around the heads of the main harness bracket mounting bolts. This locks the grille cover in position.

*Figure 7-12.*




3. Pull up on the back of the grille cover to make sure it is engaged with the casting.

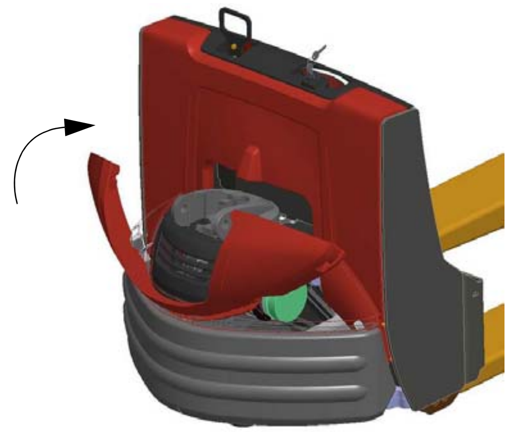
*Figure 7-13. Grille Cover Engaged with Casting*



## Lower Cover Installation

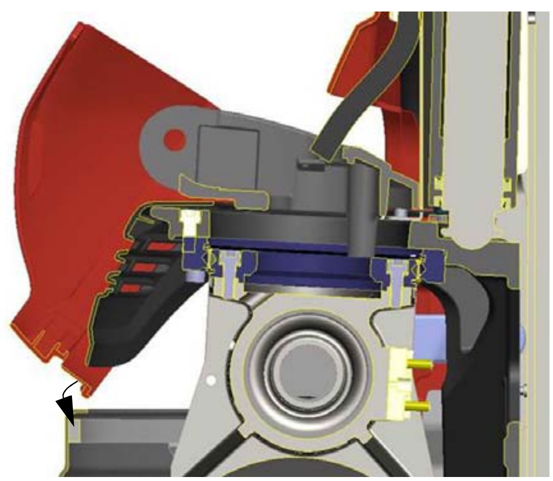
1. Place the steer tiller in the centered (straight ahead) position.
2. Place the center of the lower cover between the grille cover, drive unit, and bumper assembly.

*Figure 7-14.*



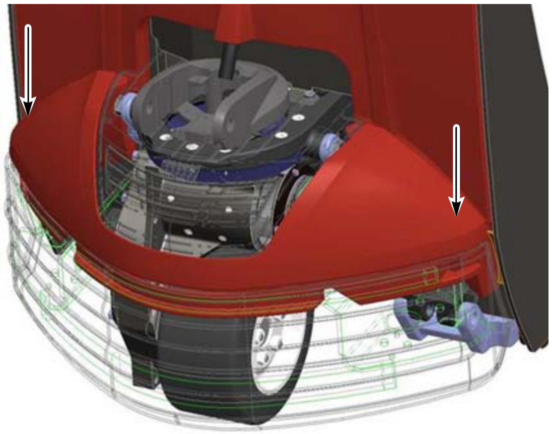
3. Seat the notch of the lower cover under the bumper beam.

*Figure 7-15. Seat lower cover in bumper*



4. Press downward on the cover corners to make sure the bumper spring clips are engaged.

*Figure 7-16. Press to engage bumper spring clips*



5. Insert the lower cover tabs (first on one side and then the other) into the opening created by the grille cover.
6. Inspect the gaps between the grille cover and lower cover and the gaps between the lower cover and the bumper to make sure they are uniform.

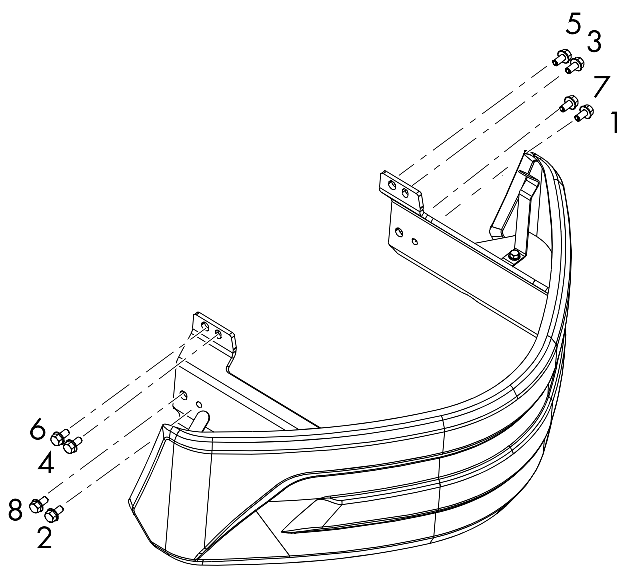
# Tractor Bumper

The tractor bumper is designed to be easily removed and re-installed for service and maintenance.

## Bumper Removal

1. Remove the tractor covers. See "Tractor Covers" on page 7-8.
2. Remove the eight bolts securing the bumper to the pivot frame.

*Figure 7-17. Bumper Removal*




## Bumper Installation

To install the bumper:

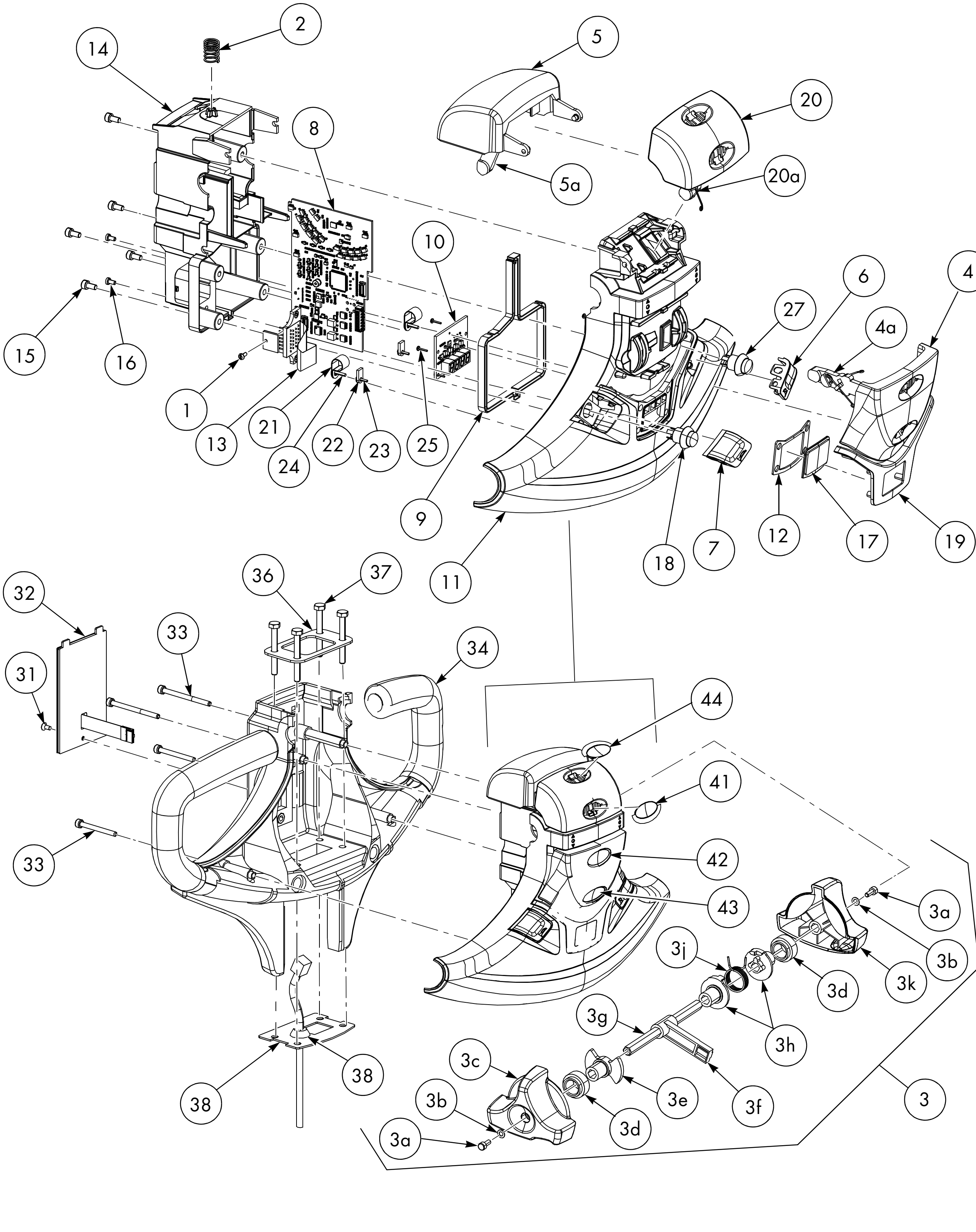
1. Attach the bumper to the pivot frame.
2. Install the bumper mounting bolts in the order identified in Figure 7-17.
3. Torque bolts to 27 ft. lb. (37 Nm).
4. Install the tractor covers. See "Tractor Covers" on page 7-8.

# Steering and Controls

# Control Handle Head

**Note:** For replacement parts information refer to the Parts Manual.

Figure 7-18.Control Handle Head



| Item No. | Description | Item No. | Description | Item No. | Description |
|---|---|---|---|---|---|
| 1 | Screw | 7 | Horn, Push Button, Left | 24 | Screw |
| 2 | Spring | 8 | Vehicle Manager | 25 | Screw |
| 3 | Throttle Control (includes 3a-3k) | 9 | Gasket | 26 | Holder (Not Shown) |
| 3a | Screw | 10 | Display Card | 27 | Push Button |
| 3b | Washer | 11 | Top Cover | 31 | Screw |
| 3c | Directional/Speed Control Thumb Wheel (Right) | 12 | Gasket | 32 | Keypad (Option) |
| 3d | Sliding Bearing | 13 | Vehicle Manager Flex Circuit Card | 33 | Screw, 2 x M5 x 45 - 2 x M5 x 60 |
| 3e | Control Stop | 14 | Lower Housing | 34 | Handle |
| 3f | Arm | 15 | Screw, M4 x 40 | 35 | Cover, keypad |
| 3g | Axle | 16 | Screw, Pan Head | 36 | Top Plate |
| 3h | Spring Holder | 17 | Display Glass | 37 | Screw |
| 3j | Spring | 18 | Push Button, Horn | 38 | Bottom plate |
| 3k | Directional/Speed Control Thumb Wheel (Left) | 19 | Display Cover | 39 | Grommet |
| 4 | Push Button, Lift/Lower | 20 | Push Button (Click-to-Creep) | 41 | Symbol Plate, Blank |
| 4a | Arm, Lift/Lower | 20a | Arm | 42 | Symbol Plate, Lift |
| 5 | Emergency Reverse Button | 21 | Cable Clamp | 43 | Symbol Plate, Lower |
| 5a | Arm, Emergency Reverse Button | 22 | Retainer | 44 | Symbol Plate, Turtle |
| 6 | Horn Push Button, Right | 23 | Screw | | |

## Control Handle Disassembly

1. Remove screw [31], then unplug and remove the keypad [32].
2. Remove the four screws [33] that secure the top cover assembly to the handle. Hold the top cover [11] firmly while removing the screws.
3. Disconnect the cable connected to the vehicle manager [8].
4. For access to the vehicle manager [8], remove the screws [15] securing the lower housing [14].

**⚠CAUTION**

**Static electricity! Risk of static discharge that can damage the electronics. Make sure to take the necessary precautions before working with the electronics.**

5. Carefully lift off the lower housing [14].

**NOTE:** Place your finger between the lower housing [14] and button [5] to hold the directional/speed control thumb wheel [3c, 3k] in place.

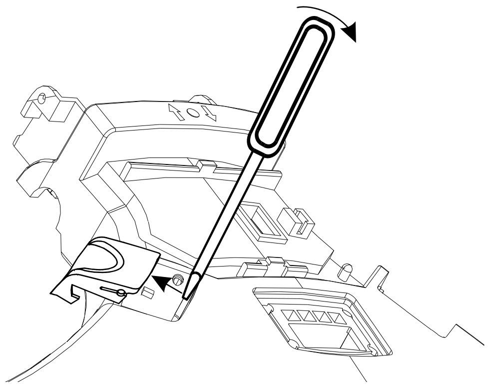
## Horn Button/Switch Replacement

1. Remove the horn button [7, 18]. See Figure 7-19.

**⚠CAUTION**

**Do *not* use excessive force when prying on the button or you can damage the locking tabs.**

Figure 7-19. Horn Button Removal

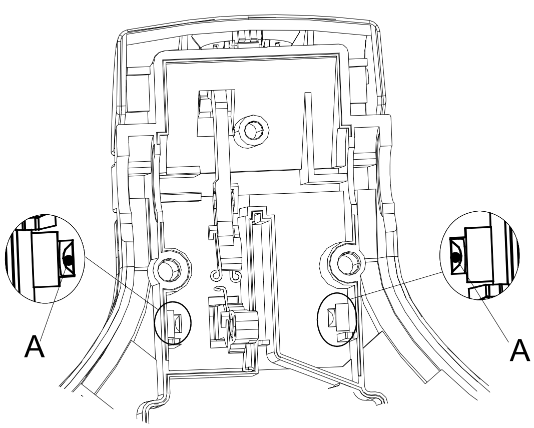


2. Disconnect the connection for the switch on the vehicle manager [8].
3. Press out the switch from its mounting in the top cover assembly.

## Lift/Lower Button Replacement

1. Remove the lift/lower button [4] by placing a screwdriver in the hole [A]. See Figure 7-20.

Figure 7-20. Lift/Lower Button Removal

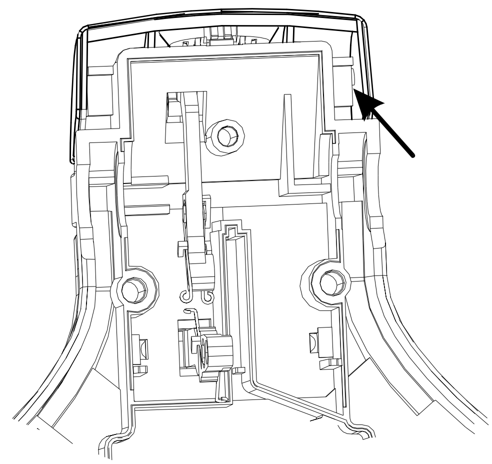



2. Unscrew the holder [26] so that the arm [4a] comes loose.

## Push Button Replacement

1. Press the push button [20] sideways.
2. Insert a screwdriver and carefully pry the button loose. See Figure 7-21.

Figure 7-21. Push Button Removal

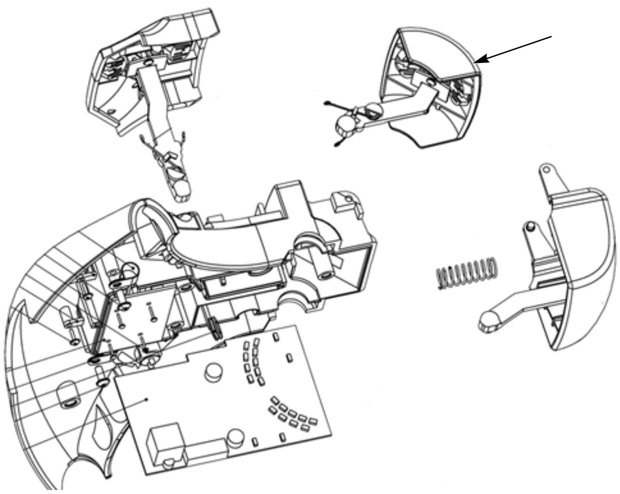


3. Unscrew the button's holder and arm [20a].

## Emergency Reverse Replacement

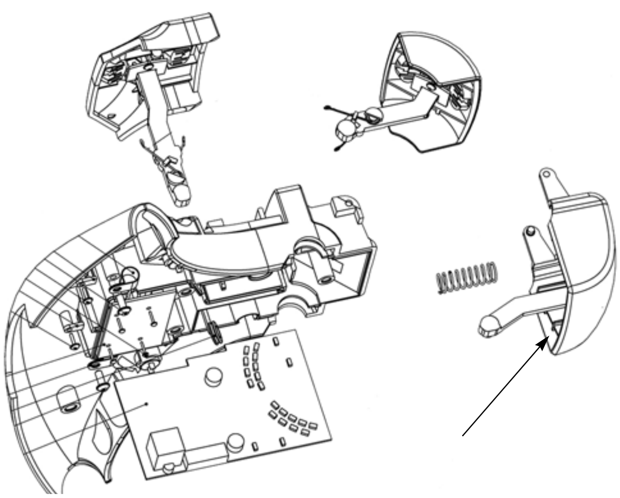
1. Remove the push button following steps 1 and 2 in the Push Button Replacement procedure. See Figure 7-22.

*Figure 7-22. Remove the Push Button*



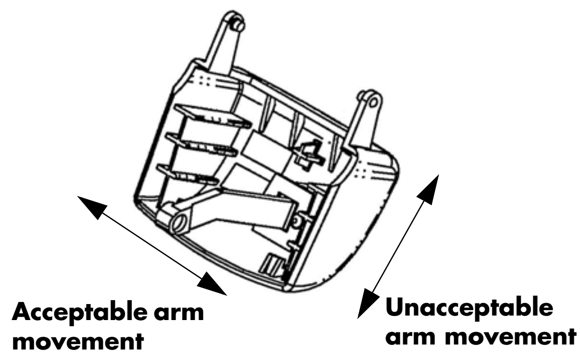
2. Insert a screwdriver and carefully pry the Emergency reverse button loose. See Figure 7-23.

*Figure 7-23. Emergency Reverse Button Removal*



**NOTE:** If the Emergency reverse button is removed for any reason, verify that the magnet arm is not loose in the up and down direction. (It is normal for the arm to move slightly from side to side.) See Figure 7-24. A loose arm in the up and down direction can result in "odd" truck behavior, such as unexpected travel.

*Figure 7-24.*




3. Install the Emergency reverse button in the control head.
4. Install the push button in the control head.

# Keypad

A keypad is offered as optional equipment. The keypad allows access to the lift truck through a personalized PIN number. Keypads may be added in the field to lift trucks originally manufactured without one. Factory authorization must be granted before the addition of any equipment on the lift truck.

## Keypad Removal

1. Remove the keypad retaining screw.

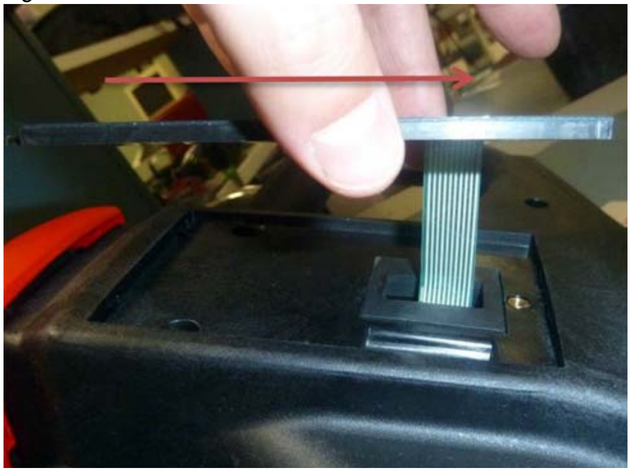
*Figure 7-25.*

2. Slowly lift the keypad from the retaining screw edge.

*Figure 7-26.*

3. Carefully move the keypad toward the handle stem to release the keypad mounting tabs from the control handle head.

*Figure 7-27.*



4. Tilt the keypad to gain access to the ribbon cable connector.

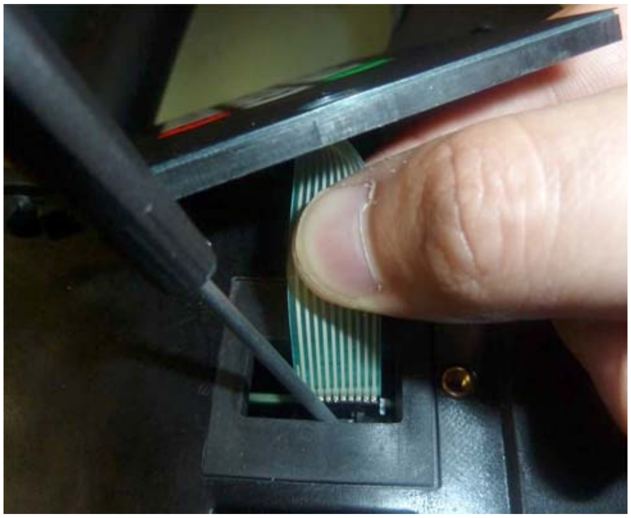
*Figure 7-28.*

5. Use a fine tip flat head screwdriver to carefully press the connector's latch toward the circuit board (VM).

*Figure 7-29.*

6. With the connector's latch pressed, gently pull on the ribbon cable until the connector is released. Connector removal requires very little force when the latch is correctly pressed.

*Figure 7-30.*



## Keypad Installation

1. Install the ribbon cable connector into the VM. The connector will only install in one orientation. You will hear the latch click when the connector is fully inserted.

*Figure 7-31. Align Cable Connector with VM Connector*

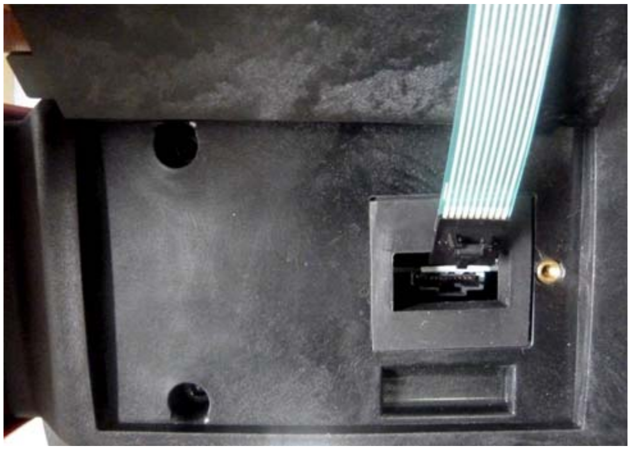
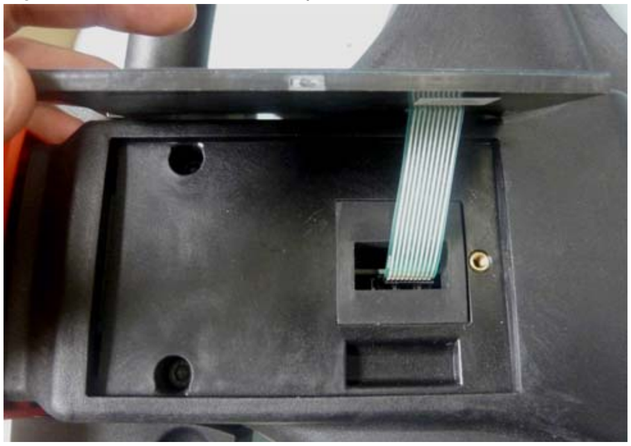


*Figure 7-32. Connector Fully Installed*

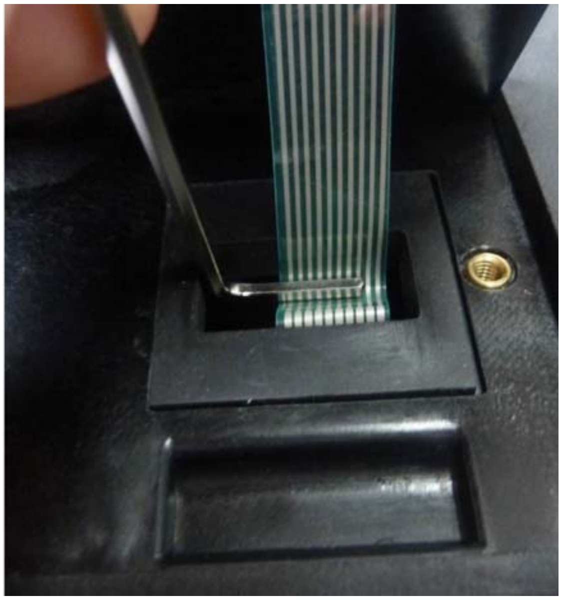


2. Insert the keypad mounting tabs into the control handle head.

*Figure 7-33.*

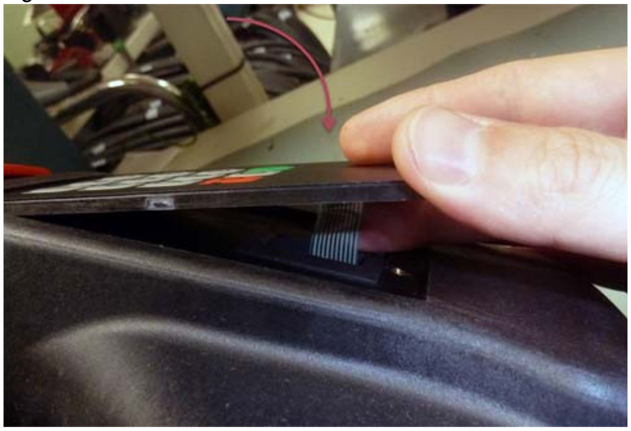
3. Use the short end of an Allen key to gently bend the excess ribbon cable into the handle head away from the connector latch. Use extreme care when performing this step so that the ribbon cable is not creased.

*Figure 7-34.*



4. Carefully lower the keypad into the control handle head. Use a finger to help guide the excess ribbon cable into the handle head.

*Figure 7-35.*

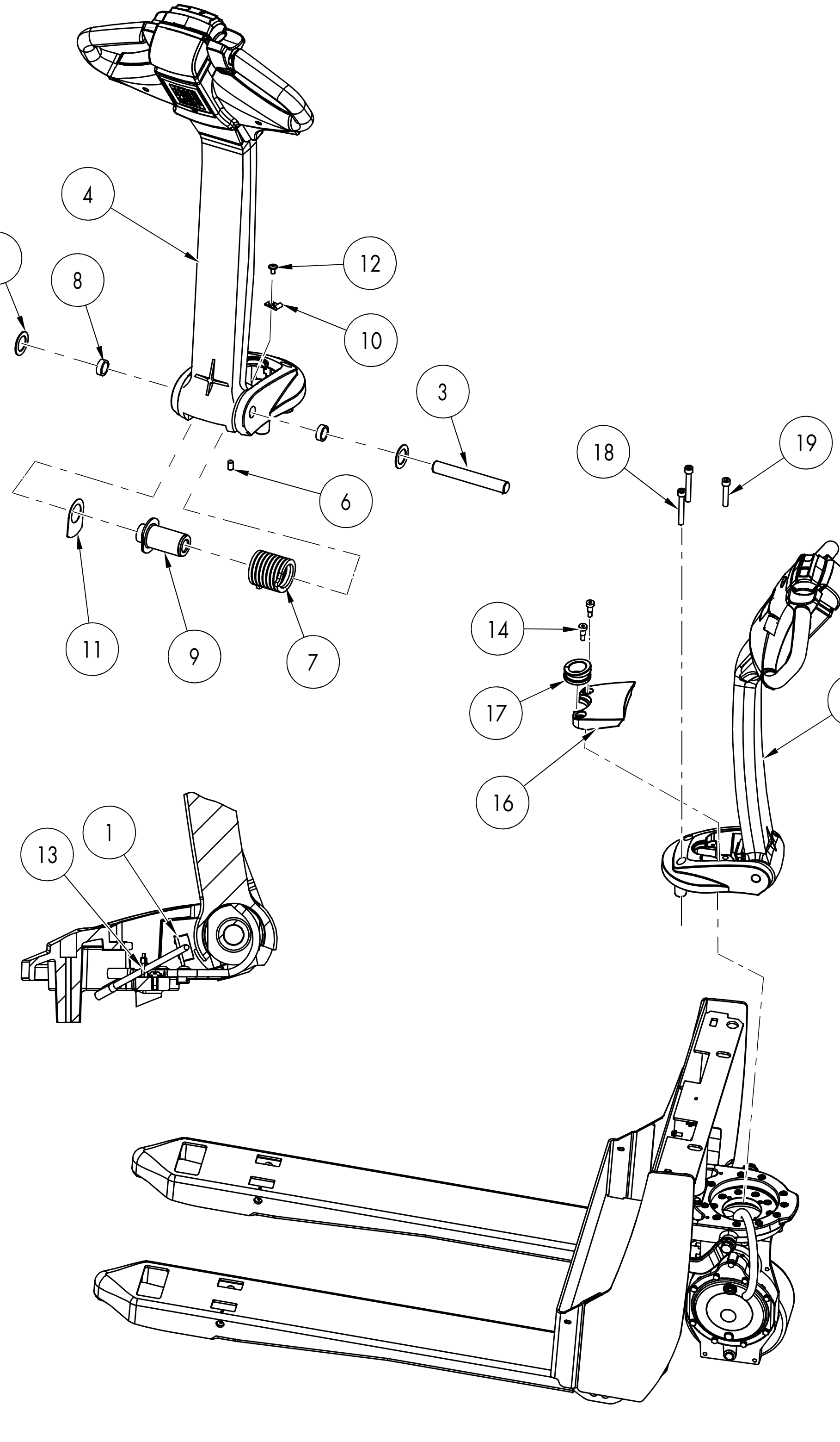


5. Once the keypad is fully seated. Install the retaining screw. Torque the screw to 4.4 in. lb. (0.5 Nm).

*Figure 7-36.*

# Control Arm Assembly

*Figure 7-37: Control Arm Assembly/Installation*



| Item | Description | Item | Description |
|---|---|---|---|
| 1 | Arm Angle Proximity Switch Assembly | 10 | P-Clamp |
| 2 | Stem Mount | 11 | Plastic Washer |
| 3 | Handle Pivot Pin | 12 | Screw, Button Socket Head |
| 4 | Handle Stem | 13 | Cable Tie |
| 5 | Bearing Race | 14 | Bolt, Shoulder |
| 6 | Set Screw | 16 | Bumper Stop |
| 7 | Torsion Spring | 17 | Grommet |
| 8 | Bearing | 18 | Screw |
| 9 | Spring Sleeve | 19 | Screw |

# Control Arm Removal

**⚠ WARNING**

**Park the truck on a level surface and make sure all wheels are blocked to prevent accidental movement.**

**NOTE:** Callouts in this procedure refer to the illustration on page 7-23.

1. Release pressure in the hydraulic system by pressing the lowering button on control head.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
3. Remove the lower and grille covers.
4. Remove the screws [14] securing the control handle bumper stop [16].
5. Remove the control handle bumper stop [16] and lay to one side.
6. Disconnect the wiring harness connector.
7. Remove the screw [12] and P-Clamp [10].
8. Remove the set screw [6] and remove the pin [3].
9. Carefully remove the control arm handle stem [4].

# Control Arm Installation

**NOTE:** Callouts in this procedure refer to the illustration on page 7-23.

1. Position the control arm handle stem [4] on the stem mount [2].
2. Slide pin [3] through the stem mount [2], and the handle stem [4] assembly. Make sure to align the hole in the pin with the hole in the stem mount [2].
3. Install the set screw [8] into the hole in pin [3]. Tighten the set screw, then tap the end of the pin to make sure it is seated correctly.
4. Route the handle cable between the spring sleeve flange [9] and the plastic washer [11]. Reconnect the handle harness connector.
5. Install screw [12] and the P-Clamp [10] to secure the handle harness. Torque screw to 11 ft. lbs. (15 Nm).
6. Adjust the angle arm proximity switch (see page 7-81).
7. Install cable tie [13] to secure angle arm proximity switch cable to handle harness.
8. Install the control handle bumper stop [16]. Secure with screws [14].
9. Install the lower and grille covers.
10. Remove the blocks securing the truck.

11. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
12. Test the operation of the truck.

# Drive and Brake

# Drive Unit Assembly

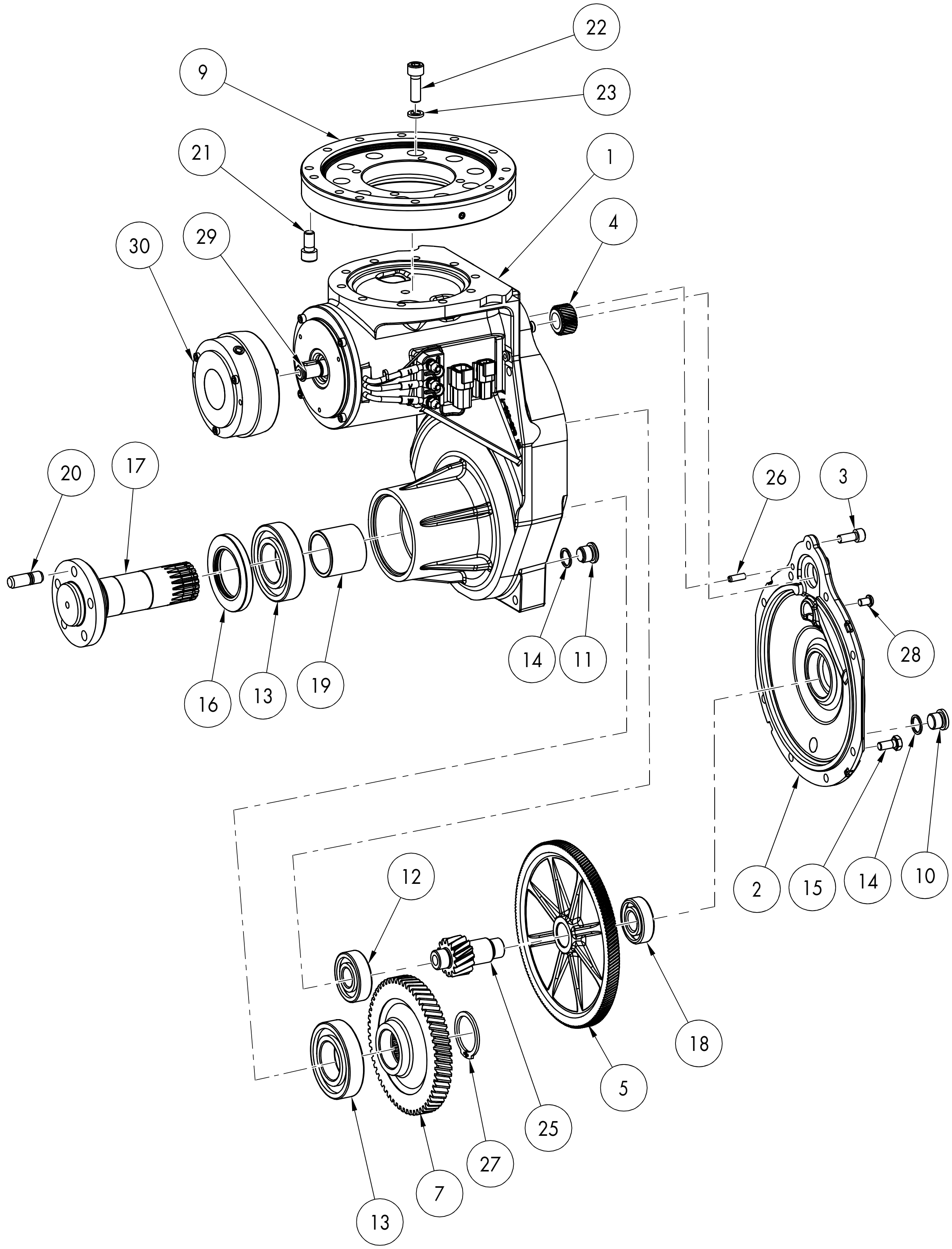
The drive unit assembly includes the traction motor, electromagnetic brake assembly, drive wheel, and drive unit (transmission).

**NOTE:** For replacement parts information refer to the Parts Manual.

*Table 7-1. Legend for "Drive Unit, Exploded View" on page 7-29*

| Item No. | Description | Item No. | Description |
|---|---|---|---|
| 1 | Housing, Gear Case | 17 | Drive Axle |
| 2 | Cover, Gear Case | 18 | Bearing, Sealed |
| 3 | Cap Screw, Socket Head | 19 | Spacer, Bearing |
| 4 | Pinion, Drive 1$^{st}$ Stage | 20 | Stud, Wheel |
| 5 | Gear Set, 2$^{nd}$ Stage | 21 | Cap Screw, Socket Head |
| 7 | Gear Set, Output | 22 | Cap Screw, Socket Head |
| 9 | Bearing, Steering | 23 | Washer, Lock |
| 10 | Plug, Fill/Level | 25 | Axle, 2$^{nd}$ Stage |
| 11 | Plug, Drain | 26 | Sleeve (roll pin) |
| 12 | Bearing, Sealed | 27 | Ring, Retaining |
| 13 | Bearing Set, Cup and Cone | 28 | Vent |
| 14 | Seal, Ring | 29 | Motor, Traction |
| 15 | Screw, Hex Head | 30 | Brake (with Boot) |
| 16 | Seal, Shaft | 31 | Brake Hub Mounting Location |

*Figure 7-38. Drive Unit, Exploded View*



# Drive Unit

## Drive Unit Removal

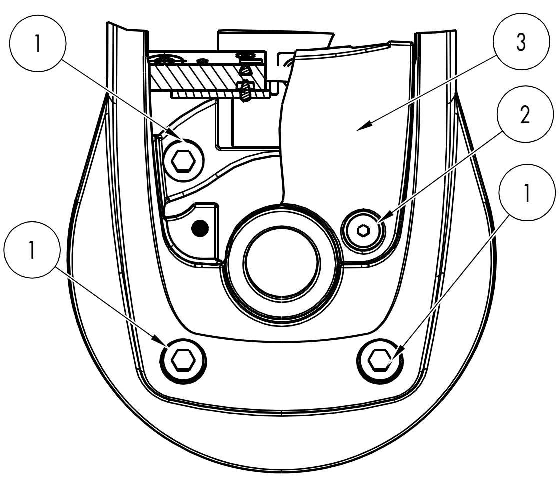
1. Lower the forks. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector.
2. Remove the lower and grille covers (page 7-8).
3. Remove the bumper. See “Bumper Removal” on page 7-13.

**⚠ WARNING**

**Use extreme care whenever the truck is jacked up. Keep hands and feet clear from the vehicle while jacking the truck. After the truck is jacked, put solid blocks beneath it to support it. DO NOT rely on the jack alone to support the truck. For details, see “Jacking Safety” on page 2-10.**

4. Jack the truck and block the frame.
5. Remove the drive wheel. See “Drive Wheel” on page 7-34.
6. Remove the screws [2] that secure the control arm stop [3]. See Figure 7-39.

*Figure 7-39. Control Handle Assembly Removal*

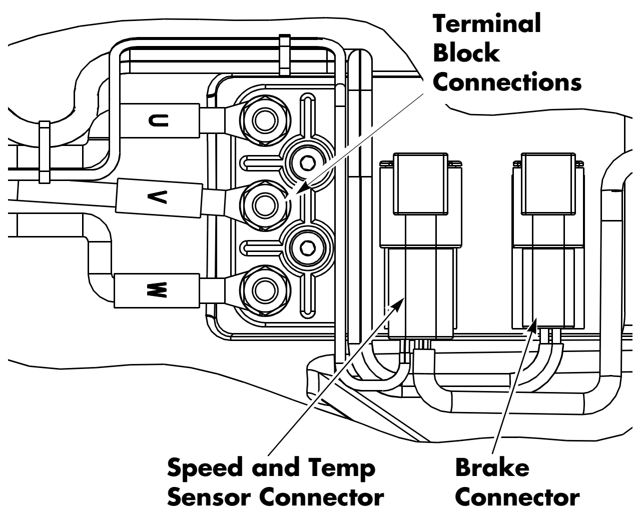



7. Disconnect the handle harness and arm angle proximity switch.

*Figure 7-40. Angle Arm Proximity Switch Disconnected*

8. Disconnect the cables/wire from the traction motor to the terminal block on the drive unit housing. See Figure 7-41.

*Figure 7-41. Terminal Block*



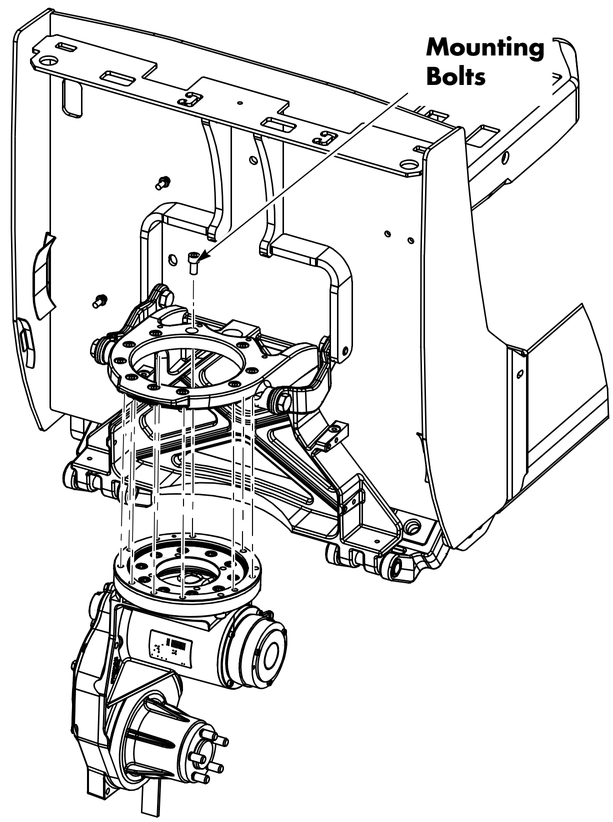

9. Disconnect the brake connector and the speed and temperature sensor connector on the drive unit housing.
10. Feed the control wiring harness and power cables through the drive unit pivot mount.
11. Remove the screws [1] that secure the control handle assembly to the drive unit assembly. See Figure 7-39.

**⚠ WARNING**

**The drive unit assembly may fall free from its mounting after the bolts are removed.**

12. Secure the drive unit to prevent it from falling. Remove the ten bolts that secure the drive unit and bearing assembly to the frame.

*Figure 7-42. Drive Unit Mounting Bolts*




**NOTE:** Remove the ten Allen head bolts to remove only the transmission but not the steering bearing.

13. Remove the drive unit and steer bearing assembly by lowering them through the pivot-frame mount.

## Drive Unit Disassembly

**NOTE:** The drive unit capacity is 0.4 qt. (0.4 liters). Make sure to have a correct size waste container for collecting the removed gear case lubricant.

1. Remove the drain plug [11]. Drain the gear case lubricant. After draining the gear case lubricant, thoroughly clean the outside with solvent or other non-corrosive cleaning fluid. Air dry all parts, and proceed with the disassembly of drive unit.
2. Remove the screws [40] securing the speed sensor cover assembly [21] to the gear case cover. Move the speed sensor cover assembly to the side to prevent damage to the cable. Refer to Figure 7-102 on page 7-86.
3. Remove the nine hex head cap screws [3] securing the gear case cover [2]. Pull the gear case cover away from the drive unit to disengage it from the roll pin [26] in the housing.
4. Remove the hex screw [19], washer [18], speed sensor pinion gear [17], and shaft seal [16] from the end of the armature. Refer to Figure 7-102 on page 7-86.
5. Remove the pinion nut [20, Figure 7-102 on page 7-86] that secures the top (pinion) gear [4].
6. Remove the top (pinion) gear [4] attached to the traction motor armature shaft.
7. Remove the traction motor if required. See “Traction Motor” on page 7-86.
8. Remove the traction motor bearing [10, Figure 7-102 on page 7-86], retaining ring [9], and seal [8] from the gear case if required.
9. Remove the 2$^{nd}$ stage gear set [5] from the gear case.

**NOTE:** The 2$^{nd}$ stage axle [25] has an M8 tapped hole in the end. It can be removed with the gear [5] in place.

10. Remove the retaining ring [27] from the drive axle [17] then remove the output gear [7].
11. Press the drive axle [17] through the inner and outer gear case bearings [13].
12. Remove the old bearings [13], spacer [19], and seal [16] from the gear case. Thoroughly clean the case.

## Drive Unit Assembly

1. Install the new bearings [12 and 18] in the gear case [1] and cover [2] for the 2$^{nd}$ stage gear set.
2. Install the new bearings [13], spacer [19], and seal [16] in the gear case for the drive axle.

3. Place the case in a fabricated assembly base and carefully press the drive axle [17] in the gear case [1].
4. Position the drive unit on the work bench with the gear case side up. Install the output gear [7] in the case and secure with the retaining ring [27].
5. Install the 2$^{nd}$ stage gear set [5].
6. Install the traction motor bearing [10, Figure 7-102 on page 7-86], retaining ring [9], and seal [8] in the gear case if removed previously.

**NOTE:** The seal must be installed correctly with its open side and seal ring positioned towards the bearing.

7. Carefully slide the traction motor armature shaft into the gear case if removed previously. See "Traction Motor" on page 7-86.

**⚠CAUTION**

**Make sure that the motor end cap is installed before proceeding to the next step.**

8. Install the pinion gear [4] and the pinion nut [20] on the armature shaft. Torque the pinion nut to 25.8 ft. lb. (35 Nm).
9. Install the shaft seal [16], speed sensor pinion gear [17], washer [18], and hex screw [19] to the end of the armature. Refer to Figure 7-102 on page 7-86,
10. Use gasket compound on the mating surface and install the gear case cover [2].
11. Torque the nine cover bolts [3] to 16.9 ft. lb. (23 Nm).
12. Install the two screws [40] to secure the speed sensor cover assembly [21] to the gear case cover. Torque the screws to 23.8 in. lb. (2.7 Nm).

**NOTE:** Make sure the speed sensor cable is correctly seated in the grommet to prevent damage from contact with the linkage when the drive unit turns.

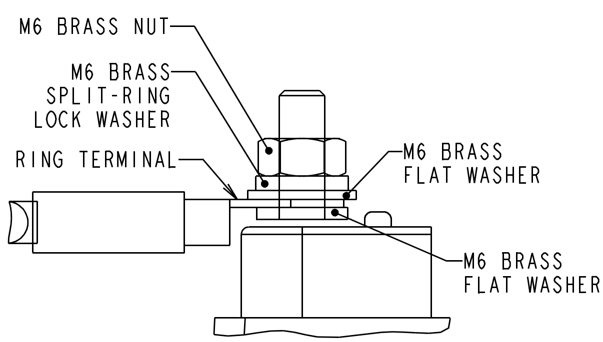
## Drive Unit Installation

**⚠CAUTION**

**Make sure the main ON/OFF switch is OFF and the battery connector is disconnected before you start.**

1. Install the steer bearing assembly using ten Allen head bolts, if it was removed previously. Torque bolts to 48.6 ft. lb. (66 Nm).
2. Install the drive unit assembly to the pivot frame. Feed the harness and cables through the drive unit housing to the locations where they were previously removed.
3. Secure the assembly with the ten Allen head bolts removed previously. Torque bolts to 336 to 384 in. lb. (38.0 to 43.4 Nm).
4. Feed the wire harness and power cables through the pivot tube.
5. Reconnect the cables/wire to the terminal block on the drive unit. See Figure 7-43 for correct hardware order. Torque the brass nuts to 26.5 in. lb. (3 Nm) to secure the phase cables.

*Figure 7-43. Hardware Stack for Phase Cables*




6. Reconnect the brake connector and the speed and temperature sensor connector on the drive unit housing.
7. Install the drive wheel. See "Drive Wheel" on page 7-34.
8. Unblock the truck.
9. Make sure the drive unit is filled to the correct level with gear oil. See "Lubrication Equivalency Chart" on page A-2. The drive

unit capacity is 0.4 qt. (0.4 liters). Install the fill/level plug. Torque the fill/level plug to 292 in. lb. (33 Nm).

10. Install the bumper. See "Bumper Installation" on page 7-13.
11. Install the control handle (page 7-24).
12. Reconnect the wiring harness.
13. Install the lower and grille covers (page 7-10).
14. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
15. Check the controls for correct operation.

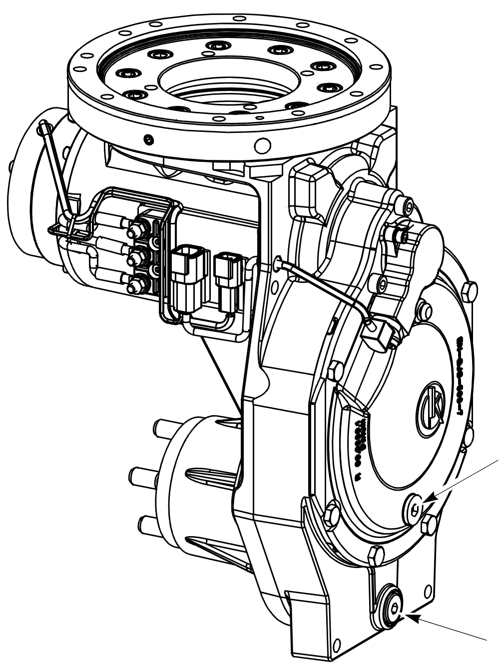
# Drive Unit Housing Lubrication

## Gear Oil Level Check

The drive unit capacity is 0.4 qt. (0.4 liters).

1. Remove the fill/level plug. See Figure 7-44.

*Figure 7-44. Drive Unit Fill/Drain Plugs*




2. Check the oil level. When the truck is level, the oil must be up to the bottom of the fill/level plug opening.
3. If necessary, add the specified oil through the fill/level plug opening, but do not overfill. See "Lubrication Equivalency Chart" on page A-2.
4. Install the fill/level plug securely. Torque the fill/level plug to 292 in. lb. (33 Nm).

## Changing Gear Oil

1. Remove the drain plug from the bottom of the drive unit housing and permit the oil to drain. See Figure 7-44.
2. After the oil has drained completely, flush the housing with a suitable solvent and allow it to drain.

**NOTE:** Make sure all solvent is drained from drive unit before adding new gear oil.

3. Clean the magnet and install the drain plug. Torque to 292 in. lb. (33 Nm).
4. Fill the drive unit housing with the specified gear oil through the fill/level plug opening. See "Lubrication Equivalency Chart" on page A-2.
5. Install the fill/level plug securely. Torque the fill/level plug to 292 in. lb. (33 Nm).

# Drive Wheel

**NOTE:** For replacement parts information refer to the Parts Manual.

## Drive Wheel Removal

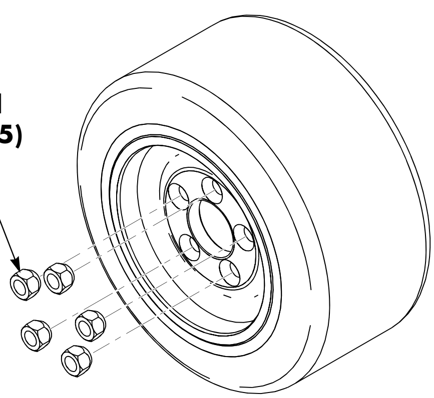
1. Park the truck on a level surface.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.

**⚠ WARNING**

**Use extreme care whenever the truck is jacked up. Keep hands and feet clear from the vehicle while jacking the truck. After the truck is jacked, put solid blocks beneath it to support it. DO NOT rely on the jack alone to support the truck. For details, see “Jacking Safety” on page 2-10.**

3. Remove the bumper. See “Bumper Removal” on page 7-13.
4. Jack and block the truck under the tractor frame. See “Jacking Safety” on page 2-10.
5. Remove the five drive wheel mounting nuts. See Figure 7-45.

*Figure 7-45. Drive Wheel Nuts*




6. Remove the drive wheel.

## Tire Replacement

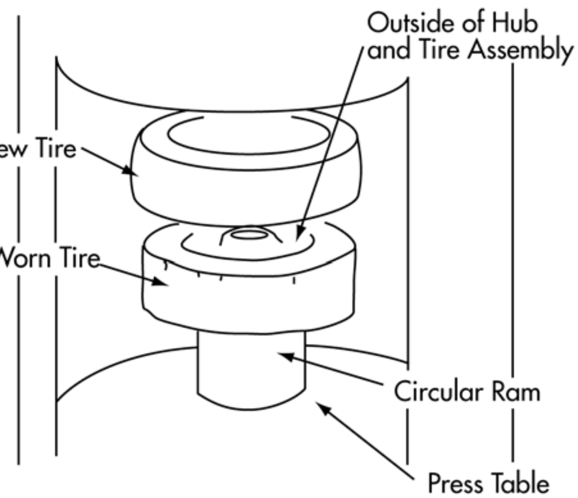
Any misalignment of the tire and hub while the tire is being pressed on the hub can cause damage to the hub. For this reason, chamfers are provided on the outside edge of the hub and on the end of the inside diameter of the tire’s metal insert. The chamfers help to center the hub and tire during pressing and to reduce the possibility of misalignment.

**⚠CAUTION**

**To prevent damage, the hub must be installed on the circular ram with the chamfered side up.**

1. Check the inside surface of the metal insert on the new tire. Remove any scaling or rust with sandpaper. Clean the inside of the metal insert.
2. Place a circular ram on the press table. The length of the ram must be longer than the width of the old tire to permit complete removal of the old tire. The outside diameter of the ram must be small enough to fit loosely in the insert of the tire but must be large enough to rest squarely on the flat surface on the outer edge of the hub. See Figure 7-46.

*Figure 7-46. Tire Replacement*




3. If the outside edge of the hub is not flush with the edge of the metal insert in the old tire, measure how far the hub is recessed inside the tire. The new tire must be

placed in the same position the old tire was installed on the hub. You can use a spacer (slightly smaller in diameter that the inside diameter of the tire insert and the same thickness as the depth of the recess) to obtain the correct amount of recession.

4. Position the hub assembly with the old tire on top of the circular ram so the outside of the wheel is positioned upward. The outside edge of the hub has a chamfer to help guide the new tire on the wheel. The chamfered edge must always be the leading edge when a tire is pressed on the hub.
5. Center the hub assembly on top of the ram and make sure they mate squarely.
6. Position the new tire with its chamfered insert facing the hub. Align the new tire and the hub so they are concentric.
7. Begin pressing the new tire on the hub and the old tire off the wheel. Run the press slowly for the first few inches of travel because this is the critical step of the operation. If the tire starts to cock, stop the press immediately and realign the tire. Use a soft-headed mallet to realign the tire on the hub.

**NOTE:** If the new tire does not press on with a minimum pressure of 5 tons (68,947 kPa), replace the hub.

8. Release the press. Remove the wheel, tire assembly, and the old tire from the press table. Inspect the wheel and tire assembly.

## Drive Wheel Installation

1. Inspect the drive wheel studs for damage or looseness. Replace any damaged studs. Use the following procedure to tighten loose studs.
   a. Remove and clean the studs.
   b. Install two lug nuts, jammed together, to the end of the stud. This allows tightening of the studs using a torque wrench.
   c. Apply thread-locking primer (P/N 990-666) to the studs.
   d. Apply thread-locking compound (P/N 990-571) to the studs.
   e. Install the studs and torque to 23 to 28 ft. lbs. (32 to 38 Nm).

**NOTE:** Do not over-tighten the studs. If tightened to more than 30 ft. lbs. (41 Nm), the studs start to cut new threads and go in too far, hitting the lip seal. If the lip seal is damaged due to studs being installed too far, the axle must be replaced.

   f. Remove the nuts on the studs. Make sure the studs do not rotate.
2. Install the wheel and tire assembly on the drive axle of the truck.
3. Install the five drive wheel mounting nuts. Torque the nuts in at least three steps, starting with a low torque [5 ft. lbs. (7 Nm)]. Torque nuts in a star pattern to make sure that the lugs are centered in the mounting holes. Torque to 55 ft. lbs. (75 Nm) on the final pass.
4. Remove the blocks and lower the lift truck.
5. Install the bumper. See “Bumper Installation” on page 7-13.
6. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
7. Test the operation of the truck.

# Electromagnetic Brake Assembly

**NOTE:** For replacement parts information refer to the Parts Manual.

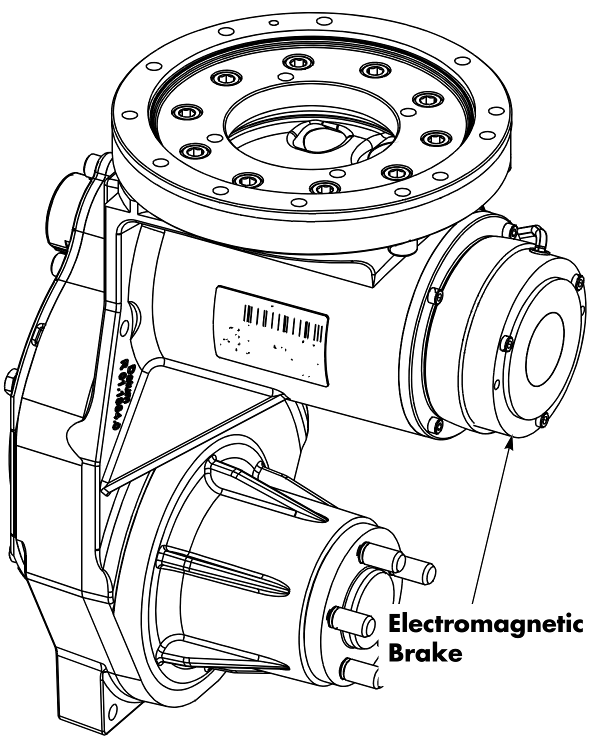
**⚠CAUTION**

**Allow the brake to cool completely before servicing.**

## Brake Disc Location

The electromagnetic brake assembly is mounted directly to the traction motor. See Figure 7-47.

*Figure 7-47. Brake Location*




## Air Gap Inspection

The nominal gap between the coil housing and the pressure plate in the applied position is 0.001 to 0.013 in. (0.03 to 0.33 mm).

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Block the drive wheels to prevent any movement while the brake repairs are in process.
3. Remove the lower and grille covers. See "Tractor Cover Removal" on page 7-8.
4. Remove the rubber brake boot to access the brake.
5. With the brake applied, measure the air gap. If the gap is greater than 0.0157 in. (0.40 mm), replace the brake.
6. Install the rubber brake boot to protect the brake from moisture and dirt.
7. Install the lower and grille covers. See "Tractor Cover Installation" on page 7-10.
8. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.
9. Test the operation of the truck.

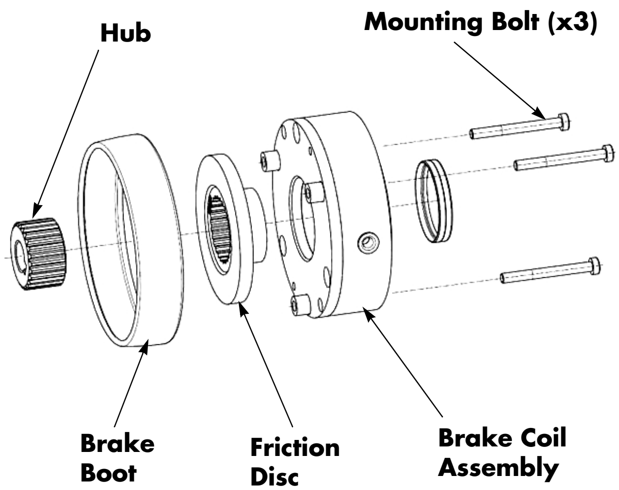
## Friction Disc Replacement

Replace the friction disc when the air gap is greater than 0.0157 in. (0.40 mm).

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Block the drive wheels to prevent any movement while the brake repairs are in process.
3. Remove the lower and grille covers.
4. Remove the rubber brake boot to access the brake.

5. Loosen and remove the three 4 mm mounting bolts and remove the brake coil assembly. See Figure 7-48.
6. Replace the friction disc on the hub. Inspect friction plate for wear.
7. Install the brake coil assembly on the motor end. Torque mounting bolts to 47.8 in. lb. (5.4 Nm).

*Figure 7-48. Friction Disc Replacement*




8. Install the rubber brake boot to protect the brake from moisture and dirt.
9. Install the lower and grille covers.
10. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.
11. Test the operation of the truck.

## Electric Brake Release

An Electric Brake Release is standard equipment on these trucks. The switch to activate the brake release is located in the console cover to the left of the operator manual. When the switch is activated, the Traction Amplifier releases the brake.

## Mechanical Brake Release

If the brake cannot be electrically released, perform the following:

1. Remove the lower and grille covers.
2. Remove the rubber brake boot to access the brake.
3. Loosen the three 4 mm Allen head mounting bolts evenly until the clamping force on the friction disc is low enough to allow the truck to roll.
4. Move the truck to the desired location.
5. Inspect the gap and torque mounting bolts to 47.8 in. lb. (5.4 Nm). See “Air Gap Inspection” on page 7-36.
6. Install the rubber brake boot to protect the brake from moisture and dirt.
7. Install the lower and grille covers.

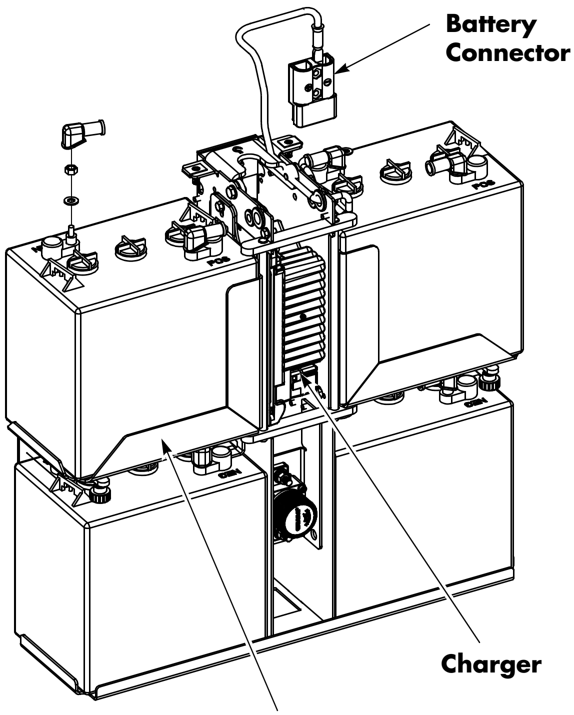
# Electrical Components

# Lead Acid Battery

The pallet truck may be equipped with one of several different types of batteries:

- a Raymond swing-out battery pack and charger with either a wet cell or a maintenance-free battery

*Figure 7-49.*




- a lead acid industrial battery
- a swing-out battery pack (vendor other than Raymond)

Refer to the battery manufacturer's battery and battery charger manuals for battery specific recommendations.

**⚠CAUTION**

**Before working on the battery, see "Lead Acid Battery Safety" on page 2-5. Industrial batteries and assembled battery packs can weigh more than 500 pounds (227 kg). Use extreme care during replacement. Use a suitable battery replacement device or hoist for lifting.**

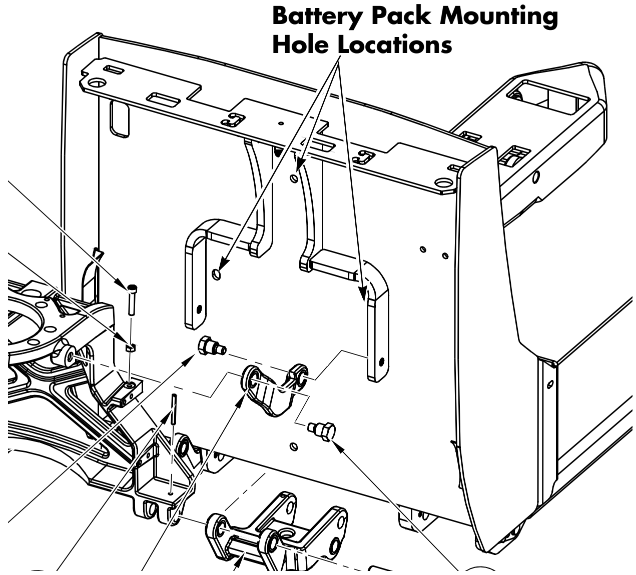
## Swing-out Battery Pack

### Removal (entire battery pack)

1. Park truck on a level surface and make sure the parking brake is applied.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position.
3. Remove the load backrest.
4. Disconnect the battery connector found at the top of console cover.
5. Remove all battery pack retention hardware.

**NOTE:** The battery pack may be secured to the tractor frame with two screws part way down the compartment or one screw at the top of the compartment.

*Figure 7-50. Battery Pack to Frame Mounting Locations*




6. Position the battery replacement device above the battery and attach it to the battery manufacturer's designated lift points.
7. Remove the battery with the lifting device.

**NOTE:** Attach the lifting hook to the designated lifting area. Do **not** attach the lifting hook to the charger cord storage handle.

8. Put the discharged battery on the charging stand.

## Installation (entire battery pack)

1. With a fully charged and tested battery on the lifting device, position the lifting device in accordance with the manufacturer's recommendations.

**NOTE:** Attach the lifting hook to the designated lifting area. Do **not** attach the lifting hook to the charger cord storage handle.

2. Position the battery in the compartment. Make sure the battery has no more than 1/2 in. (13 mm) of "free play" movement in the battery compartment in any direction.
3. Remove the lifting device from the area.
4. Install the mounting hardware through the battery retainer bracket.
5. Install the load backrest.
6. Reconnect the battery connector.

**NOTE:** When installing a battery pack, make sure the CAN bus is connected, if equipped.


7. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
8. Configure the truck parameters 21, 24, and 44 for the battery type installed.

## Removal (single battery)

1. Park the truck on a level surface and make sure the parking brake is applied.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position.
3. Disconnect the battery connector found at the top of the console cover.
4. Unwind the AC power cord.
5. Remove the battery pack cover.
6. Disconnect the CAN connector, if equipped.
7. Release the latch and swing out the top battery tray.
8. Remove the battery retainer bracket.
9. The cables may now be disconnected from the battery terminals and the batteries may be serviced or replaced if necessary. See Figure 7-52.

**NOTE:** When replacing batteries, make sure all four batteries are the same type.

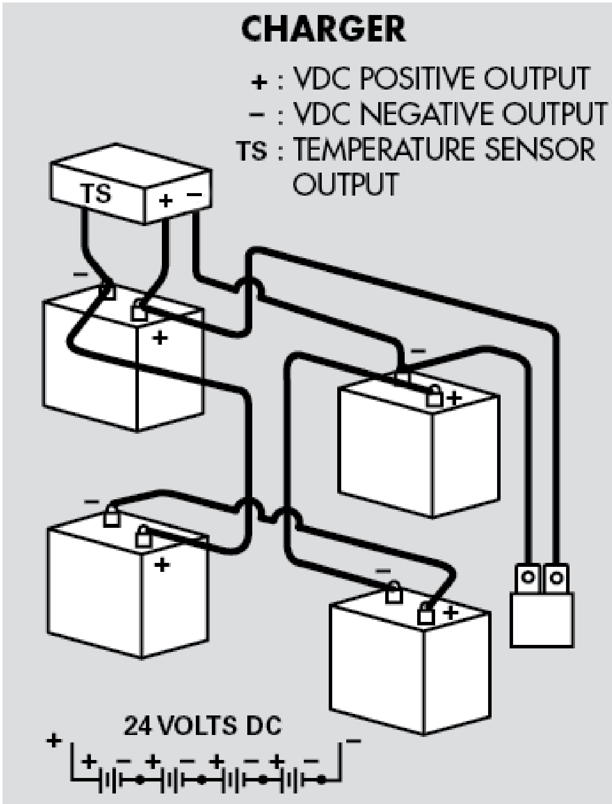
## Installation (single battery)

1. Position the fully-charged and tested batteries in the battery pack.
2. Install the battery retention bracket.

*Figure 7-51. Battery Retention Bracket Installed*


3. Reconnect the battery cables to the battery terminals in accordance with the battery wiring hookup shown in Figure 7-52.

*Figure 7-52. Multiple Battery Wiring Hookup*




**NOTE:** Make sure there is sufficient cable to allow battery swing-out movement.

4. Install the battery cover,
5. Rewind the charger cord on the charger cord storage handle.
6. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
7. Configure the truck parameters 21, 24, and 44 for the battery type installed.

## Charger Replacement

1. Park truck on a level surface and make sure the parking brake is applied.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position.
3. Disconnect the battery connector found at the top of console cover.
4. Remove the battery pack cover.
5. Swing out the top battery trays.
6. Disconnect all cables and wires from the charger assembly to the batteries. Note the orientation and location for reconnection.
7. Remove the screws and nuts securing the charger unit to the battery tower.
8. Slide the charger unit out of the battery tower.
9. Disconnect all cables and wires from the charger assembly. Note the orientation and location for reconnection.
10. Connect cables and wires to the new charger. Make sure the orientation and location is the same as previously noted.
11. Place the new charger unit in the battery tower in the same orientation as the one that was removed.
12. Secure the charger to the battery tower frame with bolts and nuts. Use a 10mm wrench to hold the nuts while torquing the bolts to 89 in. lb. (10 Nm).
13. Connect the cables from the charger to the batteries. Follow the battery wiring hookup shown in Figure 7-52. Make sure to loop cables and secure with cable ties.

*Figure 7-53. Battery Cables Looped and Secured*

14. Torque terminal nuts to 101 in. lb. (11.4 Nm) for wet cell batteries. Torque nuts to 89 in. lb. (10 Nm) for AGM batteries.
15. Install terminal caps.
16. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter

your PIN-key code and then press the green ON ( **|** ) key.

17. Configure the truck parameter 21 for the battery type installed. See “Programming Service Parameters” on page 3-8.
18. Test the charger operation.
    a. Plug the AC power cord into an outlet.
    b. Wait approximately 1 minute. The yellow lightening bolt on the charger will light up to confirm correct charger operation.

*Figure 7-54. Yellow Lightening Bolt Location*

    c. Unplug the AC power cord.

19. Install the battery cover.

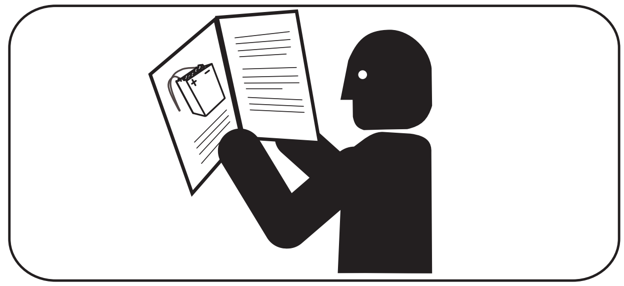
*Figure 7-55.*

20. Wind the charger cord around the storage handle on the battery cover.
21. Install the lower and grille covers.

# Battery Maintenance

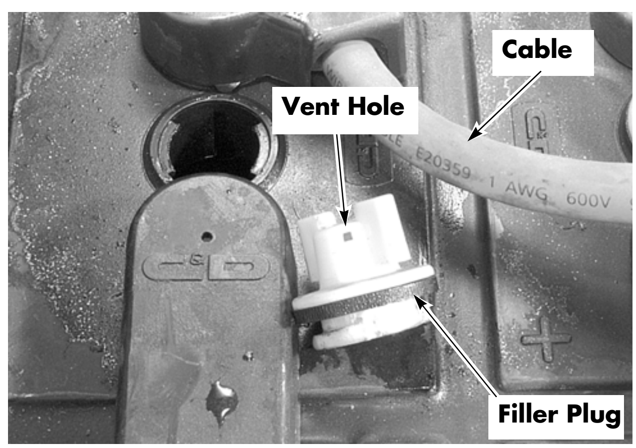
## Battery Exterior Cleaning

1. Read, understand, and follow procedures, recommendations, and specifications in the battery and battery charger manufacturer’s manuals.



2. Wear personal protective equipment. See “Lead Acid Battery Safety” on page 2-5.
3. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck. Remove the battery from the lift truck.
4. Inspect the battery cables to make sure they are not frayed or loose. Inspect the battery connector to make sure there is no foreign material inside the connector.
5. Make sure the filler plugs are tight and the vent holes in the filler plugs are open. See Figure 7-56.

*Figure 7-56. Battery Filler Plugs and Vent Holes*




6. Keep the top of the battery clean and dry. Corrosion, dust, and moisture provide a

conducting path to short-circuit cells or create grounds.

**CAUTION**

**Do *not* use soda solution to clean the top of the battery while it is installed in the truck. Water can seep into the electrical compartments and cause serious damage.**

7. Tighten the battery filler plugs.
8. Wash dirty batteries (or any that have had electrolyte spilled on them) with a solution of 1 lb. (0.45 kg) of baking soda added to 1/2 gal (1.9 liters) of hot water.
9. Use a brush with flexible bristles to clean the entire top of the battery with the soda solution. Wait until all foaming stops, indicating that the battery exterior is neutralized.
10. Rinse the battery with clean water.
11. Dry the battery completely before re-installing it.
12. Reinstall the battery in the truck, using a suitable battery moving device. Reconnect the battery connector.
13. Apply a thin coat of petroleum jelly to the battery posts and cable terminals.

## To Charge a Battery

To charge a battery, direct current is passed through the battery cells in the direction opposite to that of discharge. Charging time is 5% to 20% longer than discharge time.

The most important element in battery service and prolonging battery life is correct charging. Make sure you follow the approved method for each application, following the battery and battery charger manufacturer's instructions.

**CAUTION**

**The vent holes in the filler plugs must be open to permit hydrogen gas to escape from the cells. When you charge the battery, make sure the polarity connections are correct. The positive lead of the charger must be connected to the positive terminal, and the negative lead must be connected to the negative terminal.**

1. Familiarize yourself with the following:
   - Charging rate, starting rate, and finish rate
   - Time available for charge
   - Overheating, excessive gassing, or overcharging
   - Variations between cell voltage
2. Battery performance varies depending on the type of application. If you use more than one truck, one battery, and one charger, keep the same batteries and charger assigned to each truck. This makes it easier to diagnose any battery or charging problem.
3. Consult your battery and battery charger manufacturer's manual for specific charging procedures.
4. Wear personal protective equipment. See "Lead Acid Battery Safety" on page 2-5.
5. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
6. Inspect the battery.
7. For wet cell batteries, examine the electrolyte level in each cell. Electrolyte must cover the top of the battery plates. If electrolyte level is below the plates, add just enough water to cover the plates. Do not fill to level yet. Electrolyte level increases during charging. See "To Add Water to a Battery" on page 7-45.
8. Using a hydrometer, measure and record the specific gravity of each cell. See "Battery Specific Gravity" on page 7-46.
9. Measure and record the voltage of each cell.
10. To get the maximum use out of each battery, recharge only when effectively discharged. Routinely recharging batteries when only partially discharged decreases battery life. At maximum recommended

discharge, the specific gravity must read 1.140 to 1.160 or less.

11. Make sure that the filler plugs are clean and the vent holes are open. Tighten all filler plugs.

**⚠CAUTION**

**NEVER plug the battery charger into the truck. This severely damages the truck's electrical system. Plug the charger ONLY into the connector from the battery.**

12. Charge the battery, following specific instructions in your battery and battery charger manufacturer's manual.
13. Using a hydrometer, measure the specific gravity of each cell. No amount of charging increases battery capacity or increases specific gravity above its fully charged level (specific gravity 1.280 to 1.300).
14. If, after charging, the electrolyte level is below the fill port, add water to move it to the fill level. *Do not overfill the cells* or permit electrolyte to spill on the battery case. See "To Add Water to a Battery".

In general, a lead acid battery may be charged at any rate in amperes that does not cause excessive gassing or produce electrolyte temperatures above 110°F (43°C). Temperatures of 120°F (49°C) are permitted only for short periods of time.

## To Charge a Raymond Battery Pack (Option)

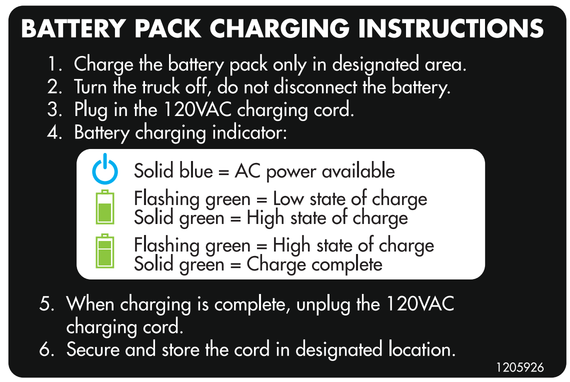
**NOTE:** Make sure to follow battery manufacturer's instructions included with the battery charger.

A battery/charger pack that uses a combination of four 6-volt batteries is optional on this lift truck. The battery pack includes a "self-contained" battery charger.

To charge the battery pack, do the following:

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position ( O ).
2. Make sure the lift truck handle is in the upright position.
3. Plug the battery charger's extension cord into a 120-volt outlet.
4. Charge the batteries.
5. The battery pack charging instruction decal defines the various status indicator lights on the battery pack.

*Figure 7-57.*




After the battery pack is charged:

1. Unplug the battery charger's extension cord from the outlet.
2. Return the battery charger's extension cord to the storage space provided.

## To Add Water to a Battery

Water must be added to battery cells periodically. Frequency and quantity depend on the water level above the plates and the amount of gassing during charge.

Guidelines:

- Wear personal protective equipment. Use only distilled water or water approved by the battery manufacturer. See "Lead Acid Battery Safety" on page 2-5.
- Before charging, make sure battery plates are covered with electrolyte. Add water, if necessary, only enough to cover the plates.
- Electrolyte level increases during charging. To prevent overflow, add water to fill port, if necessary, after charging, and after hydrometer and voltmeter readings are taken.

- Keep the outside of the battery case clean and dry.

## Battery Specific Gravity

Specific gravity readings for lead acid batteries with electrolyte temperatures at 80°F (27°C) change with the state-of-charge as follows:

| Charge | Specific Gravity |
|---|---|
| 100% | 1.2540 |
| 80% | 1.2340 |
| 60% | 1.2128 |
| 40% | 1.1906 |
| 20% | 1.1674 |

Most industrial deep-cell discharge batteries are considered discharged when 20% of the charge remains (specific gravity 1.1674).

**NOTE:** Maintenance-free batteries do not require specific gravity measurements.

## Specific Gravity Check

1. Wear personal protective equipment. See “Lead Acid Battery Safety” on page 2-5.
2. Insert the nozzle of the hydrometer into the battery cell and draw enough electrolyte into the tube to permit the float to ride free.
3. Leave the hydrometer nozzle in the battery and read the specific gravity of the cell.
4. If there is not enough electrolyte in the battery to get a hydrometer reading then:
   a. Add just enough water to cover the battery plates.
   a. Charge the battery, then take readings.
5. Record the specific gravity reading on the battery maintenance chart.

## Battery Voltage Check

Perform a voltage check on each battery in the battery pack (or each cell of an industrial battery) after performing the specific gravity check.

1. Attach a voltmeter to the positive and negative terminals of each cell. Voltage readings change with the state-of-charge. See “Battery Specific Gravity” on page 7-46.

| Charge | Voltage Reading |
|---|---|
| 100% | 2.14 |
| 75% | 2.07 |
| 50% | 1.95 |
| 25% | 1.80 |

2. Record the voltage reading for each cell on the battery maintenance chart. There must be a difference of less than 0.2V between cells.

## Battery Storage

Before you store a battery, make sure the electrolyte is at the correct level in all cells, the filler plugs are tight, and the battery is fully charged.

Store the battery in a clean, cool, dry location away from radiators and other sources of heat. Refer to the storage instructions in your battery manufacturer’s manual.

Measure the electrolyte level and the specific gravity every 30 days during storage. Whenever the specific gravity is less than 1.230, charge the battery.

# Lithium-Ion Battery

*Figure 7-58. Lithium-ion Battery Components*

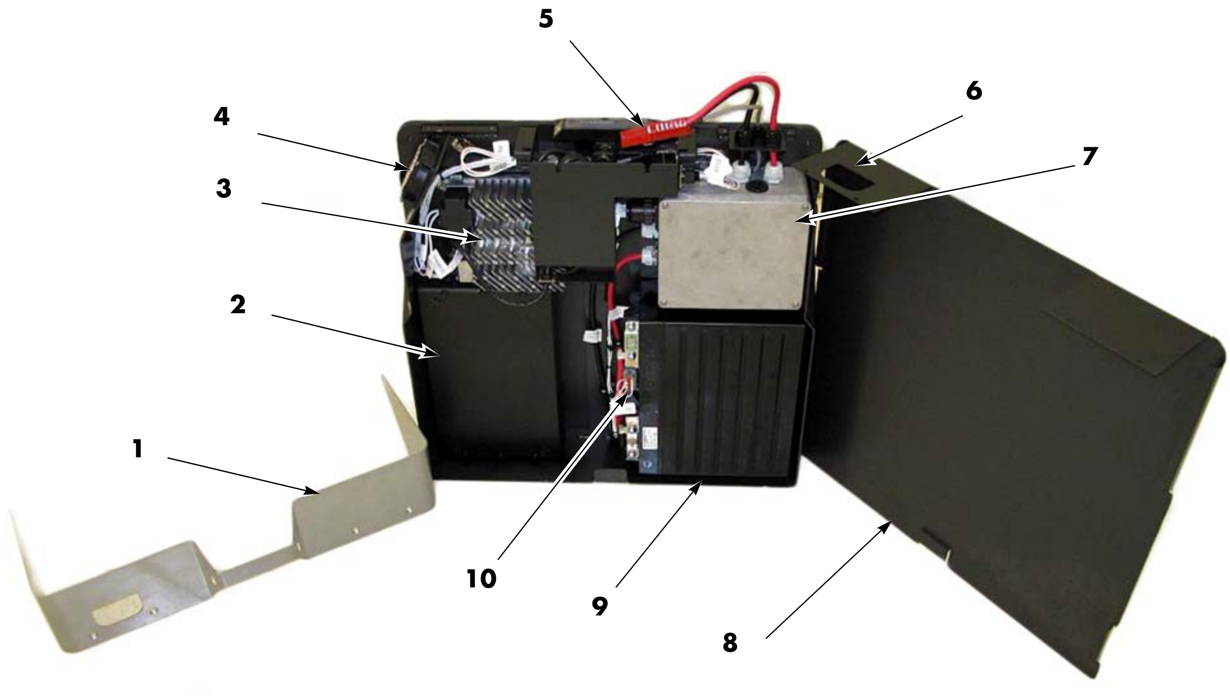



*Table 7-2. Lithium-ion Battery Component Legend*

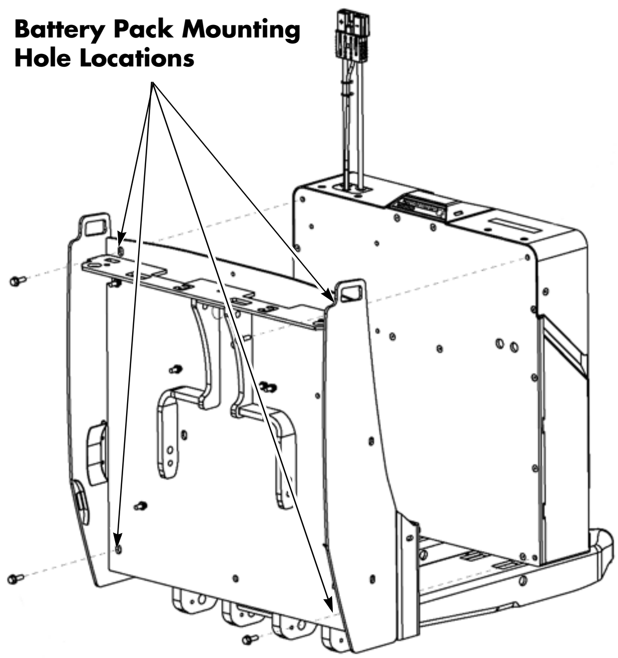
| Item Number | Description |
|---|---|
| 1 | Top Cover |
| 2 | Counterweight (Cell Module Assembly in high capacity models) |
| 3 | Charger |
| 4 | Fan |
| 5 | SBX Battery Connector |
| 6 | D-Shaped cutout in front cover |
| 7 | Battery Management Controller (BMC) |
| 8 | Front Cover |
| 9 | Cell Module Assembly (CMA) |
| 10 | Battery Monitoring Unit (BMU) |

## Battery Removal

1. Park truck on a level surface and make sure the parking brake is applied.
2. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position.
3. Remove the load backrest.
4. Disconnect the battery connector found at the top of console cover.
5. Remove all battery pack retention hardware.

**NOTE:** The battery pack is secured to the tractor frame with four screws.

*Figure 7-59. Battery Pack to Frame Mounting Locations*




6. Position the battery replacement device above the battery and attach it to the battery manufacturer's designated lift points.
7. Remove the battery with the lifting device.
8. Attach the lifting hook to the designated lifting rod. Do not attach the lifting hook to the ventilation slots.
9. Put the removed battery in a designated storage area.

## Battery Installation

1. With a fully charged and tested battery on the lifting device, position the lifting device in accordance with the manufacturer's recommendations.
2. Attach the lifting hook to the designated lifting area. Do not attach the lifting hook to the charger cord storage handle.
3. Position the battery in the compartment. Make sure the battery has no more than 1/2 in. (13 mm) of "free play" movement in the battery compartment in any direction.
4. Remove the lifting device from the area.
5. Install the mounting hardware through the battery retainer bracket.
6. Install the load backrest.
7. Reconnect the battery connector.
8. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
9. Configure truck parameters 56 through 66 for the lithium-ion battery.

# Lithium-Ion Battery Covers

## General Information

To replace or install internal components, the top and front lithium-ion battery covers must be removed. One exception is the lithium-ion battery display only needs the top cover removed. Details of the procedures to remove these two items are included here. Both processes can be performed with the lithium-ion battery either mounted in the truck or located outside of it.

If the lithium-ion battery is mounted in the truck, it is in the correct orientation for removing the top and front cover.

If the lithium-ion battery is outside of the truck, orient the lithium-ion battery so that it is not at risk of tipping over.

**CAUTION**

**If the lithium-ion battery is not supported, the lithium-ion battery risks tipping over. This can cause harm to body parts. This can be avoided if the lithium-ion battery is resting on the back plate (see Figure 2 for back plate location).**

**CAUTION**

**The process to remove the front cover is best performed with two people; one to remove cover, and the other person to guide the SBX connector through the D-shaped hole in the covers.**

**CAUTION**

**Disconnect the AC charge cord from any energized electrical outlet to remove the chances of coming into contact with outlet power upon removal of any cover piece.**

# Top Cover

## Top Cover Removal

Perform the following steps to remove the top cover.

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Verify the lithium-ion battery is not in a position to tip over.
3. Disconnect the SBX connector from the truck.
4. Verify the AC charge cord will not interfere.
5. Remove the six screws keeping the top cover mounted in place. Two are located on the sides, four are located on the top.

*Figure 7-60. Top Cover Mounting Screws (left side shown)*

6. Remove the top cover by lifting the cover up, while guiding the SBX connector through the D-shaped cutout.

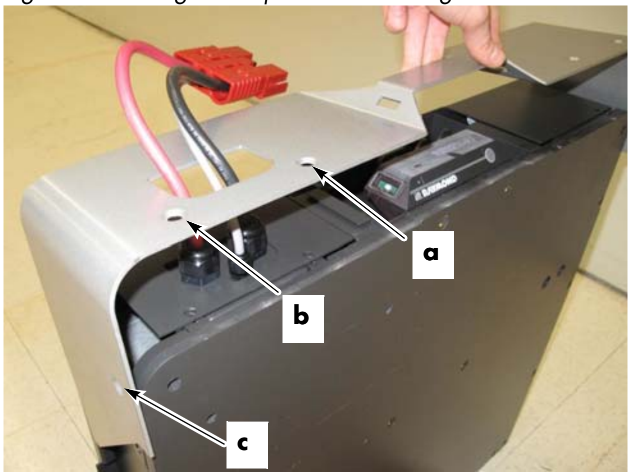
## Top Cover Installation

Perform the following steps to install the top cover.

1. Place the top cover over the previously installed front cover. See “Front Cover Installation” on page 7-50. Align the mounting holes of the top cover with the screw holes on the lithium-ion battery.

2. Prime the six screws by partially screwing in all screws.
3. Once all screws have been aligned and partially screwed, torque fasteners to 5.8 ft. lb. (7.8 Nm).
   a. Start by tightening the two screws on the top closest to the lithium-ion battery user interface (HUD).
   b. Continue by tightening the two remaining screws on the top cover.
   c. Finish by tightening the two screws on the side of the lithium-ion battery.

*Figure 7-61. Tighten Top Cover Mounting Screws*




4. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
5. Test the operation of the truck.

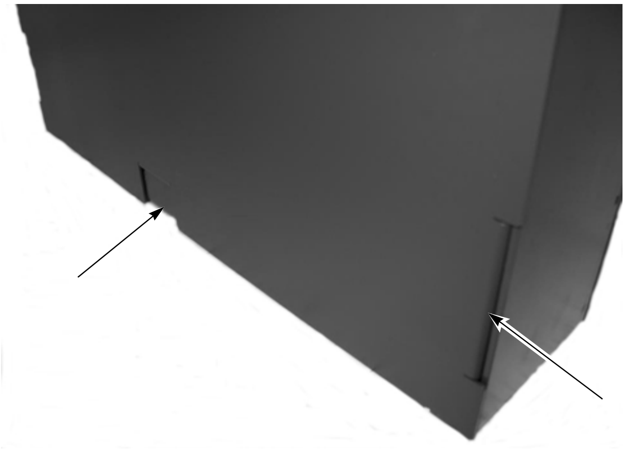
# Front Cover

## Front Cover Removal

Perform the following steps to remove the front cover.

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Lift the front cover up, guiding the SBX connector through the D-shaped cutout.
3. If needed, rock the front cover from side to side to dislodge the cover tabs from the side walls.

*Figure 7-62. Center and Side Cover Tab Locations*



4. The enclosure might require the service technician to rock the part from side to side to dislodge the tabs, depending on how tight the fit is. Do not perform this action in an aggressive manner to avoid knocking the lithium-ion battery over.

## Front Cover Installation

Perform the following steps to install the front cover.

1. Secure the lithium-ion battery from tipping over.
2. Place the AC charge cord in a position that does not interfere with installation.
3. Guide the front cover into the lithium-ion battery.
4. The tabs on the front cover go inside of the lithium-ion battery walls.
5. The SBX connector must be guided through the D-shaped opening on the cover.
6. Verify the top four holes on the cover are aligned with the lithium-ion battery.
7. Install the top cover. See “Top Cover Installation” on page 7-49.
8. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter

your PIN-key code and then press the green ON ( | ) key.

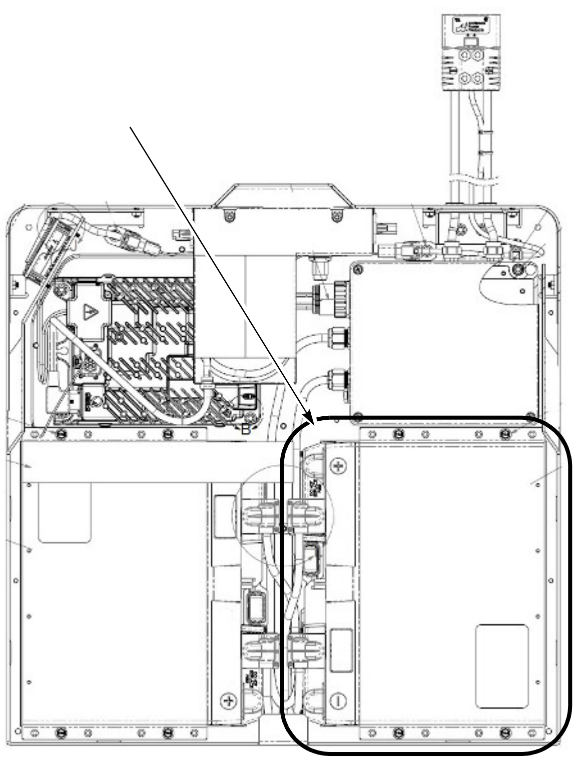
9. Test the operation of the truck.

## Cell Module Assembly, Battery Monitoring Unit, and Cell Module Assembly Fuse

**NOTE:** When replacing any internal components of the Lithium-ion battery, make sure the software in the components (if applicable) is correct. Refer to FlashWare Help.

The Cell Module Assembly (CMA), Battery Monitoring Unit (BMU) and Cell Module Assembly Fuse should be installed or removed together. Do not separate these components. The installation and removal instructions in this section are for both the standard capacity lithium-ion battery as well as the high capacity lithium-ion battery. The Cell Module Assembly comes equipped with a Cell Module Assembly cover that should remain with the Cell Module Assembly, unless instructed otherwise.

*Figure 7-63. CMA and BMU Location*

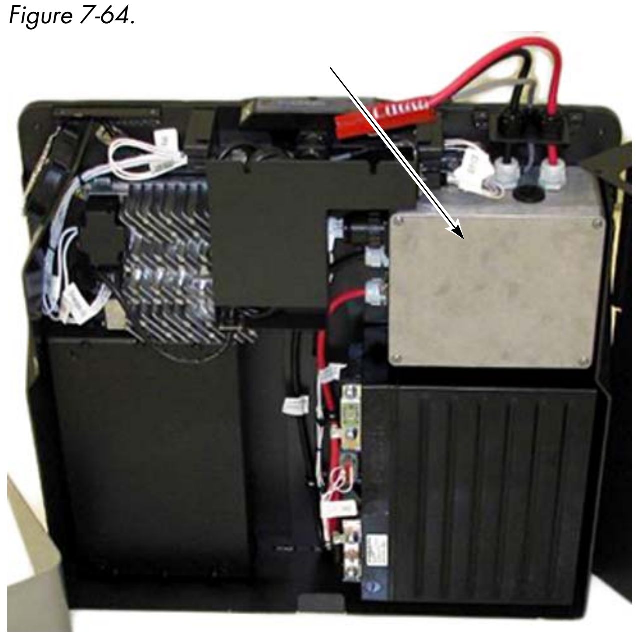


## Cell Module Assembly, Battery Monitoring Unit, and Cell Module Assembly Fuse Removal

Perform the following steps to remove the Cell Module Assembly, Battery Monitoring Unit, and Cell Module Assembly Fuse (it is recommended to perform these steps with the back plate of the lithium-ion battery laying down on a level surface).

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the lithium-ion battery from the truck. See “Battery Removal” on page 7-48.
3. Remove the top cover. See “Top Cover Removal” on page 7-49.
4. Remove the front cover. See “Front Cover Removal” on page 7-50.
5. Remove the Battery Management Controller (BMC) cover with a screwdriver.

*Figure 7-64.*



6. Disconnect the BMC F1 Fuse.

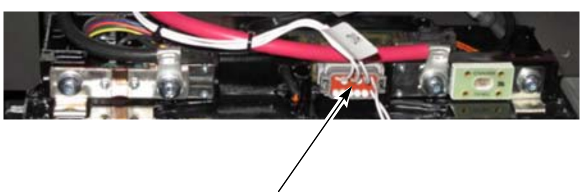
*Figure 7-65. BMC FI Fuse Location*

7. Remove the B+ and B– terminal covers.
8. Disconnect the B+ and B– cables from the Cell Module Assembly.

*Figure 7-66. CMA Connections*


9. Remove the Battery Monitoring Unit communication connection by pinching the two connector tabs and pulling outward.

*Figure 7-67. BMU Communication Connector Location*



10. Gently rocking the connector as you pull out is a preferred method of extraction. The connection between the connector and the Battery Monitoring Unit might be tight and pulling directly outward might cause damage to components.
11. Do not pull the wires to remove the connector.
12. Remove the four mounting fasteners (on the Cell Module Assembly.
13. Remove the Cell Module Assembly Drip Protector, if one is present.
14. Carefully dislodge the Cell Module Assembly by removing the Cell Module Assembly directly outward from the lithium-ion battery.
15. For a high capacity lithium-ion battery, repeat steps 8 through 14 on the other Cell Module Assembly.

**⚠CAUTION**

**If the Cell Module Assembly is being replaced because of a blown fuse, do not attempt to replace the fuse. The Cell Module Assembly with a blown fuse should be sent to The Raymond Corporation for service.**

## Cell Module Assembly, Battery Monitoring Unit, and Cell Module Assembly Fuse Installation

Perform the following steps to install the Cell Module Assembly, Battery Monitoring Unit, and Cell Module Assembly Fuse:

1. Remove the BMC F1 Fuse. See Figure 7-65.

**⚠CAUTION**

**If the BMC F1 fuse is not removed, the technician might observe sparks when unscrewing the bolts on the Cell Module Assembly terminals. This is expected, but steps should be taken to avoid this.**

2. Carefully align the Cell Module Assembly to be installed with the four fastener holes on the lithium-ion battery.
3. Be careful of body parts when placing the Cell Module Assembly.

**⚠CAUTION**

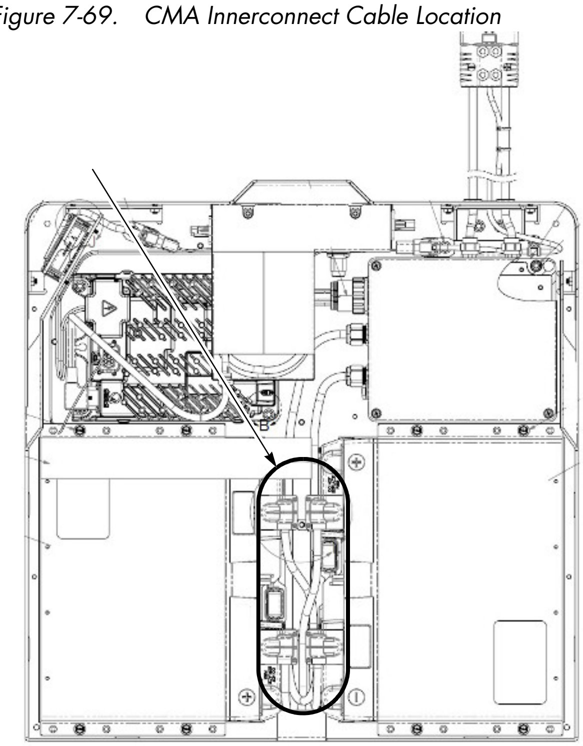
**The Cell Module Assembly is heavy and is fitted in a tight space when installed in the lithium-ion battery. This physical constraint will put body parts at risk of being pinched under the Cell Module Assembly.**

4. Remove any cables/plugs/debris from the area before installing.
5. Install the Cell Module Assembly Drip Protector, if installing the second Cell Module Assembly in the high capacity lithium-ion battery. (This is the Cell Module Assembly located below the charger).
6. Install the four Cell Module Assembly mounting bolts. Torque the bolts to 9.9 ft. lb. (13.4 Nm).
7. Install the B+ and B- terminals onto the Cell Module Assembly. Torque fasteners to 6.6 ft. lb. (9 Nm).

*Figure 7-68. CMA Connections*

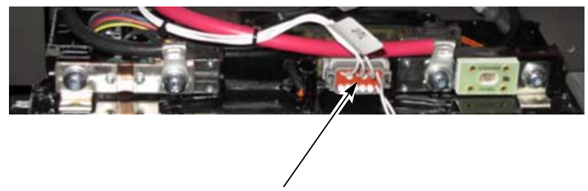

8. Install the Cell Module Assembly Interconnecting Cables, if installing a second Cell Module Assembly in the high capacity lithium-ion battery.

*Figure 7-69. CMA Innerconnect Cable Location*



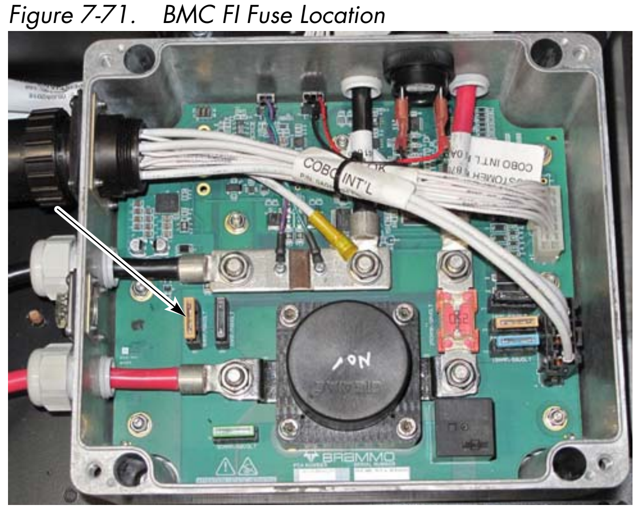
9. Install the Cell Module Assembly terminal covers. Torque fasteners to 6.7 ft. lb. (10.9 Nm).
10. Plug in the Battery Monitoring Unit communication connector into the Battery Monitoring Unit.

*Figure 7-70. BMU Communication Connector Location*



11. Install the BMC F1 Fuse.

*Figure 7-71. BMC FI Fuse Location*



**NOTE:** The lithium-ion battery might wake up.

12. Install the BMC cover by screwing in the four BMC cover screws. Torque fasteners to 1.3 ft. lb. (1.8 Nm).
13. Install the front cover. See “Front Cover Installation” on page 7-50.
14. Install the top cover. See “Top Cover Installation” on page 7-49.
15. For a high capacity lithium-ion battery, repeat steps 1 through 10).
16. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
17. Test the operation of the truck.

# Battery Management Controller

The Battery Management Controller assembly (BMC) includes the BMC enclosure and power cables connecting to the Cell Module Assembly and outside world.

*Figure 7-72. BMC Component Locations*

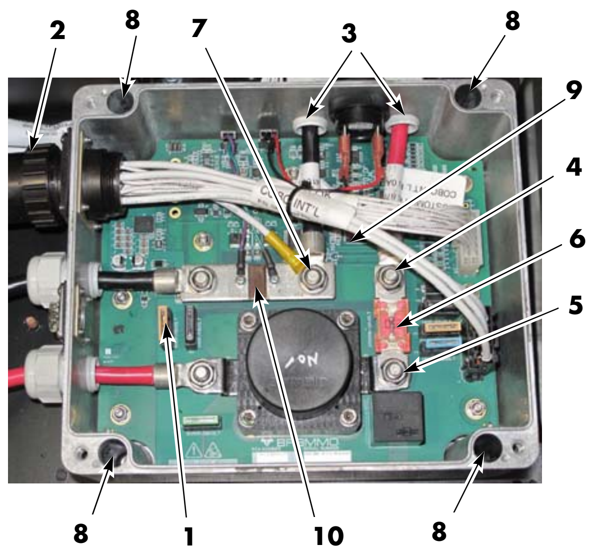



*Table 7-3. BMC Component Legend*

| Item | Description |
|---|---|
| 1 | BMC F1 Fuse |
| 2 | BMC I/O Power Connector |
| 3 | Power Cord Cable Glands (external) |
| 4 | Truck Side B+ and BMC Fuse Terminal Nut |
| 5 | Contactor Side BMC Fuse Terminal Nut |
| 6 | BMC Fuse |
| 7 | Truck Side B– Terminal |
| 8 | BMC Mounting Nut Location |
| 9 | SD Card |
| 10 | BMC Shunt |

## BMC Removal

Perform the following steps to remove the BMC.

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the top cover. See "Top Cover Removal" on page 7-49.
3. Remove the front cover. See "Front Cover Removal" on page 7-50.
4. Unscrew the four BMC cover screws with a screwdriver and remove the BMC cover.
5. Remove the BMC F1 Fuse [1].

**⚠CAUTION**

**If the BMC F1 fuse is not removed, the handler might observe sparks when unscrewing the bolts on the Cell Module Assembly terminals. This is expected, but not desired.**

6. Unplug the BMC I/O and Power Connector [2] on the outside wall of the BMC enclosure.
7. Loosen the two Power Cord Cable Glands [3] for the power cables going to the outside world.
8. Unscrew the Truck side B+ and BMC Fuse Terminal nut [4].
9. Unscrew the Contactor side BMC Fuse Terminal nut [5].
10. Remove the BMC Fuse [6].
11. Unscrew the Truck side B- Terminal nut [7].
12. Remove the BMC I/O Cable Bundle Ground ring terminal from the Truck side B- Terminal.
13. Remove power cable ring terminals from all screw posts and pull through circular cutouts through BMC enclosure wall.
14. Cover the Cell Module Assembly side B+ and B- power cable ring terminals with non-conductive material.
15. Verify all cords/cables are disconnected from the BMC.

16. Unscrew the four M6 BMC mounting nuts [8].
    - Located in the top left BMC mounting hole.
    - Located in the bottom left BMC mounting hole.
    - Located in the top right BMC mounting hole.
    - Located in the bottom right BMC mounting hole.
17. Remove the BMC enclosure with the BMC mounted within, outwards from the lithium-ion battery.
18. Optional: Remove the SD Card [9] and store in a safe place.

**NOTE:** The installer should remove the SD card from the old BMC and install it into the new BMC.

19. Replace the BMC Shunt and BMC Fuse and tighten the nuts to the terminals.

**NOTE:** These should be kept with the BMC assembly.

20. Install the BMC cover.

## BMC Installation

Perform the following steps to install the BMC. Refer to Figure 7-72.

1. Open the BMC cover with a screwdriver and remove the BMC F1 Fuse [1].
2. Align the four BMC Mounting Holes with the four welded standoffs located on the lithium-ion battery.
3. Screw the BMC Mounting Nuts into the four BMC mounting holes [8].
    - Located in the top left BMC mounting hole.
    - Located in the bottom left BMC mounting hole.
    - Located in the top right BMC mounting hole.
    - Located in the bottom right BMC mounting hole.
4. Install fasteners. Torque fasteners to 6.6 ft. lb. (9 Nm).
5. Guide the power cables through the BMC wall cutouts.
6. Install the BMC Fuse [6].
7. Place the power cable ring terminals on the appropriate screw terminal.
8. Truck side B– power cable ring terminal goes on the top of shunt on the Truck side B– Terminal [7].
9. Truck side B+ power cable ring terminal goes on top of the BMC Fuse on the Truck side B+ and BMC Fuse Terminal [4].
10. Place the BMC I/O Cable Bundle Ground ring terminal onto the Truck side B– Terminal [7] (on top of shunt and Truck side B- Terminal).
11. Install the screw terminal washers and nuts onto the appropriate screw terminals.
12. Install fasteners. Torque fasteners to 6.6 ft. lb. (9 Nm).
13. Install the cable glands [3] into the side of the BMC walls.
14. Install fasteners. Torque fasteners to 2.8 ft. lb. (3.75 Nm).
15. Install the BMC I/O and Power Connector [2].
16. Plug in the shunt [10] or speaker connector, if not presently connected.
17. Attach the Cell Module Assembly power cables to the Cell Module Assembly using the pre-installed screws on the Cell Module Assembly.
18. Install fasteners. Torque fasteners to 6.6 ft. lb. (9 Nm).
19. Install the BMC F1 Fuse [1] and replace the cover.
20. Optional: Replace the SD card [9] with the old SD card if this is a replacement.
21. Place the BMC cover on the BMC enclosure and tighten the cover using the four screws attached to the cover.
22. Install fasteners. Torque fasteners to 1.3 ft. lb. (1.8 Nm).
23. Install the front cover. See “Front Cover Installation” on page 7-50.
24. Install the top cover. See “Top Cover Installation” on page 7-49.

25. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
26. Test the operation of the truck.

# BMC Fuse

## BMC Fuse Removal

Perform the following steps to remove the BMC Fuse. Refer to Figure 7-72.

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the top cover. See “Top Cover Removal” on page 7-49.
3. Remove the front cover. See “Front Cover Removal” on page 7-50.
4. Unscrew the four BMC screws on the cover and remove the cover.
5. Remove the BMC F1 Fuse [1].
6. Unscrew the two nuts [4 and 5] mounted to the two shanks on both sides of the BMC Fuse [6].
7. Remove washers and other cable terminations.
8. Remove the BMC Fuse.

## BMC Fuse Installation

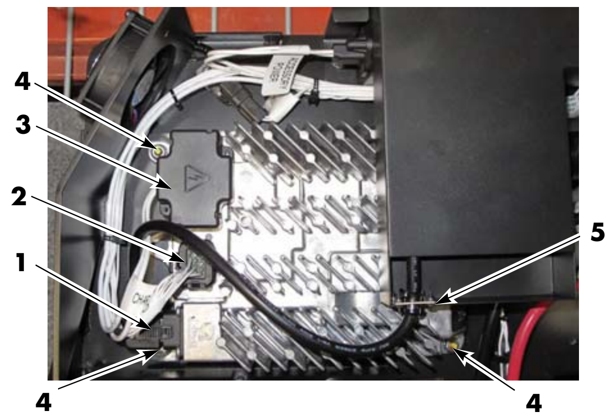
Perform the following steps to install the BMC Fuse.

1. Place a known good BMC fuse between the truck side BMC fuse terminal [4] and the contactor side BMC fuse terminal [5].
2. The fuse should rest physically on top of the truck side B+ cable and the contactor.
3. Install washers and nuts. Torque nuts to 6.6 ft. lb. (9 Nm).
4. Replace the BMC cover.
5. Install fasteners. Torque fasteners to 1.3 ft. lb. (1.8 Nm).

6. Install the front cover. See “Front Cover Installation” on page 7-50.
7. Install the top cover. See “Top Cover Installation” on page 7-49.
8. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
9. Test the operation of the truck.

# Charger

*Figure 7-73.*




## Charger Removal

Perform the following steps to remove the Charger. Refer to Figure 7-73.

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the top cover. See “Top Cover Removal” on page 7-49.
3. Remove the front cover. See “Front Cover Removal” on page 7-50.
4. Disconnect the Charger AC input connector [1].
5. Disconnect the Charger I/O connector [2].
6. Pinch the two tabs on the Charger DC output housing and remove the cover [3].
7. Unscrew and remove the B+ and B– terminals.

8. Unscrew the four charger mounting bolts [4] on the four corners of the charger.

### Charger Installation

Perform the following steps to install the Charger. Refer to Figure 7-73.

1. Align the charger mounting holes to the studs welded to the lithium-ion battery and position the charger in place.
2. Install the four charger mounting nuts [4].
3. Install fasteners. Torque fasteners to 6.6 ft. lb. (9 Nm).
4. Install the B+ and B- ring connectors on the appropriate terminal in the DC output location.
5. Install fasteners. Torque fasteners to 3.3 ft. lb. (4.5 Nm).
6. Place the DC output housing over the DC output terminals [3].
7. Verify two tabs have locked into place.
8. Install the charger I/O connector [2] into the I/O socket.
9. Install the charger AC input connector [1] into the charger.

**⚠CAUTION**

**Do not reverse the polarity on the DC output side during installation. This could result in sparks, fire, or explosion; and damage to property and/or injury to personnel are possible.**

10. Install the front cover. See “Front Cover Installation” on page 7-50.
11. Install the top cover. See “Top Cover Installation” on page 7-49.
12. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
13. Test the operation of the truck.

## Charger Cord

### Charge Cord Removal

Perform the following steps to remove the Charge Cord. Refer to Figure 7-73.

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the top cover. See “Top Cover Removal” on page 7-49.
3. Remove the front cover. See “Front Cover Removal” on page 7-50.
4. Pinch the tabs on the charger AC input connector [1] and disconnect the plug.
5. Slide the split cord grip [5] out of the cubby.
6. Guide the AC charge cord assembly through the cubby.

### Charge Cord Installation

Perform the following steps to install the Charge Cord. Refer to Figure 7-73.

1. Guide the AC charge cord through the cubby.
2. Install the split cord grip [5] into the notch at the bottom of the cubby.
3. Install fasteners. Torque fasteners to 0.5 ft. lb. (0.63 Nm).
4. Plug the charger AC input connector [1] into the AC socket.
5. Install the front cover. See “Front Cover Installation” on page 7-50.
6. Install the top cover. See “Top Cover Installation” on page 7-49.
7. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
8. Test the operation of the truck.

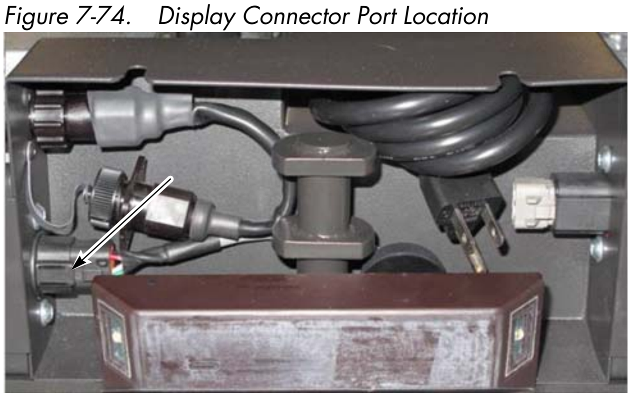
# Lithium-ion Battery User Interface (HUD)

## Lithium-ion Battery Display Removal

Perform the following steps to remove the lithium-ion battery display. Refer to Figure 7-74.

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the top cover. See “Top Cover Removal” on page 7-49.
3. Loosen the two screws keeping the lithium-ion battery display mounted in place.
4. Unplug the lithium-ion battery display from the lithium-ion battery display connector port.

*Figure 7-74. Display Connector Port Location*



## Lithium-ion Battery Display Installation

Perform the following steps to install the lithium-ion battery display (this section assumes the top cover is already removed). Refer to Figure 7-74.

1. Plug the lithium-ion battery display into the lithium-ion battery display connector port.
2. Prime the lithium-ion battery display mounting screws by partially screwing in the screws.
3. Align the two notches on the lithium-ion battery display with the two partially installed screws.
4. Tighten the lithium-ion battery display mounting screws. Do not overtighten as that might damage the plastic.
5. Install fasteners. Torque fasteners to 5 ft. lb. (6.8 Nm).
6. Install the top cover. See “Top Cover Installation” on page 7-49.
7. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
8. Test the operation of the truck.

# Wire Harness

The harness will be different in a single-Cell Module Assembly harness versus a two-Cell Module Assembly harness. The process to install and remove is the same, with the only exception that there are two Cell Module Assembly communication connectors in the two-Cell Module Assembly lithium-ion battery.

## Wire Harness Removal

Perform the following steps to remove the Harness:

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the top cover. See “Top Cover Removal” on page 7-49.
3. Remove the front cover. See “Front Cover Removal” on page 7-50.
4. Use snips to cut the cables attached to zip ties throughout the lithium-ion battery.
5. Avoid cutting the cables.
6. Disconnect the BMC I/O and Power Connector from the outside of the BMC.
7. Disconnect the communication connector from the Cell Module Assembly.

8. For a high capacity lithium-ion battery, disconnect the communication connector from the second Cell Module Assembly.
9. Do not pull on the wires to remove the connector.
10. Disconnect the Charger I/O connector by pinching the tabs and pulling the connector out.
11. Do not pull on the wires to remove the connector.
12. Pinch the two tabs on the charger DC output housing and remove the cover.
13. Unscrew and remove the charger DC output B+ and B– terminals.
14. Remove the lithium-ion battery display port from the cubby (male end).
15. Remove the Accessory Power connector from the cubby (male end).
16. Remove the Service CAN port from the cubby (female end).
17. Remove the Truck CAN connector from the lithium-ion battery (male end).
18. Remove the Cooling Fan connector from the lithium-ion battery (male end).
19. Remove harness assembly.

## Wire Harness Installation

Perform the following steps to install the Harness:

1. Plug the BMC I/O and Power Connector to the BMC.
2. Plug the Cell Module Assembly communication connector to the Cell Module Assembly. If a second Cell Module Assembly is present, plug the Cell Module Assembly communication connector to the Cell Module Assembly.
3. Plug the Charger I/O Connector into the charger.
4. Install the DC output ring terminals into the appropriate terminal on the charger.
5. Install fasteners. Torque fasteners to 3.3 ft. lb. (4.5 Nm).
6. Snap the DC output housing over the terminals.
7. Install the cooling fan connector onto the lithium-ion battery.
8. Install fasteners. Torque fasteners to 6.6 ft. lb. (9 Nm).
9. Place the wire clips on the standoff on the lithium-ion battery.
10. Install the truck CAN connector onto the lithium-ion battery.
11. Install the Accessory Power connector (male end) into the cubby.
12. Install fasteners. Torque fasteners to 1.7 ft. lb. (2.3 Nm).
13. Install the lithium-ion battery display connector into the cubby wall.
14. Install fasteners. Torque fasteners to 1.7 ft. lb. (2.3 Nm).
15. Install the service connector into the cubby wall.
16. Install fasteners. Torque fasteners to 0.7 ft. lb. (0.9 Nm).
17. Add zip ties to the cable mounting posts and the harness.
18. Install the front cover. See “Front Cover Installation” on page 7-50.
19. Install the top cover. See “Top Cover Installation” on page 7-49.
20. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
21. Test the operation of the truck.

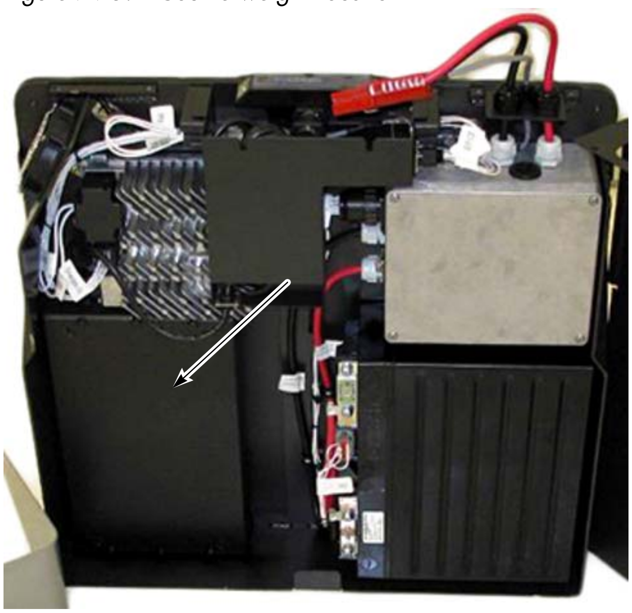
**CAUTION**

**Do not reverse the polarity on the DC output side on the AC charger during installation. This could result in sparks, fire, or explosion; and damage to property and/or injury to personnel could occur.**

## Counterweight

The Counterweight is a heavy object that must be installed and/or removed with caution. If the lithium-ion battery is outside of the truck, perform the following procedures with the lithium-ion battery back plate laying down on a flat surface.

*Figure 7-75. Counterweight Location*



### Counterweight Removal

Perform the following steps to remove the counterweight:

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the top cover. See "Top Cover Removal" on page 7-49.
3. Remove the front cover. See "Front Cover Removal" on page 7-50.
4. Unscrew the six fasteners keeping the counterweight attached to the lithium-ion battery.
5. Lift the counterweight out of the lithium-ion battery.

### Counterweight Installation

Perform the following steps to install the counterweight:

1. Align the Counterweight between the roll pins and holes for the mounting fasteners.
2. Screw six fasteners into the six counterweight mounting locations.
3. Install fasteners. Torque fasteners to 9.9 ft. lb. (13.4 Nm).
4. Install the front cover. See "Front Cover Installation" on page 7-50.
5. Install the top cover. See "Top Cover Installation" on page 7-49.
6. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
7. Test the operation of the truck.

# Cooling Fan

## Cooling Fan Removal

Perform the following steps to remove the Cooling Fan.

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the top cover. See "Top Cover Removal" on page 7-49.
3. Remove the front cover. See "Front Cover Removal" on page 7-50.
4. Unplug the cooling fan from the connector.
5. Unscrew the four cooling fan screws mounting the fan to the lithium-ion battery.

*Figure 7-76. Fan Mounting Screws*

6. Remove the cooling fan and all related hardware.

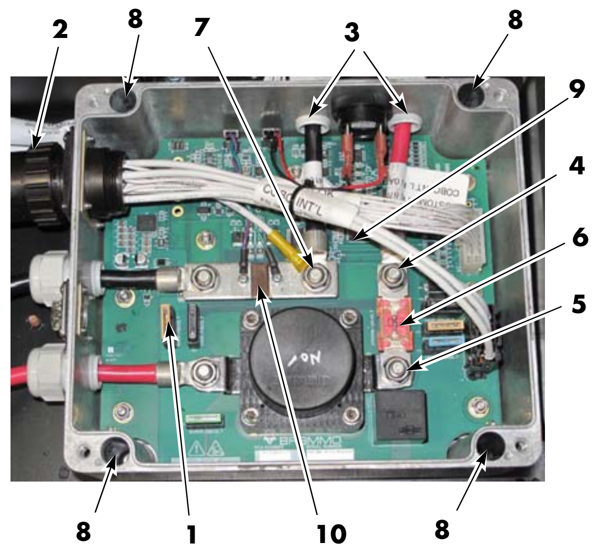
# Cooling Fan Installation

Perform the following steps to install the cooling fan.

1. Align the cooling fan to the four mounting screw holes on the lithium-ion battery.
2. Guide the four screws into the fan and into the lithium-ion battery,
3. Place the four cooling fan spacers on the screws once the screws are guided all the way through.
4. Screw the four cooling fan mounting screws.
5. Install fasteners. Torque fasteners to 0.9 ft. lb. (1.2 Nm).
6. Connect the cooling fan to the cooling fan connector inside the lithium-ion battery.
7. Install the front cover. See "Front Cover Installation" on page 7-50.
8. Install the top cover. See "Top Cover Installation" on page 7-49.
9. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
10. Test the operation of the truck.

# SBX Cable Assembly

*Figure 7-77. BMC Component Locations*




## SBX Cable Assembly Removal

Perform the following steps to remove the SBX Cable Assembly. Refer to Figure 7-77.

1. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the top cover. See "Top Cover Removal" on page 7-49.

3. Remove the front cover. See "Front Cover Removal" on page 7-50.
4. Remove the BMC cover with a screwdriver and remove the BMC F1 Fuse [1].
5. Remove the Truck side B– Terminal nut [7] and remove the B– cable from the termination.
6. Remove the Truck side B+ and BMC Fuse Terminal nut [4] and remove the B+ cable from the termination.
7. Loosen the B+ Power Cord Cable Gland [3].
8. Loosen the B– Power Cord Cable Gland [3].
9. Disconnect the SBX communication connector from the lithium-ion battery.
10. Remove the SBX Cable Assembly mount from the lithium-ion battery.
11. Guide the power cords through the cable holes in the wall of the BMC enclosure.

## SBX Cable Assembly Installation

Perform the following steps to install the SBX Cable Assembly. Refer to Figure 7-77.

1. Open the BMC cover with a screwdriver and remove the BMC F1 Fuse.
2. Guide the power B+ and B- power cords through the power cord holes in the wall of the BMC enclosure.
3. Install the B+ power cord ring terminal to the Truck side B+ and BMC Fuse Terminal [4] and the B– power cord ring terminal the Truck side B– Terminal [7].
4. Install washers and nuts to both terminals.
5. Install fasteners. Torque fasteners to 6.6 ft. lb. (9 Nm).
6. Install the Power Cord Cable Gland [3].
7. Install fasteners. Torque fasteners to 2.8 ft. lb. (3.75 Nm).
8. Connect the SBX communication connector to the lithium-ion battery.
9. Install the SBX Cable Assembly to the lithium-ion battery.
10. Install fasteners. Torque fasteners to 6.6 ft. lb. (9 Nm).
11. Install the front cover. See "Front Cover Installation" on page 7-50.
12. Install the top cover. See "Top Cover Installation" on page 7-49.
13. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
14. Test the operation of the truck.

# Specification and Warning Labels

## Label Removal

Perform the following steps to remove the lithium-ion battery label:

1. Peel the label from the lithium-ion battery
2. Do not operate the lithium-ion battery without a specification or warning label.

## Label Installation

Perform the following steps to install the lithium-ion battery Label:

1. Clean the surface with an appropriate cleaner.
2. Dry the area off.
3. Apply label as required.

# Battery Connector (Lead Acid Battery)

The truck battery connector is located at the top of the console cover. The lead acid battery and charger have similar battery connectors. It is necessary to connect the battery to the truck before powering ON.

*Figure 7-78. Battery Connector Location*


## Inspection

Visually inspect cables and connectors. Negative and positive cables should be visually inspected for breaks and wear. Check connectors for damage. Make sure cable wires are in good condition and connector has correct connections to cables.

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Check battery connectors for damaged cables.
3. Check to see if cables are pulling out of the connectors.
4. Check cables at battery terminals. Connections should be tight, with no corrosion.
5. Look inside the connector. Check all internal contacts for damage, dirt, or corrosion. Do not use a metal object to clean the connector.

**CAUTION**

**Some degreasers and parts cleaners will cause the connector shell to disintegrate. Avoid contacting battery connectors with solvents.**

6. Connector housings must not be cracked or broken.

## Battery Connector Access and Removal

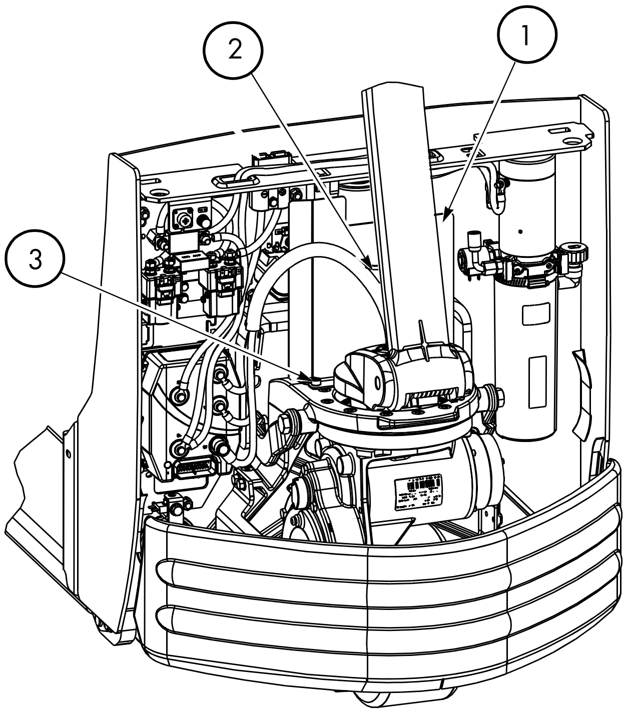
The battery connector may be accessed in two ways for removal or inspection.

1. To remove the battery connector by raising the truck:
   a. Jack the truck and block the forks. See “Jacking Safety” on page 2-10.
   b. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
   c. Remove the tractor covers.
   d. Disconnect the battery connector cables at the main contactor and traction amplifier.
   e. Remove the screws securing the battery connector to the electrical panel.
   f. Remove the battery connector from the truck.
2. To remove the battery connector by removing the main harness bracket:
   a. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.

b. Remove the tractor covers.

c. Remove the two screws and U-clamp [2] securing the harness to the main harness bracket [1]. See Figure 7-79.

*Figure 7-79. Main Harness Bracket and Harness Mounting Hardware Locations*




d. Remove the two screws [3] securing the base of the main harness bracket [1] to the pivot frame. See Figure 7-79.

e. Disconnect the battery connector cables at the main contactor and traction amplifier.

f. Remove the screws securing the battery connector to the electrical panel.

g. Remove the battery connector from the truck.

## Cable Removal, Replacement, and Installation

The cables to either half of the connector have a lip on their forward end. This lip snaps over a spring-loaded retainer that is part of the connector.

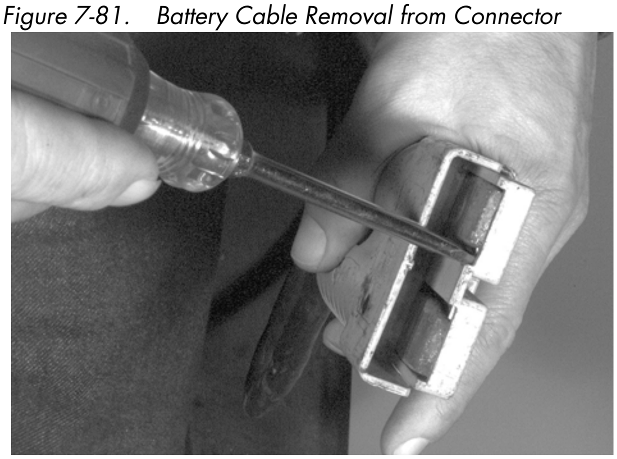
*Figure 7-80. Battery Cable and Connector*

**⚠ WARNING**

**When replacing battery cable ends, remove only one end at a time from the connector to avoid the cable ends touching and causing a short circuit. Do not allow the metal cable end to touch the battery. Use insulated tools and avoid contact with battery case or cable ends.**

1. To remove a cable from the connector, push the retainer down while pulling the battery cable towards the rear and out of the connector. See Figure 7-81 and Figure 7-80.

*Figure 7-81. Battery Cable Removal from Connector*



2. Do not attempt to repair battery cables by crimping new terminals. Replace the cable.

# Power Cables

Check the power cables for damage:

- Evidence of overheating
- Burned spots in the cable
- Nicks in the insulation
- Damaged or overheated terminal lugs
- Damaged mounting hardware or brackets

Replace damaged cables or mounting hardware as necessary.

Power cables are marked on the insulation near the terminal lug with the location where they belong. If the marking is missing or is not readable, remark the cable with the correct information.

**NOTE:** Replace terminal lugs in the field using the appropriate crimping tools. Crimping tool, lugs, and heat-shrink are available through the Parts Distribution Center. Failure to use correct cables, terminal hardware, and torque values can result in overheating and damage to components.

## Power Cable Repair

Traditional lug crimping techniques for power cables on DC motors may not meet the higher current requirements of AC motors. Use Manual Crimp Tool (P/N 1069861) when crimping power cables for all AC motors. This tool may also be used to repair power cables for DC motors, yielding enhanced repair.

1. Remove the failed cable(s) from the truck.
2. Using the removed cable(s) for reference, cut an appropriate amount of replacement cable of the same gauge.
3. Set the adjustment screw on the manual crimp tool head to match the cable gauge.
4. Referring to Table 7-4 and Table 7-5, strip the cable jacket to fit the terminal to be crimped.

*Table 7-4. Cable Jacket Strip Length - Short barrel*

| AWG | Lug P/N | Strip Length |
|---|---|---|
| 1/0 | 1002215/001-004 | 11/16 in. (17 mm) |
| 2/0 | 1002215/005-008 | 13/16 in. (21 mm) |
| 3/0 | 1002215/009-012 | 1 in. (25 mm) |
| 4/0 | 1002215/013-015 | 1-1/16 in. (27 mm) |

*Table 7-5. Cable Jacket Strip Length - Long barrel*

| AWG | Lug P/N | Strip Length |
|---|---|---|
| 1/0 | 1002217/001-003 | 1-9/16 in. (40 mm) |
| 2/0 | 1002217/004-006 | 1-9/16 in. (40 mm) |
| 3/0 | 1002217/007-009 | 1-9/16 in. (40 mm) |
| 4/0 | 1002217/010-011 | 1-11/16 in. (43 mm) |

5. Insert the cable into the terminal.
6. Place the cable and terminal into the tool die and crimp. Refer to Figure 7-82 and Figure 7-83 for what completed crimps should look like.

*Figure 7-82. Short Barrel Terminal*

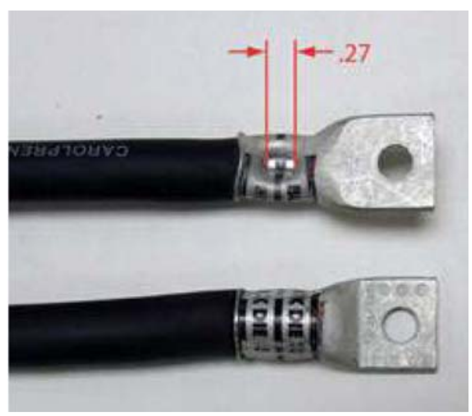
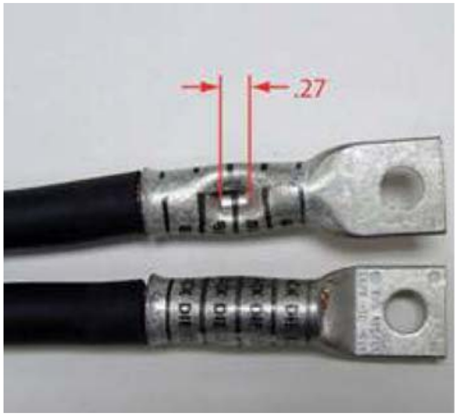


*Figure 7-83. Long Barrel Terminals*




7. Apply heat-shrink tubing (P/N 611-035) to the terminal after crimping.
   a. For short barrels, use 1.75 in. (45 mm) of tubing. Apply 1 in. (25 mm) over the jacket extending 0.75 in. (19 mm) over the terminal barrel.
   a. For long barrels, use 2.5 in. (64 mm) of tubing. Apply 1 in. (25 mm) over the jacket extending 1.5 in. (38 mm) over the terminal barrel.

# Wiring Harness

**⚠CAUTION**

**Use correct electrostatic discharge precautions. See "Static Safety" on page 2-13.**

**NOTE:** For replacement parts information refer to the Parts Manual.

The wiring harness is designed to connect the electrical components of the pallet truck.

All wires in the harness are marked with a wire identification number. These numbers correspond to wire identification given in the Electrical Schematics.

## Wiring Harness Terminology

The term "connector JPx" means a mated connector consisting of two connector halves. One half contains male connectors, or pins (P); the other half contains female connectors, or jacks (J).

When you disconnect a mated JP connector, you have two connector halves. The individual connector halves are designated by "Jx" and "Px."

For example, connector JP2 is the mated connector in the control handle. J2 represents the jack connections. P2 represents the pin connections.

To find electrical connectors on the truck, refer to the Electrical Schematics.

## Wiring Harness Inspection

Make sure the truck is OFF and the battery connector disconnected before working on harnesses. Whenever working on the truck, use care around wiring harnesses.

- Do not pull on wires.
- Carefully connect and disconnect all connectors.
- Do not pry apart connectors with unspecified tools.
- Examine and maintain any added materials used to dress or protect the wire. This includes spiral wrap, brackets, cable ties, fasteners, flexible conduit, and so forth.
- Check harness wires for abrasions, scrapes, nicks in the wire, damage from overheating or burns, or other general insulation damage.
- Replace terminations with exposed wire visible at the connectors. Damaged terminations, exposed wires, or damaged connectors can cause operational failure of the truck.

During troubleshooting and repairs, it is sometimes necessary to unmate a connector, move a harness, cut a cable tie, or remove the wire from a bracket. Note carefully the location of the wire and all protective or securing attachments before moving the harness.

After repair, return or replace all protective and/or securing hardware to its original condition. Protective materials are necessary to provide reliable performance of the interconnect system.

In addition to the wire identification number printed every 1 to 1.5 in. (25.4 to 38.1 mm) on the wire, there is a wire marker at each termination. If the marker is missing or unreadable, remark the wire to permit easier identification.

**NOTE:** It is normal to find unused connectors for uninstalled options that have had heat shrink applied over them and have been strapped to the harness.

## Wiring Harness Repair

Make sure the truck is OFF and the battery connector disconnected before working on harnesses. When pulling a wire out through a bundle, cut off the pin or socket so it does not snag. When replacing wires, in some cases you can tape or solder one end of a new wire to one end of the failed wire. Then you can pull the old wire out of the bundle and pull the new wire into the bundle, all at the same time. Make sure to disconnect the old wire from the new wire.

In other cases, it is easier to secure a new wire to the outside of the existing wire bundle with cable ties of an appropriate size. The failed wire can be left in the bundle, or can be pulled by one end to remove it from the bundle.

When replacing wires, follow these guidelines:

- Use the appropriate tools to remove and insert terminations at each connector.
- Remove damaged terminations and discard. Never reuse a termination from a wire.
- Do *not* cut away a terminal lug and reuse the wire strands that were crimped into the original lug.
- When stripping wire, use new wire strands for new terminations. Make sure to use a new wire with extra length to allow for cutting and stripping of the ends to install new terminations.
- Use a new wire that is the same gauge (typically gauge 18 AWG), size, type, and color as the wire it is replacing.
- Use a hand stripper capable of stripping by wire gauge number. Use care not to nick or cut any of the wire strands. Discard and replace a wire with damaged strands. Insert the correct length wire strands into the termination before crimping.

## Wiring Harness Soldering Procedures

To prevent damage from excessive heat when soldering small components in assemblies, follow these guidelines:

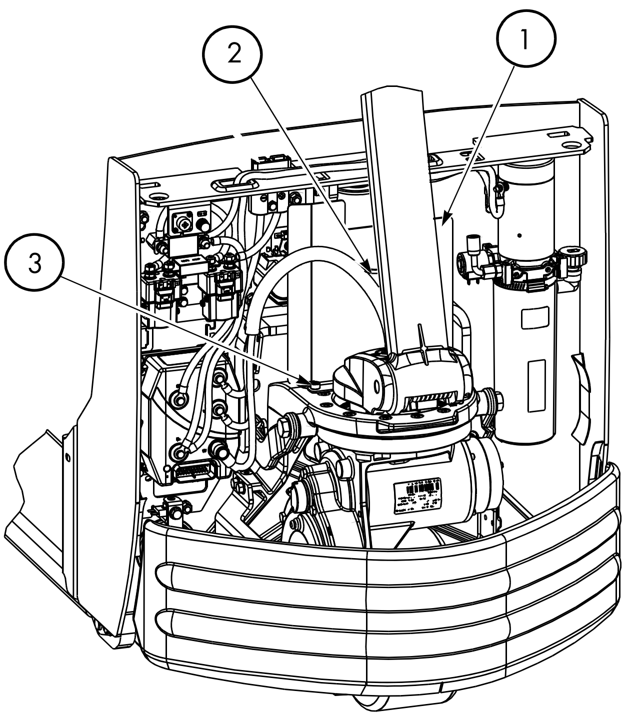
- Make sure the soldering tip is clean. A dirty tip does not transfer heat well and encourages long dwell time and greater pressure. Apply light pressure on the terminal.
- Flux: rosin base
- Solder: 60/40 rosin core or equivalent
- Solder Iron: 15-20 watt “pencil-type” maximum
- Tip Size: 3 mm (0.118 in.) diameter x 30 mm (1.182 in.) long screwdriver tip. Make sure the tip is clean.
- After soldering, clean the terminals with a brush dampened with an alcohol-based cleaner (P/N 990-600/FOF). Do *not* allow any cleaner to seep into the switches or potentiometers, or contact contamination may occur.

## Main Harness Bracket Removal

The main harness bracket should be removed to allow access to many of the hydraulic and electrical components. To remove the main harness bracket:

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the tractor covers.
3. Remove the two screws and U-clamp [2] securing the harness to the main harness bracket [1]. See Figure 7-79.

*Figure 7-84. Main Harness Bracket and Harness Mounting Hardware Locations*




4. Remove the two screws [3] securing the base of the main harness bracket [1] to the pivot frame. See Figure 7-84.

## Main Harness Installation

It is important to make sure that the harness is installed correctly to prevent premature failure. When installing the main harness, make sure that:

- the main harness bracket to pivot frame mounting bolts are torqued to 31 ft. lbs. (42 Nm).
- the harness is secured to the bracket near the traction amplifier. Make sure the harness conduit is 0.25 to 0.5 in. below the cable tie.

*Figure 7-85.*

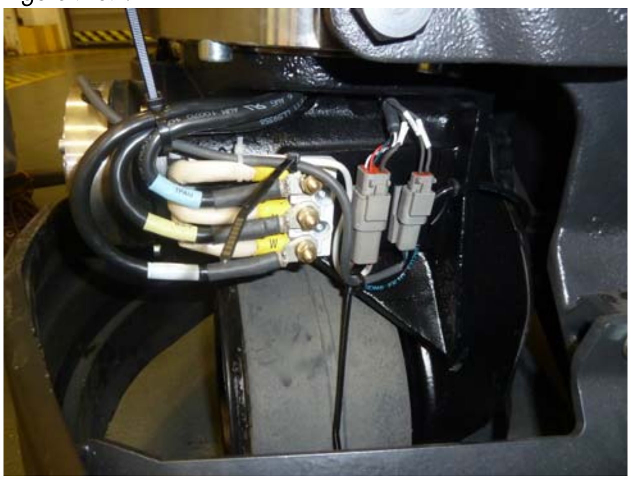
- the distance from the top of the main harness bracket and the highest point of the arched harness shall be 2.5 ±0.25 in. with the lift truck fully lowered.

*Figure 7-86.*

**NOTE:** The harness loop must not be allowed to be pulled over the main harness bracket when the lift truck is fully raised.

- the phase cables at the traction motor terminal block should be looped enough to allow full turning from left to right without contact with the lift linkage.

*Figure 7-87.*

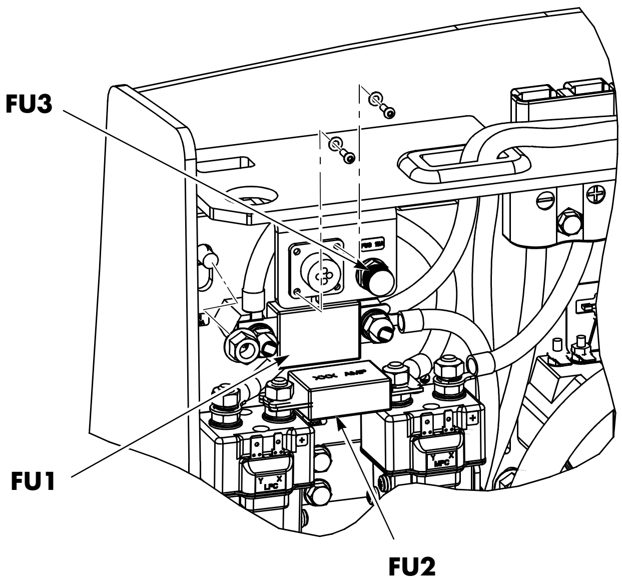


# Fuses

**NOTE:** For replacement parts information refer to the Parts Manual.

Under extreme operating conditions, fuses (FU1, FU2, and FU3) protect the electrical circuits and components from excessive current or voltage overloads. See Figure 7-88.

*Figure 7-88. Fuse Locations*




## Fuse Test/Inspection

Examine each fuse for signs of overheating, discoloration, cracking, or other physical damage. Replace the fuse if you find damage.

To test a fuse, remove it or isolate it from the electrical circuit. Do this by removing the fuse from the truck or by removing all the connections from one side of the fuse.

Use an ohmmeter set to Rx1 scale and measure the resistance across the fuse. The resistance must be less than 1 ohm.

# Horn

**NOTE:** For replacement parts information refer to the Parts Manual.

## Horn Removal

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the tractor covers.
3. Disconnect the wires from the horn terminals. See Figure 7-89.

*Figure 7-89. Horn Location*

4. Remove the horn mounting bolt.

## Horn Installation

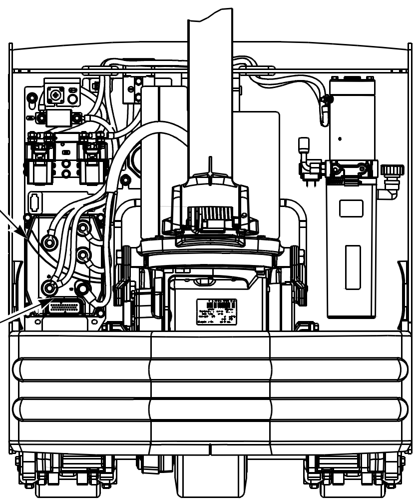
1. Attach the horn to the electrical panel and tighten the mounting bolt.
2. Attach the wires to the horn terminals.
3. Reconnect the battery connector.
4. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key. Test the operation of the horn.
5. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
6. Install the tractor covers.
7. Reconnect the battery connector.

# Traction Amplifier

**NOTE:** For replacement parts information refer to the Parts Manual.

The traction amplifier is installed in the left side electrical compartment of the tractor. See Figure 7-90.

*Figure 7-90. Traction Amplifier Location*




**CAUTION**

**Do *not* open the traction amplifier. Opening the traction amplifier could cause damage to it and voids the warranty.**

## To Clean the Traction Amplifier

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the tractor covers.
3. Discharge the capacitors in the traction amplifier by connecting a load (such as a contactor coil) across the amplifier's TA B+ and TA B– terminals.
4. Clean the traction amplifier with a moist rag. Let it dry before connecting the battery.
5. Make sure the power cable connections are tight. Do not stress the cable connections or put strain on internal components.

## Traction Amplifier Removal

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the tractor covers.
3. Discharge the capacitors in the traction amplifier by connecting a load (such as a contactor coil or a horn) across the amplifier's TA B+ and TA B– terminals.
4. Push the locking tab and disconnect the J1 connector from the traction amplifier. See Figure 7-90.
5. Disconnect the power cables (terminals TA B+ and TA B–) on the traction amplifier. Do not stress the power cable connections. Disconnect the traction motor cables (terminals U, V, and W) at the traction amplifier.
6. Remove the hex flange head screws securing the traction amplifier to the electrical panel. Remove the traction amplifier.

## Traction Amplifier Installation

**NOTE:** All traction amplifiers are preset to factory default specifications.

1. Secure the traction amplifier to the electrical panel with the previously removed hex flange head screws. Torque screws to 11 ft. lbs. (14.9 Nm).
2. Connect the power cables (terminals TA B+ and TA B–) on the traction amplifier. Torque screws to 80 to 100 in. lb. (9.1 to 11.3 Nm) maximum. Do not stress the cable connections and put strain on internal components.
3. Connect the J1 connector and traction motor connectors (terminals U, V, and W) to the traction amplifier.
4. Connect the battery connector.

5. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
6. Test the operation of the truck.
7. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position.
8. Install the tractor covers.

# Contactors

**NOTE:** For replacement parts information refer to the Parts Manual.

The main contactor (MPC) controls battery power to both the traction and lift motors. The lift pump contactor (LPC) provides power to the lift motor to control the hydraulics.

Visually inspect the contactors for any signs of burning or physical damage. These are sealed contactors; therefore, the entire contactor must be replaced if there are any signs of damage.

## Resistance Testing

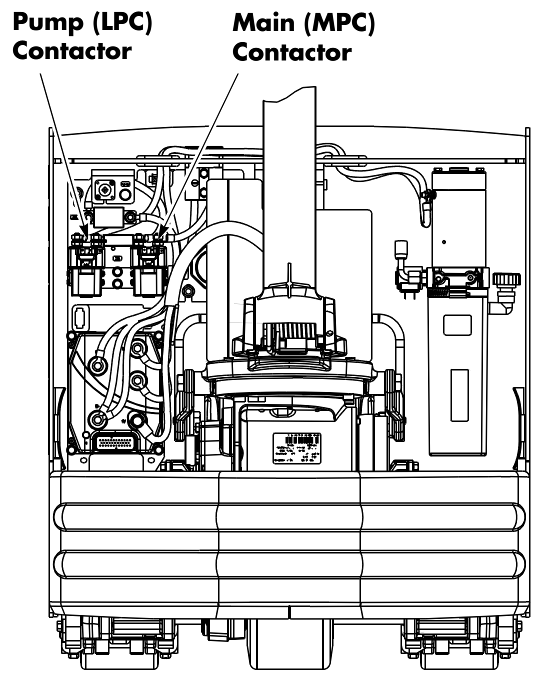
1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the tractor covers to access the contactor panel.
3. Disconnect the wires from the contactor coil.
4. Connect the ohmmeter to the wire connectors on the contactor coil.
5. The reading should be 22 to 24.2 Ohms for standard contactor (76.5 to 93.5 Ohms for cold storage contactor). If resistance is outside that range, replace the contactor.

**NOTE:** Replace the main contactor (MPC) and/or lift pump contactor (LPC) if coil resistance is not 22 to 24.2 Ohms for standard contactor (76.5 to 93.5 Ohms for cold storage contactor).

## Contactor Removal

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the tractor covers to access the contactor panel.
3. Disconnect the wires from the MPC-X and MPC-Y or LPC-X and LPC-Y terminals on the contactor. See Figure 7-91.

*Figure 7-91. Main Contactor (MPC) and Pump Contactor (LPC)*




4. Remove the nuts securing the cables and fuse (FU2) to the contactor, noting the locations for assembly later. See Figure 7-91.
5. Remove screws and flat washers holding the contactor to the bracket.

## Contactor Installation

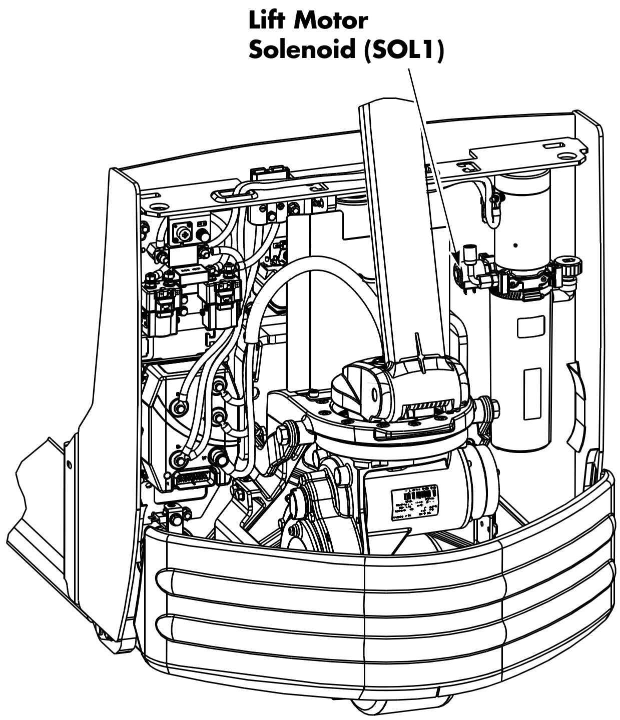
1. Secure the contactor to the bracket with flat washers and screws. Torque the screws to 39 in. lb. (4.4 Nm).
2. Secure the cables and fuse to the contactor with jam nuts. Torque the nuts to 71 to 84 in. lb. (8 to 9.5 Nm).
3. Connect the wires to the MPC-X and MPC-Y or LPC-X and LPC-Y terminals.
4. Install the tractor covers.
5. Reconnect the battery connector to the truck.
6. Place the Main ON/OFF Switch in the ON position. If equipped with the optional

keypad, enter your PIN-key code and then press the green ON ( | ) key.

7. Test the truck for correct operation before returning to service.

# Lift Motor Solenoid

*Figure 7-92. Lift Motor Solenoid SOL1*




See "Lift Motor" on page 7-91.

# Switches (General)

**NOTE:** For replacement parts information refer to the Parts Manual.

Refer to the Electrical Schematics to identify the electrical circuit location of the switch.

## Testing/Inspecting Switches

Examine the switch for signs of arcing, overheating, discoloration, cracking, or other physical damage. Replace the switch if you see such damage.

1. To test a switch, isolate it from the electrical circuit. Do this by removing all the connections from the switch, making sure all wires are labeled and identified for later reconnection.
2. Use an ohmmeter set to a low resistance scale to measure the resistance across the switch. In the closed position, the switch must have a resistance of less than 1 ohm. In an open position, the switch must show a resistance greater than 10 megohms.

## Main ON/OFF Switch

A 2-position keyless (rocker) Main ON/OFF switch (SW1) is standard equipment. When in the OFF ( **O** ) position, battery power to all control functions is interrupted. When in the ON ( **|** ) position, battery potential is provided to all control functions via the traction amplifier. See Figure 7-93.

*Figure 7-93. Main ON/OFF Switch (SW1) - Standard*


## Main ON/OFF Switch Inspection

1. With the battery plugged in and the Main ON/OFF switch in the ON position, battery voltage B+ must be present on the two terminals of the switch.
2. Test the Main ON/OFF switch with an ohmmeter *after* disconnecting the battery and removing the wires from the switch terminals.

In the OFF position, the ohmmeter must read greater than 1 megohm, and in the ON position, the ohmmeter must read less than 1 ohm. If not, replace the switch.

## Main ON/OFF Switch Removal

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the tractor covers.
3. Disconnect the wires from the switch.
4. Remove the Main ON/OFF switch.

## Main ON/OFF Switch Installation

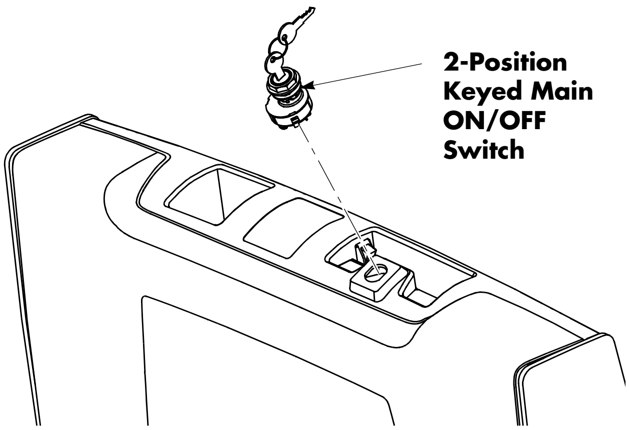
1. Install the new Main ON/OFF switch.
2. Connect the wires to the new Main ON/OFF switch terminals.
3. Reconnect the battery connector and verify operation.
4. Install the tractor covers.

## 2-Position Keyed Switch (Optional)

A 2-position keyed Main ON/OFF switch is available as an option.

### Replacement

*Figure 7-94. Main ON/OFF Switch - Keyed Option*

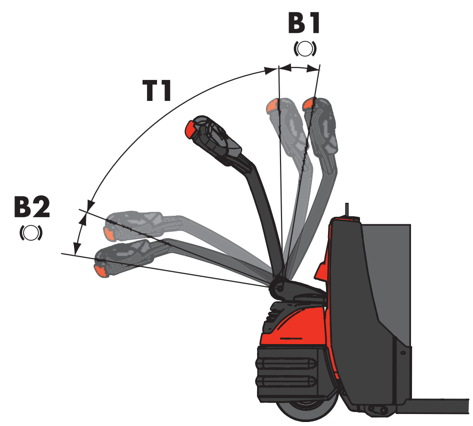



1. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the truck covers.
3. Disconnect the wires from the switch terminal.
4. Remove the key switch.
5. Install the new key switch.
6. Install the truck covers.
7. Reconnect the battery connector.

## Arm Angle Proximity Switch

The arm angle proximity switch is found at the base of the tiller arm stem. SW2 determines brake activation and truck travel. The switch is activated by the control handle position. The control handle must be in the T1 position in order for the switch to be closed, permitting travel. See Figure 7-95.

*Figure 7-95. Brake Actuation*




### Replacement

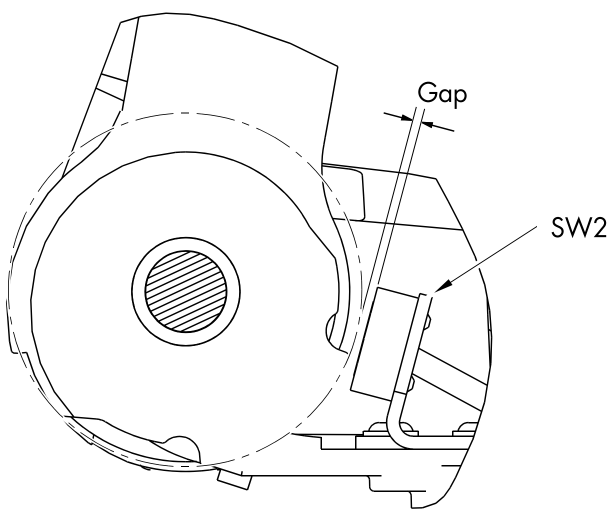
1. Lower the forks completely.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
3. Remove the truck covers.
4. Disconnect the wires from the switch.
5. Remove the screws and carefully remove the bracket and switch.
6. Remove the screws holding the switch on the mounting bracket. Remove the switch.
7. Mount the new switch to the mounting bracket.
8. Install the bracket and screws.
9. Connect the wires to the switch.
10. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. Press the green ON ( **|** ) key on the keypad, or turn the optional key switch ON.
11. Test the operation of the truck.

12. Adjust the switch if necessary.
13. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
14. Install the truck covers.
15. Reconnect the battery connector.

## Arm Angle Proximity Switch Adjustment

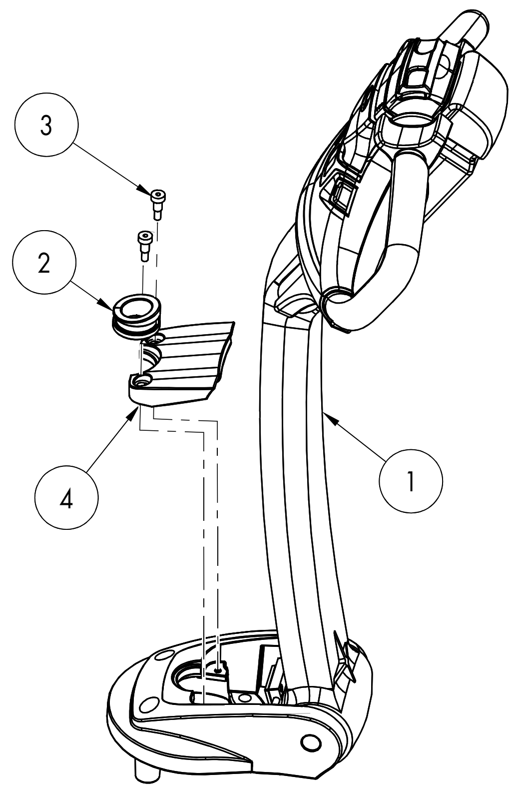
1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the screws and control handle bumper stop to access the angle arm proximity switch.
3. Loosen the screws securing the proximity switch bracket to the tiller arm base. Slide the bracket toward or away from the cam surface of the raised tiller arm to adjust the gap. Correct gap between the switch sensing face and the raised tiller cam surface is 0.070 to 0.086 in. (2.0 ±0.20 mm). See Figure 7-96.

*Figure 7-96. Arm Angle Proximity Switch Adjustment*




4. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.
5. Test the operation of the truck.
6. Install the control handle bumper stop [4] and grommet [2] using socket head locking screws [3]. Torque screws to 13 ft. lb. (17.6 Nm). See Figure 7-97.

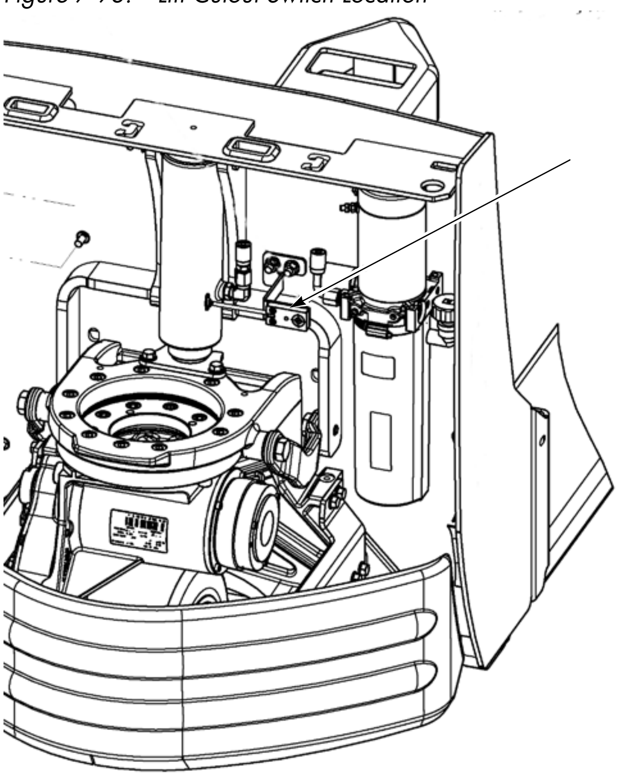
*Figure 7-97. Control Handle Bumper Stop Installation*



## Lift Cutout Proximity Switch

The lift cutout proximity switch identifies when the lift cylinder has reached full extension (maximum lift height).

*Figure 7-98. Lift Cutout Switch Location*



## Replacement

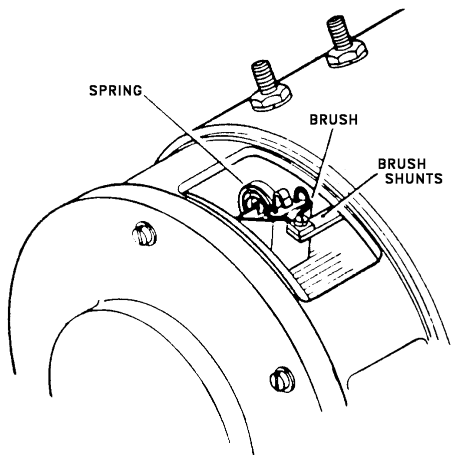
1. Lower the forks completely.
2. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
3. Remove the truck covers.
4. Disconnect the wires from the switch.
5. Remove the hex nuts and carefully remove the bracket and switch.
6. Remove the screws holding the switch on the mounting bracket. Remove the switch.
7. Mount the new switch to the mounting bracket.
8. Install the bracket and nuts. Torque nuts to 115 to 124 in. lbs. (13 to 14 Nm).
9. Connect the wires to the switch.
10. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. Press the green ON ( | ) key on the keypad, or turn the optional key switch ON.
11. Test the operation of the truck.
12. Adjust the switch bracket up or down at the tractor frame for correct switch activation.
13. If equipped with the optional keypad, press the red OFF ( O ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
14. Install the truck covers.
15. Reconnect the battery connector.

# DC Motors, General

**NOTE:** For replacement parts information refer to the Parts Manual.

## Motor Brush Inspection

*Figure 7-99. Motor Brush, Typical Location*




Conduct a partial inspection of the motor after every 1,000 hours of truck operation. If you work in an abnormally severe or caustic environment or if you have a rigorous duty cycle, inspect the motor more frequently.

Set up and rigidly adhere to a strict inspection schedule to obtain the maximum efficiency from the electrical equipment.

Each partial inspection of the motor must include the following:

1. Inspect the brushes for wear and for correct contact with the commutator. Record the level of wear on the brushes. This history gives you an indication of whether a brush must be changed or if it can wait until the next inspection. Refer to page 7-84 for acceptable brush length and general motor information.

**NOTE:** Overloading a unit is ultimately reflected in the motor and brush wear; therefore, you must take this into account when considering brush replacement.

2. Check brush spring tension. See "Motor Brush Spring Tension" on page 7-84.
3. Clean brushes and holders. Wipe the commutator with a dry, lint-free cloth. DO NOT USE lubricants of any kind on or around the commutator.
4. Check brush holders for solid connection to the mounting support. Tighten the mounting screws as necessary.
5. Check the cap screws holding the brush cross connectors to the brush holder body.
6. Make sure the motor terminals are secured tightly to the motor frame. Be careful not to strip the threads or crush the insulating parts.
7. Check all the cap screws around the frame for tightness.
8. Keep the outside frame of the motor clean and free from dirt. Maintain a free air passage around the motor to permit heat radiation.

## Motor Brush Replacement

If one brush needs replacement, always replace the entire set of brushes.

Use only genuine *Raymond* brushes. Using another type of brush could damage the commutator or cause excessive brush wear.

If the end of the brush is not already contoured to fit the commutator, use the following procedure to seat the brush to the commutator.

**NOTE:** If the motor commutator is not accessible, form the brush contour using a brush seating stone.

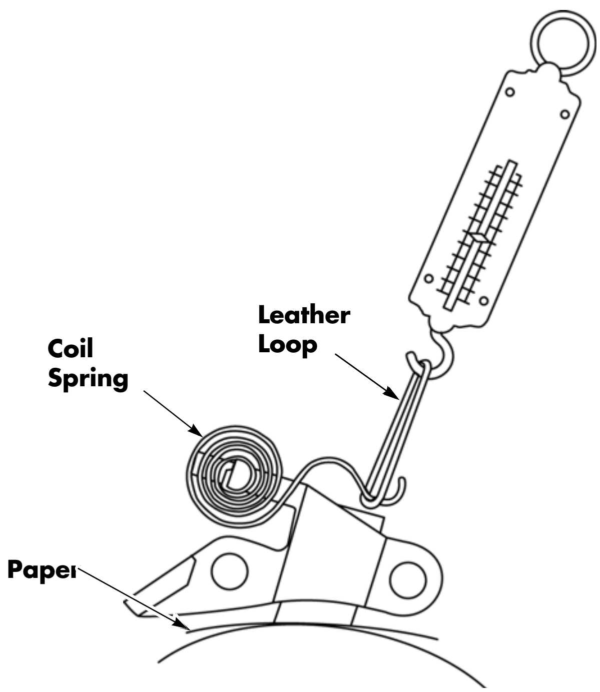
1. Move the motor brush springs out of your way.
2. Wrap a piece of 00 sandpaper around the commutator. **DO NOT use emery cloth to seat brushes.**
3. Move the brushes back down in their holders so that the face of the brushes matches the curve of the commutator.
4. Remove the sandpaper.
5. Blow any dust out of the motor with clean, compressed air at a maximum of 30 psi (207 kPa).

# Motor Brush Spring Tension

## Brush Spring Tension Inspection

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the lift motor cover.
3. Slide the brush up slightly in its holder.
4. Insert a paper strip between the brush face and the commutator. See Figure 7-100.

*Figure 7-100. Motor Brush Spring Tension Inspection*




5. Place a small leather loop around the coil spring for the brush. If the brush spring has a loop at the brush, hook the spring scale directly to the spring.
6. Attach a 5 lb. (2.27 kg) spring scale to the leather loop.
7. While gently pulling the scale outward, slowly pull the paper strip in the direction that the commutator normally rotates.
8. When the paper strip starts to move freely, the spring scale reads the spring brush tension.
9. Refer to the table below for correct spring tension.

| Motor | Minimum Brush Length | Spring Tension |
|---|---|---|
| Lift Motor | 0.35 in. (9 mm) | 32 to 40 oz. (908.8 to 1136 grams) |

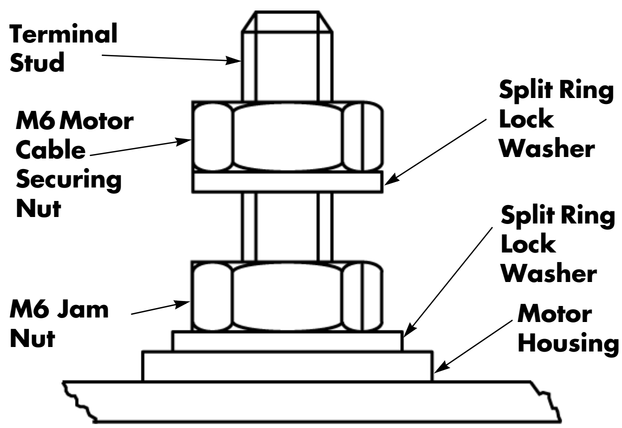
10. Repeat steps 3 through 9 for the remaining brushes.

## Terminal Nuts

**CAUTION**

**Do not attempt to repair power cable terminal lugs without approved tools. If a power cable has worn or damaged lugs, either replace the cable or replace the lugs using the procedure described in “Power Cable Repair” on page 7-66. Do not substitute other kinds of nuts for the flanged nuts. Failure to use correct cables, flanged nuts, and torque values can result in overheating and damage to components.**

*Figure 7-101. Lift Motor Terminal Nuts*




Whenever you disconnect and reconnect any power cables to the motors or traction amplifier, always tighten the cable securing nuts with a torque wrench to prevent over-tightening them and damaging the motor or traction amplifier.

Check these torques each time you check the motors and the traction amplifier. Use two wrenches to avoid twisting the terminal stud. See Figure 7-101.

| Stud Size | Motor Cable Securing Nut |
|---|---|
| M6 | 41 to 47 in. lb. (4.7 to 5.3 Nm) |

## Polishing the Commutator

**NOTE:** Wrap the stone with narrow (1/2 to 3/4 inch wide) masking tape in a spiral configuration. Leave only enough stone exposed to make contact with the commutator. As the length of the stone becomes shorter with use, peel back and remove portions of the tape.

**WARNING**

**Make sure loose clothing and hair are tied back. Do not wear jewelry. Wear insulated gloves to protect your hands during this procedure.**

1. If the motor has four or more brushes, remove an accessible brush from the motor with the motor still installed in the truck. If the motor only has 2 brushes or the commutator is not accessible with the stone, the armature must be installed in a lathe or replaced. If so, go to step 4.
2. Activate the motor.
3. With the motor turning, carefully insert the commutator stone into the brush box area.
4. Using minimal pressure, run the stone back and forth across the commutator until it is polished. Do not pass the stone over the ends of the commutator segments.
5. Thoroughly blow out the motor with compressed air.
6. Inspect the commutator. Reinstall the armature or brush and any removed covers.
7. Check operation.

# Traction Motor

*Figure 7-102. Traction Motor*

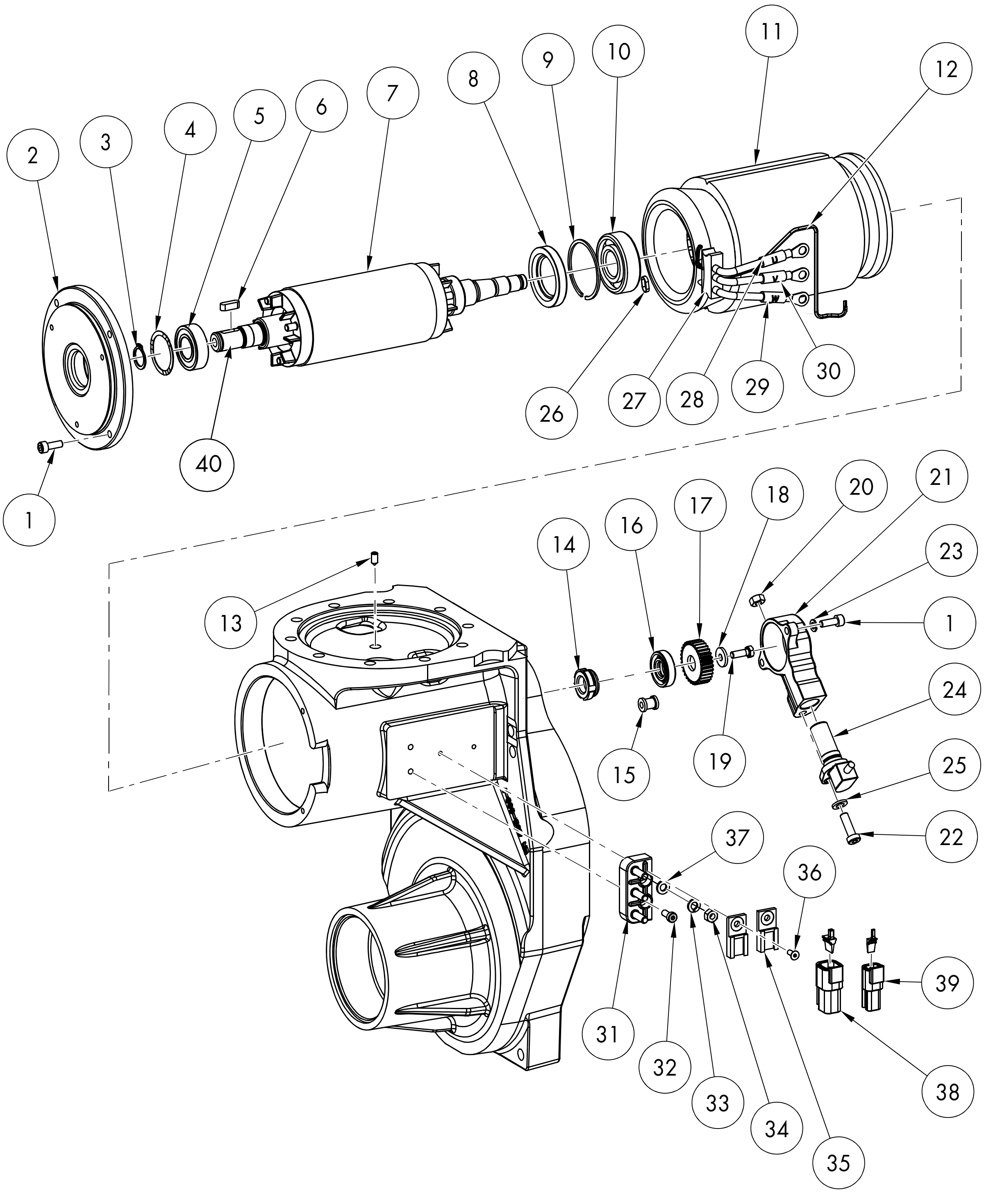


*Table 7-6. Traction Motor Component Legend*

| Item No. | Description | Item No. | Description |
|---|---|---|---|
| 1 | Cap Screw, Socket Head | 21 | Cover, Speed Sensor |
| 2 | Cover, Motor | 22 | Cap Screw, Socket Head |
| 3 | Ring, Retaining | 23 | Washer, Lock |
| 4 | Spacer | 24 | Sensor, Speed |
| 5 | Bearing, Ball | 25 | Washer |
| 6 | Key, Fitted | 26 | Cable Tie |
| 7 | Armature | 27 | Sealing Strip |
| 8 | Seal, Shaft | 28 | 'U' Identification Label |
| 9 | Ring, Retaining | 29 | 'W' Identification Label |
| 10 | Bearing, Ball | 30 | 'V' Identification Label |
| 11 | Stator | 31 | Terminal Board |
| 12 | Insulating Sleeve | 32 | Cap Screw, Socket Head |
| 13 | Pin, Threaded | 33 | Washer, Lock |
| 14 | Nut, Pinion | 34 | Nut |
| 15 | Grommet | 35 | Plug Holder |
| 16 | Seal, Shaft | 36 | Screw |
| 17 | Pinion, Speed Sensor | 37 | Washer |
| 18 | Washer | 38 | Plug and Wedge Lock |
| 19 | Screw, Hex Head | 39 | Plug and Wedge Lock |
| 20 | Nut | 40 | Brake Hub Mounting Location |

## Traction Motor Disassembly

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the lower and grille covers.
3. Disconnect the power cables from the terminal block on the traction motor, noting their correct position.
4. Follow steps 1 through 6 in "Drive Unit Disassembly" on page 7-31 to remove the top (pinion) gear attached to the motor armature shaft.
5. Remove the electromagnetic brake assembly by removing the three mounting bolts that secure the assembly to the motor end bell cover [2].
6. Remove the brake disc from the brake hub.
7. Remove the retaining ring [3] that secures the brake hub. Remove the hub and woodruff key [6].
8. Remove the four screws [1] holding the motor end bell cover [2] to the housing (stator) [11].
9. Separate the motor end bell cover [2] from the motor housing.
10. If needed, press off the bearing [5] and replace.

## Traction Motor Assembly

1. Thoroughly clean all parts with solvent or other non-corrosive cleaning fluid. Air dry all parts.

**NOTE:** Assemble the traction motor in a dirt-free area.

2. If previously removed, install the traction motor bearing, bearing shim, and seal in the gear case.

**NOTE:** The seal must be replaced any time the motor is removed from the housing.

The seal must be installed correctly with its open side and seal ring positioned *towards* the bearing.

3. Reinstall the motor end bell cover [2] on the motor armature shaft.
4. Install the brake hub, woodruff key [6], and the retaining ring [3] on the armature shaft.
5. Carefully slide the traction motor armature in the motor housing.
6. Attach the motor end bell cover [2] to the motor housing (stator) with the four cap screws [1]. Torque screws to 47.8 in. lb. (5.4 Nm).
7. Follow steps 8 through 12 in "Drive Unit Assembly" on page 7-31.
8. Make sure the drive unit is filled to the correct level with gear oil. Install the fill/level plug. See "Drive Unit Housing Lubrication" on page 7-33.
9. Install the brake friction disc.
10. Install the brake coil assembly and attach with three hex head cap screws. Torque screws to 47.8 in. lb. (5.4 Nm).
11. Before testing operation, move the handle from extreme left to right and down into the operating position and back several times. Check for any wire and cable interference. Repair or adjust as necessary.

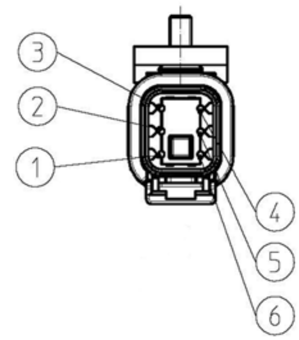
**NOTE:** Check the traction motor cable routing. Make sure the wires are not near the drive tire. Secure wires with cable ties to prevent damage.

12. Install the lower and grille covers.
13. Reconnect the battery connector, and the place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.
14. Test the operation of the truck.

## Speed Sensor (Encoder) Replacement

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the lower and grille covers.
3. Remove the bumper.
4. Disconnect the connector from main harness to the speed sensor and the temperature sensor.
5. Remove the connector [38] from the bracket [35] on the drive unit.
6. De-pin the connections for the speed sensor cable.
7. Remove the screw [22] securing the speed sensor [24] to the speed sensor cover [21].
8. Remove the speed sensor and carefully pull the speed sensor cable through the grommet.
9. Install the socket head cap screw [22] to secure the replacement speed sensor [24] to the speed sensor cover assembly [21]. Torque the screws to 39.8 in. lb. (4.5 Nm).
10. Route the cable of the replacement speed sensor through the grommet.
11. Re-pin the connector. Refer to Figure 7-103 for pin locations and wire colors.

*Figure 7-103. Speed and Temperature Sensor Connector Pin and Wire Identification*




| Pin | Wire Color | Component |
|---|---|---|
| 1 | Red | Speed Sensor |
| 2 | Blue | |
| 3 | White | |
| 4 | Black | |
| 5 | Red | Temperature Sensor |
| 6 | Blue | |

12. Install the connector [38] on the bracket [35] on the drive unit.

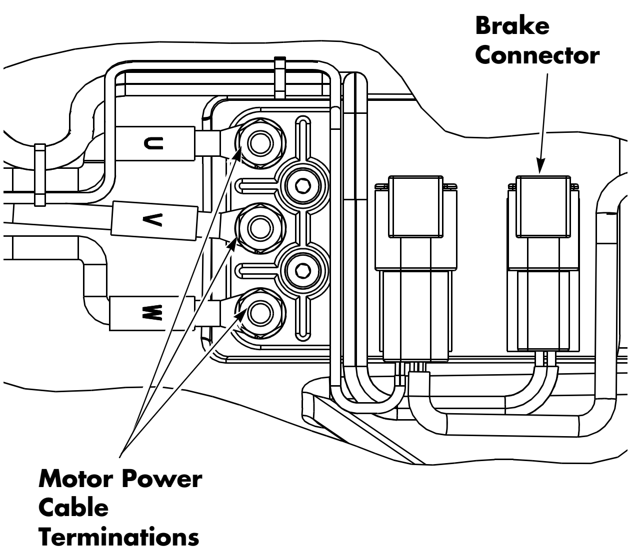
**NOTE:** Make sure the speed sensor cable is correctly seated in the grommet to prevent damage from contact with the linkage when the drive unit turns.

13. Reconnect the main harness to the speed sensor.
14. Install the bumper.
15. Install the lower and grille covers.
16. Reconnect the battery connector, and the place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.
17. Test the operation of the truck.

## Terminal Block Replacement

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the lower and grille covers.
3. Disconnect the power cables from the traction motor to the terminal block [31] on the drive unit, noting their correct position.
4. Remove the screws [32] and washers [37] retaining the terminal block [31] to the drive unit housing.
5. Secure the replacement terminal block [31] to the drive unit housing with washers [37] and screws [32]. Torque screws to 47.8 in. lb. (5.4 Nm).
6. Reattach the traction motor power cables and electromagnetic brake wiring connections to the terminal block. Torque power cable securing nuts to 26.5 in. lb. (3 Nm).

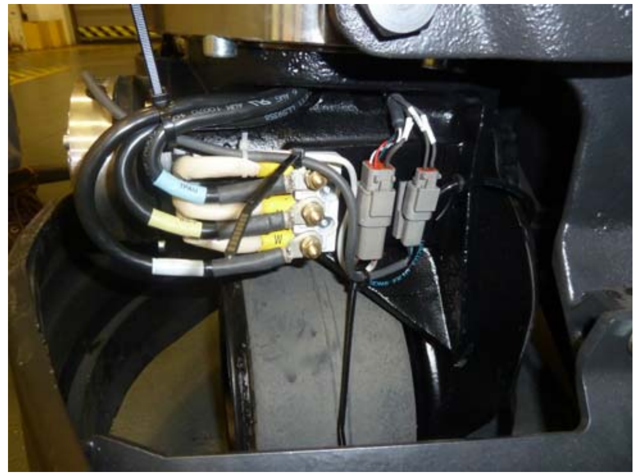
*Figure 7-104. Terminal Block*




7. Before testing operation, move the handle from extreme left to right and down into the operating position and back several times. Check for any wire and cable interference. Repair or adjust as necessary.

**NOTE:** Check the traction motor cable routing. Make sure the wires are not near the drive tire. Secure wires with cable ties to prevent damage.

*Figure 7-105.*



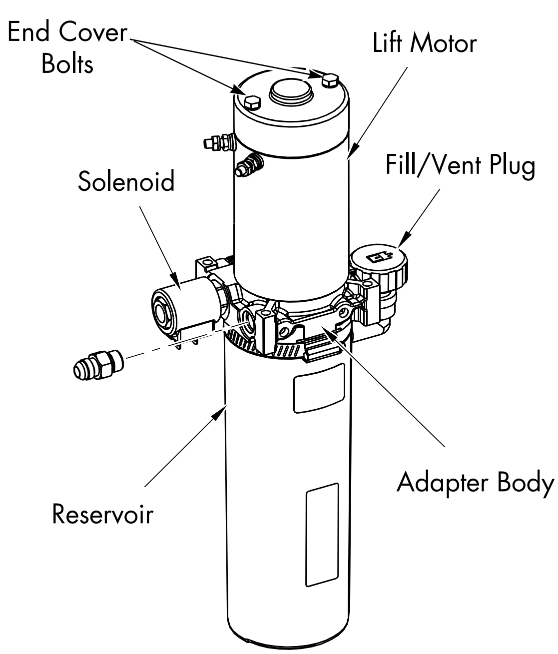
8. Install the lower and grille covers.
9. Reconnect the battery connector, and the place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.

# Lift Motor

## Lift Motor Removal

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the tractor covers.
3. Remove the hydraulic unit from the truck. See “Hydraulic Unit Removal” on page 7-98.
4. Remove the two bolts from the end cover that attach the motor to the adapter body.

*Figure 7-106. Motor End Cover Bolts*




5. Separate the motor and the adapter body.

## Lift Motor Installation

1. Stand the pump assembly on end, with the adapter body facing up.
2. Set aside the pump drive coupling for later reuse.

**NOTE:** The coupling is the mechanical connection between the pump shaft and the electric motor armature shaft. It may have been removed with the motor.

3. Insert the pump drive coupling on the end of the pump shaft. Fill the coupling cavity with anti-seize compound (P/N 990-638).
4. Rotate the pump or the motor shaft to align the motor shaft correctly with the coupling.
5. Install the new motor on the adapter body. Rotate the pump or the motor shaft if necessary to allow the motor to contact the adapter body.
6. Insert the screws into the motor end plate, through the motor and into the adapter body.
7. Make sure the motor is mating flush with the adapter body. Torque screws to 40 to 45 in. lb. (4.5 to 5.1 Nm) maximum.
8. Install the hydraulic unit. See “Hydraulic Unit” on page 7-97.

## Lift Motor Brush Replacement

1. Remove the lift motor from the hydraulic assembly before inspecting the brushes.
2. Remove the two bolts from the motor end cover. Remove the end cover to access the brushes. See Figure 7-106.
3. Push each brush back into the holder to disengage the springs. Remove the nuts and washers from the side of the drive end head. Remove the brush assembly, then remove each brush and replace with a new one.

# Hydraulic Components

# Hydraulic Components

**NOTE:** For replacement parts information refer to the Parts Manual.

## General Guidelines

To prolong the life of your *Raymond* pallet truck, follow these guidelines:

- Keep all fittings and connections tight to prevent leaks. Use care when tightening fittings. Over-tightening can cause damage or distortion.
- Before you remove any component from the hydraulic system, wash the component and surrounding area with cleaning solvent to prevent foreign matter from entering the system. Cap and plug all openings immediately.
- Whenever you remove a fitting with a pipe thread, use a sealing compound on the outside of the threads before you reinstall the fitting. (Do *not* use Teflon® tape.) Make sure all parts are clean.
- When you install a hose assembly, make sure it is not twisted when the fittings are tightened. Always use two wrenches on a swivel-type fitting—one to hold the fitting and the other to tighten the hose.
- Keep the hose clamps tight to prevent the hoses from chafing and to avoid leaks.
- Route hoses so they do not rub on sharp edges.
- Thoroughly clean up any spilled hydraulic fluid.
- Store and handle hydraulic fluid with care to prevent moisture and foreign matter from entering the hydraulic system.
- Contaminated hydraulic fluid is a major cause of hydraulic system failures. Keep any service equipment, such as containers, funnels, and hand pumps, clean at all times. Cover them when not in use.
- Dispose of waste fluid correctly.
- Do not overtighten cable ties used to secure hoses. Hoses expand when in use.

## Hydraulic Unit Torque Specifications

The following table provides torque specifications for the hydraulic unit components.

| **Component** | **Torque in. lb. (Nm)** |
|---|---|
| Solenoid Valve | 216 to 240 in. lb. (24.4 to 27.1 Nm) |
| Relief Valve | 144 to 180 in. lb. (16.3 to 20.3 Nm) |
| Coil Retaining Nut | 48 to 60 in. lb. (5.4 to 6.8 Nm) |
| Motor Terminal Nuts | 41 to 47 in. lb. (4.7 to 5.3 Nm) |
| Motor to Adapter Body Mounting Bolts | 40 to 45 in. lb. (4.5 to 5.1 Nm) |
| Pump to Adapter Body Mounting Bolts | 60 to 70 in. lb. (6.8 to 7.9 Nm) |
| Pump to Reservoir Mounting Clamp Screw | 48 to 60 in. lb. (5.4 to 6.8 Nm) |

# Hydraulic Fluid

## Hydraulic Fluid Level Check

**NOTE:** Always check the hydraulic fluid level with the forks fully lowered and when the hydraulic fluid is cold.

To prevent air from entering the lines:

1. Verify that the fluid level (with the forks fully lowered) is filled with hydraulic fluid to 6.5 ±0.25 in. (165 ±6 mm) from the bottom of the reservoir. See Figure 7-114. The usable reservoir capacity is 0.9 qt. (0.85 liters).
2. Add the specified fluid if necessary. See “Lubrication Equivalency Chart” on page A-2.

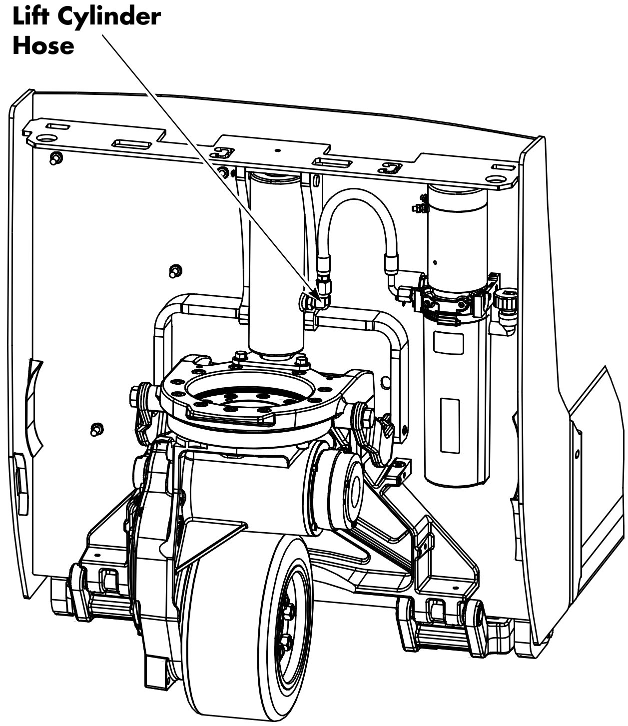
## Changing Hydraulic Fluid

**NOTE:** Store and handle the hydraulic fluid with extreme care to prevent moisture and foreign matter from entering the hydraulic system.

**NOTE:** Contaminated hydraulic fluid is the major cause of hydraulic system failures. Keep any service equipment, such as containers, funnels, and hand pumps, clean at all times. Cover them when not in use.

1. Lower the forks.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
3. Remove the tractor covers. See “Tractor Cover Removal” on page 7-8.
4. Remove the main harness bracket. See “Main Harness Bracket Removal” on page 7-70.
5. Disconnect the lift cylinder hose at the hydraulic cylinder.

*Figure 7-107. Hydraulic Connections*




6. Insert the hose in a waste fluid container.

**⚠ WARNING**

**Fluid is under pressure and may spray or splash. Point the hose away from your body and hold firmly in a waste fluid container.**

7. Connect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
8. Run the hydraulic pump by pressing the lift button until all the hydraulic fluid is pumped from the system.
9. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
10. Clean the filter screen and magnet at the bottom of the inlet tube with a suitable solvent.

11. Reconnect the lift cylinder hose. Refer to Table A-10 on page A-12 for correct torque values.
12. Remove the vent plug and fill the reservoir to within 1 in. (25.4 mm) below the filler port elbow with the recommended hydraulic fluid. See “Lubrication Equivalency Chart” on page A-2.
13. Replace the vent plug.
14. Connect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
15. Raise and lower the forks several times to purge air from the system.
16. Return the forks to the lowest position and check the fluid level. Add fluid if necessary.
17. Install the main harness bracket and the tractor covers.
18. Dispose of the waste fluid correctly.

# Hydraulic Unit

*Figure 7-108: Hydraulic Unit Component Identification*

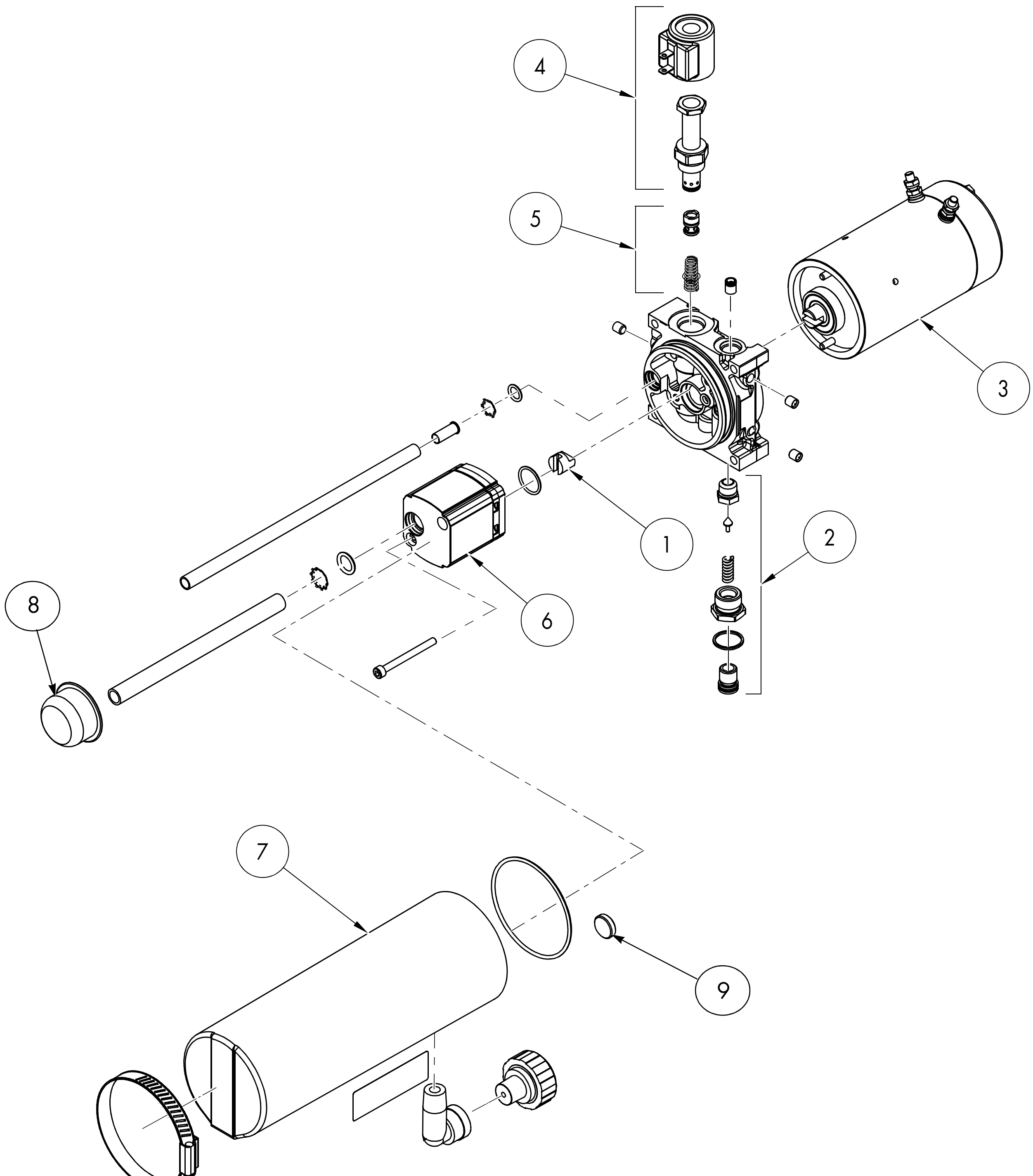
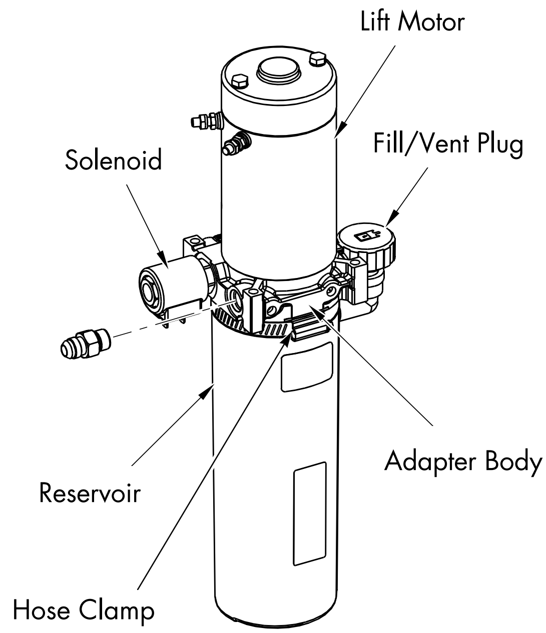


*Table 7-7. Hydraulic Unit Component Legend*

| Item No. | Description |
|---|---|
| 1 | Coupling |
| 2 | Relief Valve |
| 3 | Lift Pump Motor, DC |
| 4 | Solenoid Valve |
| 5 | 2.2 GPM Valve |
| 6 | Pump |
| 7 | Reservoir |
| 8 | Filter Screen |
| 9 | Magnet |

*Figure 7-109. Hydraulic Unit*




## Hydraulic Unit Removal

Remove the hydraulic unit from the truck in order to separate the pump from the motor.

1. Lower the forks completely.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
3. Remove the tractor covers. See "Tractor Cover Removal" on page 7-8.
4. Remove the main harness bracket. See "Main Harness Bracket Removal" on page 7-70.
5. Remove and tag all wires and cables connected to the lift pump motor. See Figure 7-109.
6. Remove and cap the hydraulic line from the hydraulic unit. Disconnect the solenoid control wires.
7. Use a suitable lifting device to remove the battery from the lift truck. This will allow access to the hydraulic unit mounting bolts.

**NOTE:** If the lift truck is equipped with a swing-out battery pack, the battery does not need to be removed, just swing it out of the way to access the hydraulic unit mounting bolts.

8. Remove the bolts that attach the hydraulic unit to the tractor frame.
9. Remove the hydraulic unit from the truck.

## Hydraulic Unit Installation

1. Install the hydraulic unit on the tractor frame and attach with the previously removed screws. Torque the mounting bolts to 9.5 to 10.3 ft. lb. (13 to 14 Nm).
2. Install the hydraulic line on the hydraulic unit.
3. Fill the reservoir with the specified hydraulic fluid. See “Lubrication Equivalency Chart” on page A-2. Use a funnel with a flexible neck. Fill the reservoir to within 1 in. (25.4 mm) below the filler plug elbow. Use up to 0.9 qt. (0.85 liters). See Figure 7-109.
4. Install the fill/vent plug. See Figure 7-109.
5. Connect all wires and cables to the pump motor and hydraulic solenoid. Refer to “Hydraulic Unit Torque Specifications” on page 7-94 for correct tightening torques.
6. If the battery was removed, use a suitable lifting device to install the battery in the lift truck. If the lift truck is equipped with a swing-out battery pack, return the swing-out trays to their correct position.
7. Connect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
8. Raise and lower the forks. Check all hoses and fittings for leaks.
9. Lower the forks.
10. Check the hydraulic fluid level in the reservoir.
11. Install the main harness bracket and the tractor covers.

# Hydraulic Reservoir

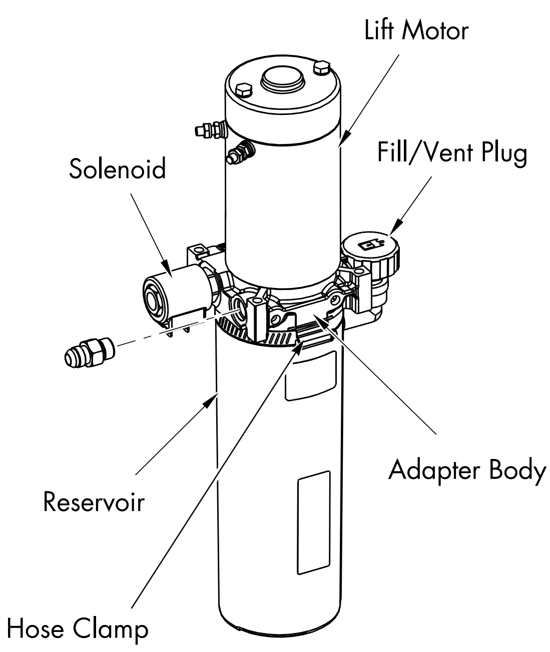
Draining and cleaning the hydraulic reservoir is important to control the accumulation of condensation and contamination that could damage the hydraulic system.

Condensation is caused by the repeated heating and cooling of hydraulic fluid during normal operation. Contaminants include dirt, rust, scaling, and products of fluid deterioration.

## Reservoir Removal

1. Remove the hydraulic unit from the truck. See "Hydraulic Unit Removal" on page 7-98.
2. Remove the hose clamp that secures the hydraulic reservoir to the adapter housing. See Figure 7-110.

*Figure 7-110. Hydraulic Unit*




3. Tap the reservoir lightly to loosen. Wiggle the reservoir sideways while pulling up on the pump and motor at the same time to remove.
4. Remove the reservoir and correctly dispose of the old hydraulic fluid.
5. After the fluid has drained, flush the inside of the reservoir with a suitable cleaning solvent.
6. Dry the inside of the reservoir with clean, dry compressed air.

## Reservoir Installation

1. Inspect the reservoir O-ring and the inside leading edge for burrs. Replace if necessary.
2. Lubricate the O-ring (with applicable hydraulic fluid) and install the O-ring.
3. Carefully install the reservoir on the adapter housing and attach with a hose clamp. Torque the hose clamp to 48 to 60 in. lb. (5.4 to 6.8 Nm).
4. Install the hydraulic unit in the truck. See "Hydraulic Unit Installation" on page 7-99.

# Filter Screen and Inlet Tube

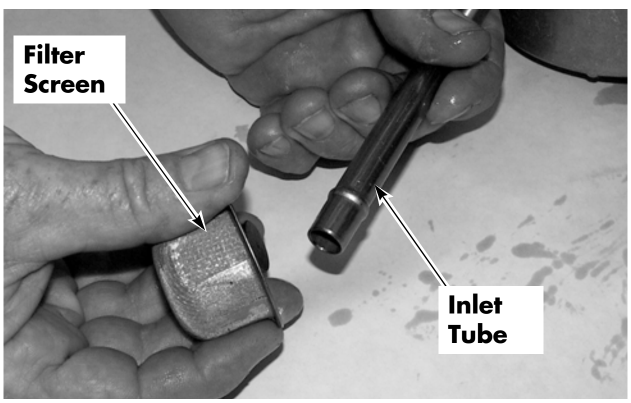
## Filter Screen and Inlet Tube Removal

1. Remove the reservoir. See “Reservoir Removal” on page 7-100.
2. With the pump assembly inverted, remove the inlet tube fitting and sealing washer from the pump housing.

**NOTE:** Use a slight twisting motion when removing the inlet tube.

3. Clean the filter screen and magnet at the bottom of the inlet tube with a suitable solvent. See Figure 7-111.
4. From the pump side, blow dry with clean, dry compressed air.

*Figure 7-111. Cleaning Filter Screen*




## Filter Screen and Inlet Tube Installation

1. Inspect the filter screen, magnet, and inlet tube for damage. Replace if necessary.
2. Install the inlet tube, sealing washer, and fitting in the pump housing.
3. Install the hydraulic reservoir. See “Reservoir Installation” on page 7-100.

# Hydraulic Pump

**NOTE:** For replacement parts information refer to the Parts Manual.

## Hydraulic Pump Removal

1. Remove the hydraulic unit from the truck and put on a clean bench for disassembly. See "Hydraulic Unit Removal" on page 7-98.
2. Remove the reservoir to gain access to the pump. See "Reservoir Removal" on page 7-100.
3. Remove the inlet tube with the filter screen from the pump housing.
4. Remove the two pump mounting bolts then remove the pump from the adapter body.

## Hydraulic Pump Installation

1. Stand the motor assembly on end, with the adapter body facing up.
2. Lubricate the O-ring with applicable hydraulic fluid and install on the adapter body facing up.
3. Insert the pump drive coupling on the end of the motor armature shaft and fill the cavity with anti-seize compound (P/N 990-638).
4. Align the motor drive and pump shaft. Gently push the pump pilot on the adapter.
5. Secure the pump with the two pump mounting bolts and torque to 60 to 70 in. lb. (6.8 to 7.9 Nm).
6. Check the inlet screen for clogging. Clean with a suitable solvent. Install the inlet tube fitting into the pump.
7. Install the reservoir. See "Reservoir Installation" on page 7-100.

# Hydraulic Pump Pressure Relief Valve Adjustment

The hydraulic system is protected by a relief valve installed in the adapter body. The relief valve is set by the manufacturer to open at a specified pressure.

## Relief Valve Settings

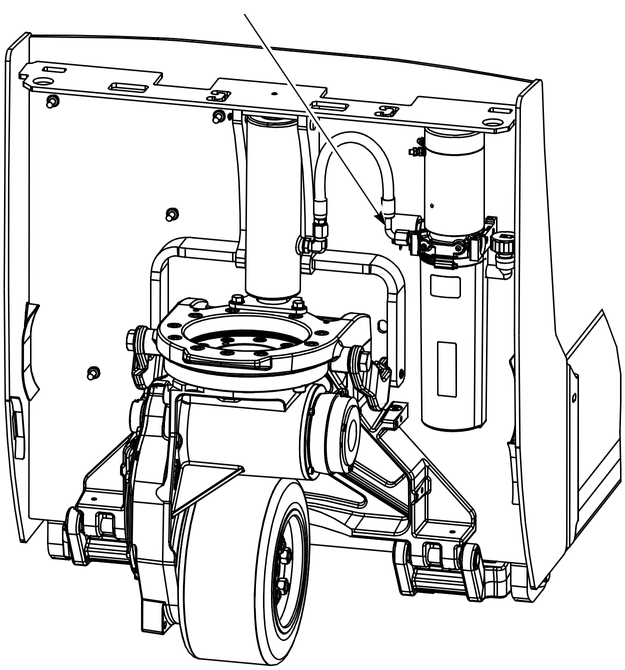
*Table 7-8. Truck Pressure Levels*

| Truck Capacity | Pressure Settings | |
|---|---|---|
| | PSI | kPa |
| 4,500 lb. (2,041 kg) | 3,000 | 20,684 |

## Relief Valve Setting Check

1. Retract the lift cylinder completely. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position.
2. Connect a pressure gauge capable of 3,500 psi (0 to 24131 kPa) into the hydraulic line. See Figure 7-112.

*Figure 7-112. Attach Gauge in Hydraulic Line*



3. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key. Leave the control handle in the neutral (vertical) position.
4. Press the lift button and observe the indicator on the pressure gauge. When the forks reach the upper lift-limit, the pump makes a high pitched squeal indicating that the relief valve is opening.

## Relief Valve Setting Adjustment

If the relief valve opens at a lower or higher pressure than the specified pressure, adjust the relief valve as follows:

1. Remove the hex cap covering the relief valve adjustment screw.
2. As you depress the lift button, turn the adjusting screw *inward* to increase the pressure setting or *outward* to decrease the pressure setting.
3. Check the indicator on the pressure gauge. After the relief valve is correctly set, release the lift button.
4. Reinstall the hex cap. Torque to 144 to 180 in. lb. (16.3 to 20.3 Nm).
5. Recheck the relief valve setting.
6. Lower the forks completely. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position.
7. Remove the pressure gauge.
8. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key. Raise the forks. Inspect all hoses and fittings for leaks.

# Hydraulic Cylinder

Slight fluid leakage indicates that the seal is becoming worn. Replace the seal, wiper, and retaining ring (Seal Kit P/N 939-028) when fluid leakage becomes excessive.

## Cylinder Removal

1. Remove the tractor covers. See "Tractor Cover Removal" on page 7-8.
2. Remove the main harness bracket. See "Main Harness Bracket Removal" on page 7-70.
3. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key. Leave the control handle in the neutral (vertical) position.
4. Press the lift button, raise the forks to maximum height and block them.

**⚠ WARNING**

**Use extreme care whenever the truck is jacked up. Keep hands and feet clear from the vehicle while jacking the truck. After the truck is jacked, put solid blocks beneath it to hold it. Do *not* rely on the jack alone to hold the truck. For details, see "Jacking Safety" on page 2-10.**

5. Lower the forks on the blocking to release the pressure on the hydraulic cylinder.
6. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.

**⚠ WARNING**

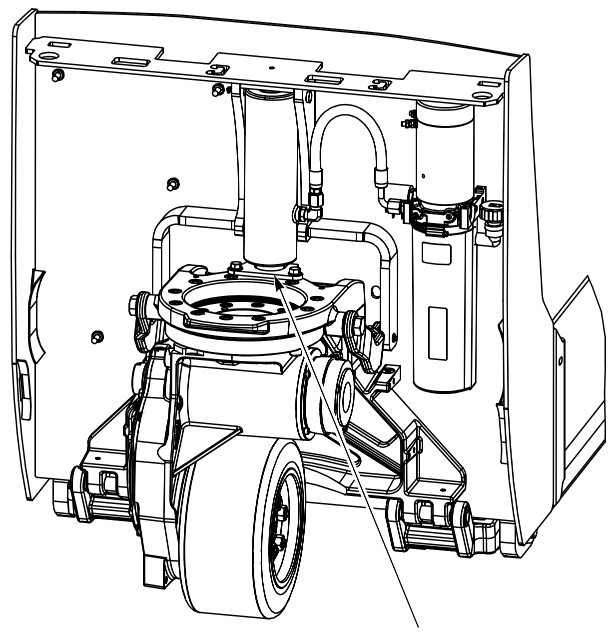
**Hot oil under pressure is present. Make sure the truck is safely blocked and the pressure is released.**

7. Thoroughly clean the outside of the cylinder before removing the hydraulic hose. Remove the hose from the cylinder. Cap and plug the fittings.

**NOTE:** Catch the hydraulic fluid under the hydraulic cylinder in a suitable container.

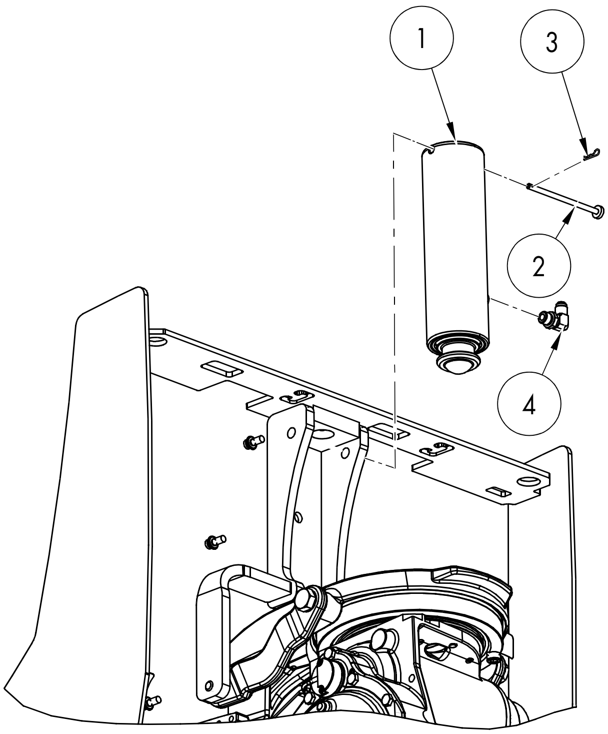
8. Remove the bolts and retaining bracket attached to the pivot frame at the base of the cylinder. See Figure 7-113.

*Figure 7-113.*




9. Remove the cotter pin [3] and retaining pin [2] securing the top of the cylinder [1] to the tractor frame. See Figure 7-114.

*Figure 7-114. Hydraulic Cylinder*

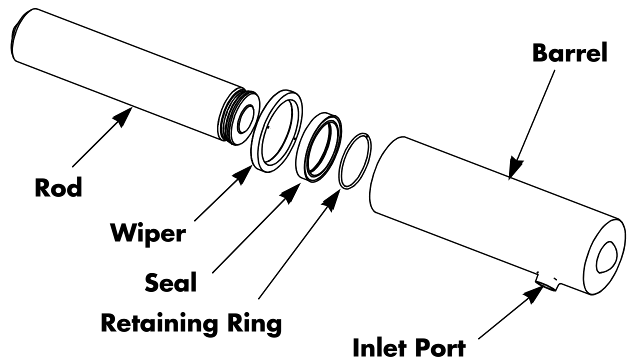



10. Press down on the top of the cylinder to compress the cylinder.
11. Lift the cylinder up and out of the tractor frame. Some spillage of hydraulic oil may occur.

## Cylinder Disassembly

1. Thoroughly clean the outside of the cylinder.
2. Pull the rod out until the end of the rod and the retaining ring are visible through the inlet port. See Figure 7-115.
3. Insert a screwdriver into the inlet port and slide the retaining ring into the deep groove in the piston rod assembly.
4. Remove the rod assembly from the barrel.
5. Remove the retaining ring from the piston rod.
6. Remove the wiper and seal from inside the barrel assembly.

*Figure 7-115. Hydraulic Cylinder Disassembly*




## Cylinder Inspection

1. Thoroughly clean all parts and remove all nicks and burrs with emery cloth.
2. Inspect the inside surface of the barrel assembly for excessive wear or scoring.
3. Inspect the outside surface of the rod for nicks, scratches, or scoring.

## Cylinder Assembly

1. Make sure all parts are cleaned and dried thoroughly. Lightly oil all metal parts before reassembly.
2. Install a new seal in the barrel. Install a new wiper in the barrel with the lips facing inward.
3. Install a new retaining ring in the deep groove of the piston rod end.
4. Oil the outside of the piston rod and carefully insert the rod into the barrel.
5. Push the rod into the barrel until the retaining ring is seen through the inlet port.
6. Insert a screwdriver into the inlet port and slide the retaining ring into the lock position.
7. Extend the rod to the full out position to make sure the retaining ring is locked in place.

## Cylinder Installation

1. Place the new cylinder into position in the tractor frame.

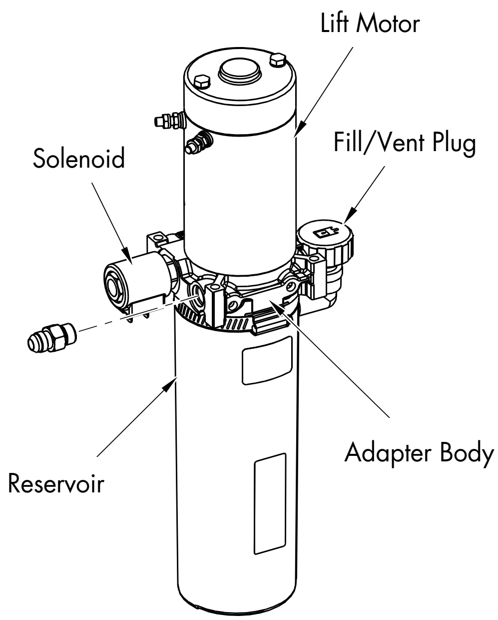
**NOTE:** To prevent air from filling the cylinder, install the cylinder with the rod retracted.

2. Align the cylinder mounting holes the top of the cylinder with the holes in the tractor frame. Install the retaining pin through the tractor frame and the cylinder from the right side. Install the cotter pin through the retaining pin.
3. Connect the hydraulic hose.
4. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
5. Press the lift button to extend the cylinder. Make sure to align the cylinder with the seat in the pivot frame.
6. Install the retaining bracket to the pivot frame with the bolts previously removed. Torque the bolts to 27 ft. lb. (37 Nm).
7. Check the hydraulic reservoir fluid level. See "Hydraulic Fluid Level Check" on page 7-95. If necessary, add fluid. See "Lubrication Equivalency Chart" on page A-2.
8. Remove the blocking.
9. Lift and lower the forks several times. Look for leakage around the fittings and cylinder seal.
10. Recheck the hydraulic reservoir fluid level. If necessary, add fluid.
11. Install the main harness bracket and the tractor covers.

# Hydraulic Solenoid

The hydraulic solenoid activates the hydraulic solenoid release valve that releases pressure on the lift cylinder, lowering the forks. See Figure 7-116.

*Figure 7-116. Hydraulic Solenoid*




## Solenoid Removal

1. Make sure that the forks are fully lowered.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
3. Remove the tractor covers. See "Tractor Cover Removal" on page 7-8.
4. Remove the main harness bracket. See "Main Harness Bracket Removal" on page 7-70.
5. Remove the two wires from the terminals on the solenoid.
6. Remove the nut holding the solenoid coil to the solenoid valve.
7. Lift the solenoid coil off the solenoid valve stem.
8. Remove the solenoid valve stem from the manifold.

## Solenoid Installation

1. Inspect the seat where the solenoid valve stem seals against the manifold. The sealing surface must be completely clean and free from any nicks or damage.
2. Thread the solenoid valve body into the manifold. Torque the valve body to 216 to 240 in. lb. (24 to 27 Nm).
3. Install the solenoid coil and tighten the nut. Torque the nut to 48 to 60 in. lbs. (5.4 to 6.8 Nm).
4. Connect the wires to the terminals.
5. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key. Test the functions of the truck.
6. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
7. Install the main harness bracket and the tractor covers.

# Mast

# Pallet Forks and Load Wheels

## Load Wheels - Single

Load wheels are installed on each pallet fork and are designed to support the rated load at the forks fully raised height. Single load wheels are standard equipment.

### Removal and Replacement

1. Jack the forks. See “Jacking Safety” on page 2-10.

*Figure 7-117.*

2. Drive out the spiral pin in the wheel fork pivot shaft.

*Figure 7-118.*

3. Place blocks under the pull rod, then drive out the wheel fork pivot shaft.

**⚠CAUTION**

**The pull rod and wheel fork will drop to the floor if not blocked prior to removing the pivot shaft.**

*Figure 7-119.*

4. Make sure the load wheels are slightly off the floor with the wheel fork pivot shaft removed.

*Figure 7-120.*

5. Remove the axle spiral pin. See Figure 7-121.

*Figure 7-121. Removing Spiral Pin*

6. Drive the axle out of the wheel fork with a hammer and brass drift pin.
7. Push the wheel out.
8. If the bearings are to be reused, insert a brass drift pin into each end of the load wheel and knock the bearings out.
9. Install the bearings in the new load wheel with the shields facing out.
10. Place the new load wheel upright on a flat surface. Position the bearing over the load wheel. Put a flat metal plate over the bearing, and hammer the bearing in position with a plastic mallet. Make sure the bearing is fully seated in the load wheel.
11. Align the wheel with the holes in the wheel fork.
12. Drive the axle through the wheel fork and wheel. Align the “dimple” in the axle with the spiral pin hole.
13. Install the spiral pin.
14. Install the wheel fork pivot shaft.
15. Install the spiral pin in the wheel fork pivot shaft.
16. Remove the blocks under the pull rods and lower the forks to the floor.
17. Add grease if greaseable load wheel.

# Load Wheels - Dual

Dual load wheels are available as an option.

## Removal and Replacement

1. Jack the forks. See “Jacking Safety” on page 2-10.

*Figure 7-122.*

2. Drive out the spiral pin in the wheel fork pivot shaft.

*Figure 7-123.*

3. Place blocks under the pull rod, then drive out the wheel fork pivot shaft.

**CAUTION**

**The pull rod and wheel fork will drop to the floor if not blocked prior to removing the pivot shaft.**

*Figure 7-124.*

4. Make sure the load wheels are slightly off the floor with the wheel fork pivot shaft removed.

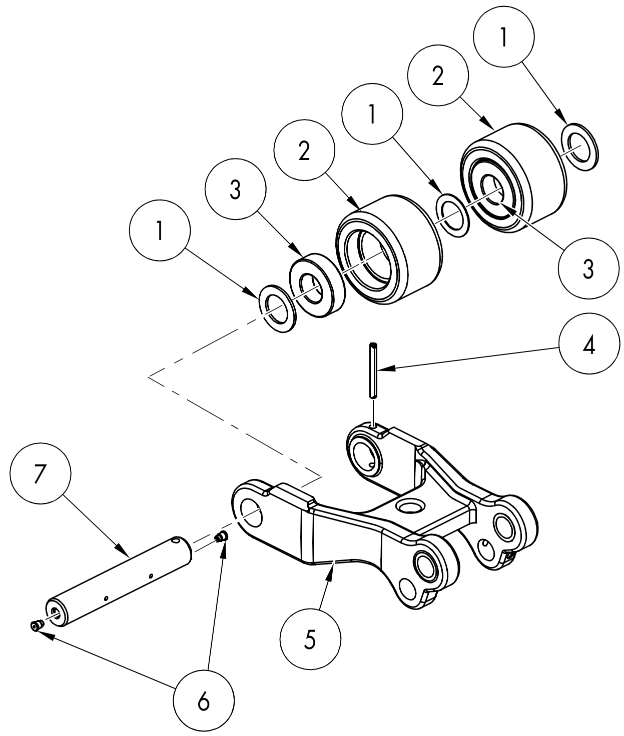
*Figure 7-125.*

5. Remove the axle spiral pin. See Figure 7-121.

*Figure 7-126. Removing Spiral Pin*

6. Drive the axle [7] out of the wheel fork [5] with a hammer and brass drift pin.

*Figure 7-127. Dual Load Wheel Disassembly*



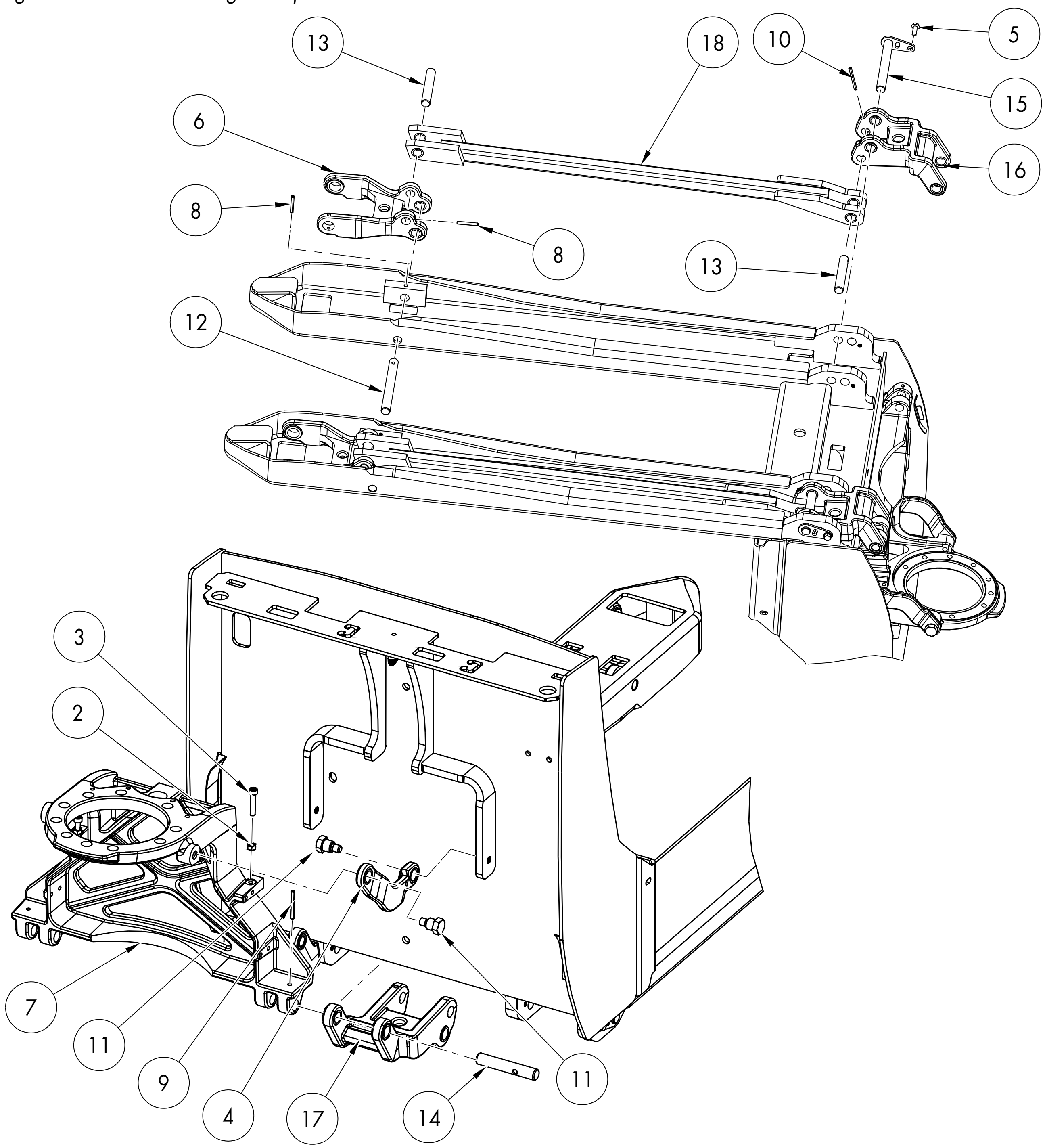

7. Remove the load wheels [2] and spacer washers [1].
8. If the bearings [3] are to be reused, insert a brass drift pin into each end of each load wheel [2] and knock the bearings out.
9. Install the bearings in the new load wheels with the shields facing out.
10. Place the new load wheel upright on a flat surface. Position the bearing over the load wheel. Put a flat metal plate over the bearing, and hammer the bearing in position with a plastic mallet. Make sure the bearing is fully seated in the load wheel.
11. Align the wheel with the holes in the wheel fork [5]. Place at least one spacer washer [1] between the load wheels [2]. Place at least one spacer washer [1] between each load wheel [2] and the wheel fork [5]. Maximum side-to-side play should be less than 0.04 in. (1 mm). Add additional spacer washers if necessary.
12. Drive the axle through the wheel fork, washers, and wheels. Align the “dimple” in the axle with the spiral pin hole.
13. Install the spiral pin [4].

14. Install the wheel fork pivot shaft.
15. Install the spiral pin in the wheel fork pivot shaft.
16. Remove the blocks under the pull rods and lower the forks to the floor.
17. Add grease if greaseable load wheel.

## Load Wheel Pull Rod

Load wheel pull rods are installed on each fork and are designed to transfer the hydraulic cylinder lifting action at the tractor end to the load wheel fork.

*Figure 7-128: Undercarriage Components*



### Removal and Replacement

**NOTE:** It may be easier to work with the truck upside down. Remove the battery before turning the truck over. Some oil may leak from the hydraulic reservoir.

1. Extend the load wheel forks [6]. Remove the spiral pin [8] and shaft [13] connecting the wheel fork to the pallet fork. See Figure 7-127.
2. Remove the pivot pin locking bolt [5] and drive the pivot pin [15] out with a brass drift pin.
3. Remove the spiral pin [10] and pull rod pin [13].
4. Remove the pull rod [18] from the lower link assembly.
5. Press new bushings in the pull rods, if necessary.
6. Reassemble the pull rods on the lower links. Install the pull rod pin [13] and the spiral pin [10].
7. Reinstall the pivot shaft [15] and the retaining bolt [5] in the lower links. Torque the retaining bolt to 27 ft. lb. (36 Nm).

**NOTE:** A spacer is available (if needed) to keep the bushings from walking out. Refer to RSI PLT-17-007 for details.

8. Reconnect the pull rod to the wheel fork. Install the shaft [13] and the spiral pin [8].
9. Place the truck upright and install the battery. Connect the battery. Check the reservoir fluid level. Test the operation of the lift frame.

## Carrier Frame Linkage

**NOTE:** For replacement parts information refer to the Parts Manual.

The bushings on the lower links can be replaced by removing the links from one side at a time and pressing in new bushings.

1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.

**⚠ WARNING**

**Use extreme care whenever the truck is jacked up. Keep hands and feet clear from the vehicle while jacking the truck. After the truck is jacked, put solid blocks beneath it to support it. DO NOT rely on the jack alone to support the truck. For details, see “Jacking Safety” on page 2-10.**

2. Jack and block the truck under the tractor frame.
3. To separate the lift and carrier frames, remove all battery connections before removing the links.
4. Remove the upper and lower linkages and lift the lift frame off the carrier frame. See Figure 7-127.

# Downstops

The downstops control the height of the fork heels above the floor when the truck is fully lowered. Downstop screws contact the fork frame arch to prevent the heels of the forks from coming too close to the floor.

Excessive floor clearance may hinder pallet entry and exit.

**NOTE:** Adjusting the downstops below the height specified in the adjustment procedure will not improve pallet entry or exit.

## Lower Link Fork Heel Inspection

It is important to inspect the lower links and fork heels on a regular basis (see "Scheduled Maintenance" for recommended intervals).

**NOTE:** It is acceptable to have slight scuff marks at the fork heels and lower links. Any sign of metal removal indicates the downstops must be adjusted.

Downstop adjustment after inspection is key in getting the maximum life from your truck.

*Figure 7-129. Inspect these surfaces (truck lifted for clarity)*

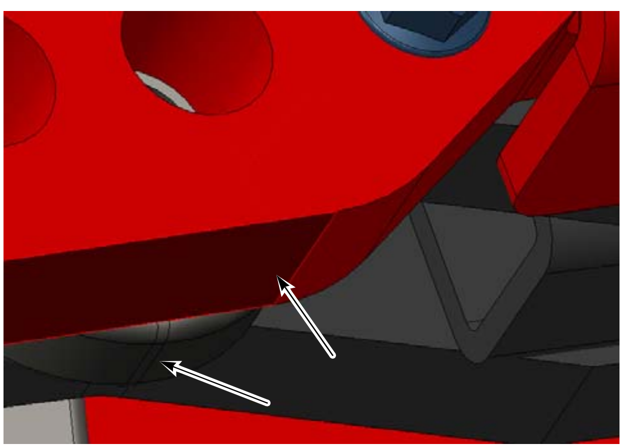
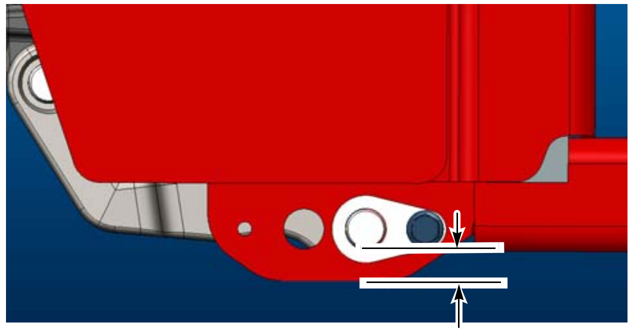


*Figure 7-130. Fork Heel Surface*




**NOTE:** There should be 0.6 in. (14.9 mm) of metal between the hole in the fork heel block and the edge of the fork heel. The lower links should be flush with the bottom of the fork heel.

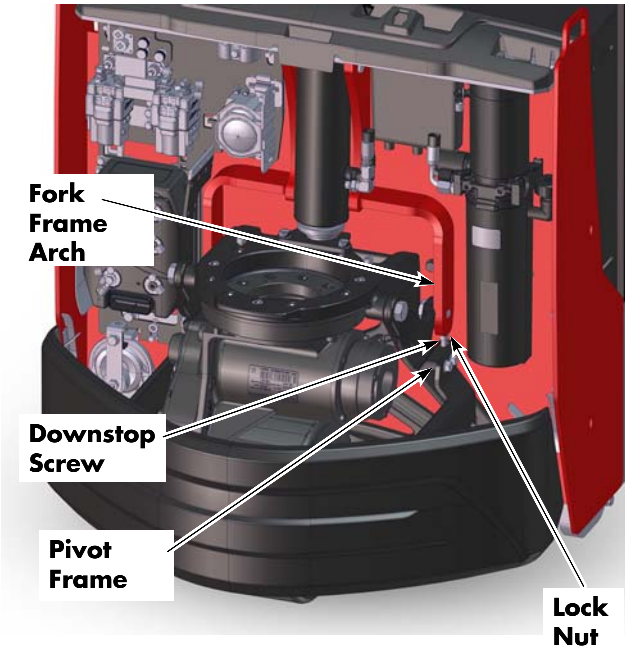
## Downstop Adjustment

Adjustment of the downstops is necessary to accommodate drive tire and linkage wear. Correct adjustment will help to prevent excessive wear of the lower links and fork heels.

1. Park the truck on a smooth, level floor.
2. Lower the forks completely.
3. Turn the truck OFF and disconnect the battery connector.
4. Remove the tractor covers.
5. Verify that the downstop adjustment screws contact the fork frame arch, see Figure 7-131. If the screws do not contact the frame, loosen the lock nut then turn the adjustment screw out of the pivot frame.

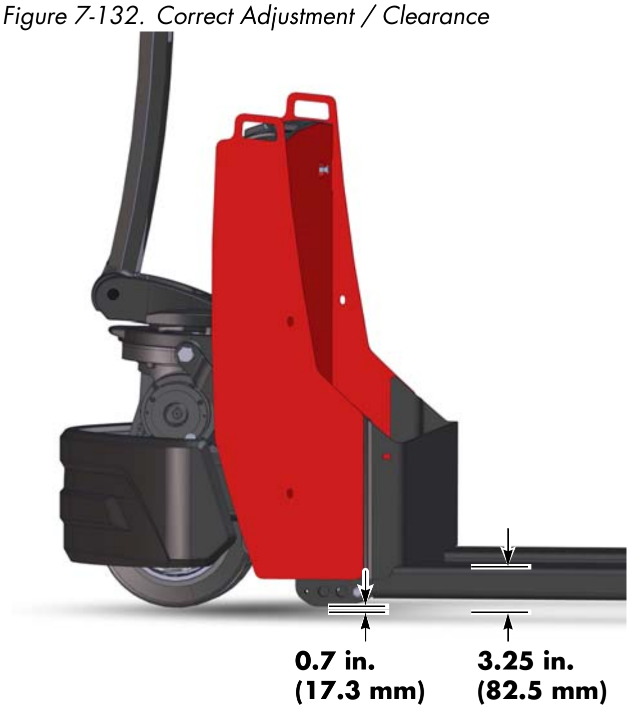
**NOTE:** Make sure there is sufficient thread engagement of the downstop adjustment screw in the pivot frame when adjustment is complete.

*Figure 7-131. Downstop Adjustment*




6. Measure the clearance of the fork heels from the floor. Verify there is a clearance of 0.7 in. (17.3 mm). Repeat steps 5 and 6 until the correct measurements are achieved.

*Figure 7-132. Correct Adjustment / Clearance*




**NOTE:** The top of the forks should be 3.25 in. (82.5 mm) (with standard forks) or 3 in. (76.2 mm) (with optional low profile forks) from the floor when the downstops are correctly adjusted.

7. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.
8. Press the lift button to lift the fork frame. Block the forks.
9. Hold the downstop screw in the correct adjustment position with an Allen wrench and firmly tighten the lock nut.

**NOTE:** Lock nuts that are not securely tightened will not hold the adjustment.

10. Install the tractor covers.
11. Remove the blocks from the forks and test the truck for correct operation.

**NOTE:** When downstop adjustment is performed, it is recommended to inspect the lower links and fork heels each day when performing the daily checklist.

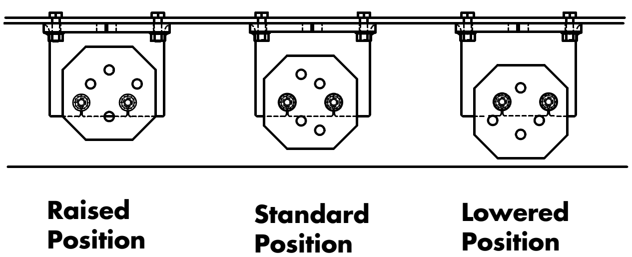
# Skid Shoe Blocks

Some trucks are supplied with skid shoe blocks to provide clearance from the floor to the bottom of the truck. If your truck is equipped with skid shoe blocks, they must be inspected for wear and adjusted to prevent wear of undercarriage components.

## Skid Shoe Adjustment

1. Lower the forks completely.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
3. Jack and block the truck under the tractor frame.
4. Adjust the skid blocks to the correct position on the mounting bolts.
5. Torque the nut to 27 ft. lb. (37 Nm).

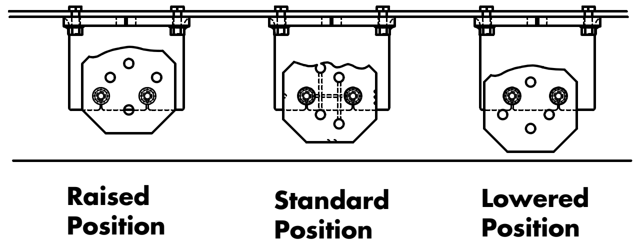
*Figure 7-133. Skid Blocks Installed*




**NOTE:** The skid pad must be rotated 90 degrees from standard to adjust to the raised or lowered position.

6. Worn skid blocks can be rotated 90 degrees for additional life.

*Figure 7-134. Skid Blocks Reversed to Restore Clearance*




**NOTE:** The worn portion is shown irregular for illustration purposes only. Typically, a worn skid pad is nearly flat.

# Options

# Cold Storage Conditioning

New *Raymond* Walkie Pallet Trucks supplied with a standard conditioning package are designed to operate in an environment where temperatures range from 32 to 110°F (0 to 43°C) in a continuous duty cycle.

Trucks supplied with a cold storage conditioning package are designed to operate to the following temperatures:

- 5°F (–15°C) in a continuous duty cycle (generally stored in freezers with short ambient temperature exposure)
- –20 (–29°C) for in and out operation (stored at ambient temperature with short trips in and out of the freezer condition)

**NOTE:** Trucks ordered as cold storage from the factory are shipped with cold storage hydraulic fluid; there is no need to replace the fluid when installing the truck.

**⚠CAUTION**

**When a truck is to be converted for use in a cold storage environment, change the fluid in the hydraulic reservoir referring to the recommendations on the Appendix.**

See "Lubrication Equivalency Chart" on page A-2 for recommended fluids in the hydraulic reservoir and the drive unit.

## Cold Storage Hydraulic Fluid

When changing from any other type of cold storage hydraulic fluid to fluid P/N 1266010:

1. Drain the reservoir completely.
2. Fill the reservoir with fluid.
3. Install decal P/N 1074515 on the hydraulic unit.

**⚠ WARNING**

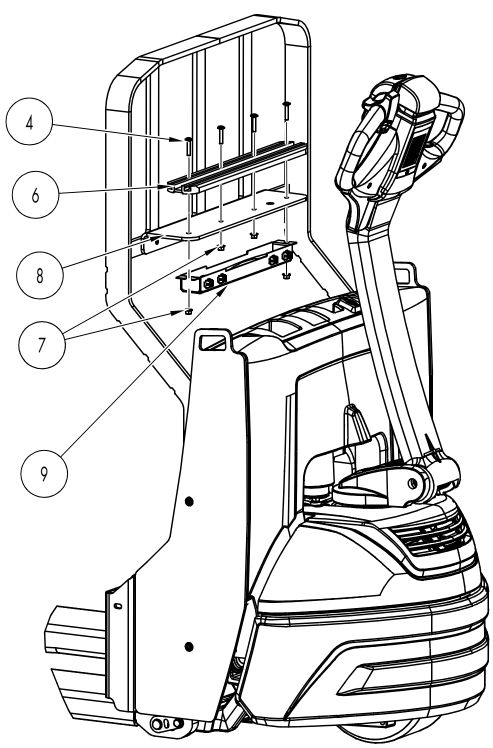
**Mixing with other types of cold storage fluids can cause "gelling" of the fluid and lead to premature pump failure.**

# Accessory Bar, Load Backrest Mounted

A load backrest mounted accessory bar is offered as optional equipment and may be added in the field to lift trucks originally manufactured without one. The accessory bar provides a mounting location and power for operator fans, auxiliary lights, or the iWAREHOUSE system operator display unit. Factory authorization must be granted before the addition of any equipment on the lift truck.

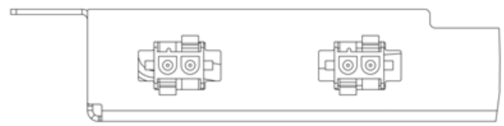
1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the battery to access the tractor frame bulkhead.
3. Remove the lower, grille, and upper covers.
4. Install accessory bar frame [8] (refer to Figure 7-135) on the load backrest with button head screws [5] (refer to Figure 7-137). Torque screws to 27 to 29 ft. lb. (37 to 39 Nm).

*Figure 7-135.*



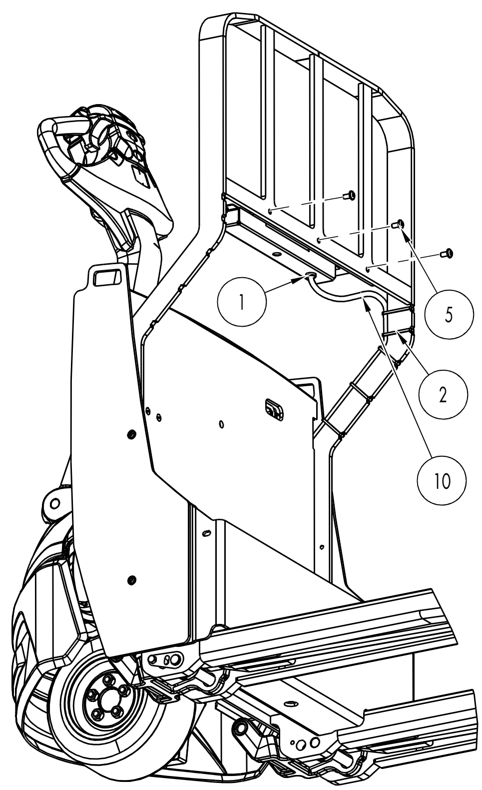

5. Route the accessory bar harness [10] through the left hole in the connector bracket [9]. Install the connectors in the connector bracket. Make sure the connectors are oriented with the flat surfaces facing towards the opening of the connector bracket (see Figure 7-136).

*Figure 7-136. Connector Orientation*



6. Install the accessory bar [6] and the connector bracket [9] to the accessory bar frame [8] using bolts [4] and nuts [7] as shown in Figure 7-135. Torque the bolts [4] to 11 ft. lb. (15.1 Nm).
7. Route the accessory bar harness [10] down the load backrest and into the tractor frame as shown in Figure 7-137. Secure using cable ties [2].

*Figure 7-137. Wire Harness Routing*



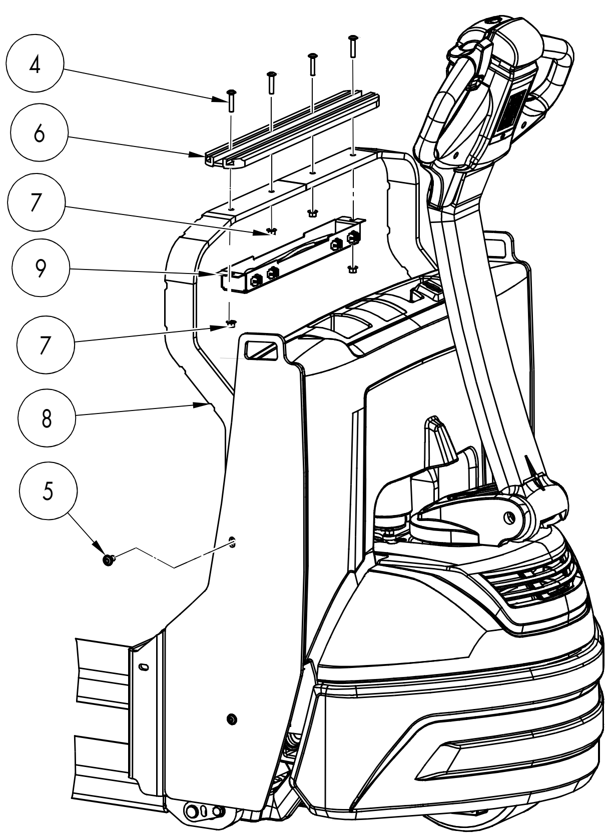
8. Continue routing the accessory bar harness [10] inside tractor frame along the existing main harness. Secure the accessory bar harness to the existing main harness and tie points using cable ties [2]. Connect the accessory bar harness to B– on the traction amp and MPC-(+) on the main contactor.
9. Install the battery.
10. Install the upper, grille, and lower covers.
11. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key. Test the functions of the truck and options.

# Accessory Bar

A side plate mounted accessory bar is offered as optional equipment and may be added in the field to lift trucks originally manufactured without one. The accessory bar provides a mounting location and power for operator fans, auxiliary lights, or the iWAREHOUSE system operator display unit. Factory authorization must be granted before the addition of any equipment on the lift truck.

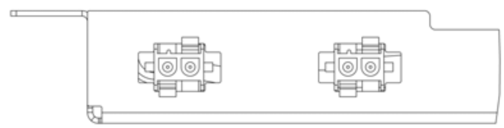
1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.
2. Remove the battery to access the tractor frame bulkhead.
3. Remove the lower, grille, and upper covers.
4. Install accessory bar frame [8] to the truck side plates with button head screws [5] (see Figure 7-138). Torque screws to 27 ft. lb. (37 Nm).

*Figure 7-138.*



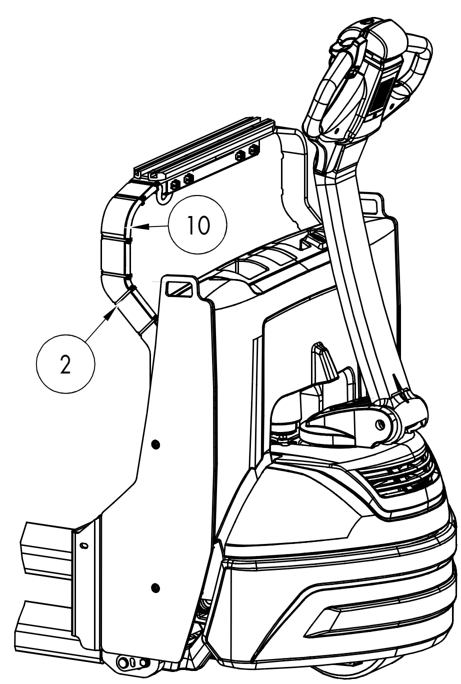

5. Route the accessory bar harness [10] through the left hole in the connector bracket [9]. Install the connectors in the connector bracket. Make sure the connectors are oriented with the flat surfaces facing towards the opening of the connector bracket (see Figure 7-139).

*Figure 7-139. Connector Orientation*



6. Install the accessory bar [6] and the connector bracket [9] to the accessory bar frame [8] using bolts [4] and nuts [7] as shown in Figure 7-138. Torque the bolts [4] to 11 ft. lb. (15.1 Nm).
7. Route the accessory bar harness [10] down the accessory bar frame and into the tractor frame as shown in Figure 7-137. Secure using cable ties [2].

*Figure 7-140. Wire Harness Routing*



8. Continue routing the accessory bar harness [10] inside tractor frame along the existing main harness. Secure the accessory bar harness to the existing main harness and tie points using cable ties [2]. Connect the accessory bar harness to B– on the traction amp and MPC-(+) on the main contactor.
9. Install the battery.
10. Install the upper, grille, and lower covers.
11. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key. Test the functions of the truck and options.

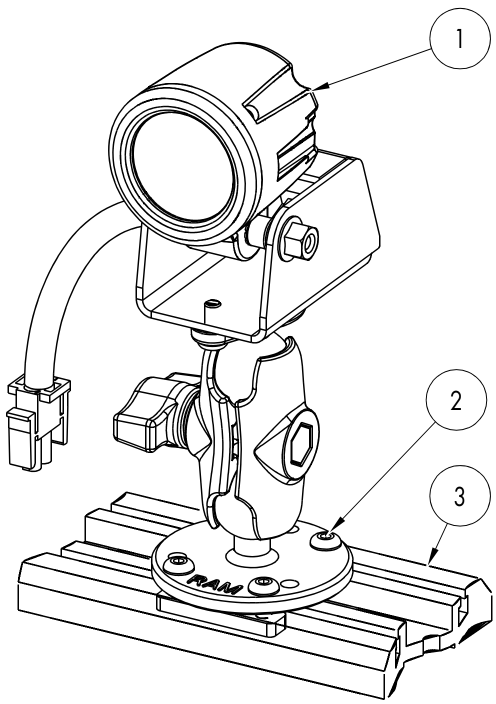
# LED Work Light Installation

An auxiliary light is offered as optional equipment. Auxiliary lights may be added in the field to lift trucks originally manufactured without one. Factory authorization must be granted before the addition of any equipment on the lift truck.

**NOTE:** The auxiliary light requires the use of an accessory bar. See "Accessory Bar, Load Backrest Mounted" on page 7-121 or "Accessory Bar" on page 7-123.

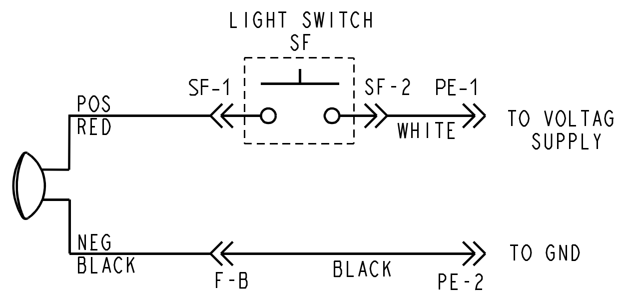
1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.

*Figure 7-141.*




2. Slide the LED work light [1] into the accessory bar [3]. Tighten the screws [2] to secure the ram mount to the accessory bar.
3. Attach the light cable to the connector on the accessory bar.

*Figure 7-142.*




4. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key. Test the functions of the lift truck and options.

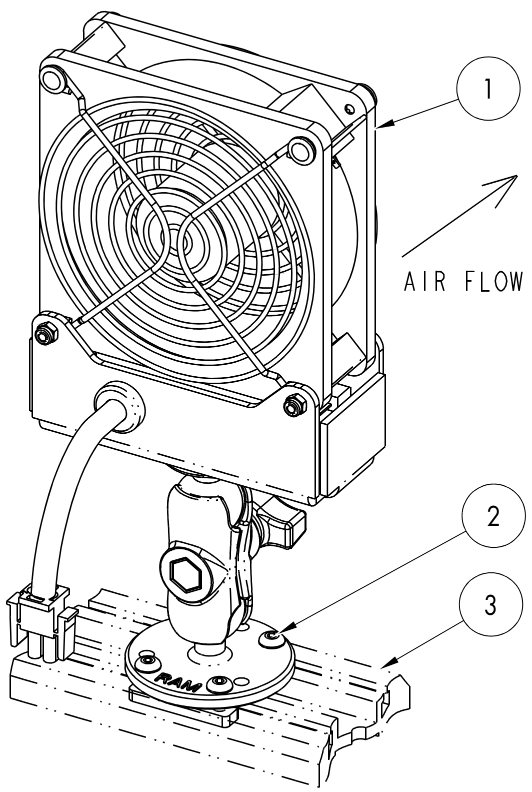
# Operator Fan Installation

An operator fan is offered as optional equipment. Operator fans may be added in the field to lift trucks originally manufactured without one. Factory authorization must be granted before the addition of any equipment on the lift truck.

**NOTE:** The operator fan requires the use of an accessory bar. See "Accessory Bar, Load Backrest Mounted" on page 7-121 or "Accessory Bar" on page 7-123.

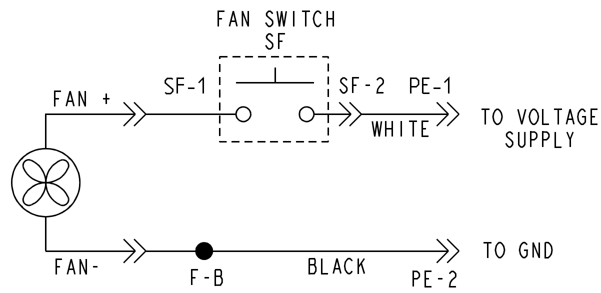
1. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.

*Figure 7-143.*




2. Slide the operator fan [1] into the accessory bar [3]. Tighten the screws [2] to secure the ram mount to the accessory bar.
3. Attach the fan cable to the connector on the accessory bar.

*Figure 7-144.*




4. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key. Test the functions of the lift truck and options.

# Casters

Some trucks are supplied with casters to provide clearance from the floor to the bottom of the truck. If your truck is equipped with casters, they must be inspected for wear and correctly maintained to prevent wear of undercarriage components.

## Caster Removal

1. Lower the forks completely.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector from the truck.

**⚠ WARNING**

**Use extreme care when the truck is jacked up. Keep hands and feet clear from the vehicle while jacking the truck. After the truck is jacked, put solid blocks beneath it to hold it. Do *not* rely on the jack alone to hold the truck. For details, see** **"Jacking Safety" on page 2-10****.**

3. Jack and block the truck under the tractor frame.

**⚠ WARNING**

**The casters are very heavy. Do *not* put hands below the casters when removing the four mounting bolts. Be cautious of falling height adjustment shims.**

4. Remove the four bolts [6] securing the caster assembly [5] to the bumper.

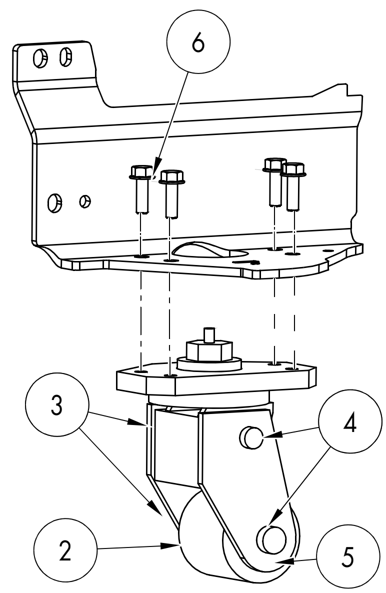
## Wheel Replacement

1. Remove the caster assembly if required. See "Caster Removal".
2. Disassemble the caster assembly by removing the hex head mounting bolts [3] and lock nuts [4] from the caster assembly.
3. Remove the wheel [2] and shims.
4. Align the new wheel with the shims on each side.
5. Install hex head lock nuts [4] on the mounting bolts [3].
6. Securely tighten the lock nuts.

## Caster Installation

1. Fasten the caster assembly [5] to the bumper frame with four bolts [6].

*Figure 7-145.*




2. Torque the mounting bolts [6] to 27 ft. lb. (37 Nm).
3. Remove the block and jack.
4. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( | ) key.
5. Check truck stability and braking in a forks-trailing direction only with the battery/battery pack installed.

# Static Strap Replacement

Lift trucks with the EE option have a static strap attached to the drive unit. Replacement of the strap is necessary when it becomes worn.

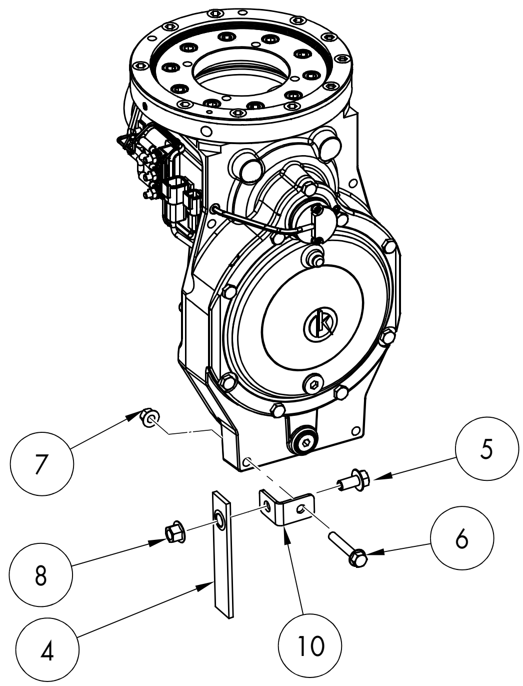
1. Lower the forks.
2. If equipped with the optional keypad, press the red OFF ( **O** ) key. Place the Main ON/OFF Switch in the OFF position. Disconnect the battery connector.
3. Remove the lower and grille covers. See "Tractor Cover Removal" on page 7-8.
4. Remove the bumper. See "Bumper Removal" on page 7-13.

**⚠ WARNING**

**Use extreme care whenever the truck is jacked up. Keep hands and feet clear from the vehicle while jacking the truck. After the truck is jacked, put solid blocks beneath it to support it. DO NOT rely on the jack alone to support the truck. For details, see "Jacking Safety" on page 2-10.**

5. Jack the truck and block the frame.
6. Remove the nut [8] and bolt [5] securing the static strap [4] to the bracket [10].

*Figure 7-146.*




7. Check the bracket [10] for damage. Replace if necessary. If the bracket is replaced, torque the bracket to drive unit mounting bolt [6] to 27 ft. lbs (37 Nm).
8. Attach a new static strap [4] to the bracket [10] using the nut [8] and bolt [5] removed previously. Torque the bolt to 56 ft. lbs. (76 Nm).
9. Unblock and lower the truck.
10. Install the bumper. See "Bumper Installation" on page 7-13.
11. Install the lower and grille covers. See "Tractor Cover Installation" on page 7-10.
12. Reconnect the battery connector. Place the Main ON/OFF Switch in the ON position. If equipped with the optional keypad, enter your PIN-key code and then press the green ON ( **|** ) key.
13. Check the lift truck for correct operation.

# Section 8. Theory of Operation

# Definitions

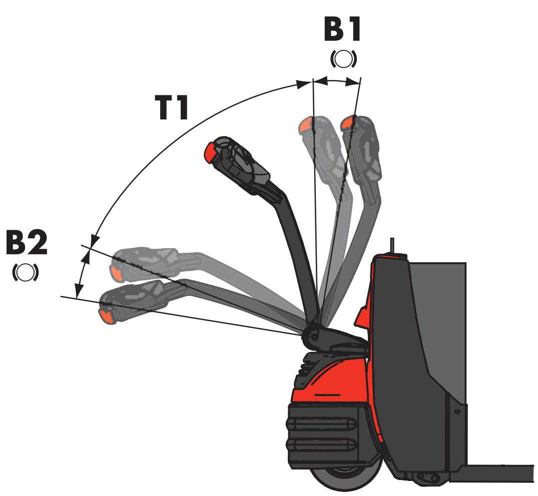
## Acceleration

The process where the truck's acceleration characteristic is determined when the truck starts from a stop. This is a driver parameter (Acceleration). The range is from 30% to 90% in increments of 5%. Default is 80%. A lower number gives less aggressive acceleration.

## Arm Angle Proximity Switch

The Arm Angle Proximity Switch sends a signal to the Traction Amplifier which then in turn determines brake activation and truck travel. The proximity switch determines control handle position by sensing the presence or absence of metal at the handle base. The control handle must be positioned between B1 and B2 of the operating range to permit travel. See Figure 8-1.

*Figure 8-1. Arm Angle Proximity Switch and Brake Actuation*




When the control handle is in position B1, the parking brake engages. When the control handle is moved to position T1, the brake disengages and enables travel. When the control handle is placed in position B2, the parking brake is activated.

## Click-to-Creep

Click-to-creep enables creep speed travel with the control arm in the fully raised or lowered position: "0" disables the option and "1" enables the option. The default setting is "0".

Activate click-to-creep by pressing the button on the control head with the tiller in the brake position. "SLO" blinks on the display when in click-to-creep mode.

Deactivate by pressing the button on the control head again. Click-to-creep mode deactivates automatically after 10 seconds or immediately if the Emergency Reverse (belly button) switch is pushed.

## Continuity

A continuous and uninterrupted path between two or more locations in an electrical circuit, typically having a resistance of less than 1 ohm.

## Controller Area Network (CAN)

Motor controller functions performed by the electronic circuitry (VM) and Traction Amplifier (TA) are communicated in the truck through this network.

## Creep Speed

Creep speed defines speed at low throttle positions. Creep speed range is adjustable from –10 to 10. The factory default value is –5. A negative number enhances low speed maneuverability. A positive number makes the truck more responsive.

## Current Limiting

A protective function of the traction amplifier that prevents excessive current levels from damaging drive components.

## Deceleration (Neutral Braking)

Deceleration is the process where the truck's braking (regenerative) characteristic is determined by any reduction in the throttle, including neutral. This feature provides some plugging effect when the throttle is returned to the neutral position. This is a Driver parameter (Deceleration). The range is from 40 to 90% in increments of 5%. Default is 65%. A lower number gives less aggressive braking. See "Programming Truck Parameters" on page 3-6.

## Emergency Reverse

Emergency reverse function is activated when the brake is released, truck is ON, and the emergency reverse switch is pressed. After the switch is released, normal controller operation is not resumed until neutral (no direction) is selected and until the brake is cycled (brake applied, then brake released). However, repeatedly pressing the emergency reverse switch restarts the emergency reverse function each time.

## Fault Codes

The VM provides fault information by displaying fault codes on the LED display built into the control handle head. See the "List of Messages and Codes" on page 6-2 for a complete list of Caution and Error Codes. The TA provides fault information by flashing fault codes through two status LED indicators built into the TA cover.

## High Pedal Disable (HPD)

The HPD feature prevents the truck from traveling while the throttle is applied. The system is programmed to give the HPD warning if there is a throttle request before the key is turned ON or brake release input.

## Open Circuit

An open circuit is the lack of a continuous path between two or more electrical connections; usually greater than one megohm resistance.

## Overvoltage Cutoff

Overvoltage Cutoff occurs when there is very high voltage (greater than 34V) at the traction amplifier TA B+ connection. Code C42 or E142 may be displayed and the truck must be restarted.

## PIN-Key Code

Trucks produced with an optional handle mounted keypad are protected from unauthorized operation by Personal Identification Number (PIN) keypad entry codes. Up to 10 operators can be assigned their own individual PIN-key code (of one to four digits) for access to the truck. Each operator can also set individual driver parameters. The default operator PIN-key code is number 1. See "Setting Individual PIN-key Codes" on page 3-8.

## Pulse Width Modulation

Pulse Width Modulation (PWM), also called "chopping", controls the speed of the motor by switching the battery voltage applied to the motor ON and OFF very quickly.

## Regenerative Braking

Regenerative Braking occurs when current, generated by the motor during plugging, is permitted to flow back into the batteries. Regenerative braking results in less motor heating. Regenerative braking also provides some return of energy to the battery pack.

## Short Circuit or "Short"

A short circuit is an unspecified path in a circuit that provides unwanted full or partial continuity between two or more locations in an electrical circuit.

Example 1: Two insulated wires are physically next to each other and the insulation has worn off each of the wires. Because the conductors are now touching each other, there is a short circuit.

Example 2: A power cable from the battery to a junction post in the truck has had the insulation worn away. Because the wire conductors are touching the tractor frame, there is a short circuit of the battery cable.

## Speed Limiting

The Vehicle Manager limits the maximum speed. The "Max speed, Walking Mode" parameter determines the maximum speed value sent by the VM to the traction amplifier (max. PWM) for the two directions of travel while in travel mode. This is a programmable parameter.

## Static Return to Off (SRO)

The SRO feature prevents starting the truck (direction selected) prior to the release of the brake. SRO examines sequencing of the brake (deadman) switch input and key switch input relative to the direction input. The switch must be activated before the directional/speed control is rotated. If a direction is selected before or simultaneously (within 50 ms) with the brake input, the TA does not command travel.

## Thermal Cutback (Traction Amplifier)

Thermal cutback of the traction amplifier is below –13°F (–25°C) or above 185°F (85°C). At 185 to 203°F (85 to 95°C), the drive current limit is linearly decreased from full set current down to zero. At –13 to –40°F (–25 to –40°C), the current limit is reduced to approximately half the set current, resulting in reduced travel speed. Below –40°F (–40°C) or above 203°F (95°C), the allowed travel speed is zero percent. This function sets code C43 on the display when a thermal cutback condition occurs. It may also be a factor when codes C44 and C46 occur.

## Tractor

The body section of the truck that contains the motors, drive unit, controls, and handle.

## Truck Off Delay (Keypad only)

This delay sets the truck OFF delay time (energy saving feature). If this time delay passes while the truck is in a brake position and the truck is idle (no inputs requested), the Vehicle Manager powers the truck OFF. This is a programmable Driver parameter setting that ranges from 0 to 20 minutes. When the parameter is set to 0, or the truck has the key switch, the truck never powers OFF. See "Programming Truck Parameters" on page 3-6.

## Undervoltage Cutoff

Undervoltage protection automatically occurs when there is very low voltage (less than 13V) at the traction amplifier TA B+ connection. Code C41 is displayed and the truck must be restarted.

## Vehicle Manager

Programmable motor control functions are performed by the electronic circuitry (VM) housed within the control handle. See "Vehicle Manager" on page 8-6.

# Truck Starting

To start the truck, the battery is plugged in. This supplies battery positive and negative to the Traction Amplifier (TA) at the TA B– and TA B+ connections. This allows the internal capacitors of the TA to precharge. This voltage is also used by the TA at J1-1 to power the internal circuitry of the TA. The TA verifies the correct voltage is at the capacitors prior to closing the main contactor (MPC). With the Main ON/OFF switch in the ON position, the display lights. If the optional keypad is present on the control handle, enter the PIN key code and then press the green ON (**|**) key.

# Traction System

## Vehicle Manager

The VM is also sometimes referred to as the Electronic Tiller Arm Card (ETAC). The programmable motor control functions are performed by the electronic circuitry housed within the VM in the control handle. The basic functions of the VM include:

- Monitor relative positioning of the thumb controls to eighteen hall effect sensors for speed and direction control.
- Monitor the position of the emergency reverse button for emergency reversing of the truck direction.
- Monitor truck performance and provide input data and commands to the operator display.
- Provide lift/lower input commands to the traction amplifier.
- Monitor horn switches S18-1 and S18-2 for horn operation.
- Monitor and determine operation based on truck performance parameter settings.

## Traction Amplifier

The basic functions of the traction amplifier include:

- The direction control section controls the traction motor direction by manipulating the traction motor polarity according to the directional inputs from the Vehicle Manager.
- The speed control section controls the traction motor speed and torque by monitoring the information received from the Vehicle Manager via the CAN Bus.
- The control circuit section that controls all truck inputs and outputs (excluding handle head inputs).

## Direction/Speed Control

The following descriptions assume the battery is charged and connected, and the truck is ON.

### Control Handle Positioning

When the control handle is in the upright position, the parking brake is engaged. When the control handle is within 6 to 78 degrees of the upright position, the Arm Angle Proximity Switch closes and enables travel, the main contactor closes, and the brake is allowed to disengage (when travel is requested). When the control handle is placed in the bottom 8 degrees of its operating range, the Arm Angle Proximity Switch opens, engaging the parking brake. When the control handle is returned to the full upright position, the main contactor remains closed for approximately 20 seconds then opens.

### Travel Request

When the thumb controls are rotated from neutral:

- A forward travel signal is detected via hall effect sensors, by the Vehicle Manager (VM). The VM determines if the request is for forward (tractor-first) direction or reverse (forks-first) direction based on which set of hall effect sensors are activated.
- The VM verifies that the emergency reverse button is not activated.
- The VM verifies the state of the Arm Angle Proximity Switch via a CAN message from the TA. Based on its position, the VM sends the travel request via the CAN to the TA.
- If the main contactor (MPC) is not closed, the TA supplies B– to pin J1-6, closing the main contactor (MPC).
- The TA releases the brake.
- The TA then controls the voltage at U, V, and W of the traction motor. Voltage will vary from 24V to –24V. The voltage is proportional to the position of the thumb controls.

- The drive wheel starts to rotate in the requested direction at a speed proportional to the voltage at the traction motor.

## Emergency Reverse

If the emergency reverse button (belly button) hall effect switch is closed with the control arm lowered in travel mode (SW2 closed), the control system activates an immediate, rapid acceleration in the reverse (forks-first) direction. The following occurs:

- The emergency reverse travel instruction is transmitted through the CAN-bus to the traction amplifier.
- Travel and speed inputs to the traction amplifier are ignored.
- The traction amplifier provides maximum torque inputs to the traction motor immediately after the emergency reverse button is activated.
- Travel in reverse (forks-first) direction continues until the emergency reverse button is released.

When the emergency reverse mode is activated, the TA ceases to respond to the normal travel command from the directional/speed control. To reset the truck for normal travel, return the directional/speed control to the neutral position and cycle the brake ON/OFF.

## Electric Brake Release Switch

A manually activated Electric Brake Release Switch (SW3) has been added. Arm angle proximity switch SW2 is supplied B– at SW2-3 any time the battery is plugged in. SW3 is supplied with B+ from fuse FU2 any time the battery is plugged in and the Main ON/OFF switch (SW1) is in the ON position. To release the brake, SW2 must be open with low voltage at brake switch inputs, JP1-9 and JP1-24 (logic low). A rising edge at JP1-33 while JP1-9 and JP1-24 are at logic low is required to enter brake release mode. The brake release mode cannot be entered if SW3 is closed during key-on or while the brake switch inputs are at logic high (control handle is in the driving position). While in brake release mode, the arm angle proximity switch (SW2) still controls the brake. The brake is released when the brake switch inputs, JP1-9 and JP1-24, are at logic high (tiller in driving position). The brake is engaged with the brake switch inputs, JP1-9 and JP1-24, are at logic low (tiller in brake position). To exit brake release mode at any time, open the brake override switch (SW3).

# Lift/Lower System

The lift/lower system consists of an electrically operated hydraulic pump assembly and related components.

The hydraulic pump assembly consists of a positive displacement rotary gear pump with reservoir mounted to an adapter. A DC electric motor is mounted to the opposite side of the pump adapter. An adjustable relief valve, check valve, and a solenoid operated load holding valve are installed within the adapter.

With the forks elevated, the normally closed solenoid valve and the check valve prevent hydraulic fluid from returning to the reservoir.

## Lift

When the battery is connected, the truck is ON, and the lift button is pressed:

- A lift request is detected by the VM via hall effect sensors. The VM checks the battery state-of-charge to verify lift is allowed. The request for lift is then converted into a CAN message and sent to the Traction Amplifier.
- The TA checks the state of the Main Contactor (MPC) and closes it if required.
- The TA supplies B– to the lift pump contactor (LPC) and it closes, applying B+ to the lift pump motor (MP).
- Hydraulic fluid is drawn into the lift pump.
- As the pump rotates, oil is forced out the pressure port through the lift hose to the lift cylinder. Oil cannot return to the reservoir because of the closed check valve and closed load holding solenoid valve.
- Hydraulic pressure in the lift cylinder lifts the forks.
- The relief valve opens if the hydraulic pressure exceeds the preset limit while lifting.
- Once maximum lift is reached and the lift button remains pressed, the motor pump will continue to pump hydraulic fluid through the relief valve and back into the reservoir until the lift timer expires. The lift timer starts when lift is requested and lasts until the lift request is removed or six seconds passes, whichever occurs first. When lower is requested, the lift timer counts down for 1.8 seconds. If lower is requested for 1.8 seconds or less the lift timer counts back up to six seconds starting at the remaining time (6 seconds minus the time lowered). If lower is requested for more than 1.8 seconds, the lift timer resets at zero and allows a lift request for the full 6 seconds.

When the lift button is released:

- The pump contactor coil (LPC) is de-energized. This stops the lift motor and pump. The forks are held in position by hydraulic fluid trapped in the cylinder by the check valve and the closed load holding solenoid valve (SOL1).

## Lower

When the battery is connected, the truck is ON, and the lower button is pressed:

- A lower request is detected by the VM via hall effect sensors. The request for lower is converted into a CAN message and sent to the TA.
- Load holding solenoid valve (SOL1) opens.
- Hydraulic fluid in the lift cylinder returns to the hydraulic reservoir through the load holding valve and the flow control valve. The lowering speed is regulated by the flow control valve.
- The forks lower.

# Pinout Matrix

The pinout matrix chart lists functions and normal voltages of terminals and harness connector pins. The matrix columns have the following meanings:

1. **Item #:** sequential number to aid in reference.
2. **Connection:** the actual wire numbers or component abbreviations on the electrical schematic. Refer to the schematic package for these trucks.
3. **Function Description:** brief definition of the connection.
4. **Theory of Operation:**
   a. A detailed description of the connection. If the voltage is variable, it indicates the state of a related component causing the voltage to vary.
   b. Identifies possible causes for lack of correct voltage.
5. **Normal Level:** the approximate voltage that should be seen on the wire for the state indicated. Unless otherwise indicated, voltages are measured with respect to (wrt) B– at TP4.
6. **Voltage Source:** the device or connection that supplies the voltage directly to the wire.
7. **Voltage User:** the device or connection that the wire directly delivers the voltage to.

| Item | Connection | Functional Description | Theory of Operations | Normal Level | Voltage Source | Voltage User |
|---|---|---|---|---|---|---|
| 1 | TA B+ | B+ Power to Traction Amplifier | B+ from MPC to Traction Amp used to produce the AC phases U, V, and W when the MPC contactor is closed to power the Traction Motor. | B+ | M1 Contactor | TA |
| 2 | TA B– | B– from Battery | B– from the battery. It is used for all the internal power and control circuits on the TA, VM, Pump Motor, SW2, Service Port, and Battery Charger. | < 0.5VDC with respect to B– | Battery | VM, SW2, TA, Pump Motor, Service Port, and Battery Charger |
| 3 | JP1-1 | B+ Key Input | Provides B+ through fuse FU2 and SPL4 when the Key Switch (SW1) is closed. It is used to power the control circuits of the TA. It is also used by the TA to produce 5 and 12VDC. | With Key Switch OFF: 0VDC<br>With Key Switch ON: 24VDC | Key Switch (SW1) | TA |
| 4 | JP1-2 | Prop Lower | Control path for the Proportional Lower Solenoid SOL-2. Voltage present during lower depends on the lower request from VR2 (Model 6210 only). | De-energized: B+<br>Lower Request: Varies proportionally from 24VDC to 0VDC at full lower. | TA | SOL-2 |

Pinout Matrix

| Item | Connection | Functional Description | Theory of Operations | Normal Level | Voltage Source | Voltage User |
|---|---|---|---|---|---|---|
| 5 | JP1-3 | Horn | The TA sends B– to activate the horn. | Activated: 0VDC Deactivated: 24VDC | TA | Horn |
| 6 | JP1-4 | Pump Contactor | The TA sends B– to activate the LPC (Lift Pump Contactor) coil. | Activated: 0V Deactivated: 24V | TA | LPC Contactor |
| 7 | JP1-5 | Brake | The TA sends B– to activate the Brake. | Activated: 0VDC Deactivated: 24VDC | TA | Brake |
| 8 | JP1-6 | Main Contactor | The TA sends B– to activate the MPC (Main Power Contactor). | Activated: 0VDC Deactivated: 24VDC | TA | MPC Contactor |
| 9 | JP1-7 | I/O GND | B– for Traction Motor Encoder (TME) and it's shield and the Traction Motor Temperature Sensor (TMT). | < 0.5 VDC with respect to B– | TA | TME, TMT |
| 10 | JP1-8 | Motor Temp | Analog voltage that varies with the traction motor temperature. used by the TA to adjust motor performance when a motor over-temperature condition exists. | 75°F or 24°C: Approx. 2VDC | Traction Motor Temp Sensor | TA |
| 11 | JP1-9 | Brake Switch | Monitored by the TA to activate the MPC (Main Power Contactor). Open between 0 and 10 degrees from vertical or horizontal. The brake (deadman) switch acts to protect against travel while the brake is applied. The handle must be between 10 degrees horizontal and vertical in order for the brake (deadman) to be closed, permitting travel. | Switch ON: 12VDC Switch OFF: 0VDC | SW2 | TA |
| 12 | JP1-13 | B+ Supply | Supplies B+ from the TA through SPL3 to Brake, LPC, MPC, SOL-1, SOL-2 (Model 6210 only), and Horn. | 24VDC | TA | Brake, LPC, MPC, SOL-1, SOL-2 (Model 6210 only), and Horn |
| 13 | JP1-19 | Travel Alarm | The TA sends B– to activate the optional Travel Alarm. | Activated: 0VDC Deactivated: 24VDC | TA | Travel Alarm |
| 14 | JP1-20 | Load Hold Valve | The TA sends B– to SOL1 to activate the Load Holding Solenoid when lower is requested. | Deactivated: 26VDC Activated: 0VDC | TA | SOL1 |

| Item | Connection | Functional Description | Theory of Operations | Normal Level | Voltage Source | Voltage User |
|---|---|---|---|---|---|---|
| 15 | JP1-21 | CAN TERM H | CAN Bus terminating resistor installed inside the TA. If not connected, intermittent communication codes will be displayed. | 2.5VDC | TA | CAN Bus |
| 16 | JP1-23 | BUS + | Carries the positive component of the digital communications between the TA, VM, Service Port, and the Battery Charger through SPL2. If this connection is missing, the TA cannot communicate with the other devices. Troubleshoot using resistance checks. No useful information can be gained by measuring the voltage on this wire. | 2.4VDC | TA | VM, Service Port, and Battery Charger |
| 17 | JP1-25 | (+) 12V OUT | Supplies 12VDC to the Traction Motor Encoder (TME) and switch SW2 through SPL6. | 12VDC | TA | SW2 and TME |
| 18 | JP1-31 | T Vel Phase A | Generated when there is movement of the Traction Motor. It is a square wave that is either high (>4VDC) or low (<1VDC). Frequency varies directly with the speed of the Traction Motor and the TA uses it to determine travel speed. Identical formation to T Vel Phase B except for the phasing between the two. The VM uses the quadrature phase relationship between Phases A and B to determine travel direction. | At Rest: Either 4.7VDC or 0.03VDC During Motion: Approx. 2.5VDC | Traction Motor Encoder | TA |
| 19 | JP1-32 | T Vel Phase B | Generated when there is movement of the Traction Motor. It is a square wave that is either high (>4VDC) or low (<1VDC). Frequency varies directly with the speed of the Traction Motor and the TA uses it to determine travel speed. Identical formation to T Vel Phase A except for the phasing between the two. The VM uses the quadrature phase relationship between Phases A and B to determine travel direction. | At Rest: Either 4.7VDC or 0.03VDC During Motion: Approx. 2.5VDC | Traction Motor Encoder | TA |
| 20 | JP1-33 | Brake Override | Input from the Brake Override Switch SW3 when closed. When the switch is closed, the TA will release the brake and not allow travel. | Open: 0VDC Closed: B+ | SW3 | TA |
| 21 | JP1-34 | CAN Term L | CAN Bus terminating resistor installed inside the TA. If not connected, intermittent communication codes will be displayed. | 2.5VDC | TA | CAN Bus |

Pinout Matrix

| Item | Connection | Functional Description | Theory of Operations | Normal Level | Voltage Source | Voltage User |
|---|---|---|---|---|---|---|
| 22 | JP1-35 | BUS – | Carries the negative component of the digital communications between the TA, VM, Service Port, and the Battery Charger through SPL2. If this connection is missing, the TA cannot communicate with the other devices. Troubleshoot using resistance checks. No useful information can be gained by measuring the voltage on this wire. | 2.2VDC | TA | VM, Service Port, and Battery Charger |

# Section A. Appendix

# Lubrication Equivalency Chart

| Approved Raymond Lubricants | | | |
|---|---|---|---|
| **Where Used** | **Type** | **Specification** | **Raymond Part Number** |
| Drive Unit Gear Case (all applications) | Gear Lubricant | 75W - 90 Synthetic Gear Oil API Service Classification GL-5 | 990-655/001 (1 quart/0.9 liter) 990-655/003 (1 gal./3.8 liters) |
| Hydraulic Reservoir (non-cold storage trucks) (+35 to +120°F) (+1.6 to +48.8°C) | Hydraulic Fluid | ISO 46 | 990-616/04 (1 quart/0.9 liter) 990-616/01 (1 gal./3.785 liters) 990-616/03 (5 gal./18.9 liters) |
| Hydraulic Reservoir (for high performance/ high temperature applications) | Hydraulic Fluid | 10W-30 Motor Oil API Service CE,CD,CC,SG,SF | 990-603/04 (1 quart/0.9 liter) (12/ case) 990-603/03 (5 gal./18.9 liters) |
| Hydraulic Reservoir (Cold Storage) * (–0 to +80°F) (–18 to +27°C) | Hydraulic Fluid | ISO VG32 | 1266010 |
| Bearings and so forth | Grease | NLGI Grade 2 | 1012992/01 (10 cartridges per case) 1012992/02 (5 gal./18.9 liters) |
| Bearings and so forth (Cold Storage) | Grease with Teflon | NLGI 2 | 990-652/001 (Spray) 990-652/002 (Cartridge) |

**NOTE:** * Do not mix with any other type of hydraulic fluid. Install decal (P/N 1074515) on hydraulic unit.

# Thread Adhesives, Sealants, and Lubricants

*Table A-1. Thread Adhesives, Sealants, and Lubricants*

| Application | Raymond P/N | *Loctite** Number/Color |
|---|---|---|
| Thread-locking 1/4 in. and below, 10 ml | 990-403/01 | 222/Purple |
| Thread-locking 1/4 in. and below, 50 ml | 990-403/02 | 222/Purple |
| Thread-locking 1/4 to 3/4 in. | 990-536 | 242/Blue |
| Thread-locking 1/4 to 3/4 in. contamination tolerant | 990-462 | 243/Blue |
| Thread-locking 1 in. and under | 990-544 | 271/Red |
| Thread-locking 1 in. and under, contamination tolerant, 10 ml | 990-463/01 | 603/Green |
| Thread-locking 1 in. and under, contamination tolerant, 50 ml | 990-463/02 | 603/Green |
| Thread-locking 1 in. and over | 990-571 | 277/Red |
| Primerless Thread-locking compound, 250 ml | 990-669/01 | 2760 |
| Primerless Thread-locking compound, 10 ml | 990-669/02 | 2760 |
| Thread-locking compound, solid | 1056414 | 268/Red |
| Thread-locking Primer | 990-538 | 707 |
| Thread-locking Primer | 990-533 | T7471 |
| | 990-666 | 7649 (19269) |
| | 1147803 | 7090 |
| Thread-locking 1 1/2 in. and below, 0.02 oz. | 1013829/01 | 2440 |
| Thread-locking 1 1/2 in. and below, 10 ml | 1013829/02 | 2440 |
| Thread-locking 1 1/2 in. and below, 50 ml | 1013829/03 | 2440 |
| Gasket Sealant | 990-411 | 51813 |
| Hydraulic Sealant, 50 ml | 990-552/01 | 569 |
| Hydraulic Sealant, 250 ml | 990-552/02 | 569 |
| RTV Silicone Sealant | 990-659 | 5900/Black |
| | | Other Manufacturer designation |
| Molybdenum Anti-Seize Compound (Molykote) | 990-638 | /Silver |
| Corrosion Inhibitor Coating | 990-644 | |
| Cold Storage silicone sealing compound | 990-445 | |
| Alcohol-based cleaner | 990-600/FOF | |

**NOTE:** **Loctite* is a registered trademark of the Loctite Corporation. Brand endorsement is not implied here, but listed only as a commonly identified product.

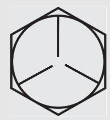
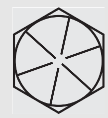
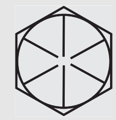
# Component Specific Service/Torque Chart

*Table A-2. Component Specific Service/Torque Chart*

| Component | Sub-Component(s) | Thread-Locking Compound P/N | Torque to: /Notes |
|---|---|---|---|
| Brake | air gap | | 0.008 in. (0.2 mm) minimum |
| Brake | friction disc | | 0.20 in. (5.5 mm) minimum thickness |
| Brake | mounting bolts (3) | | 50 in. lb. (5.6 Nm) |
| Drive Wheel | mounting nuts (5) | | 55 ft. lbs. (75 Nm) |
| Hydraulic Reservoir | screws mounting reservoir to adaptor housing | | 49 to 60 in. lbs. (5.5 to 7 Nm) |
| Lift Motor | end plate to adapter body screws | | 36 to 45 in. lb. (4 to 7 Nm) |
| Lift Motor | solenoid mounting screws | | 60 to 69 in. lb. (6.8 to 7.9 Nm) |
| Lift Motor | motor cable securing nut | | 41 to 47 in. lb. (4.7 to 5.3 Nm) |
| Lift Pump | bolts mounting pump to adaptor body (2) | | 40 to 45 in. lb. (4.5 to 5 Nm) |
| Lift Pump | inlet tube fitting to pump housing | | 245 to 275 in. lb. (27 to 31 Nm) |
| Lower Solenoid Valve Body | mounting into manifold | | 204 to 240 in. lb. (23 to 27 Nm) |
| Nut on Solenoid Coil | | | 48 to 60 in. lb. (5 to 7 Nm) |
| Relief Valve Adjustment screw hex cap | | | 144 to 180 in. lb. (16 to 20 Nm) |
| Traction Amplifier | terminals | | 12 ft. lb. (16.3 Nm) |
| Traction Motor | brass phase cable securing nuts | | 26.5 in. lbs. (3 Nm) |
| Skid Bar to Bracket | | | 27 ft. lb. (37 Nm) |
| Skid Bar Bracket to Bumper | | | 27 ft. lb. (37 Nm) |
| Optional Caster to Bumper | | | 27 ft. lb. (37 Nm) |
| Bumper to Truck | | | 27 ft. lb. (37 Nm) |

# Torque Chart - Standard (Ferrous)

*Table A-3. Torque Chart - Standard*

| **Ferrous Bolts (The grade rating is stamped on the head of the bolt.)** | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Grade Marking** | **None** | | | |  | | | |  | | | |  | | | |
| **Size** | **SAE Grade 2 Bolts Tightening Torque** | | | | **SAE Grade 5 Bolts Tightening Torque** | | | | **SAE Grade 7 Bolts Tightening Torque** | | | | **SAE Grade 8 Bolts Tightening Torque** | | | |
| | **Dry**** | | **Oiled*** | | **Dry**** | | **Oiled*** | | **Dry**** | | **Oiled*** | | **Dry**** | | **Oiled*** | |
| | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** |
| 4-40 | 5 | 0.56 | 4 | 0.45 | 8 | 0.9 | 6 | 0.68 | 11 | 1.5 | 8 | 0.9 | 12 | 1.4 | 9 | 1.0 |
| 4-48 | 6 | 0.68 | 5 | 0.56 | 9 | 1.0 | 7 | 0.8 | 12 | 1.4 | 9 | 1.0 | 13 | 1.5 | 10 | 1.1 |
| 6-32 | 10 | 1.1 | 8 | 0.9 | 16 | 1.8 | 12 | 1.4 | 20 | 2.2 | 15 | 1.7 | 23 | 2.6 | 17 | 1.9 |
| 6-40 | 12 | 1.4 | 9 | 1.0 | 18 | 2.0 | 13 | 1.5 | 22 | 2.5 | 17 | 1.9 | 25 | 2.8 | 19 | 2.1 |
| 8-32 | 19 | 2.1 | 14 | 1.6 | 29 | 3.3 | 22 | 2.5 | 36 | 4.1 | 27 | 3.0 | 41 | 4.6 | 31 | 3.5 |
| 8-36 | 20 | 2.2 | 15 | 1.7 | 31 | 3.5 | 23 | 2.6 | 38 | 4.3 | 29 | 3.3 | 44 | 5.0 | 33 | 3.7 |
| 10-24 | 27 | 3.0 | 21 | 2.4 | 42 | 4.8 | 32 | 3.6 | 52 | 5.9 | 39 | 4.4 | 60 | 6.8 | 45 | 5.1 |
| 10-32 | 31 | 3.5 | 24 | 2.7 | 48 | 5.4 | 36 | 4.1 | 60 | 6.8 | 45 | 5.1 | 68 | 7.7 | 51 | 5.8 |
| 1/4-20 | 66 | 7.5 | 49 | 5.5 | 100 | 11.3 | 76 | 8.6 | 125 | 14.1 | 94 | 10.6 | 143 | 16.2 | 107 | 12.1 |
| 1/4-28 | 75 | 8.5 | 56 | 6.3 | 120 | 13.6 | 87 | 9.8 | 143 | 16.2 | 107 | 12.1 | 164 | 18.5 | 123 | 13.9 |
| | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** |
| 5/16-18 | 11 | 15.1 | 8 | 10.9 | 17 | 23.0 | 13 | 18 | 21 | 29 | 16 | 22 | 25 | 34 | 18 | 24 |
| 5/16-24 | 12 | 16.3 | 9 | 12.2 | 19 | 26 | 14 | 19 | 24 | 33 | 18 | 24 | 25 | 34 | 20 | 27 |
| 3/8-16 | 20 | 27 | 15 | 20 | 30 | 41 | 23 | 31 | 40 | 54 | 30 | 41 | 45 | 61 | 35 | 47 |
| 3/8-24 | 23 | 31 | 17 | 23 | 35 | 47 | 25 | 34 | 45 | 61 | 30 | 41 | 50 | 68 | 35 | 47 |
| 7/16-14 | 32 | 43 | 24 | 33 | 50 | 68 | 35 | 47 | 60 | 81 | 45 | 61 | 70 | 95 | 55 | 75 |
| 7/16-20 | 35 | 47 | 25 | 34 | 55 | 75 | 40 | 54 | 70 | 95 | 50 | 68 | 80 | 109 | 60 | 81 |
| 1/2-13 | 50 | 68 | 35 | 47 | 75 | 102 | 55 | 75 | 95 | 129 | 70 | 95 | 110 | 150 | 80 | 109 |
| 1/2-20 | 55 | 75 | 40 | 54 | 85 | 115 | 65 | 88 | 105 | 143 | 80 | 109 | 120 | 163 | 90 | 122 |
| 9/16-12 | 70 | 95 | 50 | 68 | 110 | 150 | 80 | 109 | 135 | 183 | 100 | 136 | 150 | 203 | 110 | 150 |
| 9/16-18 | 80 | 109 | 60 | 81 | 120 | 163 | 90 | 122 | 150 | 203 | 110 | 150 | 170 | 231 | 130 | 177 |
| 5/8-11 | 95 | 129 | 70 | 95 | 150 | 203 | 110 | 150 | 190 | 258 | 140 | 190 | 220 | 298 | 170 | 231 |
| 5/8-18 | 110 | 150 | 85 | 115 | 170 | 231 | 130 | 177 | 210 | 285 | 160 | 217 | 240 | 326 | 180 | 244 |
| 3/4-10 | 170 | 231 | 130 | 177 | 260 | 353 | 200 | 272 | 330 | 448 | 240 | 326 | 380 | 515 | 280 | 381 |
| 3/4-16 | 190 | 258 | 140 | 190 | 300 | 407 | 220 | 298 | 370 | 502 | 280 | 381 | 420 | 569 | 320 | 434 |
| 7/8-9 | 170 | 231 | 125 | 169 | 430 | 583 | 320 | 434 | 530 | 719 | 400 | 544 | 600 | 813 | 460 | 624 |
| 7/8-14 | 185 | 252 | 140 | 190 | 470 | 637 | 350 | 475 | 580 | 786 | 440 | 598 | 660 | 895 | 500 | 680 |
| 1-8 | 250 | 339 | 190 | 258 | 650 | 881 | 480 | 651 | 800 | 1085 | 600 | 813 | 900 | 1224 | 680 | 925 |
| 1-12 | 275 | 373 | 205 | 278 | 700 | 949 | 520 | 705 | 860 | 1166 | 660 | 895 | 1000 | 1360 | 740 | 1003 |
| 1 1/8-7 | 350 | 475 | 270 | 366 | 800 | 1085 | 600 | 813 | 1120 | 1519 | 840 | 1139 | 1280 | 1136 | 960 | 1302 |
| 1 1/8-12 | 400 | 544 | 300 | 407 | 880 | 1193 | 660 | 895 | 1260 | 1708 | 940 | 1274 | 1440 | 1952 | 1080 | 1464 |
| 1 1/4-7 | 500 | 680 | 375 | 509 | 1120 | 1523 | 840 | 1139 | 1580 | 2142 | 1200 | 1627 | 1820 | 2468 | 1360 | 1884 |
| 1 1/4-12 | 550 | 746 | 420 | 569 | 1240 | 1681 | 920 | 1247 | 1760 | 2386 | 1320 | 1790 | 2000 | 2720 | 1500 | 2034 |
| 1 3/8-6 | 650 | 881 | 490 | 664 | 1460 | 1980 | 1100 | 1491 | 2080 | 2820 | 1560 | 2115 | 2380 | 3227 | 1780 | 2413 |
| 1 3/8-12 | 750 | 1017 | 560 | 759 | 1680 | 2278 | 1260 | 1708 | 2380 | 3227 | 1780 | 2413 | 2720 | 3688 | 2040 | 2766 |
| 1 1/2-6 | 870 | 1180 | 650 | 881 | 1940 | 2630 | 1460 | 1980 | 2780 | 3769 | 2080 | 2820 | 3160 | 4298 | 2360 | 3200 |
| 1 1/2-12 | 980 | 1329 | 730 | 990 | 2200 | 2983 | 1640 | 2224 | 3100 | 4203 | 2340 | 3173 | 3560 | 4857 | 2660 | 3606 |

**NOTE:** * Use "oiled" values for bolts with liquid thread-locking compound applied at point of use.

**NOTE:** ** Use "dry" values for bolts with pre-applied, chemical thread-locking compound.

# Torque Chart - Standard (Brass)

*Table A-4. Torque Chart - Standard Brass*

| Brass MS63 Standard Bolts, Coarse Thread | | |
|---|---|---|
| **Size** | **Torque (with bolts oiled*)** | |
| | **Inch pounds** | **Newton meters** |
| 0-80 | 0.5 | 0.05 |
| 1-64 | 0.9 | 0.10 |
| 1-72 | 1.0 | 0.11 |
| 2-56 | 1.5 | 0.17 |
| 2-64 | 1.6 | 0.18 |
| 3-48 | 2.3 | 0.26 |
| 3-56 | 2.5 | 0.28 |
| 4-40 | 3.3 | 0.37 |
| 4-48 | 3.6 | 0.40 |
| 5-40 | 4.8 | 0.54 |
| 5-44 | 5.3 | 0.60 |
| 6-32 | 6.1 | 0.69 |
| 6-40 | 6.8 | 0.77 |
| 8-32 | 11 | 1.24 |
| 8-36 | 11 | 1.24 |
| 10-24 | 14 | 1.58 |
| 10-32 | 17 | 1.92 |
| 12-24 | 23 | 2.59 |
| 12-28 | 25 | 2.82 |
| 1/4-20 | 35 | 3.96 |
| 1/4-28 | 40 | 4.52 |
| 5/16-18 | 73 | 8.25 |
| 5/16-24 | 81 | 9.15 |
| 3/8-16 | 130 | 14.69 |
| 3/8-24 | 147 | 16.61 |
| **NOTE: * Use "oiled" values for bolts with thread-locking compound.** | | |

# Torque Chart - Metric (Ferrous)

*Table A-5. Torque Chart - Ferrous Metric*

| **Ferrous Metric Bolts, Coarse Thread (The grade rating is stamped on the head of the bolt.)** | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Size** | **Grade 4.8** | | | | **Grade 8.8** | | | | **Grade 10.9** | | | | **Grade 12.9** | | | |
| | **Dry**** | | **Oiled*** | | **Dry**** | | **Oiled*** | | **Dry**** | | **Oiled*** | | **Dry**** | | **Oiled*** | |
| | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** | **(in. lbs.)** | **(Nm)** |
| M1.6-0.35 | 0.8 | 0.09 | 0.6 | 0.07 | 1.6 | 0.18 | 1.2 | 0.14 | 2.2 | 0.25 | 1.7 | 0.19 | 2.6 | 0.30 | 2.0 | 0.22 |
| M2-0.40 | 1.7 | 0.19 | 1.3 | 0.14 | 3.3 | 0.37 | 2.5 | 0.28 | 4.6 | 0.52 | 3.4 | 0.39 | 5.3 | 0.60 | 4.0 | 0.45 |
| M2.5-0.45 | 3.5 | 0.40 | 2.6 | 0.30 | 6.7 | 0.76 | 5.1 | 0.57 | 9.3 | 1.1 | 7.0 | 0.79 | 11 | 1.2 | 8.2 | 0.93 |
| M3-0.5 | 6.2 | 0.70 | 4.7 | 0.53 | 12 | 1.4 | 9 | 1.0 | 17 | 1.9 | 12 | 1.4 | 19 | 2.2 | 15 | 1.7 |
| M3.5-0.6 | 9.8 | 1.1 | 7.3 | 0.83 | 19 | 2.1 | 14 | 1.6 | 26 | 3.0 | 20 | 2.2 | 31 | 3.5 | 23 | 2.6 |
| M4-0.7 | 14 | 1.6 | 11 | 1.5 | 28 | 3.2 | 21 | 2.4 | 39 | 4.4 | 29 | 3.3 | 45 | 5.1 | 34 | 3.8 |
| M5-0.8 | 29 | 3.3 | 22 | 2.5 | 57 | 6.4 | 42 | 4.8 | 78 | 8.9 | 59 | 6.6 | 91 | 10.4 | 69 | 7.8 |
| | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** | **(ft. lbs.)** | **(Nm)** |
| M6-1 | – | – | – | – | 8 | 10.9 | 6 | 8.2 | 11 | 15.1 | 8 | 11.3 | 13 | 17.6 | 10 | 13.2 |
| M8-1.25 | – | – | – | – | 19 | 26 | 15 | 20 | 27 | 37 | 20 | 27 | 31 | 43 | 24 | 32 |
| M8-1 | – | – | – | – | 21 | 29 | 16 | 22 | 29 | 39 | 22 | 30 | 34 | 6 | 25 | 34 |
| M10-1.5 | – | – | – | – | 38 | 52 | 29 | 39 | 53 | 72 | 40 | 54 | 62 | 85 | 47 | 63 |
| M10-1.25 | – | – | – | – | 41 | 56 | 30 | 41 | 56 | 76 | 42 | 57 | 66 | 90 | 49 | 67 |
| M12-1.75 | – | – | – | – | 65 | 88 | 50 | 68 | 90 | 122 | 70 | 95 | 110 | 150 | 80 | 109 |
| M12-1.25 | – | – | – | – | 75 | 102 | 55 | 75 | 100 | 136 | 75 | 102 | 120 | 163 | 90 | 122 |
| M14-2 | – | – | – | – | 105 | 143 | 80 | 109 | 150 | 204 | 110 | 150 | 175 | 238 | 130 | 177 |
| M14-1.5 | – | – | – | – | 115 | 156 | 85 | 116 | 160 | 218 | 120 | 163 | 190 | 258 | 140 | 190 |
| M16-2 | – | – | – | – | 165 | 224 | 125 | 170 | 230 | 313 | 170 | 231 | 270 | 367 | 200 | 272 |
| M16-1.5 | – | – | – | – | 175 | 238 | 130 | 177 | 245 | 333 | 185 | 252 | 280 | 381 | 215 | 292 |
| M20-2.5 | – | – | – | – | 325 | 442 | 240 | 326 | 450 | 612 | 340 | 462 | 520 | 707 | 400 | 544 |
| M20-1.5 | – | – | – | – | 360 | 490 | 270 | 367 | 500 | 680 | 375 | 510 | 580 | 789 | 440 | 598 |
| M24-3 | – | – | – | – | 560 | 762 | 420 | 571 | 780 | 1061 | 580 | 789 | 900 | 1224 | 680 | 925 |
| M24-2 | – | – | – | – | 610 | 830 | 460 | 626 | 850 | 1156 | 640 | 870 | 1000 | 1360 | 740 | 1006 |
| M30-3.5 | – | – | – | – | 1120 | 1523 | 840 | 1142 | 1550 | 2108 | 1160 | 1578 | 1800 | 2448 | 1350 | 1836 |
| M30-2 | – | – | – | – | 1240 | 1686 | 920 | 1251 | 1700 | 2312 | 1280 | 1741 | 2000 | 2720 | 1500 | 2040 |
| M36-4 | – | – | – | – | 1950 | 2652 | 1460 | 1986 | 2700 | 3671 | 2000 | 2720 | 3160 | 4298 | 2350 | 3196 |
| M36-2 | – | – | – | – | 2200 | 2992 | 1640 | 2230 | 3000 | 4080 | 2250 | 3060 | 3500 | 4760 | 2650 | 3604 |
| **NOTE: * Use "oiled" values for bolts with liquid thread-locking compound applied at point of use. NOTE: ** Use "dry" values for bolts with pre-applied, chemical thread-locking compound.** | | | | | | | | | | | | | | | | |

# Torque Chart - Metric (Brass)

*Table A-6. Torque Chart - Brass Metric*

| Brass MS63 Metric Bolts, Coarse Thread | | |
|---|---|---|
| Diameter (in millimeters) | Torque (with bolts oiled*) | |
| | Inch pounds | Newton meters |
| 2 | 1.2 | 0.14 |
| 2.5 | 2.5 | 0.29 |
| 3 | 4.4 | 0.5 |
| 3.5 | 7.0 | 0.79 |
| 4 | 10 | 1.2 |
| 5 | 19 | 2.2 |
| 6 | 34 | 3.9 |
| 8 | 79 | 9 |
| 10 | 150 | 17 |
| **NOTE:** * Use "oiled" values for bolts with thread-locking compound. | | |

# Torque Chart - Thread Forming Screws

*Table A-7. Torque Chart - Standard Thread-Forming Screws*

| Standard Thread-Forming Screws | | | |
|---|---|---|---|
| **Size** | **Nearest Plate Thickness (inch)** | **Torque** | |
| | | **Inch-pounds** | **Newton-meters** |
| 2-56 | 0.0469 | 4 | 0.5 |
| | 0.0625 | 4 | 0.5 |
| | 0.0938 | 5 | 0.6 |
| 3-48 | 0.0625 | 6 | 0.7 |
| | 0.0938 | 7 | 0.8 |
| | 0.1250 | 7 | 0.8 |
| 4-40 | 0.0312 | 6 | 0.7 |
| | 0.0625 | 9 | 1.0 |
| | 0.0938 | 11 | 1.2 |
| 5-40 | 0.0625 | 12 | 1.4 |
| | 0.0938 | 18 | 2.0 |
| | 0.1250 | 20 | 2.3 |
| 6-32 | 0.0625 | 14 | 1.6 |
| | 0.0938 | 20 | 2.3 |
| | 0.1250 | 22 | 2.5 |
| 8-32 | 0.0938 | 30 | 3.4 |
| | 0.1250 | 45 | 5.1 |
| | 0.1875 | 45 | 5.1 |
| 10-24 | 0.0938 | 35 | 4.0 |
| | 0.1250 | 45 | 5.1 |
| | 0.1875 | 55 | 6.2 |
| 10-32 | 0.0938 | 35 | 4.0 |
| | 0.1250 | 50 | 5.6 |
| | 0.1875 | 70 | 7.9 |
| 12-24 | 0.1250 | 65 | 7.3 |
| | 0.1875 | 75 | 8.5 |
| | 0.2500 | 85 | 9.6 |
| 1/4-20 | 0.1250 | 85 | 9.6 |
| | 0.1875 | 125 | 14.1 |
| | 0.2500 | 125 | 14.1 |
| 5/16-18 | 0.1875 | 160 | 18.1 |
| | 0.2500 | 225 | 25.4 |
| | 0.3125 | 250 | 28.2 |

Torque Chart - Thread Forming Screws

| Standard Thread-Forming Screws | | | |
|---|---|---|---|
| **Size** | **Nearest Plate Thickness (inch)** | **Torque** | |
| | | **Inch-pounds** | **Newton-meters** |
| 3/8-16 | 0.2500 | 350 | 39.5 |
| | 0.3125 | 400 | 45.2 |
| | 0.3750 | 450 | 50.8 |
| 7/16-14 | 0.3125 | 500 | 56.5 |
| | 0.3750 | 600 | 67.8 |
| | 0.5000 | 700 | 79.1 |
| 1/2-13 | 0.2500 | 500 | 56.5 |
| | 0.3750 | 850 | 96.0 |
| | 0.5000 | 1000 | 113.0 |

*Table A-8. Torque Chart - Metric Thread-Forming Screws*

| Metric Thread-Forming Screws | | | | |
|---|---|---|---|---|
| **Size** | **Nearest Plate Thickness (inch)** | **Nearest Plate Thickness (mm)** | **Torque** | |
| | | | **Inch-pounds** | **Newton-meters** |
| M3 x 0.5 | 0.039 | 1.0 | 9 | 1.0 |
| | 0.079 | 2.0 | 9 | 1.0 |
| | 0.118 | 3.0 | 14 | 1.6 |
| M4 x 0.7 | 0.079 | 2.0 | 16 | 1.8 |
| | 0.118 | 3.0 | 29 | 3.3 |
| | 0.157 | 4.0 | 38 | 4.3 |
| M5 x 0.8 | 0.098 | 2.5 | 25 | 2.8 |
| | 0.138 | 3.5 | 53 | 6.0 |
| | 0.197 | 5.0 | 62 | 7.0 |
| M6 x 1.0 | 0.118 | 3.0 | 44 | 5.0 |
| | 0.177 | 4.5 | 89 | 10.0 |
| | 0.236 | 6.0 | 89 | 10.0 |
| M8 x 1.25 | 0.157 | 4.0 | 177 | 20.0 |
| | 0.236 | 6.0 | 248 | 28.0 |
| | 0.315 | 8.0 | 266 | 30.0 |
| M10 x 1.5 | 0.197 | 5.0 | 266 | 30.0 |
| | 0.315 | 8.0 | 398 | 45.0 |
| | 0.394 | 10.0 | 487 | 55.0 |
| M12 x 1.75 | 0.236 | 6.0 | 531 | 60.0 |
| | 0.354 | 9.0 | 575 | 65.0 |
| | 0.472 | 12.0 | 885 | 100.0 |

# Torque Chart - Hydraulic Fittings

*Table A-9. Torque Chart - Hydraulic Fittings*

| SAE Dash Size | Thread Size | JIC (37° Flare Thread) | | SAE Straight Thread O-Ring Steel Plugs | | | |
|---|---|---|---|---|---|---|---|
| | | | | Hollow Hex Head Plug HP50N | | Hex Head Plug P50N | |
| | | ft. lbs. | Nm | ft. lbs. | Nm | ft. lbs. | Nm |
| -2 | 5/16-24 | 3 ±1 | 4 ±1 | 3 ±0.5 | 4 ±0.6 | 7.5 ±0.5 | 10 ±0.6 |
| -3 | 3/8-24 | 6 ±1 | 8 ±1 | 5 ±0.5 | 7 ±0.6 | 14 ±1 | 19 ±1 |
| -4 | 7/16-20 | 12 ±1 | 16 ±1 | 11 ±1 | 15 ±1 | 18 ±1 | 24 ±1 |
| -5 | 1/2-20 | 15 ±1 | 20 ±1 | 15 ±1 | 20 ±1 | 22 ±1 | 30 ±1 |
| -6 | 9/16-18 | 21 ±1 | 28 ±1 | 18 ±1 | 24 ±1 | 27 ±2 | 37 ±3 |
| -8 | 3/4-16 | 45 ±2 | 61 ±3 | 46 ±2 | 62 ±3 | 48 ±2 | 65 ±3 |
| -10 | 7/8-14 | 60 ±5 | 81 ±7 | 75 ±5 | 102 ±7 | 90 ±5 | 122 ±7 |
| -12 | 1 1/16-12 | 85 ±5 | 115 ±7 | 85 ±5 | 115 ±7 | 110 ±5 | 149 ±7 |
| -14 | 1 3/16-12 | 105 ±5 | 142 ±7 | 130 ±6 | 176 ±8 | 145 ±6 | 197 ±8 |
| -16 | 1 5/16-12 | 120 ±5 | 163 ±7 | 135 ±6 | 183 ±8 | 160 ±6 | 217 ±8 |
| -20 | 1 5/8-12 | 170 ±10 | 230 ±14 | 225 ±12 | 305 ±16 | 225 ±12 | 305 ±16 |
| -24 | 1 7/8-12 | 200 ±15 | 271 ±20 | 250 ±12 | 339 ±16 | 250 ±12 | 339 ±16 |
| -32 | 2 1/2-12 | 270 ±20 | 366 ±27 | 325 ±15 | 441 ±20 | 325 ±15 | 441 ±20 |

# Torque Chart - Straight Thread Face Seal O-Rings

*Table A-10. Torque Chart - Straight Thread Face Seal O-Rings*

| SAE Dash Size | Tube Side Thread Size | ft. lbs. | Nm |
|---|---|---|---|
| -4 | 9/16-18 | 18 ±1 | 25 |
| -6 | 11/16-18 | 27 ±2 | 40 |
| -8 | 13/16-16 | 40 ±2 | 55 |
| -10 | 1-14 | 63 ±3 | 80 |
| -12 | 1 3/16-12 | 90 ±4 | 115 |
| -14 | 1 5/16-32 | 95 | 130 |
| -16 | 1 7/16-12 | 120 ±8 | 150 |
| -20 | 1 11/16-12 | 140 ±8 | 190 |
| -24 | 2-12 | 165 ±8 | 245 |
| -32 | 2 1/2-12 | 360 | 490 |

# Decimal Equivalent Chart

*Table A-11. Decimal Equivalent Chart*

| 4ths | 8ths | 16ths | 32nds | 64ths | To 3 Places | To 2 Places | MM Equivalent |
|---|---|---|---|---|---|---|---|
| | | | | 1/64 | .016 | .02 | .397 |
| | | | 1/32 | | .031 | .03 | .794 |
| | | | | 3/64 | .047 | .05 | 1.191 |
| | | 1/16 | | | .062 | .06 | 1.587 |
| | | | | 5/64 | .078 | .08 | 1.984 |
| | | | 3/32 | | .094 | .09 | 2.381 |
| | | | | 7/64 | .109 | .11 | 2.778 |
| | 1/8 | | | | .125 | .12 | 3.175 |
| | | | | 9/64 | .141 | .14 | 3.572 |
| | | | 5/32 | | .156 | .16 | 3.969 |
| | | | | 11/64 | .172 | .17 | 4.366 |
| | | 3/16 | | | .188 | .19 | 4.762 |
| | | | | 13/64 | .203 | .20 | 5.159 |
| | | | 7/32 | | .219 | .22 | 5.556 |
| | | | | 15/64 | .234 | .23 | 5.593 |
| 1/4 | | | | | .250 | .25 | 6.350 |
| | | | | 17/64 | .266 | .27 | 6.747 |
| | | | 9/32 | | .281 | .28 | 7.144 |
| | | | | 19/64 | .297 | .30 | 7.540 |
| | | 5/16 | | | .312 | .31 | 7.937 |
| | | | | 21/64 | .328 | .33 | 8.334 |
| | | | 11/32 | | .344 | .34 | 8.731 |
| | | | | 23/64 | .359 | .36 | 9.128 |
| | 3/8 | | | | .375 | .38 | 9.525 |
| | | | | 25/64 | .391 | .39 | 9.922 |
| | | | 13/32 | | .406 | .41 | 10.319 |
| | | | | 27/64 | .422 | .42 | 10.716 |
| | | 7/16 | | | .438 | .44 | 11.112 |
| | | | | 29/64 | .453 | .45 | 11.509 |
| | | | 15/32 | | .469 | .47 | 11.906 |
| | | | | 31/64 | .484 | .48 | 12.303 |
| 1/2 | | | | | .500 | .50 | 12.700 |

Decimal Equivalent Chart

| 4ths | 8ths | 16ths | 32nds | 64ths | To 3 Places | To 2 Places | MM Equivalent |
|---|---|---|---|---|---|---|---|
| | | | | 33/64 | .516 | .52 | 13.097 |
| | | | 17/32 | | .531 | .53 | 13.494 |
| | | | | 35/64 | .547 | .55 | 13.891 |
| | | 9/16 | | | .562 | .56 | 14.288 |
| | | | | 37/64 | .578 | .58 | 14.684 |
| | | | 19/32 | | .594 | .59 | 15.081 |
| | | | | 39/64 | .609 | .61 | 15.478 |
| | 5/8 | | | | .625 | .62 | 15.875 |
| | | | | 41/64 | .641 | .64 | 16.272 |
| | | | 21/32 | | .665 | .66 | 16.669 |
| | | | | 43/64 | .672 | .67 | 17.065 |
| | | 11/16 | | | .688 | .69 | 17.462 |
| | | | | 45/64 | .703 | .70 | 17.859 |
| | | | 23/32 | | .719 | .72 | 18.256 |
| | | | | 47/64 | .734 | .73 | 18.653 |
| 3/4 | | | | | .750 | .75 | 19.050 |
| | | | | 49/64 | .766 | .77 | 19.447 |
| | | | 25/32 | | .781 | .78 | 19.844 |
| | | | | 51/64 | .797 | .80 | 20.241 |
| | | 13/16 | | | .812 | .81 | 20.637 |
| | | | | 53/64 | .828 | .83 | 21.034 |
| | | | 27/32 | | .844 | .84 | 21.431 |
| | | | | 55/64 | .859 | .86 | 21.828 |
| | 7/8 | | | | .875 | .88 | 22.225 |
| | | | | 57/64 | .891 | .89 | 22.622 |
| | | | 29/32 | | .906 | .91 | 23.019 |
| | | | | 59/64 | .922 | .92 | 23.416 |
| | | 15/16 | | | .938 | .94 | 23.812 |
| | | | | 61/64 | .953 | .95 | 24.209 |
| | | | 31/32 | | .969 | .97 | 24.606 |
| | | | | 63/64 | .984 | .98 | 25.003 |
| | | | | | 1.000 | 1.00 | 25.400 |

# Standard/Metric Conversions

| To Convert... | Multiply, Add, or Subtract... |
|---|---|
| **Area** | |
| Square Inches to Square Centimeters | Square Inches x 6.452 |
| Square Centimeters to Square Inches | Square Centimeters x 0.155 |
| Square Feet to Square Meters | Square Feet x 0.093 |
| Square Meters to Square Feet | Square Meters x 10.753 |
| Square Yards to Square Meters | Square Yards x 0.836 |
| Square Meters to Square Yards | Square Meters x 1.196 |
| **Distance** | |
| Inches to Millimeters | Inches x 25.4 |
| Millimeters to Inches | Millimeters x 0.039 |
| Inches to Centimeters | Inches x 2.54 |
| Centimeters to Inches | Centimeters x 0.394 |
| Feet to Meters | Feet x 0.305 |
| Meters to Feet | Meters x 3.281 |
| Yards to Meters | Yards x 0.914 |
| Meters to Yards | Meters x 1.094 |
| Miles to Kilometers | Miles x 1.609 |
| Kilometers to Miles | Kilometers x 0.621 |
| **Mass** | |
| Ounces to Grams | Ounces x 28.35 |
| Grams to Ounces | Ounces x 0.035 |
| Ounces to Kilograms | Ounces x 0.028 |
| Kilograms to Ounces | Kilograms x 35.27 |
| Pounds to Kilograms | Pounds x 0.454 |
| Kilograms to Pounds | Kilograms x 2.2 |
| **Pressure** | |
| Pounds per Square Inch (PSI) to kiloPascals | PSI x 6.894 |
| kiloPascals to Pounds per Square Inch (PSI) | kiloPascals x 0.145 |
| Speed | |
| Miles per hour to Kilometers per hour | Miles per hour x 1.609 |
| Kilometers per hour to Miles per hour | Kilometers per hour x 0.6214 |
| **Temperature** | |
| Fahrenheit to Celsius | (°F minus 32) x 0.555 |
| Celsius to Fahrenheit | (°C x 1.8) plus 32 |
| **Torque** | |
| Inch Pounds (in. lb.) to Newton Meters (Nm) | Inch Pounds x 0.113 |
| Newton Meters (Nm) to Inch Pounds (in. lb.) | Newton Meters x 8.85 |
| Foot Pounds (ft. lb.) to Newton Meters (Nm) | Foot Pounds x 1.3568 |
| Newton Meters (Nm) to Foot Pounds (ft. lb.) | Newton Meters x 0.737 |
| **Volume** | |
| Pints to Liters | Pints x 0.473 |
| Liters to Pints | Liters x 2.113 |
| Quarts to Liters | Quarts x 0.946 |
| Liters to Quarts | Liters x 1.057 |
| Gallons to Liters | Gallons x 3.785 |
| Liters to Gallons | Liters x 0.26 |

# Section I. Index

## D



## E



## F



## G



## H

## P



## R



## S

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RAYMOND

# Schematics

## Models

8210
8250

## Serial Numbers

821-15-00001 and up
825-17-00001 and up

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*Figure 1. Model 8210 Walkie Pallet Trucks*

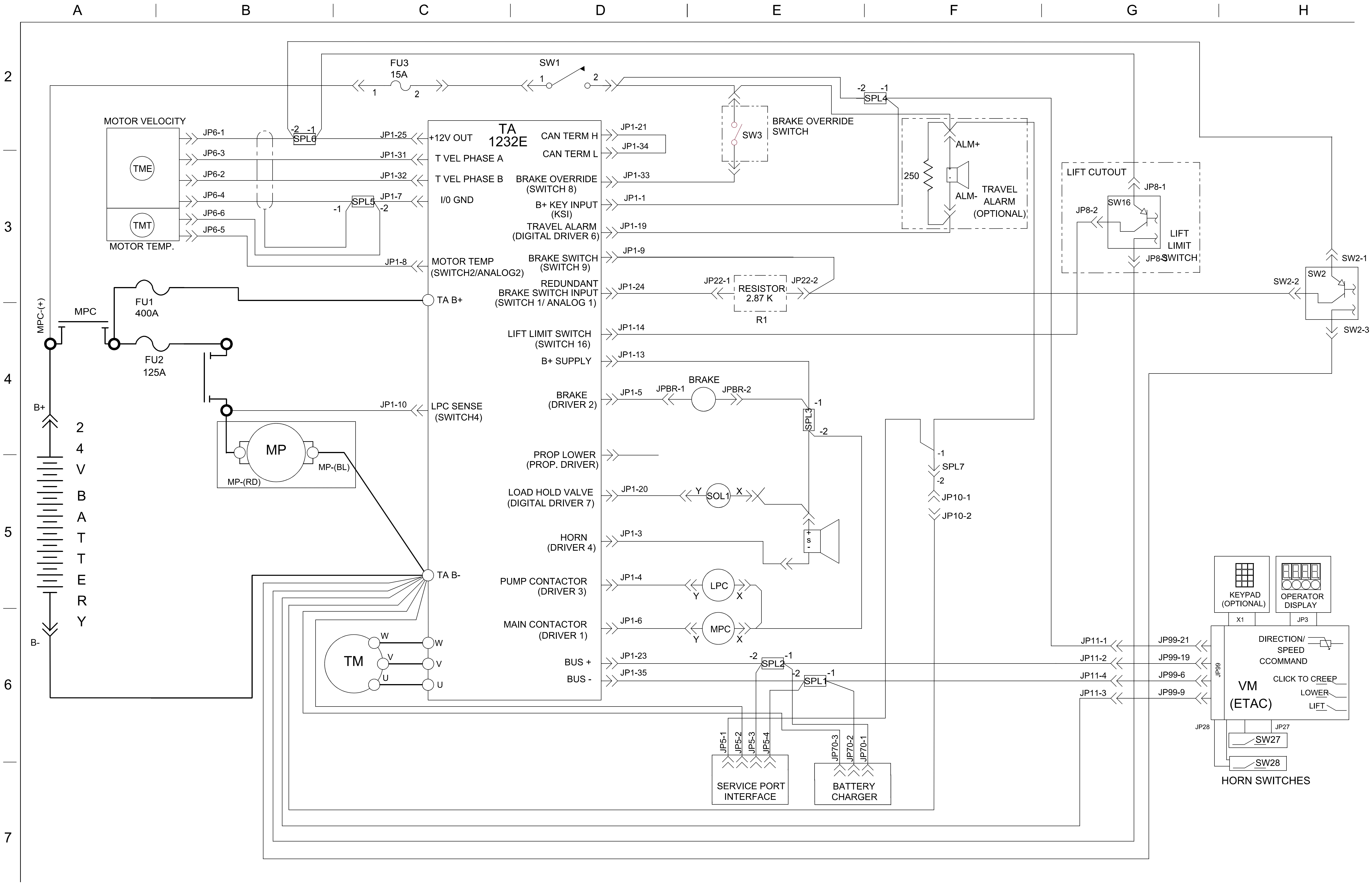

*Figure 2. Model 8210 Options*

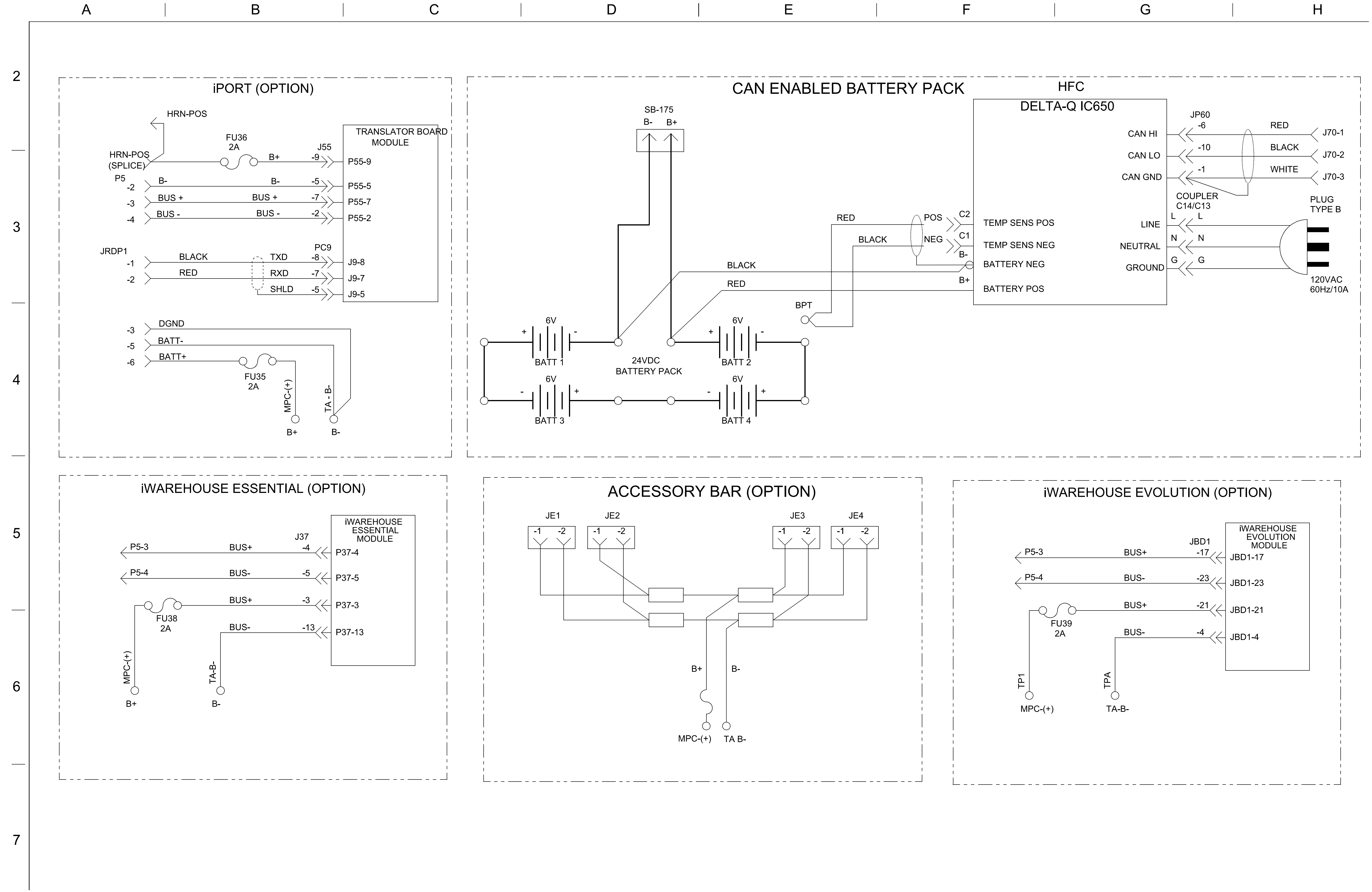

*Figure 3. Model 8210 Component Legend*

| LEGEND | | | |
|---|---|---|---|
| IDENT | SHEET | ZONE | DESCRIPTION |
| ALM | 1 | F3 | TRAVEL ALARM (OPTIONAL) |
| BATT (1&2) | 2 | D4 | 6 VOLT BATTERY CELLS |
| BATT (3&4) | 2 | D4 | 6 VOLT BATTERY CELLS |
| BPT | 2 | E4 | BATTERY PACK TEMP SENSOR |
| BRAKE | 1 | D4 | ELECTROMECHANICAL BRAKE |
| DISPLAY | 1 | H5 | OPERATOR DISPLAY |
| ETAC | 1 | G6 | ELECTRONIC TILLER ARM CARD (VEHICLE MANAGER) |
| | | | |
| FU1 | 1 | A3 | TRACTION SYSTEM FUSE (400A) |
| FU2 | 1 | B4 | 8210 LIFT PUMP MOTOR FUSE, 125A |
| FU3 | 1 | C2 | CONTROL CIRCUIT FUSE, 15A |
| FU4 | 2 | D6 | ACCESSORY BAR FUSE, 15A |
| FU35 | 2 | B5 | CONTROL FUSE FOR iPORT, 2A |
| FU36 | 2 | B4 | TRANSLATOR FUSE FOR iPORT, 2A |
| FU38 | 2 | A6 | CONTROL FUSE FOR ESSENTIAL, 2A |
| FU39 | 2 | G6 | CONTROL FUSE FOR EVOLUTION, 2A |
| | | | |
| HFC | 2 | F3 | HIGH FREQUENCY BATTERY CHARGER |
| HRN | 1 | E5 | HORN |
| JBD1 | 2 | G5 | iWAREHOUSE EVOLUTION CONNECTOR |
| JE(1&2) | 2 | D5 | ACCESSORY BAR POWER CONNECTORS |
| JE(3&4) | 2 | E5 | ACCESSORY BAR POWER CONNECTORS |
| JP1 | 1 | C3 | TA CONNECTOR |
| JP2 | 1 | G3 | BRAKE SWITCH CONNECTOR |
| JP3 | 1 | H6 | DISPLAY CONNECTOR VM |
| JP5 | 1 | E6 | SERVICE PORT |
| JP6 | 1 | B3 | TRACTION MOTOR ENCODER & TEMP SENSOR CONNECTOR |
| JP8 | 1 | G3 | LIFT CUTOUT CONNECTOR |
| JP11 | 1 | G6 | HANDLE COMMUNICATION CONNECTOR |
| JP22 | 1 | E3 | R1 CONNECTOR |
| JP27 | 1 | H6 | HORN SWITCH 27 CONNECTOR VM |
| JP28 | 1 | G6 | HORN SWITCH 28 CONNECTOR VM |
| JP37 | 2 | B6 | iWAREHOUSE ESSENTIAL CONNECTOR |
| JP55 | 2 | B3 | TRANSLATOR BD COMMUNICATION CONNECTOR |
| JP60 | 2 | G3 | BATTERY CHARGER COMMUNICATION CONNECTOR |
| JP70 | 1 | E6 | BATTERY CHARGER CONNECTOR |
| JP71 | 2 | B6 | BATTERY CHARGER CONNECTOR ON iWH ADAPTER (OPTIONAL) |
| JP99 | 1 | G6 | CAN COMMUNICATION CONNECTOR VM |

| LEGEND CONT'D | | | |
|---|---|---|---|
| IDENT | SHEET | ZONE | DESCRIPTION |
| JPBR | 1 | D4 | BRAKE CONNECTOR |
| JPC9 | 2 | B5 | TRANSLATOR BD SERIAL CONNECTOR |
| JRDP1 | 2 | A5 | CONNECTOR FOR 3RD PARTY SWM DEVICES |
| KEY PAD | 1 | G5 | KEY PAD (OPTIONAL) |
| LPC | 1 | B4 | LIFT POWER CONTACTOR |
| MP | 1 | B5 | LIFT MOTOR & PUMP POWERPACK |
| MPC | 1 | A3 | MAIN POWER CONTACTOR |
| R1 | 1 | E3 | 2.87K OHM, BRAKE SWITCH DETECTION RESISTOR |
| R2 | 1 | F3 | 250 OHM, TRAVEL ALARM LOAD RESISTOR (OPTIONAL) |
| SB-175 | 2 | D2 | BATTERY PACK POWER CONNECTOR |
| SOL1 | 1 | E5 | LOAD HOLD SOLENOID |
| | | | |
| SPL1 | 1 | E6 | CAN BUS - |
| SPL2 | 1 | E6 | CAN BUS + |
| SPL3 | 1 | E4 | B+ SUPPLY COIL RETURN |
| SPL4 | 1 | E2 | B+ KEY |
| SPL5 | 1 | C3 | I/O GND |
| SPL6 | 1 | B3 | +12V OUT |
| SPL7 | 1 | F4 | WARNING LIGHT HARNESS CONNECTOR |
| SPL8 | 2 | E6 | B+ BATTERY, ACCESSORY BAR HARNESS |
| SPL9 | 2 | E6 | B- BATTERY, ACCESSORY BAR HARNESS |
| SPL10 | 2 | D6 | B+ BATTERY, ACCESSORY BAR HARNESS |
| SPL11 | 2 | D6 | B- BATTERY, ACCESSORY BAR HARNESS |
| SW1 | 1 | D2 | MAIN ON/OFF SWITCH |
| SW2 | 1 | H3 | BRAKE SWITCH |
| SW3 | 1 | D2 | BRAKE OVERRIDE SWITCH |
| SW16 | 1 | G3 | LIFT-LIMIT SWITCH |
| SW27 | 1 | H6 | LEFT HORN SWITCH |
| SW28 | 1 | H6 | RIGHT HORN SWITCH |
| TA | 1 | D3-6 | TRACTION AMPLIFIER |
| TM | 1 | C5 | AC TRACTION MOTOR |
| TME | 1 | B3 | TRACTION MOTOR ENCODER |
| TMT | 1 | B3 | TRACTION MOTOR TEMP SENSOR |
| VM | 1 | H6 | VEHICLE MANAGER (ETAC) |
| WARNING LIGHT | 1 | F4 | WARNING LIGHT (OPTIONAL) |
| X1 | 1 | G6 | KEY PAD CONNECTOR VM |

*Figure 4. Model 8250 Walkie Pallet Trucks*

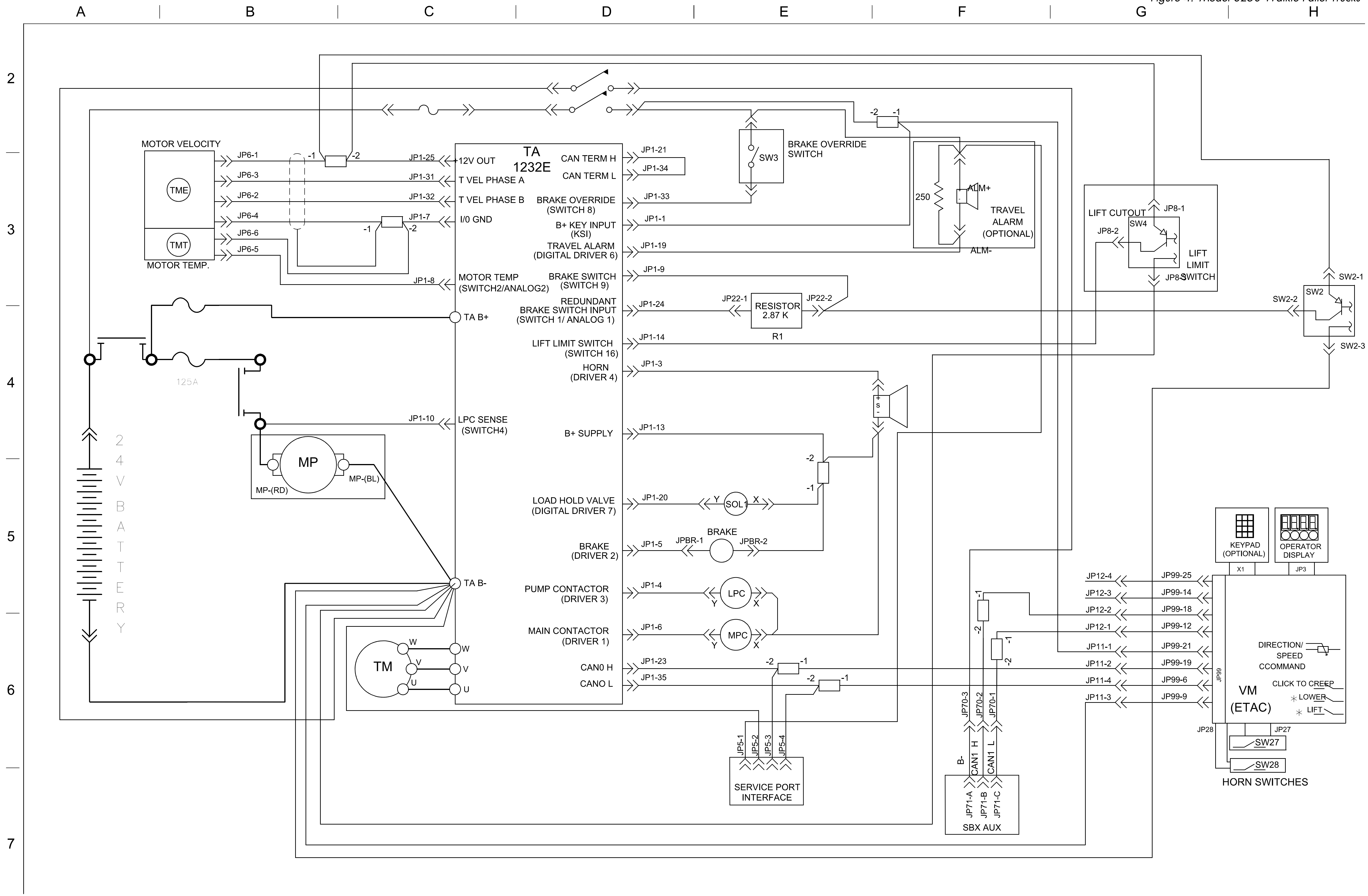

*Figure 5. Model 8250 Options*

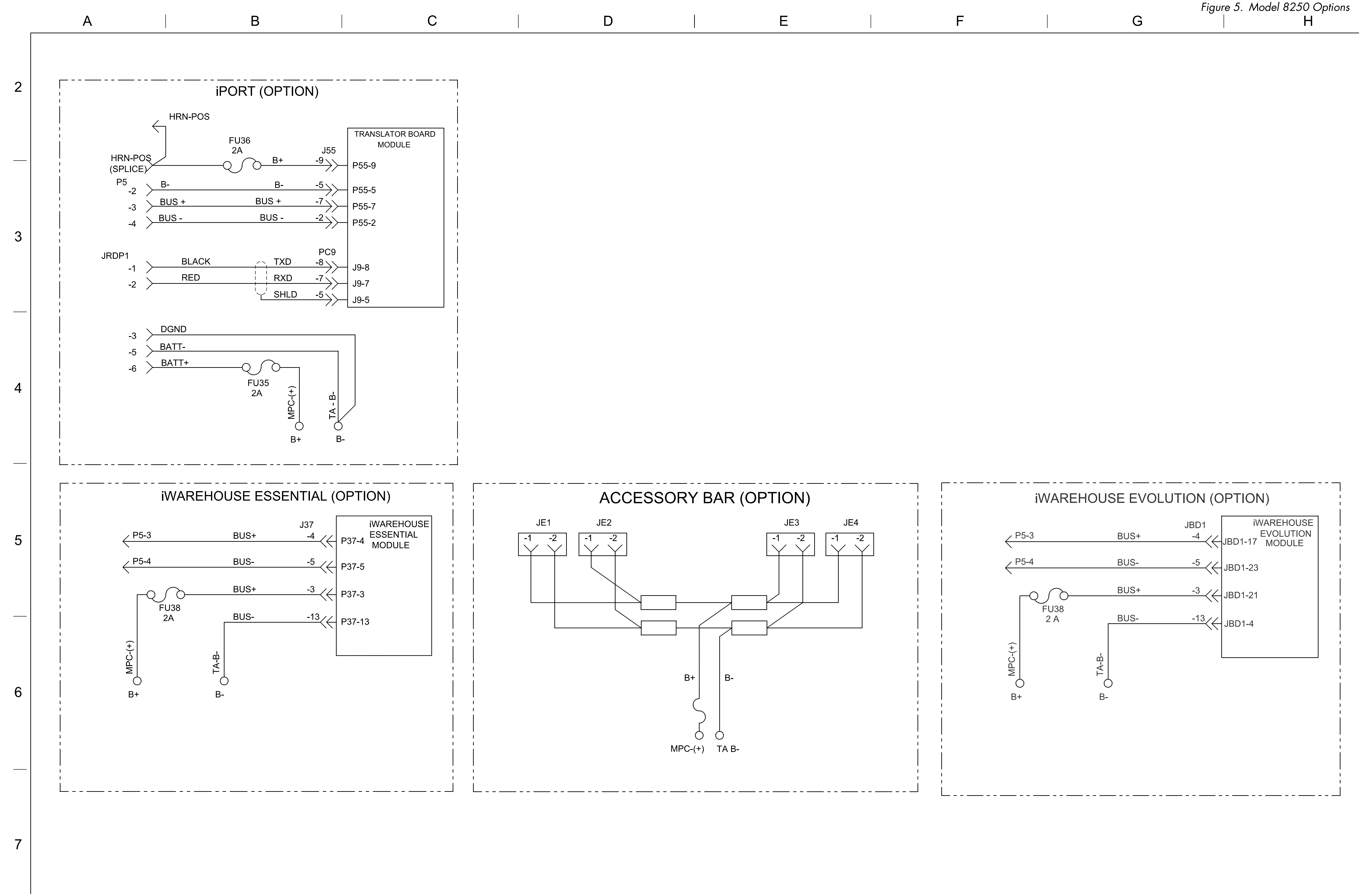

*Figure 6. Model 8250 Component Legend*

| LEGEND | | | |
|---|---|---|---|
| IDENT | SHEET | ZONE | DESCRIPTION |
| ALM | 1 | F3 | TRAVEL ALARM (OPTIONAL) |
| | | | |
| | | | |
| | | | |
| BRAKE | 1 | D4 | ELECTROMECHANICAL BRAKE |
| DISPLAY | 1 | H5 | OPERATOR DISPLAY |
| ETAC | 1 | G6 | ELECTRONIC TILLER ARM CARD (VEHICLE MANAGER) |
| | | | |
| FU1 | 1 | A3 | TRACTION SYSTEM FUSE (400A) |
| FU2 | 1 | B4 | 8210 LIFT PUMP MOTOR FUSE, 125A |
| | | | |
| FU3 | 1 | C2 | CONTROL CIRCUIT FUSE, 15A |
| FU4 | 2 | D6 | ACCESSORY BAR FUSE, 15A |
| FU35 | 2 | B5 | CONTROL FUSE FOR iPORT, 2A |
| FU36 | 2 | B4 | TRANSLATOR FUSE FOR iPORT, 2A |
| FU38 | 2 | A6 | CONTROL FUSE FOR iRED, 2A |
| | | | |
| | | | |
| HRN | 1 | E5 | HORN |
| | | | |
| JE(1&2) | 2 | D5 | ACCESSORY BAR POWER CONNECTORS |
| JE(3&4) | 2 | E5 | ACCESSORY BAR POWER CONNECTORS |
| JP1 | 1 | C3 | TA CONNECTOR |
| JP2 | 1 | G3 | BRAKE SWITCH CONNECTOR |
| JP3 | 1 | H6 | DISPLAY CONNECTOR VM |
| JP5 | 1 | E6 | SERVICE PORT |
| JP6 | 1 | B3 | TRACTION MOTOR ENCODER & TEMP SENSOR CONNECTOR |
| JP8 | 1 | G3 | LIFT CUTOUT CONNECTOR |
| | | | |
| JP11 | 1 | G6 | HANDLE COMMUNICATION CONNECTOR |
| JP22 | 1 | E3 | R1 CONNECTOR |
| JP27 | 1 | H6 | HORN SWITCH 27 CONNECTOR VM |
| JP28 | 1 | G6 | HORN SWITCH 28 CONNECTOR VM |
| JP37 | 1 | B6 | iRED CONNECTOR |
| JP55 | 2 | B4 | TRANSLATOR BD COMMUNICATION CONNECTOR |
| | | | |
| JP70 | 1 | E6 | BATTERY COMMUNICATION CONNECTOR |
| JP71 | 1 | E6 | AUXILARY BATTERY CONNECTOR FOR COMMUNICATION |
| JP99 | 1 | G6 | CAN COMMUNICATION CONNECTOR VM |

| LEGEND CONT'D | | | |
|---|---|---|---|
| IDENT | SHEET | ZONE | DESCRIPTION |
| JPBR | 1 | D4 | BRAKE CONNECTOR |
| JPC9 | 2 | B5 | TRANSLATOR BD SERIAL CONNECTOR |
| JRDP1 | 2 | A5 | CONNECTOR FOR 3RD PARTY SWM DEVICES |
| KEY PAD | 1 | G5 | KEY PAD (OPTIONAL) |
| LPC | 1 | B4 | LIFT POWER CONTACTOR |
| MP | 1 | B5 | LIFT MOTOR & PUMP POWERPACK |
| MPC | 1 | A3 | MAIN POWER CONTACTOR |
| R1 | 1 | E3 | 2.87K OHM, BRAKE SWITCH DETECTION RESISTOR |
| R2 | 1 | F3 | 250 OHM, TRAVEL ALARM LOAD RESISTOR (OPTIONAL) |
| | | | |
| SOL1 | 1 | E5 | LOAD HOLD SOLENOID |
| | | | |
| SPL1 | 1 | E6 | CANO H |
| SPL2 | 1 | E6 | CANO L |
| SPL3 | 1 | E4 | B+ SUPPLY COIL RETURN |
| SPL4 | 1 | E2 | B+ KEY |
| SPL5 | 1 | C3 | I/O GND |
| SPL6 | 1 | B3 | +12V OUT |
| SPL8 | 2 | E5 | B+ BATTERY, ACCESSORY BAR HARNESS |
| SPL9 | 2 | E5 | B- BATTERY, ACCESSORY BAR HARNESS |
| SPL10 | 2 | D6 | B+ BATTERY, ACCESSORY BAR HARNESS |
| SPL11 | 2 | D6 | B- BATTERY, ACCESSORY BAR HARNESS |
| SPL12 | 1 | F6 | CAN1 H |
| SPL13 | 1 | F6 | CAN1 L |
| SW1 | 1 | D2 | MAIN ON/OFF SWITCH |
| SW2 | 1 | G3 | BRAKE SWITCH |
| SW3 | 1 | D2 | BRAKE OVERRIDE SWITCH (8250 ONLY) |
| SW27 | 1 | H6 | LEFT HORN SWITCH |
| SW28 | 1 | H6 | RIGHT HORN SWITCH |
| TA | 1 | D3-6 | TRACTION AMPLIFIER |
| TM | 1 | C5 | AC TRACTION MOTOR |
| TME | 1 | B3 | TRACTION MOTOR ENCODER |
| TMT | 1 | B3 | TRACTION MOTOR TEMP SENSOR |
| VM | 1 | H6 | VEHICLE MANAGER (ETAC) |
| | | | |
| X1 | 1 | G6 | KEY PAD CONNECTOR VM |

*Figure 7. Hydraulic Schematic*